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ISTQB-CTFL

**ISTQB Certified Tester
Foundation Level
(CTFL v4.0)**

Version: 8.4

[Total Questions: 399]

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Question #:1

Which of the following statements about testware are correct?

- I When closing the test activities, all the testware resources can be uninstalled and released
 - II All the testware should be subject to Configuration Management
 - III. The testware. at the end of the project, should be transferred to the organization responsible for maintenance
 - IV The developers are responsible for the correct installation of the testware
- A. II, III
B. I, III
C. I, IV
D. II, IV

Answer: A

Explanation

Testware is a term that refers to all artifacts produced during the testing process, such as test plans, test cases, test scripts, test data, test results, defect reports, etc. The following statements about testware are correct:

- II) All the testware should be subject to Configuration Management. Configuration management is a process that establishes and maintains consistency among work products throughout their life cycle. Configuration management applies to all testware, as it helps ensure their quality and consistency, track their changes and defects, control their versions and access rights, and link them to other artifacts.
- III) The testware at the end of the project should be transferred to the organization responsible for maintenance. Maintenance testing is testing performed on a software product after delivery to correct defects or improve performance or other attributes. Maintenance testing requires testware from previous testing activities or phases, such as test cases, test data, test results, etc. Therefore, the testware at the end of the project should be transferred to the organization responsible for maintenance testing, such as support team or maintenance team. The following statements about testware are incorrect:
- I) When closing the test activities, all the testware resources can be uninstalled and released. This statement is incorrect, as some testware resources may still be needed for future testing activities or phases, such as maintenance testing or regression testing. Therefore, when closing the test activities, some testware resources should be archived and stored for future use, while others can be uninstalled and released.
- IV) The developers are responsible for the correct installation of the testware. This statement is incorrect, as the testers are responsible for the correct installation of the testware. The testers should

ensure that they have access to all necessary testware resources and that they are installed and configured properly before starting the test execution. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 6, page 58-61.

Question #:2

The following diagram displays the logical dependencies between requirements and the individual requirement priorities. For example, "R2->R3" means that R3 is dependent on R2. Priority is indicated by the number next to the letter "P" i.e. P1 has a higher priority than P2.

Which one of the following options best describes the test execution sequence using both requirement dependency and priority

- A. R2, R1, R3, R4, R5, R6, R7, R8.
- B. R1, R2, R3, R4, R5, R6, R7, R8.
- C. R2, R4, R8, R5, R1, R6, R3, R7.
- D. . R2, R1,R3,R7,R6,R5,R4,R8.

Answer: D

Explanation

The correct test execution sequence should consider both the dependencies between the requirements and their priorities. According to the diagram, the sequence begins with R2 (P1) as it is a prerequisite for R3 (P4). Then R1 (P3) can be tested. R3 follows as it depends on R2. Next, R7 (P4) should be tested before R6 (P3) and R5 (P2), as indicated by their dependencies. Finally, R4 (P1) and R8 (P1) can be tested. Therefore, the best sequence is R2, R1, R3, R7, R6, R5, R4, R8. Reference: ISTQB CTFL Syllabus V4.0, Section 5.1.5

Question #:3

Which of the following statements about Experience Based Techniques (EBT) is correct?

- A. EBT use tests derived from the test engineers' previous experience with similar technologies.
- B. EBT is based on the ability of the test engineer to implement various testing techniques.
- C. EBT is done as a second stage of testing, after non-experienced-based testing took place.
- D. EBT require broad and deep knowledge in testing but not necessarily in the application or technological domain.

Answer: A

Explanation

Experience based techniques (EBT) are techniques that use the knowledge, intuition and skills of the test engineers to design and execute tests. EBT use tests derived from the test engineers' previous experience with similar technologies, domains, applications or systems. EBT are not based on the ability of the test engineer to implement various testing techniques, but rather on their personal judgment and creativity. EBT are not done as a second stage of testing, after non-experience-based testing took place, but rather as a complementary or alternative approach to other techniques. EBT require broad and deep knowledge in both testing and the application or technological domain, as this can help the test engineer identify potential risks, scenarios or defects. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 5, page 48-49.

Question #:4

Consider the following testing levels:

- 1) Component Testing
- 2) Integration Testing
- 3) System Testing
- 4) Acceptance Testing

Which of the following statements is true?

A. Integration and system testing are applicable when V-model is used.

Component and acceptance testing are applicable when iterative development models are used.

B. All the testing levels are applicable to V-model for software development.

Only acceptance testing is applicable for iterative models.

C. Acceptance testing is applicable for all software development models.

Component and system testing are applicable only for the V-model.

D. All testing levels are applicable, independent of which software development life-cycle process (V-model, iterative, incremental) is used.

Answer: D

Explanation

All testing levels are applicable, independent of which software development life-cycle process (V-model, iterative, incremental) is used. Testing levels are defined based on the scope and objectives of testing, not on the software development model. Component testing, integration testing, system testing and acceptance testing are common testing levels that can be applied to any software development model, as long as they are planned and executed properly. The V-model is a software development model that emphasizes the relationship between each development phase and its corresponding testing phase. Iterative and incremental

models are software development models that divide the development process into smaller cycles or iterations, where each iteration produces a working version of the software that can be tested and evaluated. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 18.

Question #:5

Which of the following is a role that is usually responsible for documenting the findings (e.g., action items, decisions, recommendations) made by the review team as part of a typical formal review?

- A. Review leader
- B. Facilitator.
- C. Recorder.
- D. Moderator

Answer: C

Explanation

In a formal review process, the recorder's role is typically responsible for documenting the findings of the review team, including action items, decisions, and recommendations. This ensures that there is an accurate record of what was discussed and agreed upon, facilitating follow-up and continuous improvement. Therefore, statement C is correct as per the ISTQB CTFL syllabus.

Question #:6

You are testing an e-commerce system that sporadically fails to properly manage customers' shopping carts. You have stressed the urgency of this situation to the development manager and development team and they recognize the priority of resolving the underlying defect. The development team is waiting for more information, which you will include in your defect report. Given the following items of information they are included in a typical defect report:

1. The expected results
2. The actual results
3. The urgency and priority to fix this
4. The date and author of the defect report
5. A description of the defect in order to reproduce, including screenshots and database dumps

Which of these items will be MOST useful to the developers to help them identify and remove the defect causing this failure?

- A. 1, 2, 5
- B. 1, 2, 3, 4, 5

C. 1, 2, 4

D. 3, 4

Answer: A

Explanation

When developers are trying to identify and remove a defect, they need clear information on what went wrong and what was expected. The items that will be most useful to developers in this context are the expected results (item 1), the actual results (item 2), and a description of the defect including steps to reproduce, screenshots, and database dumps (item 5). This information helps developers understand the nature of the defect and provides the necessary details to reproduce and diagnose the issue effectively.

References:

ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 5.5.1.

Question #:7

A typical objective of testing is to

- A. Determine the most appropriate level of detail with which to design test cases.
- B. Verify the compliance of the test object with regulatory requirements.
- C. Plan test activities in accordance with the existing test policy and test strategy
- D. Verify the correct creation and configuration of the test environment

Answer: B

Explanation

One of the primary objectives of testing is to ensure that the software or system being tested meets all regulatory requirements. This is crucial in many industries where compliance with laws and standards is mandatory. According to the ISTQB CTFL Syllabus v4.0, testing aims to evaluate the quality of the software product and verify that it complies with specified requirements, including regulatory requirements.

Question #:8

Which ONE of the following options MOST ACCURATELY describes the activities of “testing” and “debugging”?

- A. Testing triggers a failure that is caused by a defect in the software, whereas debugging is concerned with finding the causes of this failure (defects), analyzing these causes, and eliminating them.
- B. Testing triggers a failure that is caused by a defect in the software, whereas debugging is concerned with finding the causes of this failure (defects), analyzing these causes, and reproducing them.

- C. Testing identifies a defect that is caused by an error in the software, whereas debugging is concerned with finding the causes of this defect, analyzing these causes, and eliminating them.
- D. Testing triggers a defect that is caused by an error in the software, whereas debugging is concerned with finding the causes of this defect, analyzing these causes, and eliminating them.

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

Testing and debugging are separate but related activities. Testing executes the software to identify failures that result from defects (A). Debugging is the developer's responsibility and involves finding the cause of a failure (defect), analyzing it, and fixing the defect. The ISTQB syllabus explicitly differentiates these activities. Testing does not modify the software, whereas debugging does.

Reference: ISTQB CTFL v4.0 Syllabus, Section 1.1.2 – Testing and Debugging

Question #:9

Which of the following options is correct with respect to the differentiation of black-box techniques, white-box techniques, and experience-based testing?

- A. Black-box test techniques are based on requirements, white-box tests on source code. Therefore, black-box test techniques are always performed at higher test levels (e.g. system and acceptance level) and white-box test techniques at lower test levels.
- B. Experience-based testing can complement white-box and black-box test techniques well or can be used if the basic principles for these test techniques are lacking.
- C. Experience-based testing is an extension of black-box test techniques, since experience-based test techniques also do not require knowledge about the source code.
- D. The coverage items can be specified in advance, e.g., coverage of statements, states or checklists, for all three of the categories mentioned above.

Answer: B

Question #:10

Which of the following BEST defines risk level?

- A. Risk level is determined by calculating the absolute value of the sum of all potential issues that may occur on the project
- B. Risk level is calculated by adding the probabilities of all planned risks to a project

- C. Risk level is calculated by dividing the sum of all known risks by the sum of all unknown risks
- D. Risk level is determined by the likelihood of an event happening and the impact or harm from that event

Answer: D

Explanation

Risk level is determined by the combination of two factors: the likelihood of an event occurring and the impact or harm that could result from that event. This approach allows risks to be prioritized based on their potential effect on the project or system. The likelihood represents the probability of the risk event occurring, while the impact represents the severity of the consequences if the event does happen. This concept is fundamental in risk-based testing and helps guide decision-making during the testing process.

References:

ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 1.4.1.

Question #:11

Which of the following statements about statement coverage is TRUE?

- A. Achieving 90% statement coverage ensures that 90% branch coverage is achieved.
- B. Achieving 100% statement coverage ensures that no variable within the code has been used without being initialised.
- C. Achieving 100% statement coverage ensures that 100% branch coverage is achieved
- D. Achieving 80% statement coverage ensures that 80% of all executable statements within the code have been exercised.

Answer: D

Explanation

Statement coverage measures the percentage of executable statements that have been exercised by a test suite. Achieving 80% statement coverage means that 80% of the executable code lines have been tested. This metric helps in understanding how much of the code has been covered during testing. However, it does not guarantee branch coverage, variable initialization, or detection of all possible defects. The ISTQB CTFL Syllabus v4.0 explains statement coverage as a measure of the extent to which the code has been tested, without implying other types of coverage or testing goals.

Question #:12

The following open incident report provided:

Date: 01.01.01

Description: When pressing the stop button the application status remain in "Attention" instead of "Ready".

Severity: High

Life Cycle: Integration

Which of the following details are missing in the giving incident report?

- I. Identification or configuration of the application
 - II. The name of the developer
 - III. Recommendation of the developer
 - IV The actions and/or conditions that came before the pressing of the button
- A. IV
 - B. I. IV
 - C. I. II
 - D. II, III

Answer: B

Explanation

In an incident report, essential details provide context and facilitate the investigation and resolution of the incident. The missing elements in the given incident report are:

I. Identification or configuration of the application: This detail is crucial as it specifies which version or configuration of the application is affected, helping in reproducing the issue. IV. The actions and/or conditions that came before pressing the button: Understanding the sequence of actions leading to the issue is vital for replicating and diagnosing the problem.

The name of the developer (II) and the recommendation of the developer (III) are not typically included in an incident report as they do not contribute to identifying or resolving the incident. The focus is on the incident's details, reproduction steps, and the system's state rather than on personnel or proposed solutions at this stage. Therefore, option B, which includes both I and IV, is the correct answer.

Question #:13

Out of the following, what is not needed to specify in defect report?

- A. Test environment details
- B. How to reproduce the defect
- C. How to fix the defect

D. Seventy and priority

Answer: C

Explanation

A defect report is a document that records the details of a defect found during testing. A defect report typically contains the following items:

- Identifier: A unique identifier for the defect report
- Summary: A concise summary of the defect
- Description: A detailed description of the defect, including the steps to reproduce it, the expected and actual results, and any relevant screenshots or logs
- Severity: The degree of impact that the defect has on the system
- Priority: The level of urgency for resolving the defect
- Status: The current state of the defect, such as new, open, resolved, closed, etc.
- Resolution: The action taken to resolve the defect, such as fix, workaround, reject, etc. Out of these items, the one that is not needed to specify in a defect report is how to fix the defect. How to fix the defect is a technical solution that is usually determined by the developer who is assigned to resolve the defect. How to fix the defect is not part of the defect report, but rather part of the code change or patch that is delivered to fix the defect. The other items are needed to specify in a defect report, as they provide essential information for identifying, tracking and resolving defects. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 3, page 32-33.

Question #:14

Which of the statements correctly describes when a whole team approach may NOT be suitable?

- A. When a high level of test independence may be required.
- B. When acceptance tests need to be created.
- C. When a test automation approach needs to be determined.
- D. When the team dynamics need to be improved.

Answer: A

Explanation

The whole team approach involves collaboration among all team members, including testers, developers, and business representatives, to achieve quality goals. However, this approach may not be suitable in situations where a high level of test independence is required. Test independence is essential in cases where unbiased

testing is critical, such as in regulated environments or where high-risk systems are involved. This is because team members might unintentionally influence each other's work, leading to potential bias in testing outcomes.

Question #:15

You are a test manager responsible for implementing risk-based testing in your organization which deals with software in the healthcare domain. You are writing a handbook of various product risk mitigation options. Which ONE of the following options correctly represent the correct mitigation options?

- A. Using a third party testing company to transfer the risk to that company
- B. Choosing a tool for automated unit testing to reduce the risks
- C. Increasing the number of testers to be able to take care of all the risks
- D. Selecting a tester with required knowledge related to compliance and standards

Answer: D

Question #:16

A tester created a decision table(DT) given below. While reviewing the DT some of the actions for the given conditions were found to be incorrect. The user is trying to view the profile, which of the given conditions has CORRECT action(s) listed?

Condition	C1	C2	C3	C4
Is the user logged in?	Yes	Yes	No	No
Is the user authorized to view the profile?	Yes	No	Yes	No
Action				
Display the user's profile	Yes	No	Yes	Yes
Display an error message	Yes	Yes	No	No

- A. C1
- B. C4
- C. C3
- D. C2

Answer: A

Question #:17

Which of the following characterizations applies to a test tool used for the analysis of a developer's code prior to its execution?

- A. Tool support for test design and implementation.
- B. Tool support for static testing.
- C. Tool support for test execution and logging.
- D. Tool support for performance measurement and dynamic analysis.

Answer: B

Explanation

A test tool used for the analysis of a developer's code prior to its execution falls under the category of static testing tools. Static testing involves examining the code and documentation without executing the code. These tools are used to perform static analysis, which helps in identifying potential defects and code quality issues early in the development process. The ISTQB CTFL syllabus specifies that static analysis tools are essential for finding defects that do not manifest themselves during the execution of the program.

References: ISTQB CTFL Syllabus, Section 3.1, "Static Testing."

Question #:18

Which of the following statements about traceability is FALSE?

- A. Traceability between test basis items and the test cases designed to cover them, makes it possible to determine which test basis items have been covered by the executed test cases.
- B. Traceability between test basis items and the test cases designed to cover them, enables experience-based test techniques to be applied
- C. Traceability between test basis items and the test cases designed to cover them, enables identification of which test cases will be affected by changes to the test basis items.
- D. Traceability can be established and maintained through all the test documentation for a given test level, such as from test conditions through test cases to test scripts.

Answer: B

Explanation

Traceability primarily refers to the ability to link test cases back to their sources in the test basis, such as requirements or design documents. This linkage allows for the determination of coverage, impact analysis, and maintaining consistency across test documentation. However, traceability does not directly enable the

application of experience-based test techniques, which are more about using the tester's intuition and experience. The ISTQB CTFL Syllabus v4.0 does not state that traceability enables experience-based techniques, making option B the false statement.

Question #:19

Which ONE of the following options is **NOT** a benefit of **test automation**?

- A. **Reduced test execution times**
- B. **More objective assessment**
- C. **Prevention of simple human errors**
- D. **Eliminating completely the need for manual testing**

Answer: D

Explanation

Comprehensive and Detailed In-Depth Explanation:

Test automation **improves efficiency and accuracy**, but **it does NOT eliminate manual testing (D)**.

- **(A) is correct** because automation **speeds up execution**.
- **(B) is correct** as automated tests produce **consistent and unbiased results**.
- **(C) is correct** because automation **reduces human errors in repetitive tests**.

However, **manual testing remains essential for exploratory testing, usability testing, and complex test scenarios**.

Reference: ISTQB CTFL v4.0 Syllabus, Section 6.2 – Test Automation Benefits and Risks

Question #:20

Which of the following statements best describe Behavior-Driven Development (BDD)?

- A. A collaborative approach that allows every stakeholder to contribute to how the software component must behave.
- B. Expresses the behavior of an application with test cases written in Given When Then format.
- C. Is used to develop code guided by automated test cases.
- D. A psychological technique in which the team's behavior in agile teams is evaluated.

Answer: A**Explanation**

Behavior-Driven Development (BDD) is a collaborative approach that enhances communication among project stakeholders, including developers, testers, and business analysts. It involves defining how software should behave through examples written in a common language understandable by all stakeholders, often using the Given-When-Then format.

Question #:21

Which of the following is NOT a typical debugging activity when a failure is triggered by dynamic testing?

- A. Fixing the cause
- B. Reproducing the failure
- C. Diagnosing the root cause
- D. Adding new test cases

Answer: D**Question #:22**

What role in a formal or technical review should mediate between different opinions to ensure an effective review?

- A. Scribe (or recorder)
- B. Manager
- C. Moderator (or facilitator)
- D. Reviewer

Answer: C**Question #:23**

The following decision table is used to assist a doctor in determining the drug therapy to prescribe for a patient (aged 6 to 65 years) diagnosed with acute sinusitis. The table consists of three Boolean conditions and six actions

Based only on the given information, which of the following statements is TRUE?

- A. Column 7 represents an impossible situation and thus can be deleted

- B. Columns 1 and 3 can be merged into a single column
- C. Columns 2, 4, 6 and 8 can be merged into a single column
- D. Columns 5 and 7 can be merged into a single column

Answer: B

Explanation

Decision tables are used to model complex decision logic by considering different combinations of conditions and actions. Based on the given decision table for prescribing drug therapy:

- Column 1 and Column 3 both result in the same actions (prescribing Amoxicillin).
- These columns can be merged because the actions taken do not depend on whether the patient is taking anticoagulant therapy (both are 'T' for this condition).

Thus, combining these columns simplifies the decision table without losing any information.

Reference: ISTQB CTFL Syllabus V4.0, Chapter 4.2.3, Decision Table Testing.

Question #:24

The following incident report that was generated during test of a web application.

What would you suggest as **the most important** report improvement?

Defect detected date: 15 8.2010

Defect detected by: Joe Smith

Test level System test

Test case: Area 5/TC 98

Build version: 2011-16.2

Defect description After having filled out all required fields in screen 1, t click ENTER to continue to screen 2
Nothing happens, no system response at all.

- A. Add information about which web browser was used
- B. Add information about which developer should fix the bug
- C. Add the time stamp when the incident happened
- D. Add an impact analysis

Answer: A**Explanation**

The most important report improvement for the given incident report would be to add information about which web browser was used when the defect was detected. This information is relevant for reproducing and debugging the defect, as different web browsers may have different behaviors or compatibility issues with the web application. The other options are less important or irrelevant for the incident report. The developer who should fix the bug can be assigned by the project manager or the defect tracking system, not by the tester who reports the defect. The time stamp when the incident happened is not very useful, as it does not indicate the cause or the frequency of the defect. The impact analysis is not part of the incident report, but rather of the risk assessment or prioritization process. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 140.

Question #:25

Given the following priorities and dependencies for these test cases:

Test Case	Priority	Technical dependency on:	Logical dependency on:
TC1	High	TC4	
TC2	Low		
TC3	High		TC4
TC4	Medium		
TC5	Low		TC2
TC6	Medium	TC5	

Which of the following test execution schedules BEST takes into account the priorities and technical and logical dependencies?

- A. TC1 - TC3 - TC2 - TC4 - TC6 - TC5
- B. TC3 - TC4 - TC2 - TC6 - TC1 - TC5
- C. TC1 - TC3 - TC2 - TC4 - TC5 - TC6
- D. TC2 - TC4 - TC1 - TC3 - TC5 - TC6

Answer: C**Explanation**

When scheduling test cases, priorities and dependencies must be considered. The best execution order will respect both the logical dependencies and the priorities assigned to each test case.

Given the options, the correct order considering the priorities and dependencies is:

- TC1 (Priority 1)
- **TC (ISTQB not-for-profit association)ity 2, dependent on TC1)
- TC2 (Priority 3, dependent on TC1)
- TC4 (Priority 4)
- TC5 (Priority 5)
- TC6 (Priority 6, dependent on TC4)

According to the ISTQB Certified Tester Foundation Level (CTFL) v4.0 syllabus, understanding dependencies and scheduling tests accordingly is crucial for effective test execution#6†source#.

References:

Certified Tester Foundation Level v4.0

ISTQB Foundation Level Syllabus 4.0 (2023)

Question #:26

A system computes prices for bus tickets. The price depends on

- the passenger type (baby, child, adult, senior citizen, student, military)
- the travelling type (as single or in a group)
- the distance (zone 1. 2. 3)
- the kind of transport (ordinary, express)

Which of the following test techniques is the most appropriate one for testing the price computation?

- A. Statement coverage
- B. State transition testing
- C. Equivalence partitioning
- D. Use case testing

Answer: C

Explanation

Equivalence partitioning is a technique that divides the input data and output results of a software component into partitions of equivalent data. Each partition should contain data that is treated in the same way by the component. Equivalence partitioning can be used to reduce the number of test cases by selecting one

representative value from each partition. Equivalence partitioning is suitable for testing the price computation, as it can identify different partitions based on the passenger type, the travelling type, the distance and the kind of transport. Equivalence partitioning is not statement coverage, which is a technique that measures how many executable statements in a source code are executed by a test suite. Statement coverage is not appropriate for testing the price computation, as it does not consider the input data or output results. Equivalence partitioning is not state transition testing, which is a technique that models how a system transitions from one state to another depending on events or conditions. State transition testing is not relevant for testing the price computation, as it does not involve any states or transitions. Equivalence partitioning is not use case testing, which is a technique that tests how users interact with a system to achieve a specific goal. Use case testing is not applicable for testing the price computation, as it does not focus on a single function or component. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 4, page 37-38.

Question #:27

Consider the following **user story** and its **acceptance criteria**:

User Story:

As a member of the "**Agnotoly**" association, I want to be able to **pay the association's dues through an automatic payment method** so that I can forget about payment deadlines.

Acceptance Criteria:

- The member can only pay with '**Agnesis Master**' credit card.
- The member must register their **cell phone number**.
- The member must register the **credit card** in the payment system.
- The member must select a **payment method (monthly or annual)**.
- The member must **accept the terms and conditions** for the selected method.
- If terms and conditions are accepted, the system sends an **OTP** (one-time password) to the member's phone for verification.
- The payment method is **registered only after OTP verification**.
- **Once the payment method is registered, the system will charge on the first business day of the following month.**

Test Cycle Results:

- Members successfully paid dues using the credit card.
- Payments were processed **monthly and for over a year**.

Which ONE of the following options identifies the acceptance criteria that has NOT been fully covered?

- A. Acceptance criteria 8 has not been fully covered during the above test cycle.
- B. Acceptance criteria 4 has not been fully covered during the above test cycle.
- C. All acceptance criteria have been fully covered during the above test cycle.
- D. Acceptance criteria 6 has not been fully covered during the above test cycle.

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

Acceptance criteria 8 states that **the system should charge the user on the first business day of the following month**. However, the **test cycle only verified monthly payments for over a year**, without confirming whether payments were processed **specifically on the first business day**.

- **(B) is incorrect** because the test validated **monthly and annual payment options**.
- **(C) is incorrect** because not all criteria were fully validated.
- **(D) is incorrect** because OTP verification (6) was tested as part of registration.

Ensuring **timely execution of payments** (Criteria 8) requires additional validation.

Reference: ISTQB CTFL v4.0 Syllabus, Section 5.1.5 – Test Prioritization and Acceptance Testing

Question #:28

Which of the following activities is NOT a part of the fundamental testing process?

- A. Archiving automation code
- B. Test status reporting
- C. Test process improvement
- D. Build release and maintenance

Answer: D

Explanation

The fundamental testing process includes activities that are directly related to the planning, preparation, execution, and evaluation of tests, as well as the closure activities of the testing phase. Option D, "Build release and maintenance," falls outside the scope of the fundamental testing process as it relates more to software development and operations rather than specific testing activities. Options A, "Archiving automation code," B, "Test status reporting," and C, "Test process improvement," are all activities that can be part of or

associated with the fundamental testing process. Archiving automation code is part of test closure, test status reporting is part of test monitoring and control, and test process improvement can be an outcome of test closure activities.

Question #:29

Which of the following work products cannot be examined by static analysis?

- A. Test plans
- B. Source code
- C. Compiled code
- D. Formal models

Answer: A

Explanation

Static analysis is the process of examining the work products of a software development or testing activity without executing them. Static analysis can be applied to various types of work products, such as requirements, design, code, test cases, etc. However, test plans are not suitable for static analysis, because they are high-level documents that describe the test objectives, scope, strategy, resources, schedule, and risks of a testing project. Test plans are not executable or formalized in a way that static analysis tools can analyze them. Therefore, option A is the correct answer.

: ISTQB® Certified Tester Foundation Level Syllabus v4.01, Section 2.2.1, page 20; ISTQB® Glossary v4.02, page 45.

Question #:30

Which one of the following statements relating to the benefits of static testing is NOT correct?

- A. Static testing enables early detection of defects before dynamic testing is performed.
- B. Static testing reduces testing costs and time.
- C. Static testing increases development costs and time.
- D. Static testing identifies defects which are not easily found by dynamic testing.

Answer: C

Explanation

The statement that "static testing increases development costs and time" is NOT correct. Static testing actually helps to reduce development costs and time by identifying defects early in the development process before dynamic testing is performed. Early detection of defects reduces the cost and effort required to fix them and prevents the propagation of defects to later stages, thus reducing overall testing and development costs.

References: ISTQB CTFL Syllabus, Section 3.1.2, "The Value of Static Testing."

Question #:31

In a museum there are staggered entrance fees:

- Free admission: children under 14 years
- Reduced admission: young people aged 14 and under 18, retiree, students and welfare recipients
- Regular price: all other persons

Which of the following four options describes the minimum set of test data that reaches 100% coverage of all valid equivalence classes for the admission fees?

- A. Child (12 years), student (23 years), retiree (68 years)
- B. Youth (16 years), adult (35 years), adult (60 years)
- C. Child (12 years), welfare recipient (45 years), retiree (68 years)
- D. Child (12 years), adult (35 years), retiree (68 years)

Answer: D

Question #:32

Which TWO of the following are benefits of continuous integration?

- I. Allows earlier detection and easier root cause analysis of integration problems and conflicting changes.
- II. Removes the need for manual test analysis, design and execution.
- Hi. Removes the dependency on automated regression packs when integrating larger systems, or components.
- iv. Gives the development team regular feedback on whether the code is working.

Select the correct answer:

- A. i and iv
- B. i and ii
- C. i and iii
- D. iii and iv

Answer: A

Explanation

The benefits of continuous integration include: i. Allows earlier detection and easier root cause analysis of integration problems and conflicting changes. iv. Gives the development team regular feedback on whether the code is working. These benefits help in maintaining the stability and quality of the codebase by integrating and testing changes frequently and providing quick feedback to developers.

Question #:33

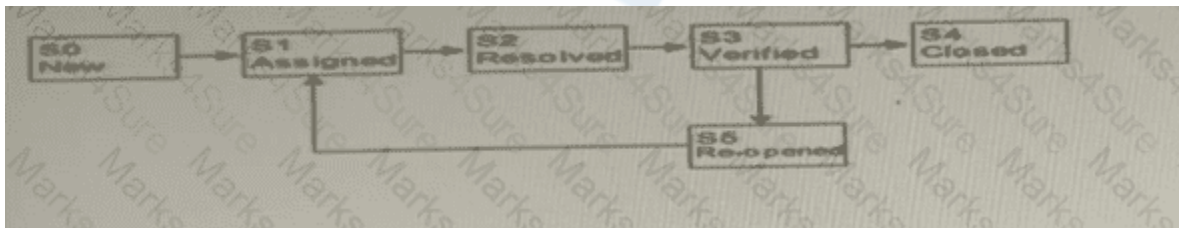
Which of the options below about statement testing is correct?

- A. 100% statement coverage ensures that there are no more defects in the test object.
- B. 100% statement coverage ensures that every transition between two statements has been executed at least once.
- C. 100% statement coverage ensures that there are no unreachable program parts (dead code).
- D. 100% statement coverage is a useful exit criterion for system testing.

Answer: C

Question #:34

Which sequence of state transition stated in the answer choices is correct in accordance with the following figure depicting the life-cycle of a defect?



- A. S0->S1->S2->S3->S4
- B. S0->S1->S2->S3->S5^>S1
- C. S0->S1->S2->S3->S5->S1->S2->S3
- D. S0->S1->S2->S3->S5->S3->S4

Answer: C

Explanation

The figure depicts the life-cycle of a defect using state transition testing. State transition testing is a technique that models how a system transitions from one state to another depending on events or conditions. The figure

shows six states (S0 to S5) and seven transitions (T0 to T6). The correct sequence of state transitions that follows the figure is S0->S1->S2->S3->S5->S1->S2->S3. This sequence represents the following scenario:

- S0: The defect is not yet detected (initial state).
- T0: The defect is detected by testing (event).
- S1: The defect is reported and registered (state).
- T1: The defect is assigned to a developer for fixing (event).
- S2: The defect is being fixed by the developer (state).
- T2: The developer fixes the defect and delivers a new version (event).
- S3: The defect is verified by testing (state).
- T5: The testing fails to confirm that the defect is fixed (event).
- S5: The defect is rejected by testing (state).
- T6: The defect is reassigned to a developer for fixing (event).
- S1: The defect is reported and registered (state).
- T1: The defect is assigned to a developer for fixing (event).
- S2: The defect is being fixed by the developer (state).
- T2: The developer fixes the defect and delivers a new version (event).
- S3: The defect is verified by testing (state). The other sequences are incorrect, as they do not follow the transitions shown in the figure. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 4, page 40-41.

Question #:35

A test manager has started a cycle of testing for an e-commerce application. The reason for testing is the change in the protocol for connecting to the payment gateway because of new regulations. Which of the following correctly names this type of testing?

- A. Beta testing
- B. Maintenance testing
- C. System integration testing
- D. Retirement testing

Answer: B

Question #36

Which of the following tasks is MOST LIKELY to be performed by the tester?

- A. Develop a test strategy and test policy for the organization
- B. Promote and advocate the test team within the organization
- C. Create the detailed test execution schedule
- D. Introduce suitable metrics for measuring test progress

Answer: C

Explanation

Testers are typically involved in creating detailed test execution schedules, among other tasks such as designing tests, executing tests, and logging defects. Creating a test strategy and test policy, promoting and advocating the test team, and introducing metrics are typically responsibilities of test managers or senior roles.

In the ISTQB Certified Tester Foundation Level (CTFL) v4.0 syllabus, the responsibilities of testers include creating test cases, setting up test (ISTQB not-for-profit association)nts, executing tests, and reporting defects, which align with creating detailed test execution schedules#6†source#.

References:

Certified Tester Foundation Level v4.0

ISTQB Foundation Level Syllabus 4.0 (2023)

Question #37

A software application incorrectly provided customers discounts of 50% off their total purchases if the purchases exceeded S100. It was discovered through an audit that the discount should have been only 5% off these purchases. A root cause analysis uncovered that the requirements Incorrectly stated 50% instead of 5% in this scenario.

Which of the following MOST accurately reflects this scenario?

- A. The audit finding Is the root cause, the incorrect calculation of 50% Is the defect, and the Incorrect requirement Is the effect
- B. The incorrect customer discount is the effect and the reason for the requirement error is the root cause
- C. The incorrect discount is the root cause, requiring a root cause analysis which led to investigating the software code, design, and requirements
- D. A defect in the code is determined to be the root cause of the incorrect calculation

Answer: B**Explanation**

According to the ISTQB CTFL syllabus, a defect is a deviation from the expected result which in this scenario is the incorrect discount applied to the customers. The root cause, as per the ISTQB definition, is the originating cause of a defect, which in this case is the incorrect requirement stating 50% instead of 5%. Therefore, the incorrect requirement is the root cause and the customer receiving the wrong discount is the effect of this root cause.

References: ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 1.4.3 "Defects, Root Causes, and Effects".

Question #:38

Which of the following statements is assigned correctly as a project or product risk?

- A. The software does not perform its intended functions according to the specification: this is a project risk.
- B. The gradual expansion of the project scope result in significant rework: this is a product risk.
- C. The system architecture does not adequately support some of the non-functional requirements: this is a product risk.
- D. User experience feedback does not meet product expectations: this is a project risk.

Answer: C**Question #:39**

Which of the following statements about the shift-left approach is true?

- A. Shift-left in testing can be implemented only in Agile/DevOps frameworks, as it relies completely on automated testing activities performed within a continuous integration process
- B. Performance testing performed during component testing, is a form of shift-left in testing that avoids planning and executing costly end-to-end testing at the system test level in a production-like environment
- C. Shift-left in testing can be implemented in several ways to find functional defects early in the lifecycle, but it cannot be relied upon to find defects associated with non-functional characteristics
- D. Continuous integration supports shift-left in testing as it can reduce the time between the introduction of a defect and its detection, thereby reducing the cost to fix it

Answer: D**Explanation**

This answer is correct because shift-left in testing is an approach that aims to perform testing activities as early as possible in the software development lifecycle, in order to find and fix defects faster and cheaper, and to improve the quality of the software product. Continuous integration is a practice that supports shift-left in testing, as it involves integrating and testing the software components frequently, usually several times a day, using automated tools and processes. Continuous integration can reduce the time between the introduction of a defect and its detection, thereby reducing the cost to fix it and the risk of accumulating defects that could affect the functionality or performance of the software product. References: ISTQB Foundation Level Syllabus v4.0, Section 3.1.1.3, Section 3.2.1.3

Question #:40

Which ONE of the following statements is correct?

- A. Static analysis can be used against a work product containing unstructured artifacts
- B. Difficult to interpret work products are best tested using static analysis.
- C. There may be legal issues in performing static testing on a commercially available executable library.
- D. Test cases can be tested using static testing techniques but not the models

Answer: B

Question #:41

Which type of software development product can undergo static testing?

- A. Any software development product can undergo static testing, including requirements specifications, design specifications and code.
- B. Static tests should be performed on the installation and user guide documents as these documents are used by the end user.
- C. Static testing is done only on the code as part of the "code review" sessions Other documents are reviewed, but not by static testing.
- D. Static testing is done only on the requirements You need to execute the software in order to find defects in the code.

Answer: A

Explanation

Static testing is a form of testing that does not involve executing the software, but rather analyzing it for defects, errors, or violations of standards. Static testing can be applied to any software development product, including requirements specifications, design specifications, code, test cases, test plans, user manuals, etc. Static testing can be done by using various techniques such as reviews, inspections, walkthroughs, checklists, static analysis tools, etc. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 7.

Question #:42

In a two-hour uninterrupted test session, performed as part of an iteration on an Agile project, a heuristic checklist was used to help the tester focus on some specific usability issues of a web application.

The unscripted tests produced by the tester's experience during such session belong to which one of the following testing quadrants?

- A. Q1
- B. Q2
- C. Q3
- D. Q4

Answer: C

Explanation

The unscripted tests produced by the tester's experience during the two-hour test session belong to the testing quadrant Q3. The testing quadrants are a classification of testing types based on two dimensions: the test objectives (whether the testing is focused on supporting the team or critiquing the product) and the test basis (whether the testing is based on the technology or the business). The testing quadrants are labeled as Q1, Q2, Q3, and Q4, and each quadrant represents a different testing perspective, such as unit testing, acceptance testing, usability testing, or performance testing. The testing quadrant Q3 corresponds to the testing types that have the objective of critiquing the product from the business perspective, such as exploratory testing, usability testing, user acceptance testing, alpha testing, beta testing, etc. The unscripted tests performed by the tester in the given scenario are examples of exploratory testing and usability testing, as they are based on the tester's experience, intuition, and learning of the web application, and they focus on some specific usability issues, such as the user interface, the user satisfaction, the user feedback, etc. The other options are incorrect, because:

- The testing quadrant Q1 corresponds to the testing types that have the objective of supporting the team from the technology perspective, such as unit testing, component testing, integration testing, system testing, etc. These testing types are usually performed by developers or testers who have access to the source code, the design, the architecture, or the configuration of the software system, and they aim to verify the functionality, the quality, and the reliability of the software system at different levels of integration.
- The testing quadrant Q2 corresponds to the testing types that have the objective of supporting the team from the business perspective, such as functional testing, acceptance testing, story testing, scenario testing, etc. These testing types are usually performed by testers or customers who have access to the requirements, the specifications, the user stories, or the business processes of the software system, and they aim to validate that the software system meets the expectations and the needs of the users and the stakeholders.
- The testing quadrant Q4 corresponds to the testing types that have the objective of critiquing the product from the technology perspective, such as performance testing, security testing, reliability

testing, compatibility testing, etc. These testing types are usually performed by testers or specialists who have access to the tools, the metrics, the standards, or the benchmarks of the software system, and they aim to evaluate the non-functional aspects of the software system, such as the efficiency, the security, the reliability, or the compatibility of the software system under different conditions or environments. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:

- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.3.1, Testing in Software Development Lifecycles
- ISTQB® Glossary of Testing Terms v4.0, Testing Quadrant, Exploratory Testing, Usability Testing, Unit Testing, Component Testing, Integration Testing, System Testing, Functional Testing, Acceptance Testing, Story Testing, Scenario Testing, Performance Testing, Security Testing, Reliability Testing, Compatibility Testing

Question #:43

Given the following state model of sales order software:

- PLACED -> IN PRODUCTION
- IN PRODUCTION -> CANCELLED
- IN PRODUCTION -> SHIPPED
- SHIPPED -> INVOICED
- INVOICED -> CANCELLED
- CANCELLED -> PLACED

Which of the following sequences of transitions provides the highest level of transition coverage for the model (assuming you can start in any state)?

- A. IN PRODUCTION -> CANCELLED -> PLACED -> IN PRODUCTION -> CANCELLED -> PLACED
- B. IN PRODUCTION -> SHIPPED -> INVOICED -> CANCELLED -> PLACED -> IN PRODUCTION
- C. PLACED -> IN PRODUCTION -> SHIPPED -> CANCELLED -> PLACED
- D. PLACED -> CANCELLED -> PLACED -> CANCELLED -> PLACED -> IN PRODUCTION -> CANCELLED

Answer: B

Explanation

To achieve the highest level of transition coverage, one must consider all the possible transitions between the states in the given state model of the sales order software. The transitions in the sequence provided in Option

B - "IN PRODUCTION -> SHIPPED -> INVOICED -> CANCELLED -> PLACED -> IN PRODUCTION" cover all the states and transitions effectively. This covers the transitions from IN PRODUCTION to SHIPPED, SHIPPED to INVOICED, INVOICED to CANCELLED, CANCELLED to PLACED, and PLACED to IN PRODUCTION, thereby maximizing the transition coverage.

References:

ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 4.3.5.

Question #:44

Who of the following has the best knowledge to decide what tests in a test project should be automated?

- A. The developer
- B. The customer
- C. The development manager
- D. The test leader

Answer: D**Explanation**

The test leader is the person who is responsible for planning, monitoring, and controlling the test activities and resources in a test project. The test leader should have the best knowledge of the test objectives, scope, risks, resources, schedule, and quality criteria. The test leader should also be aware of the test automation criteria, such as the execution frequency, the test support, the team education, the roles and responsibilities, and the devs and testers collaboration¹. Based on these factors, the test leader can decide which tests are suitable for automation and which are not, and prioritize them accordingly. The test leader can also coordinate with the test automation engineers, the developers, and the stakeholders to ensure the alignment of the test automation strategy with the test project goals and expectations. References = ISTQB Certified Tester Foundation Level (CTFL) v4.0 Syllabus, Chapter 2, Section 2.3.1, Page 152; ISTQB Glossary of Testing Terms v4.0, Page 403; ISTQB Certified Tester Foundation Level (CTFL) v4.0 Syllabus, Chapter 6, Section 6.1.1, Page 514; Top 8 Test Automation Criteria You Need To Fulfill - QAMIND1

Question #:45

Which of the following statements about independent testing is WRONG?

- A. Independent testing is necessary because developers don't know any testing.
- B. Independent testing is best suited for the system test level.
- C. A certain degree of independence makes the tester more effective at finding defects.
- D. Independent test teams may find other types of defects than developers who are familiar with the system's structure.

Answer: A

Explanation

Independent testing is testing performed by a person or group that is independent of the development team. Independent testing can have various degrees of independence, ranging from testers who are part of the same organization as developers to testers who are external contractors or consultants. Independent testing can have various benefits, such as reducing bias, increasing objectivity, improving quality, or providing different perspectives. Independent testing is not necessary because developers don't know any testing, as this is a wrong and disrespectful statement. Developers can perform various types of testing, such as unit testing, component testing, or integration testing. However, independent testing can complement developer testing by providing additional levels of verification and validation, such as system testing, acceptance testing, or non-functional testing. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 2, page 16-17.

Question #:46

Metrics can be collected during and at the end of testing activities to assess which of the following?

- i. Progress against the planned schedule and budget.
- II. Current quality of the test object.
- Hi. Adequacy of the test approach.
- iv. Effectiveness of the test activities with respect to the objectives.
- v. All the above.

Select the correct answer:

- A. Only i and ii.
- B. Only i and iii.
- C. Only I, ii and iv.
- D. Only v.

Answer: D

Explanation

Metrics can be collected during and at the end of testing activities to assess various aspects including progress against the planned schedule and budget, the current quality of the test object, the adequacy of the test approach, and the effectiveness of the test activities with respect to the objectives. Collecting these metrics helps in understanding the overall performance and quality of the testing process.

Question #47

A software development company invests heavily in tools to support the entire software development lifecycle, including testing. They have a tool that allows automated creation and installation of builds and subsequent execution of various types of automated tests. Which ONE of the following is a CORRECT statement about this type of test tool?

- A. It is an example of collaboration tool
- B. It is an example of test implementation tool
- C. It is an example of DevOps related tools
- D. It is an example of automated unit testing tool

Answer: B

Question #48

Which of the following types of tools is BEST suited for determining source code compliance with the guidelines provided by a coding standard?

- A. Containerisation tool
- B. Fault seeding tool.
- C. Static analysis tool.
- D. Test data preparation tool

Answer: C

Explanation

A static analysis tool is best suited for determining source code compliance with coding standards. These tools analyze the code without executing it and can check for adherence to coding standards, syntax errors, and other static properties of the code. The ISTQB CTFL syllabus emphasizes the role of static analysis tools in verifying that code meets predefined standards and guidelines.

References: ISTQB CTFL Syllabus, Section on static testing and tools.

Question #49

Which description reflects aspects of checklist-based testing?

- A. The checklist contains test cases that need to be executed in a certain order (e.g. defect-detecting test cases from previous projects).
- B. The checklist includes failure descriptions that have been found in the past which should be avoided.

- C. The checklist contains experience-based test conditions, mostly in the form of questions and based on causes of previous failures, which are used when creating the test.
- D. The checklist contains common testing techniques (e.g. equivalence class partitioning) and their uses to simplify the selection of the "appropriate" test techniques.

Answer: C

Question #:50

When should component integration tests be carried out?

- A. Integration tests should always be done after system tests
- B. Integration tests should be done at the customer's site, after acceptance tests
- C. Integration tests can be done before or after system tests
- D. Integration tests should always be done before system tests

Answer: D

Explanation

Component integration tests are designed to verify the interactions and interfaces between integrated components. These tests should be carried out after component testing (where individual components are tested in isolation) but before system testing (where the entire system is tested as a whole). This ensures that any issues arising from the integration of components are identified and resolved early in the testing process, making option D the correct answer.

Question #:51

For a mandatory input field "ZIP code" the following rules are given:

- 1 - The valid ZIP code format is 5 numeric digits.
- 2 - The code has to exist in the post office's official ZIP code list

Using equivalence classes partitioning, how many test cases are required to test this field?

- A. 8
- B. 3
- C. 6
- D. 4

Answer: D

Explanation

Equivalence classes partitioning is a technique that divides the input data and output results of a software component into partitions of equivalent data. Each partition should contain data that is treated in the same way by the component. Equivalence classes partitioning can be used to reduce the number of test cases by selecting one representative value from each partition. For the ZIP code field, there are four equivalence classes based on the given rules:

- Valid ZIP code format and valid ZIP code value (e.g., 12345)
- Valid ZIP code format and invalid ZIP code value (e.g., 99999)
- Invalid ZIP code format and valid ZIP code value (e.g., 1234)
- Invalid ZIP code format and invalid ZIP code value (e.g., ABCDE) Therefore, four test cases are required to test this field, one for each equivalence class. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 4, page 37-38.

Question #:52

Consider the following user story about the authentication functionality of an e-commerce website:

"As a logged-in user, I want to change my current password with a new one, so that I can make my account safer".

The following are some of the acceptance criteria defined for the user story:

- [a] After the logged-in user has successfully changed his password, an email confirming the change must be sent to him
- [b] To successfully change the password, the logged-in user must enter the current password, enter a new valid password, and finally confirm by pressing the 'Change Password' button
- [c] To be valid, the new password entered by the logged-in user is not only required to meet the criteria related to the length and type of characters, but must also be different from the last 5 passwords of that user
- [d] A dedicated error message must be presented to the logged-in user when he enters a wrong current password
- [e] A dedicated error message must be presented to the logged-in user when he enters the correct current password, but enters an invalid password

Based only on the given information, which of the following ATDD tests is most likely to be written first?

- A. The logged-in user enters a wrong current password and views the dedicated error message
- B. The logged-in user enters the correct current password, enters a valid new password (different from the last 5 passwords), presses the 'Change Password' button, and finally receives the e-mail confirming that the password has been successfully changed

- C. The logged-in user enters the correct current password, enters an invalid password, and finally views the dedicated error
- D. The logged-in user submits a purchase order containing ten items, selects to pay with a Visa credit card, enters credit card information of a valid card, presses the 'Confirm' button, and finally views the dedicated message confirming that the purchase has been successful

Answer: B

Explanation

ATDD stands for Acceptance Test-Driven Development, which is a collaborative approach to software development and testing, in which the acceptance criteria of a user story are defined and automated as executable tests before the implementation of the software system. ATDD tests are usually written in a Given-When-Then format, which describes the preconditions, the actions, and the expected outcomes of a test scenario. ATDD tests are intended to verify that the software system meets the expectations and the needs of the users and the stakeholders, as well as to provide feedback and guidance for the developers and the testers. Based on the given information, the ATDD test that is most likely to be written first is the one that corresponds to option B, which is:

Given the logged-in user is on the Change Password page
When the user enters the correct current password, enters a valid new password (different from the last 5 passwords), and presses the Change Password button
Then the user receives an email confirming that the password has been successfully changed

This ATDD test is most likely to be written first, because it covers the main functionality and the happy path of the user story, as well as the most important acceptance criterion [a]. It also verifies that the user can change the password with a valid new password that meets the criteria related to the length, the type of characters, and the history of the passwords, as specified in the acceptance criterion [c]. The other options are not likely to be written first, because they either cover less critical or less frequent scenarios, such as entering a wrong current password [d] or an invalid new password [e], or they are not related to the user story or the acceptance criteria at all, such as submitting a purchase order [d]. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:

- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.3.1, Testing in Software Development Lifecycles¹
- ISTQB® Glossary of Testing Terms v4.0, Acceptance Test-Driven Development, User Story, Acceptance Criterion, Given-When-Then²

Question #:53

Which one of the following is a typical entry criteria for testing?

- A. Planned tests have been executed.
- B. Availability of testable requirements.
- C. The number of unresolved defects is within an agreed limit.
- D. The number of estimated remaining defects is sufficiently low.

Answer: B**Explanation**

A typical entry criterion for testing is the availability of testable requirements. Testable requirements provide a basis for designing and executing test cases. Without clear and testable requirements, it is challenging to determine what needs to be tested and to create effective test cases. Entry criteria ensure that the necessary preconditions are met before testing begins, which helps in conducting efficient and effective testing.

References: ISTQB CTFL Syllabus, Section 5.1.3, "Entry and Exit Criteria."

Question #:54

Match the Work Product with the category it belongs to:

Work Product: 1. Risk register 2 Risk information

3. Test cases

4. Test conditions

Category of work products:

A. Test planning work products

B. Test analysis work products.

C Test design work products

C. Test monitoring and control work products

D. 1-A.2-D.3-C.4-e

E. 1-A.2-C.3-B.4-D

F. 1-C, 2-A. 3-D, 4-B

G. 1-B. 2-D, 3-A.4-C

Answer: A**Question #:55**

A new web app aims at offering a rich user experience. As a functional tester, you have run some functional tests to verify that, before releasing the app, such app works correctly on several mobile devices, all of which are listed as supported devices within the requirements specification. These tests were performed on stable

and isolated test environments where you were the only user interacting with the application. All tests passed, but in some of those tests you observed the following issue: on some mobile devices only, the response time for two web pages containing images was extremely slow.

Based only on the given information, which of the following recommendation would you follow?

- A. You should open a defect report providing detailed information on which devices and by running which tests you observed the issue
- B. The issue is related to performance efficiency, not functionality. Thus, as a functional tester, you should not open any defect report as all the functional tests passed
- C. You should not open any defect report as the problem is most likely due to poor hardware equipment on the devices where you observed the issue
- D. You should not open any defect report and inform the test manager that the devices on which you observed the issue should no longer be supported so that they will be removed from the requirements specification

Answer: A

Explanation

As a functional tester, you should open a defect report providing detailed information on which devices and by running which tests you observed the issue. A defect report is a document that records the occurrence, nature, and status of a defect detected during testing, and provides information for further investigation and resolution. A defect report should include relevant information such as the defect summary, the defect description, the defect severity, the defect priority, the defect status, the defect origin, the defect category, the defect reproduction steps, the defect screenshots, the defect attachments, etc. Opening a defect report is a good practice for any tester who finds a defect in the software system, regardless of the type or level of testing performed. The other options are not recommended, because:

- The issue is related to performance efficiency, not functionality, but that does not mean that as a functional tester, you should not open any defect report as all the functional tests passed. Performance efficiency is a quality characteristic that measures how well the software system performs its functions under stated conditions, such as the response time, the resource utilization, the throughput, etc. Performance efficiency is an important aspect of the user experience, especially for web applications that run on different devices and networks. Even if the functional tests passed, meaning that the software system met the functional requirements, the performance issue observed on some devices could still affect the user satisfaction, the usability, the reliability, and the security of the software system. Therefore, as a functional tester, you have the responsibility to report the performance issue as a defect, and provide as much information as possible to help the developers or the performance testers to investigate and resolve it.

Question #:56

Which of the following statements is CORRECT?

- A. Test cases are made up of input values, expected results and actual results developed to cover test objectives

- B. Test cases describe items or events to test that are derived from the test basis during the test analysis activity
- C. Test cases are sequences of actions for test execution specified during the test implementation activity
- D. Test cases are derived during the test design activity to cover test objectives or test conditions

Answer: C

Explanation

A test case is a set of input values, execution preconditions, expected results and execution postconditions, developed for a particular objective or test condition. A test case is a sequence of actions for test execution that can be followed by a tester or a test automation tool. A test case is specified during the test implementation activity, which is the activity that prepares the testware needed for test execution. A test case does not include actual results, as these are obtained during test execution and compared with the expected results. A test case does not describe items or events to test, as these are derived from the test basis during the test analysis activity. A test case is not derived during the test design activity, as this is the activity that specifies the test conditions or objectives that need to be tested. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 3, page 23-24; Chapter 4, page 34.

Question #:57

A bank software has a feature of locking down accounts that have an overdraft higher than \$10,000.

Due to a bug in the code, the system locks down accounts once they reached \$1000 overdraft.

Based on this context, which of the following is correct statement

- A. It is an example of a defect that if goes undetected, will cause harm to the end customer and to the company (the bank)
- B. It is an example of a defect, that if goes undetected, will cause harm to the end customer but not to the company (the bank)
- C. It is an example of a defect that if goes undetected, will not cause harm to the end customer, but will cause harm to the company (the bank)
- D. It is an example of a defect that if goes undetected, will not cause any real harm to either the end customer or to the company (the bank)

Answer: A

Explanation

A defect is a flaw in a component or system that can cause the component or system to fail to perform its required function. A defect can cause harm to the end customer, the company, or both, depending on the impact and severity of the failure. In this case, the defect in the code causes the system to lock down accounts that have an overdraft higher than \$1000, instead of \$10,000. This can cause harm to the end customer, as they may lose access to their funds and face inconvenience or financial difficulties. This can also cause harm

to the company, as they may lose customer trust and satisfaction, face legal issues or complaints, or incur additional costs for fixing the defect and restoring the accounts. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 1, page 3-4.

Question #:58

Which of the following is a possible reason for introducing a defect in software code?

- A. Rushing to meet a tight deadline to turn code over for testing
- B. Improper unit testing
- C. Improper system testing
- D. Focus on static testing over dynamic testing

Answer: A

Explanation

The ISTQB CTFL syllabus identifies several causes for defects in software. One prominent reason, as highlighted in the curriculum, is the pressure and rush to meet tight deadlines, which can lead to insufficiently reviewed or tested code being moved into further stages of testing or production. This scenario describes rushing to meet a deadline as a potential cause for defects because it may compromise the thoroughness of code development and testing.

References: ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 1.4.1 "Why is Testing Necessary?".

Question #:59

Manager responsibilities in formal review includes all except one of the following:

- A. Planning the review
- B. Determines if the review objectives have been met
- C. Decide on the execution of reviews
- D. Allocate time for review

Answer: B

Explanation

A formal review is a type of review that follows a defined process with formal entry and exit criteria and roles and responsibilities for participants. A formal review can have various roles involved, such as manager, moderator, author, reviewer and scribe. The manager responsibilities in formal review include all except one of the following:

- Planning the review (correct responsibility)
- Determines if the review objectives have been met (incorrect responsibility)
- Decide on the execution of reviews (correct responsibility)
- Allocate time for review (correct responsibility) The responsibility of determining if the review objectives have been met belongs to the moderator role, not to the manager role. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 3, page 28-29.

Question #:60

A program got 100% decision coverage in a test. Which of the following statements is then guaranteed to be true?

- A. Every executable statement Is covered.
- B. Every output equivalence class has been tested.
- C. Every input equivalence class has been tested.
- D. The "dead" code has not been covered.

Answer: A

Explanation

If a program got 100% decision coverage in a test, then it is guaranteed that every executable statement is covered. Decision coverage (also known as branch coverage) is a type of structural coverage (also known as white-box coverage) that measures how many decision outcomes have been exercised by a test suite. A decision outcome is a possible result of a decision point (such as an if-then-else statement) in a program's code. Decision coverage requires that each decision point has both true and false outcomes executed at least once by a test suite. Decision coverage implies statement coverage, which is another type of structural coverage that measures how many executable statements have been executed by a test suite. Statement coverage requires that each executable statement is executed at least once by a test suite. Therefore, if a program got 100% decision coverage in a test, then it also got 100% statement coverage in a test, which means that every executable statement is covered. The other options are not guaranteed to be true if a program got 100% decision coverage in a test. Every output equivalence class has been tested and every input equivalence class has been tested are not guaranteed to be true if a program got 100% decision coverage in a test, because equivalence classes are based on functional requirements or specifications, not on code structure or logic. Equivalence classes are used in specification-based testing (also known as black-box testing), which is a type of testing that does not consider the internal structure or implementation of the system under test. Decision coverage is used in structure-based testing (also known as white-box testing), which is a type of testing that considers the internal structure or implementation of the system under test. Therefore, achieving 100% decision coverage does not imply achieving 100% equivalence class coverage. The "dead" code has not been covered is not guaranteed to be true if a program got 100% decision coverage in a test, because dead code (also known as unreachable code) is code that can never be executed due to logical errors or design flaws. Dead code can reduce readability and maintainability of the code, as well as increase complexity and size. Decision coverage does not account for dead code, as it only considers the decision outcomes that are possible to execute. Therefore, achieving 100% decision coverage does not imply that the dead code has not

been covered. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 36.

Question #:61

During component testing of a program if 100% decision coverage is achieved, which of the following coverage criteria is also guaranteed to be 100%?

- A. 100% State transition coverage
- B. 100% Equivalence class coverage
- C. 100% Boundary value coverage
- D. 100% Statement coverage

Answer: D

Explanation

Statement coverage is a structural coverage metric that measures the percentage of executable statements in the source code that are executed by a test suite¹. Decision coverage is another structural coverage metric that measures the percentage of decision outcomes (such as branches or conditions) in the source code that are executed by a test suite¹. Decision coverage is a stronger metric than statement coverage, because it requires that every possible outcome of each decision is tested, while statement coverage only requires that every statement is executed at least once². Therefore, if a test suite achieves 100% decision coverage, it also implies that it achieves 100% statement coverage, because every statement in every branch or condition must have been executed. However, the converse is not true: 100% statement coverage does not guarantee 100% decision coverage, because some branches or conditions may have multiple outcomes that are not tested by the test suite². For example, consider the following pseudocode:

```
if (x > 0) then print("Positive") else print("Non-positive") end if
```

A test suite that executes this code with $x = 1$ and $x = -1$ will achieve 100% statement coverage, because both print statements are executed. However, it will not achieve 100% decision coverage, because the condition $x > 0$ has only been tested with two outcomes: true and false. The third possible outcome, $x = 0$, has not been tested by the test suite. Therefore, the test suite may miss a potential bug or error in the condition or the branch.

The other options, such as state transition coverage, equivalence class coverage, and boundary value coverage, are not guaranteed to be 100% by achieving 100% decision coverage. State transition coverage is a structural coverage metric that measures the percentage of transitions between states in a state machine that are executed by a test suite³. Equivalence class coverage is a functional coverage metric that measures the percentage of equivalence classes (or partitions) of input or output values that are tested by a test suite⁴. Boundary value coverage is another functional coverage metric that measures the percentage of boundary values (or extreme values) of input or output ranges that are tested by a test suite⁴. These metrics are independent of decision coverage, because they are based on different aspects of the system under test, such as its behavior, functionality, or specification. Therefore, achieving 100% decision coverage does not

imply achieving 100% of any of these metrics, and vice versa. References = ISTQB® Certified Tester Foundation Level Syllabus v4.0, Test Coverage in Software Testing - Guru99, Structural Coverage Metrics - MATLAB & Simulink - MathWorks India, Test Design Coverage in Software Testing - GeeksforGeeks.

Question #:62

Which of the following statements regarding the impact of the chosen software development lifecycle model on testing is true?

- A. The chosen software development lifecycle model has no impact on the number of regression test cases.
- B. The chosen software development lifecycle model influences when testing activities should start and end.
- C. The chosen software development lifecycle model determines the type of tests to be performed.
- D. The chosen software development lifecycle model does not affect the scope of the non-functional tests and the test tools to be used.

Answer: B

Question #:63

A company wants to reward each of its salespeople with an annual bonus that represents the sum of all the bonuses accumulated for every single sale made by that salesperson. The bonus for a single sale can take on the following four values: 3%, 5%, 7% and 10% (the percentage refers to the amount of the single sale). These values are determined on the basis of the type of customer (classified as "Basic" or "Premium") to which such sale was made, and on the amount of such sale classified into the following three groups G1, G2 and G3:

- [G1]: less than 300 euros
- [G2]: between 300 and 2000 euros
- [G3]: greater than 2000 euros

Which of the following is the minimum number of test cases needed to cover the full decision table associated with this scenario?

- A. 12
- B. 6
- C. 4
- D. 3

Answer: B

Explanation

The minimum number of test cases needed to cover the full decision table associated with this scenario is 6. This is because the decision table has 4 conditions (type of customer and amount of sale) and 4 actions (bonus percentage). The conditions have 2 possible values each (Basic or Premium, and G1, G2 or G3), so the total number of combinations is $2 \times 2 \times 2 \times 2 = 16$. However, not all combinations are valid, as some of them are contradictory or impossible. For example, a sale cannot be both less than 300 euros and greater than 2000 euros at the same time. Therefore, we need to eliminate the invalid combinations and keep only the valid ones. The valid combinations are:

Type of customer

Amount of sale

Bonus percentage

Basic

G1

3%

Basic

G2

5%

Basic

G3

7%

Premium

G1

5%

Premium

G2

7%

Premium

G3

10%

These 6 combinations cover all the possible values of the conditions and actions, and they are the minimum number of test cases needed to cover the full decision table.

: ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents,

Question #:64

Match each objective to the correct test level

Objective:

- A) Verifying whether the functional and non-functional behaviors of the system are as designed and specified.
- B) Verifying whether the functional and non-functional behaviors of the interfaces are as designed.
- C) Verifying whether the functional and non-functional behaviors of the components are as designed and specified.
- D) Establishing confidence in the quality of the system as a whole.

Test Level:

- 1. Component testing.
- 2. Integration testing.
- 3. System testing.
- 4. Acceptance testing.

- A. A3, B2, C4, D1
- B. A2, B3, C1, D4
- C. A3, B2, C1, D4

Answer: C

Explanation

The test levels and their objectives can be matched as follows:

- Verifying whether the functional and non-functional behaviors of the system are as designed and specified (A3: System testing).
- Verifying whether the functional and non-functional behaviors of the interfaces are as designed (B2: Integration testing).
- Verifying whether the functional and non-functional behaviors of the components are as designed and specified (C1: Component testing).

- Establishing confidence in the quality of the system as a whole (D4: Acceptance testing).

Question #:65

Which ONE of the following options CORRECTLY describes one of the seven principles of the testing process?

- A. The objective of testing is to implement exhaustive testing and execute as many test cases as possible.
- B. Exhaustive testing can only be carried out using behavior-based techniques.
- C. It is impossible to test all possible combinations of inputs and preconditions of a system.
- D. Automated testing enables exhaustive testing.

Answer: C

Explanation**Comprehensive and Detailed In-Depth Explanation:**

Exhaustive testing (testing all input combinations) is practically impossible except in trivial cases (C). Instead, testers focus on risk-based, prioritized, and efficient test techniques. The seven principles of testing in the ISTQB syllabus highlight that exhaustive testing is infeasible, and therefore, techniques such as equivalence partitioning, boundary value analysis, and risk-based testing are used to optimize test coverage.

Reference: ISTQB CTFL v4.0 Syllabus, Section 1.3 – Testing Principles

Question #:66

Which ONE of the following is a characteristic of exploratory testing?

- A. Effectiveness depends on the individual testers' skills
- B. Usually conducted when there is sufficient time for testing
- C. Testing without defined time-boxes
- D. Test cases are written once the specifications become available

Answer: A

Question #:67

Which of the following is not an example of a typical content of a test completion report for a test project?

- A. The additional effort spent on test execution compared to what was planned

- B. The unexpected test environment downtime that resulted in slower test execution
- C. The residual risk level if a risk-based test approach was adopted
- D. The test procedures of all test cases that have been executed

Answer: D

Explanation

This answer is correct because the test procedures of all test cases that have been executed are not a typical content of a test completion report for a test project. A test completion report is a document that summarizes the test activities and results at the end of a test project. It usually includes information such as the test objectives, scope, approach, resources, schedule, results, deviations, issues, risks, lessons learned, and recommendations for improvement. The test procedures of all test cases that have been executed are part of the test documentation, but they are not relevant for the test completion report, as they do not provide a high-level overview of the test project outcomes and performance. References: ISTQB Foundation Level Syllabus v4.0, Section 2.5.3.2

Question #:68

A team's test strategy was to invest equal effort in testing each of a system's modules. After running one test cycle, it turned out that most of the critical bugs were detected in one of the system's modules.

Which testing principal suggests a change to the current test strategy for the next test cycle?

- A. Pesticide Paradox
- B. Early testing
- C. Absence-of-errors fallacy
- D. Defect clustering

Answer: D

Explanation

Defect clustering is a testing principle that states that a small number of modules contain most of the defects detected during pre-release testing, or are responsible for most of the operational failures. Defect clustering can be explained by Pareto's principle (also known as the 80-20 rule), which states that approximately 80% of the problems are found in 20% of the modules. Defect clustering suggests a change to the current test strategy for the next test cycle, as it implies that more effort should be allocated to test the modules that have shown high defect density or criticality. Pesticide paradox is another testing principle that states that if the same tests are repeated over and over again, eventually they will no longer find any new defects. Pesticide paradox suggests a change to the current test strategy for the next test cycle, but not based on defect clustering, but rather on test diversity and coverage. Early testing is a testing principle that states that testing activities should start as early as possible in the software development life cycle and should be focused on defined objectives. Early testing does not suggest a change to the current test strategy for the next test cycle, but rather a

proactive approach to prevent defects from occurring or propagating. Absence-of-errors fallacy is a testing principle that states that finding and fixing defects does not help if the system built is unusable and does not fulfill the users' needs and expectations. Absence-of-errors fallacy does not suggest a change to the current test strategy for the next test cycle, but rather a focus on quality attributes and user requirements. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 1, page 9-10.

Question #:69

The whole-team approach:

- A. promotes the idea that all team members should have a thorough understanding of test techniques
- B. is a consensus-based approach that engages the whole team in estimating the user stories
- C. promotes the idea that all team members should be responsible for the quality of the product
- D. is mostly adopted in projects aimed at developing safety-critical systems, as it ensures the highest level of testing independence

Answer: C

Explanation

This answer is correct because the whole-team approach is a way of working in agile projects where all team members share the responsibility for the quality of the product, and collaborate on delivering value to the customer. The whole-team approach involves testers, developers, business analysts, product owners, and other stakeholders in planning, designing, developing, testing, and delivering the product. The whole-team approach fosters communication, feedback, learning, and continuous improvement within the team. References: ISTQB Glossary of Testing Terms v4.0, ISTQB Foundation Level Syllabus v4.0, Section 3.1.1.1

Question #:70

Which of the following is a test task that usually occurs during test implementation?

- A. Make sure the planned test environment is ready to be delivered
- B. Find, analyze, and remove the causes of the failures highlighted by the tests
- C. Archive the testware for use in future test projects
- D. Gather the metrics that are used to guide the test project

Answer: A

Explanation

A test task that usually occurs during test implementation is to make sure the planned test environment is ready to be delivered. The test environment is the hardware and software configuration on which the tests are

executed, and it should be as close as possible to the production environment where the software system will operate. The test environment should be planned, prepared, and verified before the test execution, to ensure that the test conditions, the test data, the test tools, and the test interfaces are available and functional. The other options are not test tasks that usually occur during test implementation, but rather test tasks that occur during other test activities, such as:

- Find, analyze, and remove the causes of the failures highlighted by the tests: This is a test task that usually occurs during test analysis and design, which is the activity of analyzing the test basis, designing the test cases, and identifying the test data. During this activity, the testers can use techniques such as root cause analysis, defect prevention, or defect analysis, to find, analyze, and remove the causes of the failures highlighted by the previous tests, and to prevent or reduce the occurrence of similar failures in the future tests.
- Archive the testware for use in future test projects: This is a test task that usually occurs during test closure, which is the activity of finalizing and reporting the test results, evaluating the test process, and identifying the test improvement actions. During this activity, the testers can archive the testware, which are the test artifacts produced during the testing process, such as the test plan, the test cases, the test data, the test results, the defect reports, etc., for use in future test projects, such as regression testing, maintenance testing, or reuse testing.
- Gather the metrics that are used to guide the test project: This is a test task that usually occurs during test monitoring and control, which is the activity of tracking and reviewing the test progress, status, and quality, and taking corrective actions when necessary. During this activity, the testers can gather the metrics, which are the measurements of the testing process, such as the test coverage, the defect density, the test effort, the test duration, etc., that are used to guide the test project, such as planning, estimating, scheduling, reporting, or improving the testing process. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:
 - ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.1, Test Planning1
 - ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.2, Test Monitoring and Control1
 - ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.3, Test Analysis and Design1
 - ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.4, Test Implementation1
 - ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.5, Test Execution1
 - ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.6, Test Closure1
 - ISTQB® Glossary of Testing Terms v4.0, Test Environment, Test Condition, Test Data, Test Tool, Test Interface, Failure, Root Cause Analysis, Defect Prevention, Defect Analysis, Testware, Regression Testing, Maintenance Testing, Reuse Testing, Test Coverage, Defect Density, Test Effort, Test Duration2

Question #:71

Which are the MAIN goals of risk management in a software project?

- A. To increase the success probability for the project regardless of costs.

- B. To increase focus on preventative processes and to increase satisfaction for the testers.
- C. To control contractual problems and minimize the impacts of company policies.
- D. To reduce the probability of undesired situations and to reduce the effect of potential impact.

Answer: D

Explanation

Risk management is a process that identifies, analyzes, evaluates and mitigates risks in a software project. Risks are factors that may negatively affect the quality, schedule, budget or scope of a software project. The main goals of risk management in a software project are to reduce the probability of undesired situations and to reduce the effect of potential impact. This can be achieved by applying various strategies, such as avoidance, transfer, reduction or acceptance. Risk management does not aim to increase the success probability for the project regardless of costs, as this may not be feasible or realistic. Risk management does not aim to increase focus on preventative processes and to increase satisfaction for the testers, as these are secondary or indirect outcomes. Risk management does not aim to control contractual problems and minimize the impacts of company policies, as these are specific types of risks that may not apply to all projects. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 2, page 14-15.

Question #:72

Atypical generic skill required for the role of tester is the ability to

- A. Take on the role of developer to meet challenging project deadlines
- B. Assume leadership aimed at imposing decisions on the rest of the team.
- C. Use tools to make the execution of repetitive testing tasks more efficient.
- D. Determine the corrective actions to get a test project on track in case of deviations from the test plan

Answer: C

Explanation

A key skill for testers is the ability to use various tools to automate repetitive tasks, enhancing the efficiency and effectiveness of testing processes. This includes tools for test execution, test management, and defect tracking. The ISTQB CTFL Syllabus v4.0 emphasizes the importance of using tools to improve productivity and reduce manual effort in repetitive testing tasks, making this a critical skill for testers.

Question #:73

A QA manager of a start-up company needs to implement within a week a low cost incident management tool. Which of the following is the best option?

- A. Document incidents on a large board in the lab

- B. Purchase and deploy an incident management tool
- C. Manage the incidents through E-mails and phone calls
- D. Manage the incidents in a spreadsheet posted on the intranet

Answer: D

Explanation

An incident is any event that occurs during testing that requires investigation. An incident management tool is a software tool that supports recording and tracking incidents throughout their life cycle. A QA manager of a start-up company needs to implement within a week a low cost incident management tool. The best option for this case is to manage the incidents in a spreadsheet posted on the intranet. This option has several advantages over other options:

- It is low cost, as it does not require purchasing any additional software or hardware.
- It is easy to implement within a week, as it does not require installing or configuring any complex software or hardware.
- It is accessible and transparent, as it can be viewed and updated by anyone who has access to the intranet.
- It is structured and organized, as it can store and display various information about incidents, such as identifier, summary, description, severity, priority, status, resolution, etc. The other options are not suitable for this case, as they have several disadvantages over the chosen option:
- Documenting incidents on a large board in the lab is not a good option, as it is not accessible or transparent to anyone who is not physically present in the lab. It is also not structured or organized, as it may not store or display all the necessary information about incidents.
- Purchasing and deploying an incident management tool is not a good option, as it is not low cost or easy to implement within a week. It may require spending a significant amount of money and time on acquiring, installing and configuring the software or hardware.
- Managing the incidents through emails and phone calls is not a good option, as it is not structured or organized. It may lead to confusion, inconsistency or loss of information about incidents. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 3, page 32-33.

Question #:74

You are a tester working on a critical project. Based on the risk analysis you execution? Priority 1 is the highest and Priority 3 is the lowest priority.

Test Case	Priority	Dependencies
Test A	3	-
Test B	1	Test D
Test C	2	Test A
Test D	3	-

- A. D-C- B-A
- B. D-B-A-C
- C. C-B-A-D
- D. D-A-B-C

Answer: B

Question #:75

Which of the following is not an example of a typical generic skill required for testing?

- A. Be able to apply test-driven development
- B. Be able to use test management tools and defect tracking tools
- C. Be able to communicate defects and failures to developers as objectively as possible
- D. Possess the necessary social skills that support effective teamwork

Answer: A

Explanation

Test-driven development is not an example of a typical generic skill required for testing, but rather an example of a specific technical skill or a development practice that may or may not be relevant for testing, depending on the context and the objectives of the testing activities. Test-driven development is an approach to software development and testing, in which the developers write automated unit tests before writing the source code, and then refactor the code until the tests pass. Test-driven development can help to improve the quality, the design, and the maintainability of the code, as well as to provide fast feedback and guidance for the developers. However, test-driven development is not a skill that is generally expected or needed for testers, especially for testers who are not involved in unit testing or who do not have access to the source code. The other options are examples of typical generic skills required for testing, which are skills that are applicable and beneficial for testing in any context or situation, regardless of the specific testing techniques, tools, or methods used. The typical generic skills required for testing include:

- Be able to use test management tools and defect tracking tools: These are tools that help testers to plan, organize, monitor, and control the testing activities and resources, as well as to record, track, analyze, and resolve the defects detected during testing. These tools can improve the efficiency, the effectiveness, and the communication of the testing process, as well as to provide traceability, metrics, and reports for the testing outcomes.
- Be able to communicate defects and failures to developers as objectively as possible: This is a skill that involves the ability to report and describe the defects and failures found during testing in a clear, concise, accurate, and unbiased manner, using relevant information, evidence, and terminology, without making assumptions, judgments, or accusations. This skill can facilitate the collaboration, the understanding, and the resolution of the defects and failures between the testers and the developers, as well as to prevent conflicts, misunderstandings, or blame games.
- Possess the necessary social skills that support effective teamwork: These are skills that involve the ability to interact, cooperate, and coordinate with other people involved in or affected by the testing activities, such as the test manager, the test team, the project manager, the developers, the customers, the users, etc. These skills can include communication, negotiation, leadership, motivation, feedback, conflict resolution, etc. These skills can enhance the quality, the productivity, and the satisfaction of the testing process, as well as to foster a positive and constructive testing culture. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.1.1, Testing and the Software Development Lifecycle
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.1.2, Testing and Quality
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.2.1, Testing Principles
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.2.2, Testing Policies, Strategies, and Test Approaches
- ISTQB® Glossary of Testing Terms v4.0, Test-driven Development, Test Management Tool, Defect Tracking Tool, Defect Report, Failure, Social Skill2

Question #:76

Which ONE of the following options identifies the perspectives through which a **collective authorship process** generates a shared vision of user stories?

- A. **Product owner, development, and testing**
- B. **Business, architecture, and testing**
- C. **Business, development, and testing**
- D. **Business, development, and acceptance testing**

Answer: C

Explanation

Comprehensive and Detailed In-Depth Explanation:

Business, development, and testing (C) are the three key perspectives in **agile user story refinement**:

- **Business (Product Owner, Stakeholders)** – Ensures the story aligns with user needs and business goals.
- **Development (Developers)** – Provides insights on feasibility and technical constraints.
- **Testing (Testers, QA)** – Ensures testability, acceptance criteria, and risk identification.
- **(A) is incorrect** because the product owner is part of business stakeholders.
- **(B) is incorrect** because architecture is part of development, but not the primary driver of user stories.
- **(D) is incorrect** because **acceptance testing is a process, not a stakeholder group**.

A **cross-functional team collaboration** ensures **well-defined, testable, and feasible** user stories.

Reference: ISTQB CTFL v4.0 Syllabus, Section 2.1.1 – Agile Team Collaboration in Testing

Question #:77

Which of the following sentences describe a product risk?

- A. The application might not be able to provide the expected responsiveness under a load of up-to 300 concurrent users
- B. Failure in acquiring an adequate and test automation tool
- C. A wrong configuration of the test environment that causes incidents related to the environment and not to the software under test
- D. The development team lacks knowledge of the technology on which the product is based

Answer: A

Explanation

This question relates to identifying product risks, which are potential problems associated with the product itself, such as software functionality, reliability, usability, and performance. Option A describes a scenario where the application might not meet performance requirements under specific conditions (up to 300 concurrent users), which directly impacts the product's ability to perform its intended function. This is a classic example of a product risk, as it concerns the product's quality and its ability to meet user needs. Options B, C, and D, on the other hand, relate to project risks, which are concerns related to the management and execution of the project, such as tool acquisition, environment configuration, and team expertise, rather than the quality of the product itself.

Question #:78

Which of the following is the most important task of a typical test leader?

- A. To automate tests.
- B. To prepare and acquire test data.
- C. To set up the test environment.
- D. To coordinate the test strategy with project managers.

Answer: D

Explanation

The most important task of a typical test leader is to coordinate the test strategy with project managers. The test strategy is a high-level document that defines the general approach and objectives of testing for a project or an organization. The test leader is responsible for defining, documenting, communicating, and implementing the test strategy in alignment with the project goals and constraints. The test leader also needs to coordinate with project managers and other stakeholders to ensure that the test strategy is feasible, effective, and efficient. The other options are not the most important tasks of a typical test leader. To automate tests is a task of a test automation engineer or a test automation specialist. To prepare and acquire test data is a task of a test analyst or a test engineer. To set up the test environment is a task of a test environment manager or a test environment specialist. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 13.

Question #:79

A state transition diagram describes a control system's behavior in different operational modes. The initial state is "**NORMAL MODE**".

Which ONE of the following test cases covers an **INVALID sequence**?

- A. NORMAL MODE # DIAGNOSTIC MODE # DEGRADED MODE # EMERGENCY MODE # DIAGNOSTIC MODE # NORMAL MODE # DIAGNOSTIC MODE
- B. NORMAL MODE # DEGRADED MODE # NORMAL MODE # DIAGNOSTIC MODE # DEGRADED MODE # EMERGENCY MODE # DIAGNOSTIC MODE
- C. NORMAL MODE # DIAGNOSTIC MODE # DEGRADED MODE # EMERGENCY MODE # DIAGNOSTIC MODE # NORMAL MODE # DEGRADED MODE
- D. NORMAL MODE # DIAGNOSTIC MODE # NORMAL MODE # DIAGNOSTIC MODE # EMERGENCY MODE # DIAGNOSTIC MODE # NORMAL MODE

Answer: D

Explanation

Comprehensive and Detailed In-Depth Explanation:

State transition testing validates **valid and invalid state transitions** based on defined rules.

- **D is invalid** because it transitions **from DIAGNOSTIC MODE directly to EMERGENCY MODE**, which is **not a valid state change** in most systems.
- Other options follow valid sequences according to **state transition rules**.

Reference: ISTQB CTFL v4.0 Syllabus, Section 4.2.4 – State Transition Testing

Question #:80

Which of the following statements is CORRECT about White-box testing?

- A. White-box testing helps find defects even when specifications are vague because it takes into account the code
- B. box testing helps find defects because it provides for requirements based coverage
- C. White-box testing helps find defects because it focuses on defects rather than failures
- D. White-box testing helps find defects because they can be used to measure statement coverage

Answer: D

Question #:81

Which one of the following statements correctly describes the term 'debugging'?

- A. There is no difference between debugging and testing.
- B. Debugging is a confirmation activity that checks whether fixes resolved defects.
- C. Debugging is the development activity that finds, analyses and fixes defects.
- D. Debugging is of no relevance in Agile development.

Answer: C

Explanation

Debugging is the development activity that finds, analyses and fixes defects. Unlike testing, which aims to identify defects in the software, debugging is the process that developers use to locate and correct the errors found during testing. This involves diagnosing the root causes of these defects and making necessary code changes to resolve them. Debugging is a critical part of the development cycle and ensures that the software functions correctly after defects are fixed.

Reference: ISTQB CTFL Syllabus V4.0, Section 1.1.2

Question #:82

ST is a Software Testing organization which utilizes a testing knowledge base. Access to ST knowledge base can be either full or limited. Access level is determined based on ST certification and testing experience as follows:

1. If ST certified, with less than 5 years testing experience - allow limited access
2. If ST certified, 5-10 years of testing experience - allow full access
3. If not ST certified with 5-10 years of testing experience - allow limited access.

What would be the results for:

- A - ST certified. 12 years of testing experience
- B - Not ST certified. 7 years of testing experience
- C - Not ST certified. 3 years of testing experience

- A. A - unknown
 - B - limited access
 - C - unknown
- B. A - full access
 - B - limited access
 - C - unknown
- C. A - full access
 - B - limited access
 - C - limited access
- D. A - unknown
 - B - full access
 - C - unknown

Answer: B

Explanation

The correct answer can be derived by applying the given rules to each case:

- A is ST certified and has 12 years of testing experience, which is more than 10 years. Therefore, A does not match any of the rules and the result is unknown.
- B is not ST certified and has 7 years of testing experience, which is between 5 and 10 years. Therefore, B matches rule 3 and the result is limited access.
- C is not ST certified and has 3 years of testing experience, which is less than 5 years. Therefore, C does not match any of the rules and the result is unknown. Verified References: This question does not require any external references, as it is based on logical reasoning.

Question #:83

Which of the following statements about re-testing and regression testing are TRUE?

- I Re-testing should be performed after a defect is fixed.
 - II Regression testing should always be performed after a defect is fixed.
 - III. Re-testing and regression testing may be performed at any test level.
 - IV Regression testing may include functional, non-functional and structural testing.
 - V. Re-testing should be included in the debugging activity.
- A. I, III, IV
B. II, v
C. I, III
D. II, IV, V

Answer: A

Explanation

The following statements about re-testing and regression testing are true:

- I) Re-testing should be performed after a defect is fixed. Re-testing is a type of testing that verifies that a defect has been successfully resolved by executing a test case that previously failed due to that defect. Re-testing should be performed after a defect is fixed and delivered to ensure that it does not cause any new failures or side effects.
- III) Re-testing and regression testing may be performed at any test level. Re-testing and regression testing are not limited to a specific test level, but can be applied at any level depending on the context and objectives. For example, re-testing and regression testing can be performed at unit level, integration level, system level or acceptance level.
- IV) Regression testing may include functional, non-functional and structural testing. Regression testing is a type of testing that verifies that previously tested software still performs correctly after changes.

Regression testing may include various types of testing depending on the scope and purpose of the changes. For example, regression testing may include functional testing to check if the software meets its requirements, non-functional testing to check if the software meets its quality attributes, or structural testing to check if the software meets its design or code standards. The following statement about re-testing and regression testing is false:

- II) Regression testing should always be performed after a defect is fixed. Regression testing is not always necessary after a defect is fixed, as some defects may have a low impact or low likelihood of affecting other parts of the software. Regression testing should be performed after a defect is fixed only if there is a risk of introducing new defects or causing existing defects due to the changes made to fix the defect. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 2, page 19; Chapter 4, page 45.

Question #:84

A Test Manager conducts risk assessment for a project. One of the identified risks is: "The sub-contractor may fail to meet his commitment". If this risk materializes, it will lead to delay in completion of testing required for the current cycle.

Which of the following sentences correctly describes the risk?

- A. It is a product risk since any risk associated with development timeline is a product risk.
- B. It is no longer a risk for the Test Manager since an independent party (the sub-contractor) is now managing it
- C. It is a object risk since successful completion of the object depends on successful and timely completion of the tests
- D. It is a product risk since default on part of the sub-contractor may lead to delay in release of the product

Answer: D

Explanation

- A product risk is a risk that affects the quality or timeliness of the software product being developed or tested¹. Product risks are related to the requirements, design, implementation, verification, and maintenance of the software product².
- The risk of the sub-contractor failing to meet his commitment is a product risk, as it could cause a delay in the completion of the testing required for the current cycle, which in turn could affect the release date of the product. The release date is an important aspect of the product quality, as it reflects the customer satisfaction and the market competitiveness of the product³.
- The other options are not correct because:
 - A. It is not true that any risk associated with development timeline is a product risk. Some risks could be project risks, which are risks that affect the management or control of the software project, such as budget, resources, schedule, or communication¹. For example, a risk of losing a key project stakeholder is a project risk, not a product risk.

- B. It is not true that the risk is no longer a risk for the Test Manager since an independent party is managing it. The Test Manager is still responsible for ensuring that the testing activities are completed according to the test plan and the quality objectives⁴. The Test Manager should monitor and control the sub-contractor's performance and communicate with him regularly to identify and mitigate any potential issues or deviations⁵.
- C. It is not clear what is meant by "object" in this option, but it could be interpreted as the software system under test or the test object⁶. In any case, the risk is not an object risk, as it does not affect the successful completion of the object, but rather the successful completion of the testing of the object. An object risk could be a risk that affects the functionality, reliability, usability, efficiency, maintainability, or portability of the software system under test². For example, a risk of the software system having a high complexity or a low testability is an object risk, not a product risk.

References =

- 1 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 97
- 2 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 98
- 3 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 99
- 4 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 100
- 5 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 101
- 6 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 102

Question #:85

The statement: "Test activities should start in the early stages of the lifecycle, adhering to the testing principle of early testing" is relevant to which of the recognized software development models?

- A. Sequential development model.
- B. Iterative development model.
- C. Incremental development model.
- D. All the above

Answer: D

Explanation

The principle of early testing is applicable to all recognized software development models, including sequential, iterative, and incremental models. Starting test activities early in the lifecycle helps in identifying

and addressing defects as soon as possible, which can save time and costs by preventing defects from propagating to later stages of development. This proactive approach enhances the overall quality and efficiency of the software development process. Reference: ISTQB CTFL Syllabus V4.0, Section 1.3

Question #:86

Which ONE of the following options is a **PRODUCT risk** and NOT a **PROJECT risk**?

- A. **Incorrect calculation logic in the software, leading to inaccurate results.**
- B. **Poor communication between team members, making project management more difficult.**
- C. **Staff shortages in the project leading to scheduling problems.**
- D. **Delays in the delivery of work products by the project team.**

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

Product risks are risks that **impact the quality or functionality of the software**, whereas **project risks** affect the **process, resources, or timeline**.

- **(A) is correct** because incorrect calculations impact software correctness.
- **(B), (C), and (D) are project risks** as they relate to **team coordination, staffing, and scheduling** rather than software defects.

Identifying **product risks** early **improves defect detection and ensures better coverage for high-risk areas**.

Reference: ISTQB CTFL v4.0 Syllabus, Section 5.2.2 – Product and Project Risks

Question #:87

Select which of the following statements describe the key principles of software testing?

- i. Testing shows the presence of defects, not their absence.
- ii. Testing everything Is possible.
- iii. Early testing Is more expensive and is a waste of time.
- iv. Defects cluster together.
- v. Testing is context dependent.

vi. Beware of the pesticide paradox.

vii. Absence of errors is a fallacy.

Select the correct answer:

- A. i, iv, v, vi and vii
- B. I, ii, v, vi and vii
- C. iii, iv, v, vi and vii
- D. ii, iii, iv, v and vi

Answer: A

Explanation

The key principles of software testing include: i. Testing shows the presence of defects, not their absence. iv. Defects cluster together. v. Testing is context dependent. vi. Beware of the pesticide paradox. vii. Absence of errors is a fallacy. These principles highlight the importance of recognizing the limitations and context of testing, as well as the potential for repeated tests to become less effective.

Question #:88

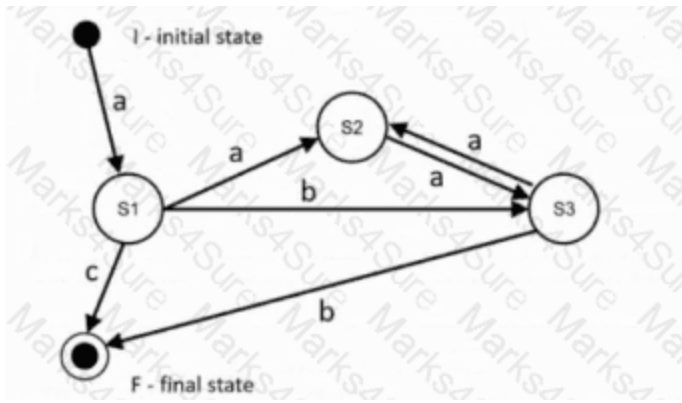
Which of the following metrics relates to the number of defects per unit size of a work product?

- A. Requirement coverage
- B. Average operating time until failure
- C. Fault density
- D. Number of executed test cases in relation to non-executed test cases

Answer: C

Question #:89

The following state diagram is given as basis for state transition testing and contains only valid transitions:



Explanation of the state diagram: States are depicted as nodes. The initial state is I, the final state is F. A state transition is depicted as a directed arrow with the initiating event as a label, e. g. from I to S1 with event a, and is also written as a triple (I,a,S1). Note: A test case is a sequence of events that initiates the corresponding sequence of state transitions.

The state diagram contains the following 7 state transitions:

(I, a, S1)

(S1, a, S2), (S1, b, S3), (S1, c, F)

(S2, a, S3)

(S3, a, S2), (S3, b, F)

What is the minimum number of test cases to achieve 100% coverage of all valid state transitions in the diagram?

- A. 2
- B. 3
- C. 4
- D. 8

Answer: C

Question #:90

Which statement best describes the key difference between a mindset for test activities and a mindset for development activities?

- A. A tester possesses professional pessimism while a developer is concerned with validating the product
- B. A tester is concerned with finding defects while a developer is interested in designing solutions
- C. A tester is interested in building solutions while a developer is concerned with verifying the product

D. A tester is concerned with verifying the product while a developer possesses professional pessimism

Answer: B

Explanation

The key difference between the mindsets for test activities and development activities lies in the objectives: a tester is primarily concerned with finding defects to ensure product quality, while a developer focuses on designing and building solutions. This distinct focus helps ensure thorough quality checks and balances within the software development life cycle (ISTQB not-for-profit association).

References:

ISTQB® Certified Tester Foundation Level Syllabus v4.0: https://istqb-main-web-prod.s3.amazonaws.com/media/documents/ISTQB_CTFL_Syllabus-v4.0.pdf

Question #:91

Which of the following BEST matches the attributes with a level of testing?

- I. Stubs and drivers are often used
- II The test environment should correspond to the production environment
- III Finding defects is not the main focus
- IV Testing can be based on use cases
- V Testing is normally performed by testers
- VI Testing for functional and non-functional characteristics

A. Component - VI

Integration - IV

System - I

Acceptance - III

B. Component - IV

Integration - I

System - VI

Acceptance - V

C. Component - I

Integration - V

System - II

Acceptance - IV

D. Component - V

Integration - II

System - IV

Acceptance - VI

Answer: D

Explanation

The relationship between impact analysis and regression testing in maintenance testing is that impact analysis is used to evaluate the amount of regression testing to be performed. Maintenance testing is a type of testing that is performed on an existing software product after it has been delivered or deployed, in order to ensure that it still meets its requirements and functions correctly after a change or a modification. Maintenance testing can be triggered by various reasons, such as corrective maintenance (fixing defects), adaptive maintenance (adapting to new environments), perfective maintenance (improving performance), preventive maintenance (avoiding future problems), etc. Impact analysis is a technique that is used to assess the extent and nature of changes introduced by maintenance activities on the software product or project. Impact analysis helps to identify which parts of the software product are affected by the changes, which parts need to be modified or updated accordingly, which parts need to be retested or verified for correctness or compatibility, etc. Regression testing is a type of testing that verifies that previously tested software still performs correctly after a change or a modification. Regression testing helps to detect any side effects or unintended consequences of maintenance activities on the software product's functionality or quality. Regression testing can be performed at various levels and scopes depending on the impact analysis results. Therefore, in maintenance testing, impact analysis is used to evaluate the amount of regression testing to be performed. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 20.

Question #:92

Which of the following coverage criteria results in the highest coverage for state transition based test cases?

- A. Can't be determined
- B. Covering all transitions at least once
- C. Covering only start and end states
- D. Covering all states at least once

Answer: B

Explanation

Covering all transitions at least once is the highest coverage criterion for state transition based test cases, because it ensures that every possible change of state is tested at least once. This means that all the events that trigger the transitions, as well as the actions and outputs that result from the transitions, are verified. Covering all transitions at least once also implies covering all states at least once, but not vice versa. Therefore, option D is not the highest coverage criterion. Option C is the lowest coverage criterion, because it only tests the initial and final states of the system or component, without checking the intermediate states or transitions. Option A is incorrect, because the coverage criteria for state transition based test cases can be determined and compared based on the number of transitions and states covered. References = CTFL 4.0 Syllabus, Section 4.2.3, page 49-50.

Question #:93

Which of the following statements about decision tables are TRUE?

- I. Generally, decision tables are generated for low risk test items.
- II. Test cases derived from decision tables can be used for component tests.
- III. Several test cases can be selected for each column of the decision table.
- IV. The conditions in the decision table represent negative tests generally.

- A. I. III
- B. I, IV
- C. II. IV
- D. II. III

Answer: D

Explanation

A decision table is a technique that shows combinations of inputs and/or stimuli (causes) with their associated outputs and/or actions (effects). A decision table consists of four quadrants: conditions (inputs), actions (outputs), condition entries (values) and action entries (results). The following statements about decision tables are true:

- II. Test cases derived from decision tables can be used for component tests. Decision tables can be used to test components that have multiple inputs and outputs that depend on logical combinations of conditions. Decision tables can help cover all possible combinations or scenarios in a systematic way.
- III. Several test cases can be selected for each column of the decision table. A column of a decision table represents a unique combination of condition entries and action entries. Several test cases can be selected for each column by varying other input values or expected results that are not part of the decision table. The following statements about decision tables are false:

- I. Generally, decision tables are generated for low risk test items. Decision tables are not related to risk level, but rather to complexity level. Decision tables are generated for test items that have complex logic or multiple conditions and actions that need to be tested.
- IV. The conditions in the decision table represent negative tests generally. The conditions in the decision table represent both positive and negative tests, depending on whether they are valid or invalid inputs for the test item. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 4, page 42-43.

Question #:94

4 equivalence classes are given for integer values:

$$0 < x < 100$$

$$100 \leq x \leq 200$$

$$200 < x < 500$$

$$x \geq 500$$

Which of the following options represent correct set of data for valid equivalence class partitions?

- A. 50; 100; 200. 1000
- B. 0. 1.99, 100.200,201.499, 500;
- C. 0.50; 100; 150.200.350.500;
- D. 50; 100; 250; 1000

Answer: C

Explanation

The correct set of data for valid equivalence class partitions should include one value from each equivalence class, and no value from outside the range. Option C satisfies this condition, as it has one value from each of the four equivalence classes (50, 100, 250, 500). Option A has two values from the same equivalence class (100 and 200), option B has values outside the range (0 and 0.99), and option D has two values from the same equivalence class (1000 and 500). Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 35.

Question #:95

To be able to define testable acceptance criteria, specific topics need to be addressed. In the table below are the topics matched to an incorrect description. Match the topics (the left column) with the correct description (the right column)

TopicDescription

A. How the system performs the specific behaviour.

8: Quality characteristicsB: A sequence of actions between an external actor and the system, to accomplish a specific goal or business task.

B. ScenariosC: The externally observable behaviour with user actions as input operating under certain configurations.

C. Business rulesD; Description of the connections between the system to be developed and the outside world.

D. External interfacesE: Activities that can only be performed in the system under certain conditions defined by outside procedures and constraints.

E. A to C, B to A, C to B. D to E and E to D

F. A to E, B to A, C to B, D to C and E to D

G. A to C, B to A, C to B, D to D and E to E

H. A TO A, B TO B, C TO B, D TO E AND E TO D

Answer: A

Explanation

The correct matching of the topics with their descriptions is as follows:

- Functional behaviour: The externally observable behaviour with user actions as input operating under certain configurations (A to C).
- Quality characteristics: How the system performs the specific behaviour (B to A).
- Scenarios: A sequence of actions between an external actor and the system, to accomplish a specific goal or business task (C to B).
- Business rules: Activities that can only be performed in the system under certain conditions defined by outside procedures and constraints (D to E).
- External interfaces: Description of the connections between the system to be developed and the outside world (E to D).

Question #:96

You are working in an agile team on the creation of software for use in robotic assistance systems that support industrial manufacturing processes. You are currently in the effort estimation phase using planning poker. In the current sprint, three user stories are to be implemented:

- A. The assistance system is to have a customizable layout. For this purpose, the user should be able to choose between 3 combinations of font size and background color at startup.
- B. In the initialization phase, the assistance system should validate the inputs of the 10 customer specific and process specific parameters against a list of 5 consistency rules and display the result to the user.
- C. The assistance system currently supports the manufacturing process with 2 robot gripper arms, this shall be extended to 5 gripper arms. In addition, the system shall perform a pre-calculation of the movements of all gripper arms in real time and automatically stop the system in case of a calculated collision of the gripper arms.

You are now asked to estimate the test effort for each of these user stories by using planning poker. The following value rules apply for the test effort estimation:

3 - low (≤ 10 test cases)

8 - medium (> 10 and ≤ 20 test cases)

13 - high (> 20 and ≤ 50 test cases)

21 - very high (> 50 and ≤ 100 test cases)

34 - too complex (number of test cases cannot be estimated from information)

For each of the three user stories, which planning poker value makes the most sense when estimating the test effort?

- D. A:3; B:21; C:8;
- E. A:21; B:13; C:13;
- F. A:3; B:13; C:34;
- G. A:13; B:13; C:13;

Answer: A

Question #:97

The ISTQB glossary defines Quality Assurance as: "Plan or quality management focused on providing confidence that quality requirements will be fulfilled. Which of the following is not one of the Quality Assurance activity?"

- A. Requirements elicitation
- B. Defect analysis
- C. Functional Testing

D. Performance Testing

Answer: C

Explanation

Quality Assurance (QA) activities are focused on providing confidence that quality requirements will be fulfilled through planned and systematic processes. These activities are preventive in nature, aimed at ensuring quality is built into the product from the beginning.

- Requirements elicitation (A) is part of the requirements engineering process and is concerned with gathering the needs and conditions to meet for a new or altered product.
- Defect analysis (B) can be part of QA activities as it involves analyzing defects to prevent them in future development cycles.
- Functional Testing (C) and Performance Testing (D) are types of dynamic testing, which are actually Quality Control activities rather than Quality Assurance. They are concerned with the identification of defects in the product, not with the processes to prevent defects.

Since the question asks for an activity that is NOT part of Quality Assurance, options A and B are incorrect because they can be part of QA activities. Between C and D, while both are dynamic testing activities, Functional Testing (C) is more directly related to verifying the functionality against specified requirements, which is more aligned with Quality Control. Therefore, C is the best answer.

Question #:98

Which of the following about typical information found within a test plan is FALSE?

- A. The need to temporarily have additional test personnel available for specific test phases and/or test activities
- B. The conditions that must be met in order for the test execution activities to be considered completed.
- C. The list of the product risks which have not been fully mitigated at the end of test execution.
- D. The conditions that must be met for part of all the planned activities to be suspended and resumed.

Answer: C

Explanation

A typical test plan includes various elements, such as resource requirements, test completion criteria, and suspension/resumption criteria. However, the list of product risks that have not been fully mitigated is generally not included in the test plan but rather in the risk management documentation.

- The test plan focuses on planning and executing tests, including resource allocation and defining criteria for test suspension and resumption.

- While risk management is crucial, unmitigated risks are typically documented in risk logs or separate risk management plans.

Reference: ISTQB CTFL Syllabus V4.0, Chapter 5.1.1, Test Planning.

Question #:99

Which of the following is a correct reason to apply test automation?

- A. When a new test automation tool is launched
- B. When there are a lot of repetitive testing tasks
- C. When it is easy to automate
- D. When it is cheap to buy test automation tools

Answer: B

Explanation

A correct reason to apply test automation is when there are a lot of repetitive testing tasks. Test automation is the use of software tools or scripts to perform or support testing activities, such as test case execution, test result comparison, test data generation, etc. Test automation can be beneficial when there are a lot of repetitive testing tasks that need to be performed frequently or consistently, such as regression testing, performance testing, load testing, etc. Test automation can help to save time and effort, increase reliability and accuracy, and improve coverage and efficiency of testing. The other options are not correct reasons to apply test automation. When a new test automation tool is launched is not a reason to apply test automation, but rather a factor for choosing a test automation tool. When it is easy to automate is not a reason to apply test automation, but rather a factor for evaluating the feasibility of test automation. When it is cheap to buy test automation tools is not a reason to apply test automation, but rather a factor for estimating the cost and benefit of test automation. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 10.

Question #:100

Which of the following statements about estimation of the test effort is WRONG?

- A. Once the test effort is estimated, resources can be identified and a schedule can be drawn up.
- B. Effort estimate can be inaccurate because the quality of the product under tests is not known.
- C. Effort estimate depends on the budget of the project.
- D. Experience based estimation is one of the estimation techniques.

Answer: C

Explanation

- Effort estimate does not depend on the budget of the project, but rather on the scope, complexity, and quality of the software product and the testing activities¹. Budget is a constraint that may affect the feasibility and accuracy of the effort estimate, but it is not a factor that determines the effort estimate. Effort estimate is the amount of work required to complete the testing activities, measured in terms of person-hours, person-days, or person-months².
- The other options are correct because:
 - A. Once the test effort is estimated, resources can be identified and a schedule can be drawn up, as they are interrelated aspects of the test planning process³. Resources are the people, tools, equipment, and facilities needed to perform the testing activities⁴. Schedule is the time frame and sequence of the testing activities, aligned with the project milestones and deadlines⁵.
 - B. Effort estimate can be inaccurate because the quality of the product under tests is not known, as it affects the number and severity of the defects that may be found and the rework that may be needed to fix them⁶. Quality is the degree to which the software product satisfies the specified requirements and meets the needs and expectations of the users and clients⁷.
 - D. Experience based estimation is one of the estimation techniques, which relies on the judgment and expertise of the testers and other project stakeholders to estimate the test effort based on similar projects or tasks done in the past. Experience based estimation can be useful when there is a lack of historical data, formal methods, or detailed information about the software product and the testing activities.

References =

- 1 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 154
- 2 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 155
- 3 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 156
- 4 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 157
- 5 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 158
- 6 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 159
- 7 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 16
- [8] ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 160
- [9] ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 161

Question #:101

As a result of the joint evaluation of a product version with the customer, it has been concluded that it would be appropriate to retrieve an earlier version of the product and carry out a benchmark. Depending on the result, further development will be carried out based on the current version or the retrieved version.

Which mechanism, process, and/or technique will allow the specific version (including the testing work products) of a given software product to be restored?

- A. Defect management
- B. Configuration management
- C. Change management
- D. Risk management

Answer: B

Explanation

Comprehensive and Detailed In-Depth Explanation:

Configuration management (B) ensures that **versions of software and test artifacts** are properly tracked, stored, and retrievable. It allows teams to:

- Restore **earlier versions of software and test work products**
- Maintain **traceability between requirements, tests, and code**
- Avoid discrepancies due to **mismanaged versions**
- **(A) is incorrect** because defect management **tracks issues but does not restore versions**.
- **(C) is incorrect** because change management **controls changes but does not track past versions**.
- **(D) is incorrect** because risk management **assesses risks but does not manage software versions**.

Effective configuration management ensures the ability to roll back changes and maintain system stability.

Reference: ISTQB CTFL v4.0 Syllabus, Section 5.4 – Configuration Management

Question #:102

You need to test the login page of a web site. The page contains fields for user name and password. Which test design techniques are most appropriate for this case?

- A. Decision table testing, state transition testing.

- B. Equivalence partitioning, Boundary value analysis.
- C. Exploratory testing, statement coverage.
- D. Decision coverage, fault attack.

Answer: B

Explanation

Equivalence partitioning and boundary value analysis are test design techniques that are most appropriate for testing the login page of a web site. The page contains fields for user name and password, which are input values that can be divided into partitions of equivalent data. Equivalence partitioning is a technique that divides the input data and output results of a software component into partitions of equivalent data. Each partition should contain data that is treated in the same way by the component. Equivalence partitioning can be used to reduce the number of test cases by selecting one representative value from each partition. Boundary value analysis is a technique that tests boundary values between partitions of equivalent data. Boundary values are values at the edge of an equivalence partition or at the smallest incremental distance on either side of an edge. Boundary value analysis can be used to detect defects caused by incorrect handling of boundary conditions. For example, for testing the user name field, we can identify two equivalence partitions: valid user name (existing and correct) and invalid user name (non-existing or incorrect). The boundary values for these partitions are the minimum and maximum length of user name allowed by the system.

Decision table testing and state transition testing are not suitable for testing the login page of a web site, as they are more applicable for testing components that have multiple inputs and outputs that depend on logical combinations of conditions or events. Decision table testing is a technique that shows combinations of inputs and/or stimuli (causes) with their associated outputs and/or actions (effects). State transition testing is a technique that models how a system transitions from one state to another depending on events or conditions.

Exploratory testing and statement coverage are not suitable for testing the login page of a web site, as they are more applicable for testing components that require learning, creativity and intuition or structural analysis. Exploratory testing is an approach to testing that emphasizes learning, test design and test execution at the same time. Exploratory testing relies on the tester's skills, creativity and intuition to explore the software under test and discover defects. Statement coverage is a type of structural testing that measures how many statements in a program have been executed by a test suite. Statement coverage can be used to assess the adequacy or completeness of a test suite.

Decision coverage and fault attack are not suitable for testing the login page of a web site, as they are more applicable for testing components that have complex logic or potential errors. Decision coverage is a type of structural testing that measures how many decision outcomes in a program have been executed by a test suite. Decision coverage can be used to assess the adequacy or completeness of a test suite. Fault attack is a type of functional testing that deliberately introduces faults into a system in order to provoke failures or errors.

Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 4, page 34-46; Chapter 5, page 47-48.

Question #:103

Which of the following statements about TDD, BDD and ATDD is TRUE?

- A. Refactonng is a practice that is an integral part of TDD and is applied both to tests and to code wntten to satisfy those tests.
- B. ATDD is a black-box test design technique that is applicable exclusively at acceptance test level.
- C. BDD is a developer practice where business stakeholders are not usually involved as the tests are directly written at unit/component test level.
- D. ATDD is the practice of running the automated acceptance tests as part of a continuous integration process.

Answer: A

Explanation

Test-Driven Development (TDD) emphasizes writing tests before code and includes refactoring as a key practice to improve both the tests and the code. This ensures that the codebase remains clean and maintainable. The ISTQB CTFL Syllabus v4.0 discusses TDD as a practice that includes writing tests first, coding to satisfy those tests, and then refactoring the code to improve its structure and readability while keeping the tests intact.

Question #:104

What is typically NOT included in a test progress report?

- A. Information about test obstacles during a test period
- B. Statistics on the success of each tester in finding bugs
- C. Changed risks within the test period
- D. Test status in relation to defined test metrics (including exit criteria)

Answer: B

Question #:105

For a given set of test-cases, which of the following is a benefit of running these tests with a test automation tool?

- A. Test coverage is increased
- B. The total cost of the test project always decreases
- C. The time spent on repetitive tasks is reduced
- D. The number of found bugs is reduced

Answer: C**Question #:106**

An application is subjected to a constant load for an extended period of time as part of a performance test. While running this test, the response time of the application steadily slows down, which results in a requirement not being met. This slowdown is caused by a memory leak where the application code does not properly release some of the dynamically allocated memory when it is no longer needed. Which of the following statements is TRUE?

- A. The slowdown is a failure while the memory leak is a defect
- B. The slowdown is a defect while the memory leak is an error.
- C. The slowdown is an error; the memory leak is a defect.
- D. The slowdown is a defect; the memory leak is a failure.

Answer: A**Explanation**

In software testing terminology, a failure is an observable deviation of the software from its expected behavior. A defect (or bug) is the cause of the failure in the software's code. In this case, the observed slowdown is the failure, while the underlying memory leak in the application code is the defect causing this failure. This distinction is clearly outlined in the ISTQB CTFL Syllabus v4.0, which differentiates between failures (observable issues) and defects (underlying issues in the code).

Question #:107

Which of the following are valid testing principles?

- I) Exhaustive testing is in general impossible.
 - II) Exhaustive testing should be executed for code intended to be reused.
 - III) Testing may guarantee that a program is correct.
 - IV) Testing cannot guarantee that a program is correct.
 - V) Defects cluster together in certain areas of the product.
- A. I, IV, V
 - B. II, IV
 - C. I, V
 - D. I, III

Answer: A**Explanation**

Statements I, IV and V are valid testing principles according to the ISTQB syllabus. Statement I states that exhaustive testing is in general impossible, because it would require testing all possible inputs, outputs and combinations of states, which is usually impractical or impossible. Statement IV states that testing cannot guarantee that a program is correct, because testing can only show the presence of defects, not their absence. Statement V states that defects cluster together in certain areas of the product, which means that some modules or functions are more likely to contain defects than others. Statements II and III are invalid testing principles. Statement II states that exhaustive testing should be executed for code intended to be reused, which contradicts statement I. Statement III states that testing may guarantee that a program is correct, which contradicts statement IV. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, pages 4-5.

Question #:108

Pariksha labs is a mature software testing company. They are TMMi level 5 certified. Their testing processes are well defined. Which ONE of the following statements is likely to be CORRECT about them?

- A. The company uses same test strategy for all the projects doing minor changes based on test manager responsible for the project because their test strategy template is very mature and do not need to change
- B. They try to perform 100% automation for every project because automation is a must for efficiency
- C. Same set of testing processes are used by them without needing any tweaks because that is the hallmark of a mature testing company.
- D. The question about the test design techniques to be used is determined based on various factors such as the domain and expectations of the stakeholders

Answer: D**Question #:109**

Shripriya is defining the guidelines for the review process implementation in her company. Which of the following statements is LEAST likely to have been recommended by her?

- A. Independent of the size of the work products, planning for the review should be performed
- B. Defect reports should be created for every review found
- C. Large sized work products should be reviewed in one go because you will have to spend too much time if you split it into multiple reviews
- D. Review initiation is the stage when the review team starts the discussion on the review comments

Answer: C

Question #:110

You are responsible for applying the correct technique for a review of the requirements document for a project to develop a new software application. You identify the reviewers and the required roles, including the meeting leader, who is the requirements document author, and a separate role for a scribe. Additionally, you decide to take a relatively informal approach to the requirements review. The goal of the review is to find defects in the requirements document, such as omissions, Inconsistencies, and duplications. Another goal of the review is to improve the software application's usability and accessibility by considering the various stakeholders' viewpoints.

Which of the following statements BEST describes this scenario?

- A. This scenario is using a pair review type and a perspective-based review technique
- B. This scenario is using a walkthrough review type and a checklist-based review technique
- C. This scenario is using a walkthrough review type and a perspective-based review technique
- D. This scenario is using a pair review type and a checklist-based review technique

Answer: C

Explanation

This scenario is using a walkthrough review type and a perspective-based review technique. In a walkthrough, the author of the document leads the meeting and it typically includes a meeting leader and a scribe, as described. This type of review is informal, focuses on discussion, and often involves scenario-based reading of the document to understand different user perspectives (ISTQB Main Web).

References:

ISTQB® Certified Tester Foundation Level Syllabus v4.0: ISTQB CTFL Syllabus v4.0 PDF

Question #:111

A test engineer finds a defect while testing. After the developer has fixed the defect, the test engineer decides to re-run a complete section of the tests. Which of the following is correct?

- A. The test engineer should not re-run the tests, as they have already been run, and results recorded.
- B. The test engineer should not re-run the tests, they should be part of the developer tests.
- C. The test engineer should re-run the tests, in order to ensure that new defects have not been introduced by the fix.
- D. The test engineer should re-run the tests, because the defect shows that the test cases need to be updated.

Answer: C

Explanation

The test engineer should re-run the tests, in order to ensure that new defects have not been introduced by the fix. This is also known as regression testing, which is a type of testing that verifies that previously tested software still performs correctly after a change. Regression testing helps to detect any side effects or unintended consequences of a fix or a modification. The other options are incorrect reasons for re-running the tests. The test engineer should not re-run the tests, as they have already been run, and results recorded, because this ignores the possibility of new defects caused by the fix. The test engineer should not re-run the tests, they should be part of the developer tests, because this assumes that developer tests are sufficient and reliable, which may not be true. The test engineer should not re-run the tests, because the defect shows that the test cases need to be updated, because this does not address the impact of the fix on other test cases or functionalities. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 41.

Question #:112

Which of the following statements is an example of testing contributing to higher quality?

- A. A test leader writes a test summary report
- B. A project manager asks to a test leader to estimate the test effort
- C. A tester installs a test ten in the lest environment
- D. A tester finds a bug which is resolved prior to release

Answer: D

- The question is about identifying an example of testing contributing to higher quality. Quality is the degree to which a component, system or process meets specified requirements and/or user/customer needs and expectations¹. Testing is the process consisting of all lifecycle activities, both static and dynamic, concerned with planning, preparation and evaluation of software products and related work products to determine that they satisfy specified requirements, to demonstrate that they are fit for purpose and to detect defects².
- Therefore, testing contributes to higher quality by verifying and validating that the software products and related work products meet the specified requirements, are fit for purpose and have no defects, or at least have a reduced number of defects. Testing also provides information about the quality of the software products and related work products to the stakeholders, who can make informed decisions based on the test results³.
- Out of the four given statements, only option D is an example of testing contributing to higher quality, as it shows that testing has detected a defect (a flaw in a component or system that can cause the component or system to fail to perform its required function⁴) and that the defect has been resolved (fixed and confirmed) prior to release (delivery of the software product to the customer or end user). This means that testing has prevented a potential failure (an event in which a component or system does not perform a required function within specified limits) from occurring in the operational environment, and thus has improved the quality of the software product.

- Option A is not an example of testing contributing to higher quality, as it is a reporting activity that summarizes the test results and evaluates the test objectives, but does not directly affect the quality of the software product or related work products. A test summary report is a document that records and communicates the outcomes of testing activities, including test completion criteria, test results, incident reports, test summary and evaluation, and lessons learned.
- Option B is not an example of testing contributing to higher quality, as it is a planning activity that estimates the resources and time needed for testing activities, but does not directly affect the quality of the software product or related work products. A test effort estimate is an approximation of the amount of work and/or the duration of time required to perform testing activities.
- Option C is not an example of testing contributing to higher quality, as it is a preparation activity that sets up the test environment (an environment containing hardware, instrumentation, simulators, software tools, and other support elements needed to conduct a test), but does not directly affect the quality of the software product or related work products. A test environment installation is a process of installing and configuring the test environment according to the test environment specification.

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- 1: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 10
- 2: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 11
- 3: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 12
- 4: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 13
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 13
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 77
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 78
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 79
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 80
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 81
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 82
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 83
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 84
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 85
- : ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 86

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 87

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 88

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 89

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 90

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 91

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 92

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 93

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 94

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 95

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 96

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 97

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 98

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 99

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 100

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 101

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 102

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 103

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 104

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 105

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 106

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 107

Question #:113

Given some statements about independence of testing

- i. Independence is a replacement for familiarity with the code
- ii. Independence helps remove the biases produced because of own work

- iii. Similar kind of failures can be detected by anyone
- iv Assumptions by different people are different revealing weaknesses

Which of the following CORRECTLY represents the benefits?

- A. ii and iv
- B. iii and iv
- C. i and iii
- D. i and iv

Answer: A

Question #:114

Which of the following is MOST likely to be an example of a PROJECT risk?

- A. A computation is not always performed correctly in some situations
- B. A system architecture may not support some non-functional requirements
- C. Team members' skills may not be sufficient for the assigned work
- D. Specific modules do not adequately meet their intended functions according to the user

Answer: C

Explanation

A project risk relates to potential issues that could affect the overall success of the project. Among the options provided, the risk that "Team members' skills may not be sufficient for the assigned work" is clearly a project risk because it pertains to the potential failure of the project due to inadequate skillsets among the team. This risk affects the entire project's ability to meet its objectives.

References:

ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 1.4.2.

Question #:115

Which of the following is an example of black-box dynamic testing?

- A. Functional Testing
- B. Code inspection

- C. Checking memory leaks for a program by executing it
- D. Coverage analysis

Answer: A

Explanation

Functional testing is an example of black-box dynamic testing. Black-box testing (also known as specification-based testing) is a type of testing that does not consider the internal structure or implementation of the system under test, but rather its external behavior or functionality. Dynamic testing is a type of testing that involves executing the system under test with various inputs and observing its outputs. Functional testing is a type of black-box dynamic testing that verifies that the system under test performs its intended functions according to its requirements or specifications. Functional testing can be performed at various levels and scopes depending on the objectives and criteria of testing. The other options are not examples of black-box dynamic testing. Code inspection is an example of white-box static testing. White-box testing (also known as structure-based testing) is a type of testing that considers the internal structure or implementation of the system under test. Static testing is a type of testing that does not involve executing the system under test, but rather analyzing it for defects, errors, or violations of standards. Code inspection is a type of white-box static testing that involves examining the source code of the system under test for quality, readability, maintainability, etc. Checking memory leaks for a program by executing it is an example of white-box dynamic testing. Memory leaks are defects that occur when a program fails to release memory that it has allocated but no longer needs. Checking memory leaks for a program by executing it requires knowledge and access to the internal structure or implementation of the program, such as memory allocation and deallocation mechanisms, pointers, references, etc. Coverage analysis is an example of white-box static testing. Coverage analysis is a technique that measures how much of the code or structure of the system under test has been exercised by a test suite. Coverage analysis requires knowledge and access to the internal structure or implementation of the system under test, such as statements, branches, paths, conditions, etc. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 7.

Question #:116

Calculate the measurement error SD for the following estimates done using three point estimation technique- Most optimistic effort (a) -120 person days Most likely effort (m) -180 person days Most pessimistic effort (b) - 240 person days

- A. 180
- B. 120
- C. 197
- D. 20

Answer: D

Question #:117

Which of the following statements about branch coverage is true?

- A. The minimum number of test cases needed to achieve full branch coverage, is usually lower than that needed to achieve full statement coverage
- B. If full branch coverage has been achieved, then all unconditional branches within the code have surely been exercised
- C. If full branch coverage has been achieved, then all combinations of conditions in a decision table have surely been exercised
- D. Exercising at least one of the decision outcomes for all decisions within the code, ensures achieving full branch coverage

Answer: D

Explanation

Exercising at least one of the decision outcomes for all decisions within the code, ensures achieving full branch coverage, which is a test coverage criterion that requires that all branches in the control flow of the code are executed at least once by the test cases. A branch is a basic block of code that has a single entry point and a single exit point, and a decision is a point in the code where the control flow can take more than one direction, such as an if-then-else statement, a switch-case statement, a loop statement, etc. The decision outcomes are the possible paths that can be taken from a decision, such as the then branch or the else branch, the case branch or the default branch, the loop body or the loop exit, etc. The other statements are false, because:

- The minimum number of test cases needed to achieve full branch coverage, is usually higher than that needed to achieve full statement coverage, which is a test coverage criterion that requires that all executable statements in the code are executed at least once by the test cases. This is because branch coverage is a stronger criterion than statement coverage, as it implies statement coverage, but not vice versa. For example, a single test case can achieve full statement coverage for an if-then-else statement, but two test cases are needed to achieve full branch coverage, as both the then branch and the else branch need to be exercised.
- If full branch coverage has been achieved, then all unconditional branches within the code have not necessarily been exercised, as unconditional branches are branches that do not depend on any decision, and are always executed, such as a goto statement, a break statement, a return statement, etc. Unconditional branches are not part of the branch coverage criterion, as they do not represent different paths in the control flow of the code. However, they are part of the statement coverage criterion, as they are executable statements in the code.
- If full branch coverage has been achieved, then all combinations of conditions in a decision table have not necessarily been exercised, as a decision table is a test design technique that represents the logical relationships between multiple conditions and their corresponding actions, in a tabular format. A decision table can have more combinations of conditions than the number of decision outcomes in the code, as each condition can have two or more possible values, such as true or false, yes or no, etc. For example, a decision table with four conditions can have 16 combinations of conditions, but the corresponding code may have only two decision outcomes, such as pass or fail. To exercise all combinations of conditions in a decision table, a stronger test coverage criterion is needed, such as

condition combination coverage, which requires that all possible combinations of condition outcomes in the code are executed at least once by the test cases. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:

- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.3.1, Test Coverage Criteria Based on the Structure of the Software
- ISTQB® Glossary of Testing Terms v4.0, Branch Coverage, Statement Coverage, Branch, Decision, Decision Outcome, Unconditional Branch, Decision Table, Condition Combination Coverage

Question #:118

A possible risk of introducing test automation is:

- A. the tool may not be fit-for-purpose.
- B. the tool may create additional development dependencies.
- C. the tool may not be compatible with the development platform.
- D. the tool will be owned and maintained by developers and replace testers.

Answer: A

Explanation

One possible risk of introducing test automation is that the selected tool may not be fit-for-purpose. This means that the tool might not meet the specific needs and requirements of the project, leading to inefficiencies and possibly failing to provide the expected benefits. It is crucial to evaluate and select the appropriate tool based on the project's context and objectives. The ISTQB CTFL syllabus highlights the importance of careful tool evaluation and selection to ensure it aligns with the testing goals and the development environment.

References: ISTQB CTFL Syllabus, Section 6.2, "Potential Benefits and Risks of Test Automation."

Question #:119

During which main group of test activity are the following tasks performed?

- Checking test results and logs against specified coverage criteria.
- Assessing the level of component or system quality based on test results and logs.
- Determining whether more tests are needed.

Select the correct answer:

- A. Test planning.
- B. Test analysis.

- C. Test design.
- D. Test monitoring and control.

Answer: D

Explanation

The activities of checking test results and logs against specified coverage criteria, assessing the level of component or system quality based on test results and logs, and determining whether more tests are needed fall under the category of test monitoring and control. This phase involves ongoing assessment and adjustment of the test activities to ensure they meet the test objectives and quality goals.

Question #:120

As the last stage of a test cycle of an embedded device, you are performing exploratory testing. You observed that some character. (A, X and Z) sent via a serial port to the device do not get registered on the device whereas they should be. You suspect that this could be due to a wrong configuration of the "bit parity" parameter.

Which of the following items of an incident report would you be UNABLE to write down based on this information?

- A. Expected result
- B. Test case identifier
- C. Test setup details
- D. Actual result

Answer: B

Explanation

An incident report is a document that records the details of an incident. An incident report typically contains the following items:

- Identifier: A unique identifier for the incident report
- Summary: A concise summary of the incident
- Description: A detailed description of the incident, including the steps to reproduce it, the expected and actual results, and any relevant screenshots or logs
- Severity: The degree of impact that the incident has on the system
- Priority: The level of urgency for resolving the incident

- Status: The current state of the incident, such as new, open, resolved, closed, etc.
- Resolution: The action taken to resolve the incident, such as fix, workaround, reject, etc. Based on the information given in the question, the tester would be able to write down all of these items except for the test case identifier. A test case identifier is a unique identifier for a test case that is used to link it to other test artifacts, such as test plans, test scripts, test results or incident reports. However, since the tester is performing exploratory testing, there is no predefined test case that can be associated with the incident. Exploratory testing is an approach to testing that emphasizes learning, test design and test execution at the same time. Exploratory testing relies on the tester's skills, creativity and intuition to explore the software under test and discover defects. Exploratory testing does not use formal test cases or scripts, but rather uses test charters or missions that guide the tester's actions and objectives. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 3, page 32-33; Chapter 5, page 47-48.

Question #:121

Which of the following statements about static testing is true?

- A. Static testing is NOT suitable for finding coding defects. For this, dynamic testing is necessary.
- B. Static testing should be done instead of dynamic testing, as both find the same defects.
- C. Static tests are NOT suitable to evaluate the quality of work results.
- D. It makes sense to perform static testing before dynamic testing, because defects can be found earlier or at lower cost.

Answer: D

Question #:122

Software was found to take much more time than the stated requirement of less than one second to save a file. Upon investigation it was found that there was an unnecessary check inside a loop which was slowing down the file-save operation. The software not being able to meet the desired response time is an example of

- A. It is not a defect
- B. Defect
- C. Error
- D. Failure

Answer: D

Explanation

A failure is an event in which a component or system does not perform a required function within specified limits. A failure is observable by the software users or other stakeholders. A failure is caused by one or more

defects in the software. In this case, the software not being able to meet the desired response time is an example of a failure, as it deviates from the stated requirement and affects the user experience. It is not a defect, which is a flaw in the software that causes the failure. It is not an error, which is a human action that produces an incorrect result. It is not a non-defect, as it clearly violates a specified requirement. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 1, page 4.

Question #:123

Which of the following statements does NOT describe good testing practice, regardless of the SDLC chosen?

- A. All development activities are subject to quality control.
- B. To avoid redundancy, different test levels have specific test objectives.
- C. To adhere to the principle of early testing, test analysis for a specific test level takes place during the corresponding phase of the SDLC.
- D. Testers are involved in the review of work results as soon as the documents have been approved by the stakeholders.

Answer: D

Question #:124

Which of the following statements about test reports are TRUE?

- I Test reports shall be approved by the test team.
 - II. Test reports shall give stakeholders information as basis for decisions.
 - III Test reports shall summarize what happened through a period of testing.
 - IV. Test reports shall be approved by the development team, the test team and the customer
 - V. Test reports shall include information about remaining risks.
- A. II, III, V
 - B. I, II, IV
 - C. I, III, v
 - D. II, III, IV

Answer: A

Explanation

Statements II, III and V are true about test reports. Test reports are documents that provide information on the results and status of testing activities for a given period or phase. Test reports should give stakeholders information as basis for decisions, such as whether to release the software product, whether to continue testing, whether to change the scope or priorities of testing, etc. Test reports should summarize what happened through a period of testing, such as what test cases were executed, what defects were found, what risks were identified, what issues were encountered, what achievements were made, etc. Test reports should include information about remaining risks, such as what defects are still open, what test cases are still pending, what functionalities are still untested, what uncertainties are still unresolved, etc. Statements I and IV are not true about test reports. Test reports do not need to be approved by the test team, the development team, or the customer, unless it is specified by the test policy or the test plan. Test reports only need to be reviewed and verified by the test leader or the test manager before being distributed to the intended recipients. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 141.

Question #:125

Which or the following would be a key difference between a peer review of code and static analysis of code using a tool?

- A. A peer reviews finds defects while static analysis finds failures.
- B. Static analysis targets the code technically whereas Peer review is applicable to further aspects.
- C. Peer reviews cannot find missing requirements whereas static analysis can
- D. A peer reviews find failures while static analysis finds defects.

Answer: B

Explanation

The key difference between a peer review of code and static analysis of code using a tool lies in their approaches and scope. A peer review is a manual inspection of the code by peers or colleagues, focusing not only on the technical aspects of the code but also on other elements such as design, compliance with standards, and maintainability. Peer reviews can identify defects, suggest improvements, and ensure that the code adheres to best practices and team standards.

On the other hand, static analysis is an automated process performed by tools designed to analyze the code without executing it. These tools can detect potential issues such as syntax errors, vulnerabilities, and code smells based on predefined rules and patterns. While static analysis is technically focused, it lacks the broader perspective that human reviewers can provide, such as evaluating the code's maintainability or adherence to project-specific standards. Therefore, static analysis targets the code technically, whereas peer review encompasses a wider range of aspects, making option B the correct answer.

Question #:126

Which of the following statements are true?

1. Early and frequent feedback helps to avoid requirements misunderstanding.

2. Early feedback allows teams to do more with less.
3. Early feedback allows the team to focus on the most Important features.
4. Early and frequent feedback clarifies customer feedback by applying static testing techniques

Select the correct answer:

- A. 3
- B. 2
- C. 1
- D. 4

Answer: C

Explanation

The statement "Early and frequent feedback helps to avoid requirements misunderstanding" is true. Early feedback from stakeholders, through reviews and other static testing techniques, helps clarify requirements and ensures that any misunderstandings are addressed promptly. This practice aligns with Agile principles and contributes to developing software that meets user needs more accurately.

References: ISTQB CTFL Syllabus, Section 2.1.1, "The Influence of Development Models on Testing" and Section 3.2.1, "The Advantages of Early Feedback."

Question #:127

Which ONE of the following options MOST ACCURATELY describes **statement testing**?

- A. In statement testing, the coverage items are control flow transfers between branches. The aim is to design test cases to exercise branches in the code until an acceptable level of coverage is achieved, expressed as a percentage.
- B. In statement testing, the coverage items are decisions and statements. The aim is to design test cases that exercise statements in the code until an acceptable level of coverage is achieved, expressed as a percentage.
- C. In statement testing, the coverage items are branches, and the aim is to design test cases to exercise branches in the code until an acceptable level of coverage is achieved, expressed as a percentage.
- D. In statement testing, the coverage items are executable statements. The aim is to design test cases that exercise statements in the code until an acceptable level of coverage is achieved, expressed as a percentage.

Answer: D

Explanation

Comprehensive and Detailed In-Depth Explanation:

Statement testing aims to execute **every executable statement** in the source code at least once.

- **(D) is correct** as statement testing ensures **maximum statement execution**.
- **(A) describes branch testing, which focuses on flow transfers.**
- **(B) and (C) incorrectly mix decision testing and branch testing concepts.**

Reference: ISTQB CTFL v4.0 Syllabus, Section 4.3.1 – Statement Testing

Question #:128

Your manager asked you when testing will be complete. In order to answer this question, you'll most likely use:

- A. Test progress reports
- B. Your colleagues advice
- C. A conversion spreadsheet
- D. A Test Oracle

Answer: A

Explanation

When a manager asks when testing will be complete, the most appropriate and informative resource to provide an answer is test progress reports (Option A). Test progress reports contain detailed information on the status of testing activities, including what has been accomplished, what remains to be done, the results of the tests conducted, and any issues or risks that might impact the completion of testing. These reports allow for an informed assessment of the testing progress and estimation of when testing might be completed. Options B, C, and D do not provide the structured, detailed, and specific information required to accurately answer the manager's question about the completion of testing.

Question #:129

Consider the following statements (I to V) about software testing:

- I. Can be applied to non-executable work products.
- II. Can be used to measure performance efficiency.
- III. Can lead to the detection of defects.

IV. Can find defects directly.

V. Can identify unexpected behaviors caused by potential defects.

Which ONE of the following options is the list that characterizes static tests the BEST?

- A. Statements I, III, and IV
- B. Statements II, IV, and V
- C. Statements I, II, and IV
- D. Statements II, III, and V

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

Static testing (**A**) applies to **non-executable work products** (I), such as requirements or code reviews. It also **leads to defect detection** (III) and **finds defects directly** (IV) without executing the software. Options **B, C, and D** incorrectly associate static testing with performance measurement (II) or dynamic testing aspects (V).

Reference: ISTQB CTFL v4.0 Syllabus, Section 3.1.2 – The Value of Static Testing

Question #:130

Which of the following options is a success factor for formal reviews (e.g. inspections)?

- A. Provide participants an adequate amount of time for preparation for the review.
- B. Evaluate participants on how they spend their time and attention to detail.
- C. Use only one type of review so that participants do not have to get used to a different type of review process as often.
- D. Conduct the review in a few large sections, each of which participants are to review as a whole.

Answer: A

Question #:131

Select the roles required in a formal review:

- A. Author, Management, Facilitator, Review Leader, Reviewers, Scribe
- B. Author, Tester, Facilitator, Review Leader, Reviewers, Scribe

- C. Author, Business analyst. Facilitator, Review Leader. Reviewers. Scribe
- D. Author. Developer, Facilitator. Review Leader. Reviewers. Scribe

Answer: A

Explanation

In a formal review, the roles involved typically include the author, management, facilitator (also known as moderator), review leader, reviewers, and scribe. Each role has specific responsibilities to ensure the effectiveness and efficiency of the review process:

- The **author** creates and refines the work product being reviewed.
- **Management** allocates resources and supports the review process.
- The **facilitator** manages the review meeting, ensuring it proceeds smoothly.
- The **review leader** plans the review and ensures it meets its objectives.
- **Reviewers** examine the work product to identify defects.
- The **scribe** records issues raised during the review meeting.

Question #:132

For the following pseudo-code determine number of tests required for 100% statement coverage

IF Gender = Boy

If Age > 3 AND Age < 5

Shoe Size = 1

ELSE IF Age >=5 AND Age < 7

Shoe Size = 2

ENDIF

ELSE

IF Age > 3 AND Age < 5

Shoe Size = 0

ELSE IF Age >=5 AND Age < 7

Shoe Size = 1

ENDIF

ENDIF

- A. 6
- B. 4
- C. 2
- D. 6

Answer: B

Explanation

To achieve 100% statement coverage, we need to design test cases that ensure every statement in the given pseudo-code is executed at least once. Analyzing the pseudo-code, we notice that there are conditions based on two variables: Gender and Age. To cover all statements, we need to consider the paths that lead to each assignment of the Shoe Size variable.

- Gender = Boy, Age ≤ 3 (Shoe Size assignment is not reached, but the condition is evaluated)
- Gender = Boy, Age > 3 AND Age < 5 (Shoe Size = 1)
- Gender = Boy, Age ≥ 5 AND Age < 7 (Shoe Size = 2)
- Gender $\neq$ Boy, Age ≤ 3 (Again, Shoe Size assignment is not reached, but the condition is evaluated)
- Gender $\neq$ Boy, Age > 3 AND Age < 5 (Shoe Size = 0)
- Gender $\neq$ Boy, Age ≥ 5 AND Age < 7 (Shoe Size = 1)

However, upon closer inspection, we see that tests 1 and 4 do not contribute to statement coverage as they do not lead to a Shoe Size assignment. Therefore, we only need 4 test cases to achieve 100% statement coverage, making option B the correct answer.

Question #:133

Scenario 1 “Happy Tomatoes” (used for questions 20-22):

An intelligent application for agricultural use incorporates temperature sensors located at different points of an enclosure. The sensors measure and record the temperature at regular intervals and extract the statistical values for these measurements. These values include the average daily temperature.

A new variety of tomatoes is currently being grown and the “**World Organization for Happy Tomatoes**” has established temperature ranges related to vegetative development.

When the system establishes that the average temperature is within a specific range, it emits a value that will be used to monitor and control the crop.

Temperature in degrees Celsius	Output Values
Temperature detrimental to the crop (too cold)	$\leq 7 \rightarrow W$
Vegetative development standstill temperature	$8-21 \rightarrow X$
Ideal temperature for vegetative development	$22-29 \rightarrow Y$
Temperature detrimental to the crop (too hot)	$\geq 30 \rightarrow Z$

Using the **equivalence partitioning** technique, identify the set of input values that provides the HIGHEST coverage.

- A. {7,10,21,40}
- B. {8,10,25,40}
- C. {7,10,25,29}
- D. {7,10,25,40}

Answer: C

Explanation

Comprehensive and Detailed In-Depth Explanation:

Equivalence partitioning (EP) divides input values into classes where test cases **represent all possible values** of each partition. The partitions in this scenario are:

- #7 (too cold # W)
- 8-21 (standstill # X)
- 22-29 (ideal # Y)
- #30 (too hot # Z)

Option C ({7,10,25,29}) covers all partitions by selecting representative values:

- 7 # Too cold (W)
- 10 # Standstill (X)
- 25 # Ideal (Y)
- 29 # Upper boundary of ideal (Y)

Other options do not cover all partitions adequately.

Reference: ISTQB CTFL v4.0 Syllabus, Section 4.2.1 – Equivalence Partitioning

Question #:134

You are testing a room upgrade system for a hotel. The system accepts three different types of room (increasing order of luxury): Platinum, Silver and Gold Luxury. ONLY a Preferred Guest Card holder is eligible for an upgrade.

Below you can find the decision table defining the upgrade eligibility:

What is the expected result for each of the following test cases?

Customer A: Preferred Guest Card holder, holding a Silver room

Customer B: Non Preferred Guest Card holder, holding a Platinum room

- A. Customer A; doesn't offer any upgrade; Customer B: offers upgrade to Gold luxury room
- B. Customer A: doesn't offer any upgrade; Customer B: doesn't offer any upgrade.
- C. Customer A: offers upgrade to Gold Luxury room; Customer B: doesn't offer any upgrade
- D. Customer A: offers upgrade to Silver room; Customer B: offers upgrade to Silver room.

Answer: C**Explanation**

According to the decision table in the image, a Preferred Guest Card holder with a Silver room is eligible for an upgrade to Gold Luxury (YES), while a non-Preferred Guest Card holder, regardless of room type, is not eligible for any upgrade (NO). Therefore, Customer A (a Preferred Guest Card holder with a Silver room) would be offered an upgrade to Gold Luxury, and Customer B (a non-Preferred Guest Card holder with a Platinum room) would not be offered any upgrade. References = The answer is derived directly from the decision table provided in the image; specific ISTQB Certified Tester Foundation Level (CTFL) v4.0 documents are not referenced.

Question #:135

The following state transition diagram describes the functionality involved in a system using fingerprint and password authentication to log onto a system.

How many distinct states of the system are visible in the above diagram?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: C**Explanation**

The state transition diagram provided shows three distinct states:

- Waiting for fingerprint
- Waiting for PIN
- Valid PIN/ask menu selection

Each state represents a different stage in the system's operation, with transitions based on user actions and system responses.

Question #:136

Which ONE of the following is a CORRECT example of the purpose of a test plan?

- A. A test lead decides to write a detailed test plan so that in future, in case of project failure responsibilities could be assigned to the right person
- B. A test manager should always create a very simple test plan because the purpose of test plan is to ensure that there is documentation for the purpose of audits.
- C. A test plan is a good document to have for the agile projects because it helps the test manager assign tasks to different people
- D. A test manager decides to create a one page test plan for an agile project for communicating the broad activities and explaining why detailed test cases will not be written as mandated by the test policy.

Answer: D**Question #:137**

Why should you choose a test technique?

- A. Because you need to match the way you test to the content of the product under test
- B. Because of the time constraints that usually accompany a test project
- C. Because this way you cover the full scope of the product's functionality
- D. Because choosing a test technique is a common practice in software testing

Answer: A**Explanation**

You should choose a test technique because you need to match the way you test to the content of the product under test. A test technique is a method or process for deriving and selecting test cases based on some criteria or rules. Different test techniques are suitable for different types of software products, depending on their characteristics, functionalities, requirements, specifications, risks, etc. Choosing a test technique helps to ensure that the test cases are relevant, effective, and efficient for the product under test. The other options are not correct reasons to choose a test technique. Time constraints are not a factor for choosing a test technique, but rather for prioritizing or optimizing testing activities. Covering the full scope of the product's functionality is not a guarantee of choosing a test technique, but rather a goal of testing. Choosing a test technique is not a common practice in software testing, but rather a professional skill and responsibility. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 31.

Question #:138

Which of the following tools is most likely to detect defects in functions or methods in source code?

- A. configuration management tool
- B. unit test framework tool
- C. test design tool
- D. monitoring tool

Answer: B

Explanation

A unit test framework tool is a tool that supports the creation, execution, and reporting of unit tests, which are tests that verify the functionality and quality of individual software components (such as functions or methods) in source code. A unit test framework tool can help to detect defects in functions or methods in source code by providing features such as test case generation, test case execution, test result comparison, test coverage measurement, etc. Some examples of unit test framework tools are JUnit, NUnit, TestNG, etc. The other options are not tools that are likely to detect defects in functions or methods in source code. A configuration management tool is a tool that supports the management and control of different versions and variants of software products or components. A test design tool is a tool that supports the design and generation of test cases based on some criteria or rules. A monitoring tool is a tool that monitors the behavior or performance of a system or component under test. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 10.

Question #:139

An alphanumeric password must be between 4 and 7 characters long and must contain at least one numeric character, one capital (uppercase) letter and one lowercase letter of the alphabet.

Which one of the following sets of test cases represents the correct outcome of a two-value boundary value analysis applied to the password length? (Note: test cases are separated by a semicolon)

- A. 1xA;aB11;Pq1ZZab;7iDD0a1x

- B. aB11;99rSp:5NnN10;7iDD0a1x
- C. 1xB: aB11: 99rSp: 5NnN10; 4NnN10T; 44ghWn19
- D. 1RhT;rSp53;3N3e10;8sBdby

Answer: D

Explanation

The correct outcome of a two-value boundary value analysis applied to the password length is the set of test cases represented by option D. Boundary value analysis is a test design technique that focuses on the values at the boundaries of an equivalence partition, such as the minimum and maximum values, or the values just above and below the boundaries. A two-value boundary value analysis uses two values for each boundary, one representing the valid value and one representing the invalid value. For example, if the valid range of values is from 4 to 7, then the two values for the lower boundary are 3 and 4, and the two values for the upper boundary are 7 and 8. The test cases in option D use these values for the password length, while also satisfying the other requirements of the password, such as containing at least one numeric character, one capital letter, and one lowercase letter. The test cases in option D are:

- 1RhT: a 4-character password that is valid
 - rSp53: a 5-character password that is valid
 - 3N3e10: a 6-character password that is valid
 - 8sBdby: an 8-character password that is invalid The test cases in the other options are incorrect, because they either use values that are not at the boundaries of the password length, or they do not meet the other requirements of the password. For example, the test cases in option A are:
 - 1xA: a 3-character password that is invalid, but it does not contain a capital letter
 - aB11: a 4-character password that is valid
 - Pq1ZZab: a 7-character password that is valid
 - 7iDD0a1x: an 8-character password that is invalid
- References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.2.1, Black-box Test Design Techniques1
 - ISTQB® Glossary of Testing Terms v4.0, Boundary Value Analysis, Equivalence Partition2

Question #:140

Which of the following is correct with regards to debugging?

- A. Debugging identifies the cause of a failure

- B. Debugging is often performed by test engineers
- C. Debugging is considered part of the testing activities
- D. Debugging is intended to find as many defects as possible in the code

Answer: A

Explanation

Debugging is the process of finding, analyzing and removing the causes of failures in software. Debugging is not considered part of testing, but rather a development activity that can involve testing. Debugging is not intended to find as many defects as possible, but rather to fix the specific failure that was observed. Debugging is usually performed by developers, not by test engineers. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 1, page 6.

Question #:141

A number of characteristics are given for impact of SDLC on the testing effort.

- i. Finishing of requirements review leading to test analysis
- ii. Both - static and dynamic testing performed at unit testing level
- iii. Frequent regression testing may need to be performed
- iv Extensive product documentation
- v. More use of exploratory testing

Which of the following statements is MOST correct?

- A. i and iv are characteristics of sequential models; ii. iii and v are characteristics of iterative and incremental models
- B. ii and iv are characteristics of sequential models; i. iii and v are characteristics of iterative and incremental models
- C. iii and iv are characteristics of sequential models and i. ii and v are characteristics of iterative and incremental models
- D. ii and iii are characteristics of sequential models; iv and v are characteristics of iterative and incremental models

Answer: A

Question #:142

Which of the following statements best describes the objective of branch testing?

- A. Branch testing is used to verify the functionality of an application on all logical branches without knowledge of its internal workings.
- B. Branch testing is used to check the interactive reachable branches of the user interface of an application.
- C. Branch testing is used to verify the performance of an application on certain branches.
- D. Branch testing is used to execute some or all branches in the code of an application at least once.

Answer: D

Question #:143

Which statement corresponds to one of the 7 principles of testing?

Select exactly one option (1 out of 4)!

- A. Exhaustive testing can show the absence of defects.
- B. To detect new defects, existing regression test cases must be updated regularly.
- C. Late testing saves time and money, especially with scarce resources.
- D. Fixing all detected defects ensures the success of the system under test.

Answer: B

Question #:144

For each test case to be executed, the following table specifies its dependencies and the required configuration of the test environment for running such test case:

Test Case	Dependencies	Configuration
TC1		CONF2
TC2	TC4	CONF2
TC3	TC4	CONF1
TC4		CONF1
TC5	TC1	CONF2

Assume that CONF1 is the initial configuration of the test environment. Based on this assumption, which of the following is a test execution schedule that is compatible with the specified dependencies and allows minimising the number of switches between the different configurations of the test environment?

- A. TC4, TC3, TC2, TC1, TC5.
- B. TC1, TC5, TC4, TC3, TC2
- C. TC4, TC3, TC2, TC5, TC1.
- D. TC4, TC1, TC5, TC2, TC3

Answer: A

Explanation

To determine the optimal test execution schedule that minimizes the number of configuration switches and respects the dependencies, we start with the initial configuration, CONF1.

- **TC4:** It has no dependencies and runs on CONF1 (initial configuration).
- **TC3:** Depends on TC4 and runs on CONF1. Since TC4 is already executed, we can proceed with TC3.
- **TC2:** Depends on TC4 and runs on CONF2. We switch to CONF2 after TC3.
- **TC1:** No dependencies and runs on CONF2. Since we are already in CONF2, we can execute TC1 next.
- **TC5:** Depends on TC1 and runs on CONF2. Since TC1 is already executed, we can proceed with TC5 without additional configuration switches.

By following this sequence (TC4, TC3, TC2, TC1, TC5), we respect the dependencies and minimize the number of configuration switches

Question #:145

Which of the statements on confirmation testing and regression testing is correct?

- A. Regression tests are performed to confirm that defects have been corrected and no longer lead to errors
- B. Confirmation tests are performed to confirm that changes have no undesired effects
- C. Regression tests are performed to confirm that errors have been corrected and no longer lead to failures
- D. Confirmation tests are performed to confirm that defects have been corrected and no longer lead to failures

Answer: D

Question #:146

You are testing the latest version of an air-traffic control system prior to production deployment using exploratory testing. After following an unusual sequence of input steps, the system crashes. After the crash, you document a defect report with the following information:

- Title: System crashes unexpectedly during input.
- Brief summary: System crashes when an unusual sequence of inputs is used.
- Version: V1.001
- Test: Exploratory testing prior to production deployment
- Priority: Urgent
- Risk: High
- References: Screenshot of crashed application

What critical Information Is missing from this report?

- A. Conclusions, recommendations, and approvals.
- B. Change history.
- C. Description of the defect to enable reproduction.
- D. Status of defect

Answer: C

Explanation

The critical information missing from the defect report is a detailed description of the defect to enable reproduction. A clear and concise description of the steps taken to reproduce the defect is essential for developers to understand the context and to be able to replicate the issue in their environment. Without this information, it can be challenging to diagnose and fix the defect. The ISTQB CTFL syllabus emphasizes the importance of providing all necessary details in a defect report to facilitate effective communication and resolution.

References: ISTQB CTFL Syllabus, Section 5.5, "Defect Management."

Question #:147

Which of the following provides the BEST description of statement coverage?

- A. A white-box test technique which covers the decision results which determine the next statement to be executed
- B. A black-box test technique which uses a state table to derive test cases

- C. A white-box test technique which focuses on the percentage of executable statements that has been executed by a test suite
- D. An experience-based test technique in which test cases are based on the tester's knowledge of past failures

Answer: C

Explanation

Statement coverage is a white-box test technique which focuses on executing all possible statements in the code at least once during testing. This helps in determining the percentage of executable statements that have been executed by the test suite, aiming to ensure that all parts of the program have been tested at least once (ISTQB Main Web).

References:

ISTQB® Certified Tester Foundation Level Syllabus v4.0: ISTQB CTFL Syllabus v4.0 PDF

Question #:148

Which ONE of the following options corresponds to an activity in the testing process in which testable features are identified?

- A. Test design
- B. Test analysis
- C. Test implementation
- D. Test execution

Answer: B

Explanation

Comprehensive and Detailed In-Depth Explanation:

Test analysis involves identifying the features to be tested and deriving test conditions. It is the phase where testers analyze the test basis (e.g., requirements, specifications) to identify testable aspects of the system. Test design (A) focuses on creating test cases, test implementation (C) involves preparing the test environment, and test execution (D) runs the tests.

Reference: ISTQB CTFL v4.0 Syllabus, Section 1.4.1 – Test Analysis

Question #:149

Which of the following is true about Oracles?

- A. Sometimes old version of a product can be used as an Oracle
- B. Oracles help in reproducing the irreproducible bugs
- C. Oracles are derived from the design
- D. Oracles can be generated automatically using data generators

Answer: A

Explanation

An oracle is a mechanism or source that can provide the expected result for a given test input or situation. Sometimes old version of a product can be used as an oracle, if it is assumed that the old version behaves correctly for the test cases that are executed on the new version. This is also known as back-to-back testing. Oracles do not help in reproducing the irreproducible bugs, as they only provide the expected results, not the actual results. Oracles are not derived from the design, but from the requirements or specifications. Oracles cannot be generated automatically using data generators, as data generators only provide test inputs, not test outputs. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 9.

Question #:150

A software company decides to invest in reviews of various types. The thought process they have is that each artifact needs to be reviewed using only one of the review methods depending on the criticality of the artifact.

- A. The thought process is incorrect. The whole company should adopt same standard for review of all artifacts.
- B. The thought process is correct. The whole company should decide or the review method based on their CMM level.
- C. The thought process is incorrect. Same artifact can be reviewed using different review methods
- D. The thought process is correct. It wastes time to review same artifact using efferent review methods

Answer: C

Explanation

The thought process of the software company is incorrect, because it assumes that each artifact can be reviewed using only one review method, and that the review method depends solely on the criticality of the artifact. This is a simplistic and rigid approach that does not consider the benefits and limitations of different review methods, the context and purpose of the review, and the feedback and improvement opportunities that can be gained from multiple reviews. According to the CTFL 4.0 Syllabus, the selection of review methods should be based on several factors, such as the type and level of detail of the artifact, the availability and competence of the reviewers, the time and budget constraints, the expected defects and risks, and the desired outcomes and quality criteria. Moreover, the same artifact can be reviewed using different review methods at different stages of the development lifecycle, to ensure that the artifact meets the changing requirements,

standards, and expectations of the stakeholders. For example, a requirement specification can be reviewed using an informal review method, such as a walkthrough, to get an initial feedback from the users and developers, and then using a formal review method, such as an inspection, to verify the completeness, correctness, and consistency of the specification. Therefore, the software company should adopt a more flexible and context-sensitive approach to selecting and applying review methods for different artifacts, rather than following a fixed and arbitrary rule. References = CTFL 4.0 Syllabus, Section 3.2.1, page 31-32; Section 3.2.2, page 33-34; Section 3.2.3, page 35-36.

Question #:151

Given the following state model of sales order software:

- PLACED -> IN PRODUCTION
- IN PRODUCTION -> CANCELLED
- IN PRODUCTION -> SHIPPED
- SHIPPED -> INVOICED
- INVOICED -> CANCELLED
- CANCELLED -> PLACED

Which of the following sequences of transitions provides the highest level of transition coverage for the model (assuming you can start in any state)?

- A. IN PRODUCTION -> CANCELLED -> PLACED -> IN PRODUCTION -> CANCELLED -> PLACED
- B. IN PRODUCTION -> SHIPPED -> INVOICED -> CANCELLED -> PLACED -> IN PRODUCTION
- C. PLACED -> IN PRODUCTION -> SHIPPED -> CANCELLED -> PLACED
- D. PLACED -> CANCELLED -> PLACED -> CANCELLED -> PLACED -> IN PRODUCTION -> CANCELLED

Answer: B

Explanation

State transition testing is a black-box testing technique where test cases are designed to cover states and transitions of a state machine.

Given the state model with the following transitions:

- PLACED -> IN PRODUCTION
- IN PRODUCTION -> CANCELLED

- IN PRODUCTION -> SHIPPED
- SHIPPED -> INVOICED
- INVOICED -> CANCELLED
- CANCELLED -> PLACED

To cover all transitions at least once, we need to create a sequence that covers all six transitions.

Option A: IN PRODUCTION -> CANCELLED -> PLACED -> IN PRODUCTION -> CANCELLED -> PLACED

- Misses SHIPPED -> INVOICED and INVOICED -> CANCELLED transitions.

Option B: IN PRODUCTION -> SHIPPED -> INVOICED -> CANCELLED -> PLACED -> IN PRODUCTION

- Covers all transitions.

Option C: PLACED -> IN PRODUCTION -> SHIPPED -> CANCELLED (ISTQB not-for-profit association) (Udemy)sses INVOICED -> CANCELLED transition.

Option D: PLACED -> CANCELLED -> PLACED -> CANCELLED -> PLACED -> IN PRODUCTION -> CANCELLED

- Misses SHIPPED -> INVOICED and INVOICED -> CANCELLED transitions.

Given these, Option B covers all the transitions.

References:

Certified Tester Foundation Level v4.0

10 Sample Exams ISTQB Foundation Level (CTFL) v4.0

Question #:152

Which ONE of the following statements about state transition testing is correct?

- A. The state transition diagram explicitly shows all invalid transitions.
- B. The size of the state table depends on the number of possible transitions between the states
- C. Usually it is not possible to create tests to cover all transitions and all states
- D. All transitions between states are explicitly shown in the state table.

Answer: D

Explanation

State transition testing is a black-box testing technique used to analyze the behavior of a system by examining the transitions between different states in response to events. In state transition testing, a state table or diagram is used to represent the states of a system and the transitions between these states triggered by events.

Option D is correct because in state transition testing, all transitions between states should be explicitly shown in the state table. This includes valid transitions that the system is expected to make under normal operation and, where relevant, invalid transitions that should be tested to ensure the system handles unexpected or erroneous inputs gracefully. The state table provides a comprehensive view of how the system should behave, making it possible to create tests that cover all defined transitions.

Question #:153

As a tester, as part of a V-model project, you are currently executing some tests aimed at verifying if a mobile app asks the user to grant the proper access permissions during the installation process and after the installation process. The requirements specification states that in both cases the app shall ask the user to grant access permissions only to the camera and photos stored on the device. However, you observe that the app also asks the user to grant access permission to all contacts on the device. Consider the following items:

[I]. Test environment

[ii]. Expected result

[iii]. Actual result.

[IV] Test level.

[V]. Root cause.

Based on only the given information, which of the items listed above, are you able to CORRECTLY specify in a defect report?

- A. [I] and [IV]
- B. [ii] and [III].
- C. [ii], [iii] and [v]
- D. [ii], [IV] and [V].

Answer: B

Explanation

When writing a defect report, the tester can specify the expected result and the actual result based on the observation. The expected result is what the requirements specify, and the actual result is what was observed during testing. These elements are crucial for clearly communicating the nature of the defect to developers and other stakeholders. The other items such as test environment, test level, and root cause may not be clear or necessary at this stage of defect reporting.

References: ISTQB CTFL Syllabus, Section on defect management and reporting.

Question #:154

Use Scenario 1 “Happy Tomatoes” (from the previous question).

When running test case **TC_59**, the actual result for **t = 35** degrees Celsius is **OUTPUT = X** instead of the expected output.

Which information should NOT be included in the defect report?

- A. Identification of the test object and test environment
- B. A concise title and a short summary of the defect being reported
- C. Description of the structure of the test team
- D. Expected results and actual results

Answer: C

Explanation

Comprehensive and Detailed In-Depth Explanation:

A defect report should contain relevant details to help developers **reproduce and fix the defect efficiently**. The essential elements include:

- **Test object & environment (A)** – To ensure reproducibility.
- **Title & summary (B)** – For quick identification.
- **Expected vs. actual results (D)** – To describe the discrepancy.

The **structure of the test team (C)** is irrelevant for defect tracking and resolution.

Reference: ISTQB CTFL v4.0 Syllabus, Section 5.5 – Defect Management

Question #:155

Consider an estimation session in which a six-member Agile team (Memb1..... Memb6) uses the planning poker technique to estimate a user story (in story points). The team members will use a set of cards with the following values: 1,2, 3,5, 8,13,21. Below is the outcome of the first round of estimation for this session:

Memb1 = 3 Memb2 = 3 Memb3 = 3
Memb4 = 21 Memb5 = 3 Memb6 = 1

Which of the following answers BEST describes how the estimation session should proceed?

- A. The final estimate of the user story in story points is determined by applying the three-point estimation technique with the following input values most optimistic estimate - 1, most likely estimate - 3, and most pessimistic estimate - 21
- B. Further estimation rounds should be performed until all team members will pick the card having the same value: this value will represent the final estimate of the user story in story points.
- C. The final estimate of the user story in story points is determined by calculating the average value between the most optimistic estimate of 21 story points (Memb4) and the most pessimistic estimate of 1 story point (Memb6)
- D. Memb6 and Memb4 which have produced the most pessimistic and the most optimistic estimates respectively, should explain the reasons of their choices to stimulate a discussion between all members before proceeding to another estimation round

Answer: D

Explanation

In Agile teams using the planning poker technique for estimating user stories, it is common practice to have further discussions and rounds of estimation if there is a significant discrepancy in the initial estimates. This helps in reaching a consensus and ensures that all team members understand the complexity and requirements of the user story. According to the ISTQB CTFL syllabus, planning poker involves discussions to clarify differences in estimates, especially when there is a wide range of values selected. By having Memb6 and Memb4, who provided the most pessimistic and optimistic estimates, explain their reasoning, it fosters a deeper understanding and encourages the team to converge towards a more accurate and agreed-upon estimate.

References: ISTQB CTFL Syllabus, Section on Agile methodologies and estimation techniques.

Question #:156

Which one of the following statements IS NOT a valid objective of testing?

- A. To build confidence in the level of quality of the test object.
- B. To find all defects in a product, ensuring the product is defect free.
- C. To find failures and defects
- D. To evaluate work products such as requirements, user stories, design, and code.

Answer: B

Explanation

Finding all defects in a product and ensuring it is defect-free is not a valid objective of testing. Testing aims to find as many defects as possible, build confidence in the quality of the product, and evaluate work products. However, it is unrealistic to expect testing to find all defects, and declaring a product completely defect-free is impractical.

Reference: ISTQB CTFL Syllabus V4.0, Section 1.1.1

Question #:157

A bank offers a savings account with various interest rates based on the current balance in the account. The balance ranges and respective interest rates are:

Up to \$100.00 = 2%

\$100.01 to \$500.00 = 4%

\$500.01 to \$1,000.00 = 5%

Above \$1,000.00 = 7%

Using two-point boundary value analysis, which of the following sets of test inputs provides the relatively highest level of boundary coverage?

- A. \$5.00,\$100.00,\$499.99,\$1,000.00,\$1,000.01
- B. \$100.00, \$100.01,\$100.02,\$500.00,\$999.99
- C. \$100.00, \$500.00,\$1,000.00,\$1,000.01
- D. \$5.00,\$100.00,\$500.00,\$1,000.01

Answer: B

Explanation

Boundary Value Analysis (BVA) is a software testing technique in which tests are designed to include values at the boundaries. The concept is to focus on the boundaries since errors tend to occur at the edges of input ranges rather than in the middle.

Given the problem statement:

- Up to \$100.00 = 2%
- \$100.01 to \$500.00 = 4%
- \$500.01 to \$1,000.00 = 5%
- Above \$1,000.00 = 7%

Two-point boundary value analysis means testing the two boundaries of each range.

For each range:

- The boundaries for "Up to \$100.00" would be \$100.00 and \$100.01.
- The boundaries for "\$100.01 to \$500.00" would be \$100.00 and \$500.00.
- The boundaries for "\$500.01 to \$1,000.00" would be \$500.00 and \$1,000.00.
- The boundaries for "Above \$1,000.00" would be \$1,000.00 and \$1,000.01.

Now, let's examine the options:

- A. \$5.00, \$100.00, \$499.99, \$1,000.00, \$1,000.01
 - Missing \$100.01 and \$500.01.
- B. \$100.00, \$100.01, \$100.02, \$500.00, \$999.99
 - Covers \$100.00, \$100.01, \$500.00, \$1000.00, and \$1000.01.
- C. \$100.00, \$500.00, \$1,000.00, \$1,000.01
 - Missing \$100.01 and \$500.01.
- D. \$5.00, \$100.00, \$500.00, \$1,000.01
 - Missing \$100.01 and \$500.01.

Given the options, B provides the highest boundary coverage (ISTQB not-for-profit association) (Udemy).

References:

Certified Tester Foundation Level v4.0

10 Sample Exams ISTQB Foundation Level (CTFL) v4.0

Question #:158

A requirement specifies that if the total amount of sales (TAS) made during the year by a corporate seller is 300,000€ or more, the bonus that must be paid to the seller is 100% of a certain amount agreed upon at the beginning of the year. The software contains a fault as it implements this requirement with the decision "IF (TAS = 300,000)" instead of "IF (TAS >= 300.000)". The application of the 3-value boundary value analysis to this problem consists of the following three test cases (TAS is an integer variable):

TC1 = 299,999 TC2=300,000 TC=300,001

Which of the following statements is TRUE?

- A. TC1 would highlight the fault
- B. TC2 would highlight the fault
- C. TC3 would highlight the fault
- D. None of the three test cases would highlight the fault.

Answer: B

Explanation

The requirement specifies that a bonus should be paid if the total amount of sales (TAS) made during the year is 300,000€ or more. The software incorrectly implements this requirement with "IF (TAS = 300,000)" instead of "IF (TAS >= 300,000)". Using boundary value analysis (BVA), which is a common technique in software testing, the three test cases provided (TC1 = 299,999, TC2 = 300,000, and TC3 = 300,001) cover the critical boundary values around the condition.

- TC1 tests just below the boundary (299,999),
- TC2 tests exactly at the boundary (300,000),
- TC3 tests just above the boundary (300,001).

Since the software incorrectly checks for TAS equal to 300,000, only TC2 will fail because the condition is exactly met and highlights the incorrect implementation of the decision logic.

Reference: ISTQB CTFL Syllabus V4.0, Chapter 4.2.2, Boundary Value Analysis (BVA).

Question #:159

Which of the following exemplifies how a software bug can cause harm to a company?

- A. "Print" prints the last page twice for a file with 1000 pages
- B. The timeout on the login page of a web site is 9 minutes, while the requirement was for 10 minutes
- C. When uninstalling the application, the uninstall dialog has a spelling mistake
- D. When calculating the final price in a shopping list, the price of the last item is not added

Answer: D

Explanation

A software bug can cause harm to a company by directly affecting its operations, reputation, user satisfaction, and financials. Option D, "When calculating the final price in a shopping list, the price of the last item is not added," describes a defect that directly impacts the core functionality of a financial transaction, potentially

leading to financial loss and customer dissatisfaction. This can have severe implications for the company's credibility and revenue. Options A, B, and C describe bugs that, while potentially annoying, do not have the same direct impact on the company's core operations and financial integrity as option D.

Question #:160

In the newest version of payroll system number of changes were made. As a tester you got a task to perform regression and confirmation tests. Which of the following project activities are related to confirmation testing?

- A. Testing due to the application of a new version of the interface
- B. Testing that fixes resolved the defects in the search function
- C. Testing if a system still works after update of an operating system
- D. Testing to ensure the adding of a new functionalities haven't broken existing functions

Answer: B

Explanation

Confirmation testing, also known as re-testing, is performed to verify that specific defects have been successfully fixed.

Option A: "Testing due to the application of a new version of the interface" would typically involve regression testing, not confirmation testing.

Option B: "Testing that fixes resolved the defects in the search function" fits the description of confirmation testing as it focuses on ensuring that specific issues have been addressed.

Option C: "Testing if a system still works after update of an operating system" is an example of regression testing, as it checks the overall system behavior after an update.

Option D: "Testing to ensure the adding of new functionalities haven't broken existing functions" is another example of re (ISTQB not-for-profit association) (Udemy) it checks for unintended consequences of new changes.

Therefore, the correct answer is B.

References:

Certified Tester Foundation Level v4.0

ISTQB Foundation Level Syllabus 4.0 (2023)

Question #:161

What is 'Component Testing'?

- A. Integration Testing

- B. Functional testing
- C. Experience-based testing
- D. A test level

Answer: D

Explanation

Component testing is a test level. A test level is a group of test activities that are organized and managed together based on some common characteristics or objectives. A test level can be defined based on various factors, such as the scope and target of testing, the phase and model of development, the stakeholders and roles involved in testing, etc. Component testing (also known as unit testing or module testing) is a test level that focuses on verifying the functionality and quality of individual software components (such as modules, classes, functions, methods, etc.). Component testing can be performed by developers or testers using various techniques and tools depending on the type and complexity of the components. The other options are not test levels. Integration testing is another test level that focuses on verifying the functionality and quality of groups of software components that interact with each other or with external systems. Functional testing is a type of black-box dynamic testing that verifies that the system under test performs its intended functions according to its requirements or specifications. Experience-based testing is a category of test design techniques that rely on the tester's knowledge and intuition to derive and select test cases based on their experience with similar systems, technologies, domains, risks, etc. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 19.

Question #:162

Which of the following statements refers to a good testing practice that applies to all software development lifecycles?

- A. Each test level should have its own specific test objectives that should be consistent with the software development lifecycle phase or type of activities it addresses.
- B. Test analysis and design for any test levels should begin as soon as coding is complete, and all system components are available for testing
- C. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.
- D. All the tests should be automated and run as part of the continuous integration process with every software change

Answer: A

Explanation

Good testing practice dictates that each test level (e.g., unit testing, integration testing, system testing) should have distinct test objectives aligned with the phase of the software development lifecycle it addresses. This ensures that testing is effective and relevant at each stage. According to the ISTQB CTFL Syllabus v4.0,

establishing clear test objectives that are consistent with the development phase helps in achieving specific goals and improving the overall quality of the software product.

Question #:163

Which of the following statements is true?

Select exactly one option (1 out of 4)!

- A. Errors are the inevitable result of software failure.
- B. Defects in the program code may be caused by environmental conditions.
- C. Defects in the program code may result in failures in the program during execution.
- D. Any deviation of the actual from the expected result represents an error.

Answer: C

Question #:164

The following requirement is given “Set X to be the sum of Y and Z”.

All the following four implementations have bugs.

Which one of the following bugs can be caught by Static Analysis?

A. `int x = 1.`

`int y = 2.`

`int y = 3.`

`X = y=z;`

B. `int x = 1.`

`int y = 2.`

`int z = 3.`

`X = z-y`

C. `int x = 1.`

`Int y = 2.`

`Int z = 3.`

`Z = x +y`

D. `int y = 2`

`Int z = 3.`

`Y = z+y`

Answer: A

Explanation

Static analysis is a technique that analyzes the source code or other software artifacts without executing them. Static analysis can detect defects such as syntax errors, coding standards violations, potential security vulnerabilities, or logical flaws. Static analysis can catch the bug in the first implementation, as it contains two syntax errors: the variable `y` is declared twice, and the assignment statement `X = y=z` is invalid. Static analysis cannot catch the bugs in the other three implementations, as they are logical errors that do not violate any syntax rules, but produce incorrect results. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 3, page 25-26.

Question #:165

Which of the following statements is true?

- A. A defect does not always produce a failure, while a bug always produces a failure
- B. A defect may cause a failure which, when occurring, always causes an error
- C. Failures can be caused by defects, but also by environmental conditions
- D. Bugs are defects found during component testing, while failures are defects found at higher test levels

Answer: C

Explanation

Failures can be caused by defects, but also by environmental conditions. A failure is an event in which the software system does not perform a required function or performs a function incorrectly, according to the expected behavior. A defect is a flaw in the software system or a deviation from the requirements or the specifications, that may cause a failure. However, not all failures are caused by defects, as some failures may be caused by environmental conditions, such as hardware malfunctions, network interruptions, power outages, incompatible configurations, etc. Environmental conditions are factors that affect the operation of the software system, but are not part of the software system itself. The other statements are false, because:

- A defect does not always produce a failure, while a bug always produces a failure. This statement is false, because a defect may or may not produce a failure, depending on the inputs, the outputs, the states, or the scenarios of the software system, and a bug is just another term for a defect, so it has the same possibility of producing a failure as a defect. For example, a defect in a rarely used feature or a hidden branch of the code may never produce a failure, while a defect in a frequently used feature or a

critical path of the code may produce a failure often. A bug is not a different concept from a defect, but rather a synonym or a colloquial term for a defect, so it has the same definition and implications as a defect.

- A defect may cause a failure which, when occurring, always causes an error. This statement is false, because an error is not a consequence of a failure, but rather a cause of a defect. An error is a human action or a mistake that produces a defect in the software system, such as a typo, a logic flaw, a requirement misunderstanding, etc. An error is not observable in the software system, but rather in the human mind or the human work products, such as the code, the design, the documentation, etc. A failure is not a cause of an error, but rather a result of a defect, which is a result of an error. For example, an error in the code may cause a defect in the software system, which may cause a failure in the software behavior.
- Bugs are defects found during component testing, while failures are defects found at higher test levels. This statement is false, because bugs and failures are not different types of defects, but rather different terms for defects and their manifestations. As mentioned before, bugs are just another word for defects, and failures are the events in which the software system does not perform as expected due to defects. Bugs and failures can be found at any test level, not only at component testing or higher test levels. Test levels are the stages of testing that correspond to the levels of integration of the software system, such as component testing, integration testing, system testing, and acceptance testing. Defects and failures can occur and be detected at any test level, depending on the test objectives, the test basis, the test techniques, and the test environment. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.1.2, Testing and Quality1
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.2.1, Testing Principles1
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.3.1, Testing in Software Development Lifecycles1
- ISTQB® Glossary of Testing Terms v4.0, Failure, Defect, Bug, Environmental Condition, Error, Test Level2

Question #:166

What type of testing measures its effectiveness by tracking which lines of code were executed by the tests?

- A. Acceptance testing
- B. Structural testing
- C. Integration testing
- D. Exploratory testing

Answer: B

Explanation

Structural testing is a type of testing that measures its effectiveness by tracking which lines of code were executed by the tests. Structural testing, also known as white-box testing or glass-box testing, is based on the internal structure, design, or implementation of the software. Structural testing aims to verify that the software meets the specified quality attributes, such as performance, security, reliability, or maintainability, by exercising the code paths, branches, statements, conditions, or data flows. Structural testing uses various coverage metrics, such as function coverage, line coverage, branch coverage, or statement coverage, to determine how much of the code has been tested and to identify any untested or unreachable parts of the code. Structural testing can be applied at any level of testing, such as unit testing, integration testing, system testing, or acceptance testing, but it is more commonly used at lower levels, where the testers have access to the source code.

The other options are not correct because they are not types of testing that measure their effectiveness by tracking which lines of code were executed by the tests. Acceptance testing is a type of testing that verifies that the software meets the acceptance criteria and the user requirements. Acceptance testing is usually performed by the end-users or customers, who may not have access to the source code or the technical details of the software. Acceptance testing is more concerned with the functionality, usability, or suitability of the software, rather than its internal structure or implementation. Integration testing is a type of testing that verifies that the software components or subsystems work together as expected. Integration testing is usually performed by the developers or testers, who may use both structural and functional testing techniques to check the interfaces, interactions, or dependencies between the components or subsystems. Integration testing is more concerned with the integration logic, data flow, or communication of the software, rather than its individual lines of code. Exploratory testing is a type of testing that involves simultaneous learning, test design, and test execution. Exploratory testing is usually performed by the testers, who use their creativity, intuition, or experience to explore the software and discover any defects, risks, or opportunities for improvement. Exploratory testing is more concerned with the behavior, quality, or value of the software, rather than its internal structure or implementation. **References** = ISTQB Certified Tester Foundation Level (CTFL) v4.0 syllabus, Chapter 4: Test Techniques, Section 4.3: Structural Testing Techniques, Pages 51-54; Chapter 1: Fundamentals of Testing, Section 1.4: Testing Throughout the Software Development Lifecycle, Pages 11-13; Chapter 3: Static Testing, Section 3.4: Exploratory Testing, Pages 40-41.

Question #:167

Which of the following statements about error guessing is true?

- A. Error guessing is a system that adopts artificial intelligence to predict whether software components are likely to contain defects or not
- B. Experienced testers, when applying error guessing, rely on the use of a high-level list of what needs to be tested as a guide to find defects
- C. Error guessing refers to the ability of a system or component to continue normal operation despite the presence of erroneous inputs
- D. Experienced testers, when applying error guessing technique, can anticipate where errors, defects and failures have occurred and target their tests at those issues

Answer: D

Explanation

This answer is correct because error guessing is a test design technique where the experience and intuition of the tester are used to anticipate where errors, defects and failures have occurred or are likely to occur, and to design test cases to expose them. Error guessing can be based on factors such as the complexity of the system or component, the known or suspected weaknesses of the system or component, the previous history of defects, or the common types of errors in the domain or technology. Error guessing can be used as a complementary technique to other more systematic or formal techniques, or when there is insufficient information or time to apply them. References: ISTQB Glossary of Testing Terms v4.0, ISTQB Foundation Level Syllabus v4.0, Section 2.3.2.5

Question #:168

Which of the following is a valid collection of equivalence classes for the following problem: An integer field shall contain values from and including 1 to and including 15.

- A. Less than 0.1 through 14. 15 and more
- B. Less than 1.1 through 14. more than 15
- C. negative numbers. 1 through 15. above 15
- D. Less than 1.1 through 15. more than 15

Answer: D

Explanation

Equivalence partitioning is a black-box test design technique where inputs to the software or system are divided into groups that are expected to exhibit similar behavior. For an integer field that should accept values from 1 to 15, the valid equivalence class is 1 through 15. The invalid equivalence classes are numbers less than 1 and numbers more than 15. Therefore, option D, "Less than 1, 1 through 15, more than 15," correctly identifies the valid equivalence class along with the two invalid classes, covering all possible input scenarios for the field. Options A, B, and C either do not accurately capture the valid range or incorrectly specify the range boundaries.

Question #:169

Consider the following table, which contains information about test cases from the test management system:

REQUIREMENT	REQUIREMENT PRIORITY	TEST CASE	STATEMENT COVERAGE	DEPENDENCIES
REQ 1	3	TC10	3%	—
		TC20	1%	—
		TC30	2%	—
REQ 2	5	TC40	5%	—
REQ 3	2	TC50	4%	—
REQ 6	2	TC60	7%	REQ 1
		TC70	2%	—
REQ 7	1	TC80	6%	REQ 2
REQ 8	5	TC90	5%	—

Which ONE of the following options **organizes the test cases based on the statement coverage strategy**, while considering practical constraints?

- A. {TC 20; TC 30; TC 10; TC 40; TC 50; TC 70; TC 60; TC 80; TC 90}
- B. {TC 10; TC 30; TC 20; TC 60; TC 40; TC 80; TC 90; TC 50; TC 70}
- C. {TC 80; TC 70; TC 50; TC 60; TC 20; TC 30; TC 10; TC 40; TC 90}
- D. {TC 60; TC 80; TC 40; TC 90; TC 50; TC 10; TC 70; TC 30; TC 20}

Answer: D

Explanation

Comprehensive and Detailed In-Depth Explanation:

Statement coverage strategy prioritizes **test cases with higher statement coverage first**, while **resolving dependencies** before execution.

- **TC60 (7%)** has the **highest coverage** but **depends on REQ 1**, so it should be executed **after its dependency is covered**.
- **TC80 (6%)** depends on **REQ 2**, so it should be prioritized **after TC40 (REQ 2)**.
- **TC40 (5%)** and **TC90 (5%)** should be executed early.
- **Lower coverage test cases (TC10, TC70, TC30, TC20)** should come last.

Thus, the correct order is {**TC 60; TC 80; TC 40; TC 90; TC 50; TC 10; TC 70; TC 30; TC 20**} (D).

Reference: ISTQB CTFL v4.0 Syllabus, Section 4.3 – White-Box Testing Techniques

Question #:170

Which of the following is the most correct statement about state testing techniques?

- A. Static techniques can be used before all code is ready for execution
- B. Static techniques find more defects than dynamic techniques.
- C. Static techniques can be used by inexperienced users.
- D. Static techniques are always cheaper than dynamic techniques.

Answer: A

Explanation

State testing techniques are a type of dynamic testing techniques that are based on the behavior of the system under test for different input conditions and events. Dynamic testing techniques require the system to be executed with test cases, whereas static testing techniques do not. Static testing techniques can be applied before the code is ready for execution, such as reviews, inspections, walkthroughs, and static analysis. Static testing techniques can help find defects early in the development process, improve the quality of the code, and reduce the cost and effort of dynamic testing. References = ISTQB Certified Tester Foundation Level (CTFL) v4.0 Syllabus, Chapter 4, Section 4.2.1, Page 281; ISTQB Glossary of Testing Terms v4.0, Page 292

Question #:171

Which ONE of the following options MOST ACCURATELY describes **branch testing**?

- A. In branch testing, the coverage items are executable statements. The aim is to design test cases that exercise statements in the code until an acceptable level of coverage is achieved, expressed as a percentage.
- B. In branch testing, the coverage items are control flow transfers between decisions, and the aim is to design test cases to exercise flow transfers in the code until an acceptable level of coverage is achieved. Coverage is measured as the number of branches exercised by the test cases divided by the total number of branches expressed as a percentage.
- C. In branch testing, the coverage items are branches, and the aim is to design test cases to exercise branches in the code until an acceptable level of coverage is achieved. Coverage is measured as the number of branches exercised by the test cases divided by the total number of branches expressed as a percentage.
- D. In branch testing, the coverage items are executable decisions. The aim is to design test cases that exercise statements in the code until an acceptable level of coverage is achieved. Coverage is expressed as a percentage.

Answer: C

Explanation

Comprehensive and Detailed In-Depth Explanation:

Branch testing is a structural testing technique that ensures each **branch (decision point)** in the control flow is executed at least once. The goal is to measure **branch coverage**, which is the number of branches exercised divided by the total number of branches.

- (A) describes statement testing, not branch testing.
- (B) and (D) introduce confusion between decisions and statements, whereas branch testing focuses on control flow branches.

In simple terms, **branch testing checks that all possible decision outcomes (true/false) are executed**, whereas statement testing only ensures that each line of code is executed.

Reference: ISTQB CTFL v4.0 Syllabus, Section 4.3.2 – Branch Testing

Question #:172

Determining the schedule for each testing activity and test milestones for a test project, using activity estimates, available resources, and other constraints is a typical task performed during

- A. Test execution
- B. Test design.
- C. Test analysis.
- D. Test planning

Answer: D

Explanation

Test planning involves defining the overall approach to testing, including scheduling, resources, and milestones. It is during this phase that the detailed schedule for each testing activity is determined based on estimates, resource availability, and constraints. The ISTQB CTFL Syllabus v4.0 outlines that test planning encompasses the creation of test plans and schedules to ensure that testing activities are properly managed and controlled.

Question #:173

A minimum of 0 and a maximum of 40 points can be achieved in a Certified Tester Foundation examination. With a score of 26 points to 36 points inclusive, the exam is considered "passed", with 37 points or higher it is "passed with distinction" and with less than 26 points the examination is "not passed".

Which of the following sets of positive integers fully covers a 2-value boundary value analysis ? (No values from invalid equivalence classes should be specified).

- A. 0, 24, 25, 35, 36, 37, 40

- B. 0, 25, 26, 36, 37, 40
- C. 0, 2, 25, 26, 27, 35, 36, 37
- D. 25, 26, 27, 39, 40, 41

Answer: B

Question #:174

Which ONE of the following options CANNOT be subjected to static analysis?

- A. COTS (Commercial off-the-shelf)
- B. Source code
- C. BPMN (Business Process Modeling and Notation) models
- D. UML (Unified Modeling Language) models

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

Static analysis requires access to **work products like source code, models, or documentation**, making it **impossible to analyze Commercial off-the-shelf (COTS) software** (A) because its source code is typically unavailable. Static analysis is applicable to source code (B), BPMN models (C), and UML diagrams (D).

Reference: ISTQB CTFL v4.0 Syllabus, Section 3.1.1 – Static Analysis

Question #:175

While repotting a defect, which attribute indicates the degree of impact that the defect has on the system?

- A. Priority
- B. Severity
- C. Status
- D. Description

Answer: B

Explanation

In defect reporting, the attribute that indicates the degree of impact that the defect has on the system is the severity. Severity reflects the seriousness of the defect in terms of its impact on the operation of the system, ranging from minor issues that do not significantly affect the system's functionality to critical defects that can cause system failure. Therefore, option B is the correct answer.

Question #:176

Which of the following statements correctly describes error guessing?

- A. Error guessing is a testing technique that uses the experience and knowledge of testers to create test cases that may reveal suspected defects.
- B. Error guessing is a testing technique in which test cases are created to exercise the items in a checklist.
- C. Error guessing is a testing technique in which the testers dynamically create test cases based on their knowledge, exploration of the test item and the results of previous tests.
- D. Error guessing is a testing technique in which the tester creates test cases based on the analysis of the specified behavior of the test object without knowing its internal structure.

Answer: A

Question #:177

Which of the following statements about the testing quadrants is TRUE?

- A. The higher the number of the testing quadrant, the more important the tests associated with this quadrant are
- B. Automated acceptance tests produced during BDD and ATDD are classified in quadrant 02.
- C. Exploratory tests are classified in quadrant Q3, and they are usually included in a continuous integration process.
- D. Automated unit tests produced during TDD are classified in quadrant Q4 as they are technology facing.

Answer: B

Explanation

The correct statement is B. According to the ISTQB CTFL syllabus, the testing quadrants help to categorize tests based on their purpose and whether they are technology-facing or business-facing, and whether they support the team or critique the product. Quadrant Q2 includes tests that are business-facing and support the team, such as automated acceptance tests produced during Behavior-Driven Development (BDD) and Acceptance Test-Driven Development (ATDD) .

Quadrant Q3 includes business-facing tests that critique the product, such as exploratory testing, usability testing, and user acceptance testing. These tests are typically manual and focus on evaluating the product from a user perspective, rather than being part of a continuous integration process.

Quadrant Q4 includes technology-facing tests that critique the product, such as performance tests, security tests, and other non-functional tests, which can be automated but are not related to unit tests produced during TDD .

Question #:178

The testers in company A were part of the development team. Due to an organizational change they moved to be part of the support team.

What are the advantages and the disadvantages of this change?

- A. Advantage: More independence in deciding what and how to test,
Disadvantage: Isolation from the development team knowledge
- B. Advantage: being closer to customer perspective,
Disadvantage less independence in perspectives
- C. Advantage: pulled to support tasks and having less time for testing,
Disadvantage less chances to move a tester to development
- D. Advantage: increased chances to move a tester to development;
Disadvantage: pulled to support tasks and having less time for testing

Answer: B

Explanation

Being part of the support team means that the testers are closer to the customer perspective, which is an advantage for testing, as they can better understand the user needs and expectations, and identify more realistic scenarios and risks. However, being part of the support team also means that they have less independence in deciding what and how to test, as they may be influenced by the customer's preferences or requests, which could compromise the objectivity and effectiveness of testing. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 6.

Question #:179

Which of the following statements best describes how configuration management supports testing?

- A. Configuration management helps reduce testing effort by identifying a manageable number of test environment configurations in which to test the software, out of all possible configurations of the environment in which the software will be released
- B. Configuration management is an administrative discipline that includes change control, which is the process of controlling the changes to identified items referred to as Configuration Items'
- C. Configuration management is an approach to interoperability testing where tests are executed in the cloud, as the cloud can provide cost-effective access to multiple configurations of the test environments
- D. Configuration management helps ensure that all relevant project documentation and software items are uniquely identified in all their versions and therefore can be unambiguously referenced in test documentation

Answer: D

Explanation

This answer is correct because configuration management is a process of establishing and maintaining consistency of a product's performance, functional, and physical attributes with its requirements, design, and operational information throughout its life. Configuration management helps ensure that all relevant project documentation and software items are uniquely identified in all their versions and therefore can be unambiguously referenced in test documentation. This supports testing by providing traceability, consistency, and control over the test artifacts and the software under test. References: : ISTQB Glossary of Testing Terms v4.0, : ISTQB Foundation Level Syllabus v4.0, Section 2.2.2.2

Question #:180

Which of the following answers describes a reason for adopting experience-based testing techniques?

- A. Experience-based test techniques provide more systematic coverage criteria than black-box and white-box test techniques
- B. Experience-based test techniques completely rely on the tester's past experience for designing test cases.
- C. Experience-based test techniques allow designing test cases that are usually easier to reproduce than those designed with black-box and white-box test techniques.
- D. Experience-based test techniques tend to find defects that may be difficult to find with black-box and white-box test techniques and are often useful to complement these more systematic techniques.

Answer: D

Explanation

Experience-based testing techniques leverage the tester's intuition and prior experience to identify defects that systematic techniques might miss. These techniques are valuable because they can uncover issues based on real-world usage and scenarios that aren't always covered by more formalized black-box and white-box

methods. The ISTQB CTFL Syllabus v4.0 highlights the complementary nature of experience-based techniques in providing a broader defect detection strategy.

Question #:181

A document describes the test procedures that have been derived for the identified test sets. Among other things, the order in which the test cases in the corresponding test set are to be executed according to the dependencies described by preconditions and postconditions is specified. This document is a typical work product produced as part of:

- A. Test design.
- B. Test analysis
- C. Test Implementation.
- D. Test monitoring and control

Answer: C**Explanation**

Test implementation involves finalizing the test procedures, including the order of execution of test cases based on their dependencies, preconditions, and postconditions. This phase ensures that all necessary test scripts, test data, and test environments are ready for execution. According to the ISTQB CTFL Syllabus v4.0, test implementation is the phase where detailed test procedures are derived and documented, making it a critical step before actual test execution.

Question #:182

Consider the following list of possible benefits of retrospectives (A to D):

- A. Team bonding and learning
- B. Increased test effectiveness/efficiency
- C. Improved quality of the test basis
- D. Better cooperation between development and testing

... and the list of possible characteristics of retrospectives (1 to 4) that enable achieving these benefits:

The possibility of implementing suggestions for process improvement

The opportunity to raise problems and propose points for improvement

The regular review and optimization of the collaboration between team members

The possibility of addressing and solving deficiencies in the extent and quality of the requirements

Which ONE of the following options presents the BEST mapping between benefits and characteristics from retrospectives?

- E. A1 - B4 - C3 - D2
- F. A2 - B1 - C4 - D3
- G. A4 - B3 - C2 - D1
- H. A3 - B2 - C1 - D4

Answer: B

Explanation

Comprehensive and Detailed In-Depth Explanation:

Retrospectives provide value by fostering improvement and collaboration. The correct mapping is:

- **A2:** Team bonding occurs when teams openly discuss issues.
- **B1:** Test effectiveness increases when process improvements are implemented.
- **C4:** Test basis improves by addressing requirement deficiencies.
- **D3:** Cooperation improves through regular review of collaboration.

Reference: ISTQB CTFL v4.0 Syllabus, Section 2.1.6 – Retrospectives

Question #:183

In which of the following test documents would you expect to find test exit criteria described?

- A. Test design specification
- B. Project plan
- C. Requirements specification
- D. Test plan

Answer: D

Explanation

Test exit criteria are the conditions that must be fulfilled before concluding a particular testing phase. These criteria act as a checkpoint to assess whether we have achieved the testing objectives and are done with

testing1. Test exit criteria are typically defined in the test plan document, which is one of the outputs of the test planning phase. The test plan document describes the scope, approach, resources, and schedule of the testing activities. It also identifies the test items, the features to be tested, the testing tasks, the risks, and the test deliverables2. According to the ISTQB® Certified Tester Foundation Level Syllabus v4.0, the test plan document should include the following information related to the test exit criteria3:

- The criteria for evaluating test completion, such as the percentage of test cases executed, the percentage of test coverage achieved, the number and severity of defects found and fixed, the quality and reliability of the software product, and the stakeholder satisfaction.
- The criteria for evaluating test process improvement, such as the adherence to the test strategy, the efficiency and effectiveness of the testing activities, the lessons learned and best practices identified, and the recommendations for future improvements.

Therefore, the test plan document is the most appropriate test document to find the test exit criteria described. The other options, such as test design specification, project plan, and requirements specification, are not directly related to the test exit criteria. The test design specification describes the test cases and test procedures for a specific test level or test type3. The project plan describes the overall objectives, scope, assumptions, risks, and deliverables of the software project4. The requirements specification describes the functional and non-functional requirements of the software product5. None of these documents specify the conditions for ending the testing process or evaluating the testing outcomes. References = ISTQB® Certified Tester Foundation Level Syllabus v4.0, Entry and Exit Criteria in Software Testing | Baeldung on Computer Science, Entry And Exit Criteria In Software Testing - Rishabh Software, Entry and Exit Criteria in Software Testing Life Cycle - STLC [2022 Updated] - Testsigma Blog, ISTQB® releases Certified Tester Foundation Level v4.0 (CTFL).

Question #:184

Which of the following is a key characteristic of informal reviews?

- A. Kick-off meeting
- B. Low cost
- C. Individual preparation
- D. Metrics analysis

Answer: B

Explanation

A key characteristic of informal reviews is low cost. Informal reviews are a type of review that does not follow a formal process or have any formal documentation. Informal reviews are usually performed by individuals or small groups of peers or colleagues who have some knowledge or interest in the product under review. Informal reviews can be done at any time and for any purpose, such as checking for errors, clarifying doubts, sharing ideas, etc. Informal reviews have low cost, as they do not require much time, effort, or resources to conduct. The other options are not key characteristics of informal reviews. Kick-off meeting is a characteristic of formal reviews, such as inspections or walkthroughs. Kick-off meeting is a meeting that is held before the review process starts, where the roles and responsibilities of the participants are defined, the

objectives and scope of the review are agreed, and the logistics and schedule of the review are planned. Individual preparation is a characteristic of formal reviews, such as inspections or walkthroughs. Individual preparation is an activity that is performed by the reviewers before the review meeting, where they examine the product under review and identify any issues or questions that need to be discussed or resolved during the review meeting. Metrics analysis is a characteristic of formal reviews, such as inspections or walkthroughs. Metrics analysis is an activity that is performed after the review process is completed, where the data and results of the review are collected and analyzed to measure the effectiveness and efficiency of the review, as well as to identify any improvement actions or lessons learned for future reviews. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 9.

Question #:185

Which of the following correctly represents a test first approach?

- A. Code is refactored first then tests are written and run
- B. Application behaviour is written in Given/When/then format and tests are then written and run
- C. Tests are derived from acceptance criteria and run to check that the code passes these tests
- D. First the tests fail then the code is written to pass the tests and this is iterated

Answer: D

Question #:186

A system has valid input numbers ranging between 1000 and 99999 (both inclusive). Which of the following inputs are a result of designing tests for all valid equivalence classes and their boundaries?

- A. 999.1000.23232.99999.100000
- B. 999.1000.50000.100000.100001
- C. 999.100000
- D. 1000,50000,99999

Answer: B

Explanation

A correct list of boundary values for the P input should include the minimum and maximum values of the valid range (15 and 350), as well as the values just below and above the boundaries (14 and 351). Boundary value analysis is a test design technique that involves testing the values at or near the boundaries of an input domain or output range, as these values are more likely to cause errors than values in the middle. Option B satisfies this condition, as it has all four boundary values (14, 15, 350, 351). Option A has two values from the same equivalence class (1000 and 99999), option C has two values outside the range (999 and 100000), and option D has no boundary values at all. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 34.

Question #:187

An e-commerce site accepts credit cards for processing the payment. The payment processing form has a field for the amount of money to be deducted. The minimum amount of money that can be processed is \$10. The credit cards have a limit of \$5,000 (Five Thousand). Assume that only integers can be accepted as inputs. Which of the following set of boundary values you will choose for EFFICIENT testing for the amount of money that can be spent?

- A. 9.5001
- B. 10.5000
- C. 10, 11.4999, 5000r 5001
- D. 9. 10.5000,5001

Answer: B

Question #:188

Use Scenario 1 “Happy Tomatoes” (from the previous question).

Using the **Boundary Value Analysis (BVA)** technique (in its two-point variant), identify the set of input values that provides the HIGHEST coverage.

- A. {7,8,21,22,29,30}
- B. {7,8,22,23,29,30}
- C. {6,7,8,21,22,29,31}
- D. {6,7,21,22,29,30}

Answer: A

Explanation**Comprehensive and Detailed In-Depth Explanation:**

Boundary Value Analysis (BVA) focuses on test cases at the edges of partitions because defects often occur at boundaries. The temperature ranges are:

- #7 (Too cold # W)
- [8-21] (Standstill # X)
- [22-29] (Ideal # Y)

➤ #30 (Too hot # Z)

A **two-point BVA** means testing both the **lower and upper boundary values** of each partition.

The correct selection {7,8,21,22,29,30} includes:

- 7 # Boundary of Too Cold (W)
- 8 # Lower boundary of Standstill (X)
- 21 # Upper boundary of Standstill (X)
- 22 # Lower boundary of Ideal (Y)
- 29 # Upper boundary of Ideal (Y)
- 30 # Lower boundary of Too Hot (Z)

This ensures maximum boundary coverage.

Reference: ISTQB CTFL v4.0 Syllabus, Section 4.2.2 – Boundary Value Analysis

Question #:189

Which of the following CORRECTLY matches the roles and responsibilities in a formal review?

- A. Facilitator - Fixes defects in the work product under review
- B. Scribe - Collates potential defects found during the individual review activity
- C. Review Leader - Creates the work product under review
- D. Author - Identify potential defects in the work product under review

Answer: B

Explanation

In formal reviews, the scribe's role is to collate potential defects and other findings during the review process. This position is crucial as it ensures all observations and defects are recorded accurately, facilitating efficient analysis and resolution of issues identified during the review.

References: ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 3.2.4 "Roles and Responsibilities in a Formal Review".

Question #:190

Consider the following examples of risks identified in different software development projects

[I]. It may not be possible to generate the expected workloads to run performance tests, due to the poor hardware equipment of the machines (load injectors) that should generate these workloads.

[ii]. A user's session on a web application is not invalidated after a certain period of inactivity (configured by the system administrator) of the user,

[iii]. The test team will not have an adequate requirements specification (since many requirements will still be missing) by the time test design and analysis activities should begin according to the test plan.

[IV]. Following a failure, the system is unable to continue to maintain its pre-failure operation and some data becomes corrupted.

Which of the following statements is TRUE?

- A. [ii] and [IV] are product risks; [i] and [iii] are project risks
- B. [ii] and [iii] are product risks. [I] and [IV] are project risks.
- C. [i], and [iv] are product risks; [ii] and [iii] are project risks
- D. [i], [II] and [iii] are product risks, [IV] is a project risk.

Answer: A

Explanation

In software testing, risks are categorized into product risks and project risks. Product risks are associated with the potential of a product to fail in meeting its quality criteria. Project risks are related to potential issues that could affect the project's ability to deliver a product.

- [i] is a project risk because it concerns the availability and adequacy of hardware resources for performance testing.
- [ii] is a product risk because it pertains to a security and functionality issue within the web application.
- [iii] is a project risk because it involves the availability of necessary requirements documentation for the testing process.
- [iv] is a product risk because it relates to the system's functionality and data integrity after a failure.

Thus, statement A correctly classifies [ii] and [iv] as product risks and [i] and [iii] as project risks.

Question #:191

Which of the following is an advantage of the whole team approach?

- A. It helps avoid the risk of tasks associated with a user story not moving through the Agile task board at an acceptable rate during an iteration.

- B. It helps team members understand the current status of an iteration by visualising the amount of work left to do compared to the time allotted for the iteration
- C. It helps the whole team be more effective in test case design by requiring all team members to master all types of test techniques.
- D. It helps team members develop better relationships with each other and make their collaboration more effective for the benefit of the project.

Answer: D

Explanation

The whole team approach, often advocated in Agile methodologies, emphasizes collaboration and collective responsibility among all team members. This approach enhances the relationships within the team and improves overall collaboration, which in turn benefits the project's success. According to the ISTQB CTFL Syllabus v4.0, the whole team approach fosters better communication and cooperation, leading to more effective problem-solving and higher-quality outcomes.

Question #:192

Decision table testing is being performed on transactions in a bank's ATM (Automated Teller Machine) system. Two test cases have already been generated for rules 1 and 4, which are shown below:

Rules		R1	R4
Conditions	Speed > 50	T	F
	School Zone	T	F
Actions	\$250 Fine	F	F
	Jail	T	F

Given the following additional test cases:

Rules		DT1	DT2	DT3	DT4
Input	Speed	55	44	66	77
	School Zone	T	T	T	F
Expected Result	\$250 Fine	F	F	F	T
	Jail	T	F	T	F

Which two of the additional test cases would achieve full coverage of the full decision table (when combined with the test cases that have already been generated for rules 1 and 4)?

- A. DT1, DT4
- B. DT3, DT4
- C. DT2, DT3
- D. DT1, DT2

Answer: C

Explanation

Decision table testing is used to analyze combinations of inputs to determine the appropriate outputs, often based on specific rules or conditions.

For the problem statement:

- Rule 1: (Withdrawal = Allowed, Balance = Sufficient, Fast Cash = True, Correct PIN = True)
 - Outcome: Transaction = Approved
- Rule 4: (Withdrawal = Allowed, Balance = Sufficient, Fast Cash = True, Correct PIN = False)
 - Outcome: Transaction = Declined

The additional test cases are:

- DT1: (Withdrawal = Allowed, Balance = Insufficient, Fast Cash = True, Correct PIN = True)
 - Outcome: Transaction = Declined
- DT2: (Withdrawal = Allowed, Balance = Sufficient, Fast Cash = False, Correct PIN = True)
 - Outcome: Transaction = Approved
- DT3: (Withdrawal = Allowed, Balance = Insufficient, Fast Cash = True, Correct PIN = False)
 - Outcome: Transaction = Declined
- DT4: (Withdrawal = Allowed, Balance = Sufficient, Fast Cash = False, Correct PIN = False)
 - Outcome: Transaction = Declined

From the given test cases, DT2 covers the scenario where Fast Cash is False, which is not covered in the initial cases. DT3 covers the case where Balance is Insufficient and PIN is incorrect.

Combining Rules 1 and 4 with DT2 and DT3 covers all the scenarios.

References:

Certified Tester Foundation Level v4.0

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Question #:193

The four test levels used in ISTQB syllabus are:

1. Component (unit) testing

2. Integration testing
3. System testing
4. Acceptance testing

An organization wants to do away with integration testing but otherwise follow V-model. Which of the following statements is correct?

- A. It is allowed as organizations can decide on men test levels to do depending on the context of the system under test
- B. It is allowed because integration testing is not an important test level arc! can be dispensed with.
- C. It is not allowed because integration testing is a very important test level and ignoring i: means definite poor product quality
- D. It is not allowed as organizations can't change the test levels as these are chosen on the basis of the SDLC (software development life cycle) model

Answer: D

Explanation

The V-model is a software development life cycle model that defines four test levels that correspond to four development phases: component (unit) testing with component design, integration testing with architectural design, system testing with system requirements, and acceptance testing with user requirements. The V-model emphasizes the importance of verifying and validating each phase of development with a corresponding level of testing, and ensuring that the test objectives, test basis, and test artifacts are aligned and consistent across the test levels. Therefore, an organization that wants to follow the V-model cannot do away with integration testing, as it would break the symmetry and completeness of the V-model, and compromise the quality and reliability of the software or system under test. Integration testing is a test level that aims to test the interactions and interfaces between components or subsystems, and to detect any defects or inconsistencies that may arise from the integration of different parts of the software or system. Integration testing is essential for ensuring the functionality, performance, and compatibility of the software or system as a whole, and for identifying and resolving any integration issues early in the development process. Skipping integration testing would increase the risk of finding serious defects later in the test process, or worse, in the production environment, which would be more costly and difficult to fix, and could damage the reputation and credibility of the organization. Therefore, the correct answer is D.

The other options are incorrect because:

- A. It is not allowed as organizations can decide on the test levels to do depending on the context of the system under test. While it is true that the choice and scope of test levels may vary depending on the context of the system under test, such as the size, complexity, criticality, and risk level of the system, the organization cannot simply ignore or skip a test level that is defined and required by the chosen software development life cycle model. The organization must follow the principles and guidelines of the software development life cycle model, and ensure that the test levels are consistent and coherent

with the development phases. If the organization wants to have more flexibility and adaptability in choosing the test levels, it should consider using a different software development life cycle model, such as an agile or iterative model, that allows for more dynamic and incremental testing approaches.

- B. It is not allowed because integration testing is not an important test level and can be dispensed with. This statement is false and misleading, as integration testing is a very important test level that cannot be dispensed with. Integration testing is vital for testing the interactions and interfaces between components or subsystems, and for ensuring the functionality, performance, and compatibility of the software or system as a whole. Integration testing can reveal defects or inconsistencies that may not be detected by component (unit) testing alone, such as interface errors, data flow errors, integration logic errors, or performance degradation. Integration testing can also help to verify and validate the architectural design and the integration strategy of the software or system, and to ensure that the software or system meets the specified and expected quality attributes, such as reliability, usability, security, and maintainability. Integration testing can also provide feedback and confidence to the developers and stakeholders about the progress and quality of the software or system development. Therefore, integration testing is a crucial and indispensable test level that should not be skipped or omitted.
- C. It is not allowed because integration testing is a very important test level and ignoring it means definite poor product quality. This statement is partially true, as integration testing is a very important test level that should not be ignored, and skipping it could result in poor product quality. However, this statement is too strong and absolute, as it implies that integration testing is the only factor that determines the product quality, and that ignoring it would guarantee a poor product quality. This is not necessarily the case, as there may be other factors that affect the product quality, such as the quality of the requirements, design, code, and other test levels, the effectiveness and efficiency of the test techniques and tools, the competence and experience of the developers and testers, the availability and adequacy of the resources and environment, the management and communication of the project, and the expectations and satisfaction of the customers and users. Therefore, while integration testing is a very important test level that should not be skipped, it is not the only test level that matters, and skipping it does not necessarily mean definite poor product quality, but rather a higher risk and likelihood of poor product quality.

References = ISTQB Certified Tester Foundation Level Syllabus, Version 4.0, 2018, Section 2.3, pages 16-18; ISTQB Glossary of Testing Terms, Version 4.0, 2018, pages 38-39; ISTQB CTFL 4.0 - Sample Exam - Answers, Version 1.1, 2023, Question 104, page 36.

Question #:194

Which of the following is NOT a common objective of testing?

- A. Finding defects in the software
- B. Preventing defects
- C. Debugging the software to find the reason for defects
- D. Providing information on the status of the system

Answer: C

Explanation

Debugging the software to find the reason for defects is not a common objective of testing, but rather a task of development or maintenance. Debugging is a process of locating and fixing errors in the software code, while testing is a process of finding and reporting defects in the software behavior or quality. Testing does not aim to fix defects, but rather to provide information on their existence and impact. The other options are common objectives of testing. Finding defects in the software is one of the main objectives of testing, as it helps to improve the quality and reliability of the software. Preventing defects is another objective of testing, as it helps to avoid rework and reduce costs and risks. Providing information on the status of the system is another objective of testing, as it helps to support decision making and risk management. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 3.

Question #:195

Consider the following user story about an e-commerce website's registration feature that only allows registered users to make purchases ; As a new user, I want to register to the website, so that I can start shopping online"

The following are some of the acceptance criteria defined for the user story

- [a] The registration form consists of the following fields: username, email address, first name, last name, date of birth, password and repeat password.
- [b] To submit the registration request, the new user must fill in all the fields of the registration form with valid values and must agree to the terms and conditions.
- [c] To be valid, the email address must not be provided by free online mail services that allow to create disposable email addresses. A dedicated error message must be presented to inform the new user when an invalid address is entered.
- [d] To be valid, the first name and last name must contain only alphabetic characters and must be between 2 and 80 characters long A dedicated error message must be presented to inform the new user when an invalid first name and/or the last name is entered
- [e] After submitting the registration request, the new user must receive an e-mail containing the confirmation link to the e-mail address specified in the registration form

Based only on the given information, which of the following ATDD tests is MOST LIKELY to be written first?

- A. The new user enters valid values in the fields of the registration form, except for the email address, where he/she enters an e-mail address provided by a free online mail service that allow to create disposable email addresses. Then he/she is informed by the website about this issue.
- B. The new user enters valid values in the fields of the registration form, except for the first name, where he/she enters a first name with 10 characters that contains a number. Then he/she is informed by the website about this issue.
- C. The user accesses the website with a username and password, and successfully places a purchase order for five items, paying by Mastercard credit card

- D. The new user enters valid values in all the fields of the registration form, confirms to accept all the terms and conditions, submits the registration request and then receives an e-mail containing the confirmation link to the e-mail address specified in the registration form

Answer: D

Explanation

Acceptance Test-Driven Development (ATDD) tests focus on verifying whether the system meets the specified acceptance criteria. The most critical path to test first would be the scenario where everything is done correctly (happy path), ensuring the basic functionality works as expected.

- The new user provides all valid data.
- This ensures the registration form works and the user receives a confirmation email.

This test covers the basic functionality and will help verify that the primary use case is handled correctly before testing invalid or edge cases.

Reference: ISTQB CTFL Syllabus V4.0, Chapter 4.5.3, Acceptance Test-Driven Development (ATDD).

Question #:196

A requirement specifies that a certain identifier (ID) must be between 5 and 10 characters long, must contain only alphanumeric characters, and its first character must be a letter. As a tester, you want to apply one-dimensional equivalence partitioning to test this ID. This means that you have to apply equivalence partitioning individually: to the length of the ID, the type of characters contained within the ID, and the type of the first character of the ID. What is the number of partitions to cover?

- A. 7.
- B. 6.
- C. 5.
- D. 3.

Answer: A

Explanation

To apply one-dimensional equivalence partitioning to the ID requirement, we need to consider each condition individually:

- Length of the ID: Valid partitions (5-10 characters), Invalid partitions (less than 5, more than 10) = 3 partitions.
- Type of characters: Valid partitions (alphanumeric), Invalid partitions (non-alphanumeric) = 2 partitions.

- First character: Valid partitions (letter), Invalid partitions (non-letter) = 2 partitions. Adding these partitions, we get a total of 3 (length) + 2 (character type) + 2 (first character) = 7 partitions. Thus, the correct answer is A.

Question #:197

A program is used to control a manufacturing line (turn machines on and off, start and stop conveyer belts, add raw materials to the flow, etc.). Not all actions are possible at all times. For example, there are certain manufacturing stages that cannot be stopped - unless there is an emergency. A tester attempts to evaluate if all such cases (where a specific action is not allowed) are covered by the tests.

Which coverage metric will provide the needed information for this analysis?

- A. Code coverage
- B. Data flow coverage
- C. Statement coverage
- D. Branch Coverage

Answer: D

Explanation

Branch coverage is a type of structural coverage metric that measures the percentage of branches or decision outcomes that are executed by the test cases. A branch is a point in the code where the control flow can take two or more alternative paths based on a condition. For example, an if-else statement is a branch that can execute either the if-block or the else-block depending on the evaluation of the condition. Branch coverage ensures that each branch is taken at least once by the test cases, and thus reveals the behavior of the software under different scenarios. Branch coverage is also known as decision coverage or all-edges coverage.

Branch coverage is suitable for testing the cases where a specific action is not allowed, because it can verify that the test cases cover all the possible outcomes of the conditions that determine the action. For example, if the program has a condition that checks if the manufacturing stage can be stopped, then branch coverage can ensure that the test cases cover both the cases where the stage can be stopped and where it cannot be stopped. This way, branch coverage can help identify any missing or incorrect branches that may lead to undesired or unsafe actions.

The other options are not correct because they are not suitable for testing the cases where a specific action is not allowed. Code coverage is a general term that encompasses various types of coverage metrics, such as statement coverage, branch coverage, data flow coverage, etc. Code coverage does not specify which type of coverage metric is used for the analysis. Data flow coverage is a type of structural coverage metric that measures the percentage of data flow paths that are executed by the test cases. A data flow path is a sequence of statements that define, use, or kill a variable. Data flow coverage is useful for testing the correctness and completeness of the data manipulation in the software, but not for testing the conditions that determine the actions. Statement coverage is a type of structural coverage metric that measures the percentage of statements or lines of code that are executed by the test cases. Statement coverage ensures that each statement is executed at least once by the test cases, but it does not reveal the behavior of the software under different scenarios. Statement coverage is a weaker criterion than branch coverage, because it does not account for the branches

or decision outcomes in the code. **References** = ISTQB Certified Tester Foundation Level (CTFL) v4.0 syllabus, Chapter 4: Test Techniques, Section 4.3: Structural Testing Techniques, Pages 51-54.

Question #:198

Your organization's test strategy states that it is desirable to use **more than one method for estimating test effort**. You are responsible for estimating test effort for the next project. Based on historical data, the **development-to-test effort ratio** is 5:3.

The initial estimate for the **development effort** is **450 person-days**.

Which ONE of the following options corresponds to the estimated **test effort** using the ratio-based method?

- A. 750 person-days
- B. 180 person-days
- C. 720 person-days
- D. 270 person-days

Answer: D

Explanation

Comprehensive and Detailed In-Depth Explanation:

The **ratio-based estimation method** relies on historical relationships between **development and testing effort**.

Using the given **5:3 ratio**:

$$\text{Test Effort} = \left(\frac{3}{5} \times \text{Development Effort} \right) = \left(\frac{3}{5} \times 450 \right) = 270 \text{ person-days}$$

Thus, the correct answer is **270 person-days (D)**.

- (A) and (C) are incorrect as they overestimate the effort.
- (B) underestimates the testing effort based on the ratio.

Ratio-based estimation **helps allocate resources effectively based on past project data**.

Reference: ISTQB CTFL v4.0 Syllabus, Section 5.1.4 – Effort Estimation in Testing

Question #:199

Which of the following is a test-first approach, where tests that express a shared understanding from stakeholders of how the application is expected to work, are first written in business-readable language (following the Given/When/Then format), and then made executable to drive development?

- A. Test-Driven Development (TDD)
- B. Acceptance Test-Driven Development (ATDD)
- C. Behavior-Driven Development (BDD)
- D. Domain-Driven Design (DDD)

Answer: C

Explanation

This answer is correct because Behavior-Driven Development (BDD) is a test-first approach, where tests that express a shared understanding from stakeholders of how the application is expected to work, are first written in business-readable language (following the Given/When/Then format), and then made executable to drive development. BDD is a collaborative approach that involves testers, developers, business analysts, product owners, and other stakeholders in defining the expected behavior of the application using scenarios that describe the preconditions, actions, and outcomes of the application. BDD scenarios are written using a domain-specific language (DSL) that can be translated into executable test cases using tools such as Cucumber or SpecFlow. BDD aims to improve communication, collaboration, and feedback among the team members, and to deliver software that meets the customer's needs and expectations. References: ISTQB Glossary of Testing Terms v4.0, ISTQB Foundation Level Syllabus v4.0, Section 3.1.1.4

, Requirements Engineering Fundamentals.

Question #:200

Test automation allows you to:

- A. demonstrate the absence of defects
- B. produce tests that are less subject to human errors
- C. avoid performing exploratory testing
- D. increase test process efficiency by facilitating management of defects

Answer: B

Explanation

Test automation allows you to produce tests that are less subject to human errors, as they can execute predefined test scripts or test cases with consistent inputs, outputs, and expected results. Test automation can also reduce the manual effort and time required to execute repetitive or tedious tests, such as regression tests, performance tests, or data-driven tests. Test automation does not demonstrate the absence of defects, as it can only verify the expected behavior of the system under test, not the unexpected or unknown behavior. Test

automation does not avoid performing exploratory testing, as exploratory testing is a valuable technique to discover new information, risks, or defects that are not covered by automated tests. Test automation does not increase test process efficiency by facilitating management of defects, as defect management is a separate activity that involves reporting, tracking, analyzing, and resolving defects, which may or may not be related to automated tests. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:

- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 3.3.1, Test Automation1
- ISTQB® Glossary of Testing Terms v4.0, Test Automation2

Question #:201

Which of the following statements refers to good testing practice to be applied regardless of the chosen software development model?

- A. Tests should be written in executable format before the code is written and should act as executable specifications that drive coding
- B. Test levels should be defined such that the exit criteria of one level are part of the entry criteria for the next level
- C. Test objectives should be the same for all test levels, although the number of tests designed at various levels can vary significantly
- D. Involvement of testers in work product reviews should occur as early as possible to take advantage of the early testing principle

Answer: D

Explanation

The statement that refers to good testing practice to be applied regardless of the chosen software development model is option D, which says that involvement of testers in work product reviews should occur as early as possible to take advantage of the early testing principle. Work product reviews are static testing techniques, in which the work products of the software development process, such as the requirements, the design, the code, the test cases, etc., are examined by one or more reviewers, with or without the author, to identify defects, violations, or improvements. Involvement of testers in work product reviews can provide various benefits for the testing process, such as improving the test quality, the test efficiency, and the test communication. The early testing principle states that testing activities should start as early as possible in the software development lifecycle, and should be performed iteratively and continuously throughout the lifecycle. Applying the early testing principle can help to prevent, detect, and remove defects at an early stage, when they are easier, cheaper, and faster to fix, as well as to reduce the risk, the cost, and the time of the testing process. The other options are not good testing practices to be applied regardless of the chosen software development model, but rather specific testing practices that may or may not be applicable or beneficial for testing, depending on the context and the objectives of the testing activities, such as:

- Tests should be written in executable format before the code is written and should act as executable specifications that drive coding: This is a specific testing practice that is associated with test-driven development, which is an approach to software development and testing, in which the developers write automated unit tests before writing the source code, and then refactor the code until the tests pass. Test-

driven development can help to improve the quality, the design, and the maintainability of the code, as well as to provide fast feedback and guidance for the developers. However, test-driven development is not a good testing practice to be applied regardless of the chosen software development model, as it may not be feasible, suitable, or effective for testing in some contexts or situations, such as when the requirements are unclear, unstable, or complex, when the test automation tools or skills are not available or adequate, when the testing objectives or levels are not aligned with the unit testing, etc.

- Test levels should be defined such that the exit criteria of one level are part of the entry criteria for the next level: This is a specific testing practice that is associated with sequential software development models, such as the waterfall model, the V-model, or the W-model, in which the software development and testing activities are performed in a linear and sequential order, with well-defined phases, deliverables, and dependencies. Test levels are the stages of testing that correspond to the levels of integration of the software system, such as component testing, integration testing, system testing, and acceptance testing. Test levels should have clear and measurable entry criteria and exit criteria, which are the conditions that must be met before starting or finishing a test level. In sequential software development models, the exit criteria of one test level are usually part of the entry criteria for the next test level, to ensure that the software system is ready and stable for the next level of testing. However, this is not a good testing practice to be applied regardless of the chosen software development model, as it may not be relevant, flexible, or efficient for testing in some contexts or situations, such as when the software development and testing activities are performed in an iterative and incremental order, with frequent changes, feedback, and adaptations, as in agile software development models, such as Scrum, Kanban, or XP, when the test levels are not clearly defined or distinguished, or when the test levels are performed in parallel or concurrently, etc.
- Test objectives should be the same for all test levels, although the number of tests designed at various levels can vary significantly: This is a specific testing practice that is associated with uniform software development models, such as the spiral model, the incremental model, or the prototyping model, in which the software development and testing activities are performed in a cyclical and repetitive manner, with similar phases, deliverables, and processes. Test objectives are the goals or the purposes of testing, which can vary depending on the test level, the test type, the test technique, the test environment, the test stakeholder, etc. Test objectives can be defined in terms of the test basis, the test coverage, the test quality, the test risk, the test cost, the test time, etc. Test objectives should be specific, measurable, achievable, relevant, and time-bound, and they should be aligned with the project objectives and the quality characteristics. In uniform software development models, the test objectives may be the same for all test levels, as the testing process is repeated for each cycle or iteration, with similar focus, scope, and perspective of testing. However, this is not a good testing practice to be applied regardless of the chosen software development model, as it may not be appropriate, realistic, or effective for testing in some contexts or situations, such as when the software development and testing activities are performed in a hierarchical and modular manner, with different phases, deliverables, and dependencies, as in sequential software development models, such as the waterfall model, the V-model, or the W-model, when the test objectives vary according to the test levels, such as component testing, integration testing, system testing, and acceptance testing, or when the test objectives change according to the feedback, the learning, or the adaptation of the testing process, as in agile software development models, such as Scrum, Kanban, or XP, etc. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.1.1, Testing and the Software Development Lifecycle1
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.2.1, Testing Principles1

- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.2.2, Testing Policies, Strategies, and Test Approaches1
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.3.1, Testing in Software Development Lifecycles1
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.1, Test Planning1
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.2, Test Monitoring and Control1
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.3, Test Analysis and Design1
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.4, Test Implementation1
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.5, Test Execution1
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.6, Test Closure1
- ISTQB® Glossary of Testing Terms v4.0, Work Product Review, Static Testing, Early Testing, Test-driven Development, Test Level, Entry Criterion, Exit Criterion, Test Objective, Test Basis, Test Coverage, Test Quality, Test Risk, Test Cost, Test Time2

Question #:202

Which of the following statements about the test pyramid is TRUE?

- A. Each layer of the test pyramid groups tests related to a single non-functional quality characteristic.
- B. The higher the layer of the test pyramid, the more production code a single automated test within the layer tends to cover
- C. The higher the layer of the test pyramid, the more maintainable a single automated test within the layer tends to be
- D. The higher the layer of the test pyramid, the more isolated a single automated test within the layer tends to be.

Answer: B

Explanation

The test pyramid concept suggests that there should be more low-level tests (unit tests) and fewer high-level tests (end-to-end tests).

- As we move higher up the pyramid (e.g., from unit tests to integration tests to end-to-end tests), each test covers more production code.
- Higher-level tests (like end-to-end) validate larger parts of the application, including multiple units and their interactions.

This aligns with the principle that higher-level tests provide broader coverage but are fewer in number and more expensive to run and maintain.

Reference: ISTQB CTFL Syllabus V4.0, Chapter 5.1.6, Test Pyramid.

Question #:203

For withdrawing money from an Automated Teller Machine (ATM), the following conditions are required:

- The bank card is valid
- The PIN code is correct
- Money is available in the user's account

The following are some possible interactions between the user and the ATM:

- The entered card is invalid The card is rejected
- The PIN code is wrong The ATM asks for another PIN code
- The requested amount is more than available in the user's account: The ATM asks for another amount
- The requested amount is available in the user's account The ATM dispenses the money

Which test design technique should be used to cover all possible combinations of the input conditions?

- A. Use case based testing
- B. Decision table
- C. Boundary value analysis
- D. Equivalence class partitioning

Answer: B

Explanation

A decision table is a technique that should be used to cover all possible combinations of input conditions for withdrawing money from an Automated Teller Machine (ATM). A decision table shows combinations of inputs and/or stimuli (causes) with their associated outputs and/or actions (effects). A decision table consists of four quadrants: conditions (inputs), actions (outputs), condition entries (values) and action entries (results). A decision table can be used to test components that have multiple inputs and outputs that depend on logical combinations of conditions. For example, for testing the ATM, we can identify three input conditions: the bank card is valid, the PIN code is correct, and money is available in the user's account. We can also identify four output actions: the card is rejected, the ATM asks for another PIN code, the ATM asks for another amount, and the ATM dispenses the money. A decision table can show all possible combinations of these conditions and actions in a systematic way.

Use case based testing is not a technique that can cover all possible combinations of input conditions for withdrawing money from an ATM. Use case based testing is a technique that verifies that a software product or system meets its specified requirements or user stories by executing realistic scenarios or workflows. Use case based testing can be used to test components that have complex or dynamic interactions with users or other systems. For example, for testing the ATM, we can identify several use cases, such as withdraw money, check balance, transfer money, etc. Each use case can have one or more scenarios that describe the steps and outcomes of the interaction. However, use case based testing may not cover all possible combinations of input conditions, as some scenarios may be omitted or overlooked.

Boundary value analysis is not a technique that can cover all possible combinations of input conditions for withdrawing money from an ATM. Boundary value analysis is a technique that tests boundary values between partitions of equivalent data. Boundary values are values at the edge of an equivalence partition or at the smallest incremental distance on either side of an edge. Boundary value analysis can be used to test components that have input values that can be divided into partitions of equivalent data. For example, for testing the ATM, we can identify boundary values for the input amount, such as the minimum and maximum amount allowed by the system or the user's account. However, boundary value analysis may not cover all possible combinations of input conditions, as some conditions may not have boundary values or may not be related to input values.

Equivalence class partitioning is not a technique that can cover all possible combinations of input conditions for withdrawing money from an ATM. Equivalence class partitioning is a technique that divides the input data and output results of a software component into partitions of equivalent data. Each partition should contain data that is treated in the same way by the component. Equivalence class partitioning can be used to test components that have input values that can be divided into partitions of equivalent data. For example, for testing the ATM, we can identify equivalence partitions for the input amount, such as valid amount (within the range allowed by the system and the user's account) and invalid amount (outside the range allowed by the system or the user's account). However, equivalence class partitioning may not cover all possible combinations of input conditions, as some conditions may not be related to input values or may have more than two partitions. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 4, page 34-46.

Question #:204

Which of the following is a typical potential risk of using test automation tools?

- A. Reduced feedback times regarding software quality compared to manual testing.
- B. Reduced test execution times compared to manual testing.
- C. Reduced repeatability and consistency of tests compared to manual testing
- D. Underestimation of effort required to maintain test scripts.

Answer: D

Explanation

One of the common risks associated with test automation tools is the underestimation of the effort required to maintain test scripts. Test scripts can become outdated or broken due to changes in the application, requiring significant effort to update and maintain them. This risk is highlighted in the ISTQB CTFL syllabus under the discussion of the benefits and risks of test automation.

References: ISTQB CTFL Syllabus, Section on test tools and automation.

Question #:205

Consider the following code

```
int premium=2500;

if (age<30)
{
    premium = premium +1500;
}
```

Which options suits for a correct combination of Boundary value and expected result. Assume first number as boundary followed by expected result.

- A. 29, 1500
30. 2500
- B. 29. 4000
30. 2500
- C. 29, 2500
30. 1500
- D. 30, 1500
31,2500

Answer: B

Explanation

In the given code snippet, the premium is increased by 1500 if the age is less than 30. Therefore, at the boundary value of age 29, the premium should be $2500 + 1500 = 4000$, and at age 30, the premium should remain at its initial value of 2500, as the condition is no longer met. Option B correctly reflects this with 29, 4000 and 30, 2500 as the boundary value and the expected results, respectively.

Question #:206

Which of the types of test tools noted below BEST describes tools that support reviews?

- A. Tools to assess data quality
- B. Tools to support usability testing
- C. Tools to support specialized testing needs
- D. Tools to support static testing

Answer: D

Explanation

Static testing refers to testing that doesn't involve executing code. It includes activities like reviews, inspections, and static analysis. Tools that support static testing help with activities such as analyzing source code, checking coding standards, and aiding in document reviews. These tools can automate or facilitate various aspects of static testing processes, such as highlighting potential issues in code or documents without executing the software.

References:

ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 6.1.1.

Question #:207

Why it is essential that defects found in a review be reported objectively?

- A. In order to facilitate easy entry of detected defects in a OTS (Defect Tracking System)
- B. In order to allow the author of reviewed work product(S) to take the feedback positively as an effort at improving the product (S) and not as a personal assault
- C. In order to allow the review moderator to easily understand them, and assign them to the right developer for fixing
- D. In order to allow augmentation of existing checklists used for reviewing the work product (S)

Answer: B

Explanation

The purpose of a review is to find defects and improve the quality of the work product, not to criticize or blame the author. Reporting defects objectively means describing them factually and constructively, without using negative or emotional language that could offend the author or damage their motivation. This way, the author can take the feedback positively as an effort at improving the product and not as a personal assault. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 138.

Question #:208

Which of the following statements is true?

- A. Functional testing focuses on what the system should do while non-functional testing on the internal structure of the system
- B. Non-functional testing includes testing of both technical and non-technical quality characteristics
- C. Testers who perform functional tests are generally expected to have more technical skills than testers who perform non-functional tests
- D. The test techniques that can be used to design white-box tests are described in the ISO/IEC 25010 standard

Answer: B

Explanation

Non-functional testing includes testing of both technical and non-technical quality characteristics. Non-functional testing is the process of testing the quality attributes of a system, such as performance, usability, security, reliability, etc. Non-functional testing can be applied at any test level and can use both black-box and white-box test techniques. Non-functional testing can cover both technical aspects, such as response time, throughput, resource consumption, etc., and non-technical aspects, such as user satisfaction, accessibility, compliance, etc. Therefore, option B is the correct answer.

: ISTQB® Certified Tester Foundation Level Syllabus v4.01, Section 1.3.1, page 13; ISTQB® Glossary v4.02, page 40.

Question #:209

Which statement is true regarding confirmation testing and regression testing?

- A. Confirmation testing confirms the quality of the test being run while regression testing ensures that the software still works after a change has been made.
- B. Confirmation testing is an optional activity whilst regression testing is not negotiable.
- C. Confirmation testing aims to verify that a defect has been resolved and regression testing ensuring that existing functionality still works after a change.
- D. Testers' involvement is essential whilst running retesting and regression testing.
- E. TESTER Involvement is essential whilst running retesting and regression testing.

Answer: C

Explanation

Confirmation testing, also known as retesting, is conducted to verify that specific defects have been fixed. Regression testing, on the other hand, is performed to ensure that recent changes have not adversely affected existing features of the software. Both types of testing are crucial for maintaining the integrity and quality of the software after modifications.

Question #:210

Which ONE of the following statements about acceptance testing is NOT correct?

- A. Testing of disaster recovery and backup/restore is usually NOT part of acceptance testing.
- B. The customers or system users are often responsible for the acceptance testing.
- C. The main goal of acceptance testing is to build confidence in the system, not find defects.
- D. Acceptance testing is the last level of testing performed prior to system release.

Answer: A

Explanation

Acceptance testing is a level of testing performed to verify that a software product meets the agreed acceptance criteria and is acceptable for delivery. Acceptance testing is often performed by the customers or system users, who are the main stakeholders of the software product. The main goal of acceptance testing is to build confidence in the system, not find defects, as defects should have been detected and fixed in earlier levels of testing. Acceptance testing is the last level of testing performed prior to system release, unless there are any changes or fixes that require re-testing. Testing of disaster recovery and backup/restore is usually part of acceptance testing, as these are important aspects of system reliability and security that affect the customer satisfaction and trust. Therefore, statement A is not correct, while statements B, C and D are correct. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 2, page 20-21.

Question #:211

Given the following review process main activities and specific review activities:

- a.Planning
 - b.Initiate review
 - c.Issue communication and analysis
 - d.Fixing and reporting
- 1.Creating defect reports
 - 2.Estimating effort and timeframe
 - 3.Recording updated status of defects

- 4. Selecting the people to participate
- 5. Distributing the work product and other material
- 6. Evaluating the review findings

Which of the following BEST matches the review process main activities with the appropriate specific review activities?

- A. 2-a, 4-a, 5-b, 6-c, 1-d, 3-d
- B. 2-a, 5-a, 1-b, 4-b, 3-c, 6-d
- C. 1-a, 4-b, 5-b, 6-c, 2-d, 3-d
- D. 2-a, 4-b, 5-c, 1-d, 3-d, 6-d

Answer: A

Explanation

Matching the main review process activities with the specific review activities, we see that planning includes estimating effort and timeframe (2) and selecting people to participate (4). Initiating a review involves distributing work products and other material (5). Issue communication and analysis includes evaluating the review findings (6). Fixing and reporting would entail creating defect reports (1) and recording the updated status of defects (3).

References: ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 3.2 "Review Process".

Question #:212

Which ONE of the following options BEST describes the **third test quadrant (Q3)**?

- A. **This test quadrant contains smoke tests and non-functional tests (except usability tests). These tests are often automated.**
- B. **This test quadrant is business-facing and criticizes the system. It contains exploratory testing, usability testing, and user acceptance testing.**
- C. **This test quadrant contains functional tests, examples, user story tests, user experience prototypes, API testing, and simulations. These tests check the acceptance criteria and can be manual or automated.**
- D. **This test quadrant is technology-facing and supports the team. These tests should be automated and included in the continuous integration (CI) process.**

Answer: B

Explanation

Comprehensive and Detailed In-Depth Explanation:

The **Agile Testing Quadrants** framework categorizes tests based on their **purpose and audience**:

- **Quadrant 1 (Q1):** Technology-facing tests (unit and component tests).
- **Quadrant 2 (Q2):** Business-facing tests supporting development (e.g., BDD tests).
- **Quadrant 3 (Q3): Business-facing tests that critique the system (B).**
 - Includes **usability testing, exploratory testing, and UAT** to ensure software meets user expectations.
- **Quadrant 4 (Q4):** Technology-facing tests that critique the system (e.g., performance, security testing).

Option B correctly defines Q3 since it focuses on **evaluating the user experience, exploring the system, and validating business expectations**.

Reference: ISTQB CTFL v4.0 Syllabus, Section 5.1.7 – Agile Testing Quadrants

Question #:213

Which of the following software development models BEST exemplifies a model that does NOT support the principle of early testing?

- A. The iterative development model
- B. The V-model
- C. The Waterfall model
- D. The incremental development model

Answer: C

Explanation

The Waterfall model exemplifies a software development model that does not support the principle of early testing. In the Waterfall model, each phase must be completed before the next begins, which delays testing until after the completion of the earlier phases like requirements gathering and design. This can often result in finding defects later in the development cycle, making them more expensive and time-consuming to fix (ISTQB not-for-profit association) (ISTQB not-for-profit association).

References:

ISTQB® Certified Tester Foundation Level Syllabus v4.0: https://istqb-main-web-prod.s3.amazonaws.com/media/documents/ISTQB_CTFL_Syllabus-v4.0.pdf

ISTQB News Release on CTFL v4.0: <https://www.istqb.org/news/posts/istqb-releases-certified-tester-foundation-level-v40-ctfl/>

Question #:214

Given the following requirement:

Requirement ID: 2 8

Requirement Description Additional Entrance Fee

Detailed Description

An additional fee of \$3 is charged during the weekend, but

- 1) Visitors aged under 7 are not charged.
- 2) Visitors aged 7 to 13 inclusive get a 20% discount off the additional fee.
- 3) Visitors aged greater than 65 get a 50% discount off the additional fee.

Age should be an integer of 0 or above.

Weekend means Friday to Sunday inclusive.

Which of the following statements is NOT correct?

- A. Thursday is a valid input boundary value.
- B. A minimum of 6 valid test cases are derived from boundary value analysis based on input age.
- C. \$3.01 is a valid output boundary value.
- D. 7 and 13 are boundary values for the equivalence partition including age 10.

Answer: A

Explanation

Boundary value analysis is a technique that tests boundary values between partitions of equivalent data. Boundary values are values at the edge of an equivalence partition or at the smallest incremental distance on either side of an edge. Boundary value analysis can be applied to both input and output values. Based on the given requirement, we can identify two input values: age and weekend. Age should be an integer of 0 or above, and weekend means Friday to Sunday inclusive. The following statement is not correct:

- A) Thursday is a valid input boundary value. This statement is not correct, as Thursday is not a boundary value for the input weekend. The boundary values for the input weekend are Friday and Sunday, as they are at the edge of the equivalence partition that represents weekend days. The following statements are correct:

- B) A minimum of 6 valid test cases are derived from boundary value analysis based on input age. This statement is correct, as we can derive six valid test cases based on input age by using the minimum and maximum values for each equivalence partition defined by the requirement. The equivalence partitions for input age are: under 7 (0 to 6), 7 to 13 inclusive (7 to 13), and greater than 65 (66 and above). The minimum and maximum values for each partition are: 0 and 6, 7 and 13, and 66 and any value above it.
- C) \$3.01 is a valid output boundary value. This statement is correct, as \$3.01 is a boundary value for the output additional fee. The additional fee can have four possible values depending on the input age: \$0 (for visitors aged under 7), \$2.40 (for visitors aged 7 to 13 inclusive with a 20% discount), \$1.50 (for visitors aged greater than 65 with a 50% discount), and \$3 (for visitors aged between 14 and 65). The boundary values for the output additional fee are \$0 and \$3, as they are at the edge of an equivalence partition or at the smallest incremental distance on either side of an edge. Therefore, \$3.01 is a valid output boundary value, as it is at the smallest incremental distance above \$3.
- D) 7 and 13 are boundary values for the equivalence partition including age 10. This statement is correct, as 7 and 13 are boundary values for the equivalence partition that represents visitors aged 7 to 13 inclusive. This partition includes age 10, which is an internal value within the partition. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 4, page 37-38.

Question #:215

Which of the following BEST explains a drawback of independent testing?

- A. Having the business organization participate as an independent test team can hurt the overall testing effort since business participants are often not trained nor experienced in testing
- B. Due to their differing backgrounds and perspectives, an independent test team may discover defects which the developers did not uncover
- C. An independent test team may be isolated from the rest of the development and project team
- D. An independent test team may possess specializations in specific test types such as usability or security which detract from the overall effectiveness of the test team

Answer: C

Explanation

Independent testing offers several advantages, such as unbiased testing and detection of different defects. However, a drawback is that an independent test team may be isolated from the development team and project team. This can lead to communication gaps, reduced collaboration, and a lack of understanding of the project context.

According to the ISTQB Certified Tester Foundation Level (CTFL) syllabus v4.0, an independent test team may not have the same level of understanding of the system as the development team, leading to potential issues in communication and integration (ISTQB not-for-profit association).

References:

Certified Tester Foundation Level v4.0

ISTQB Foundation Level Syllabus 4.0 (2023)

Question #:216

A test status report SHOULD:

- A. Specify the impediments to carrying out the planned test activities in the reporting period and the corresponding solutions put in place to remove them
- B. Be produced as part of test completion activities and report unmitigated product risks to support the decision whether or not to release the product
- C. Always be based on the same template within an organisation, as its structure and contents should not be affected by the audience to which the report is presented.
- D. Specify the lines of communication between testing, other lifecycle activities, and within the organisation that were chosen at the outset of the test project.

Answer: A

Explanation

A test status report is a document that provides a snapshot of the testing activities and their progress during a particular period. It should include information about any impediments encountered during the test execution and the actions taken to resolve them, which helps stakeholders understand the challenges and how they were addressed .

Option B describes an activity related to test completion rather than ongoing status reporting. Option C is incorrect because the structure and contents of the report may vary based on the audience's needs. Option D, while important, is not the primary purpose of a test status report, which focuses more on the current status and impediments.

Question #:217

Which ONE of the following roles is responsible for mediating, managing time, and generating a safe review environment in which everyone can speak freely?

- A. Manager
- B. Reviewer
- C. Review leader
- D. Moderator

Answer: D

Explanation

Comprehensive and Detailed In-Depth Explanation:

The **Moderator (D)** plays a key role in **formal reviews** by facilitating discussions, ensuring that reviews stay on track, and creating an open environment for feedback. The **Manager (A)** is responsible for overseeing the testing process but does not directly facilitate reviews. The **Reviewer (B)** is responsible for examining the work product, while the **Review Leader (C)** organizes reviews but does not necessarily mediate them.

Reference: ISTQB CTFL v4.0 Syllabus, Section 3.2.3 – Roles and Responsibilities in a Review

Question #:218

Which is an advantage of using testers that are independent from the development team?

- A. Independent testers support the developers with their feedback on coding rules and architecture design.
- B. Developers can delegate their sense of responsibility for quality to testers.
- C. Independent testers have a different perspective and can find defects that developers miss.
- D. Independent testers will identify themselves more with the project and the project goals and have good relations with the development team.

Answer: C

Question #:219

Which of the following statements about testing in the context of an agile (iterative-incremental) development model is correct?

- A. Unit test and acceptance test are the most important tests to make sure that the system works as expected.
- B. Each iteration of testing has to be completely finished before a new Iteration of development starts.
- C. Regression testing is necessary whenever a new Increment is added to the existing system. D. Only certain types of non-functional and explorative testing are performed.

Answer: C

Explanation

In the context of agile (iterative-incremental) development models, testing is integrated into the development process and occurs continuously throughout the lifecycle of the project. Agile testing emphasizes adaptability and the need for feedback at various stages of development.

Option C is correct because regression testing is indeed necessary whenever a new increment is added to the existing system. Agile development often involves frequent changes and additions to the codebase, which can potentially introduce new defects into previously tested code. Regression testing ensures that new changes have not adversely affected existing functionality.

Options A, B, and D present misconceptions about agile testing:

- A is incorrect because, in agile, all types of testing (unit, integration, system, acceptance) are important and occur throughout the iteration, not just unit and acceptance tests.
- B is incorrect because agile methodologies advocate for continuous integration and testing, where development and testing activities overlap and support each other throughout an iteration.
- D is incorrect because agile methodologies encourage a wide range of testing types, including both functional and non-functional, as well as exploratory testing, to ensure a comprehensive quality assessment.

Question #:220

In order to exercise the right to vote, a person must meet the following requirements:

- be able to prove citizenship of the country
- be a resident in the administrative unit concerned
- have a minimum age of 18 years
- no other reasons for exclusion exist

Create the associated decision table and minimize it so that all conditions are correctly covered.

Hint: remember to use the "no other reasons for exclusion exist" condition when creating the decision table.

How many test cases are at least required to achieve the coverage of all decision rules and what could be the concrete test cases?

- A. 5 test cases. 1 test case where all 4 prerequisites are fulfilled and 4 test cases for exactly one unfulfilled prerequisite each.
- B. 16 test cases. With 4 conditions, each of which can be true or false, there are 16 (2 to the power of 4) possible combinations to be tested.
- C. 2 test cases. 1 test case where all 4 conditions are fulfilled and 1 test case where all 4 conditions are not fulfilled.
- D. 8 test cases. With 4 conditions, each of which can be true or false, there are 8 (4×2) possible combinations to be tested.

Answer: A

Question #:221

Which ONE of the following options explains a benefit of independent testing the BEST?

- A. The testers can be isolated from the development team and thus avoid acquiring the same bias as the developers.
- B. Independent testers may lack information regarding the test object.
- C. Independent testers are likely to recognize different types of failures compared to developers.
- D. Developers may lose a sense of responsibility for the quality of the product they develop.

Answer: C

Explanation**Comprehensive and Detailed In-Depth Explanation:**

Independent testers provide a fresh perspective and are more likely to identify failures that developers might overlook due to their familiarity with the software (C). Independent testing helps avoid cognitive biases, improves defect detection, and enhances the overall quality assurance process. While A and D touch on related concepts, they do not directly define the benefit as well as C does. Option B highlights a potential challenge rather than a benefit.

Reference: ISTQB CTFL v4.0 Syllabus, Section 1.5.3 – The Independence of Testing

Question #:222

Which of the following statements about the value of maintaining traceability between the test basis and test work products is not true?

- A. Traceability can be useful for assessing the impact of a change to a test basis item on the corresponding tests
- B. Traceability can be useful for determining how many test basis items are covered by the corresponding tests
- C. Traceability can be useful for determining the most suitable test techniques to be used in a testing project
- D. Traceability can be useful to support the needs required by the auditing of testing

Answer: C

Explanation

Traceability is the ability to trace the relationships between the items of the test basis, such as the requirements, the design, the risks, etc., and the test artifacts, such as the test cases, the test results, the defects, etc. Traceability can provide various benefits for the testing process, such as improving the test coverage, the test quality, the test efficiency, and the test communication. However, not all the statements given are true about the value of maintaining traceability between the test basis and test work products. The statement that is not true is option C, which says that test objectives should be the same for all test levels, although the number of tests designed at various levels can vary significantly. This statement is false, because test objectives are the goals or the purposes of testing, which can vary depending on the test level, the test type, the test technique, the test environment, the test stakeholder, etc. Test objectives can be defined in terms of the test basis, the test coverage, the test quality, the test risk, the test cost, the test time, etc. Test objectives should be specific, measurable, achievable, relevant, and time-bound, and they should be aligned with the project objectives and the quality characteristics. Test objectives should not be the same for all test levels, as different test levels have different focuses, scopes, and perspectives of testing, such as component testing, integration testing, system testing, and acceptance testing. The other statements are true about the value of maintaining traceability between the test basis and test work products, such as:

- Traceability can be useful for assessing the impact of a change to a test basis item on the corresponding tests: This statement is true, because traceability can help to identify which tests are affected by a change in the test basis, such as a new requirement, a modified design, a revised risk, etc., and to determine the necessary actions to update, re-execute, or re-evaluate the tests. Traceability can also help to estimate the effort, the cost, and the time needed to implement the change and to verify its impact on the software system.
- Traceability can be useful for determining how many test basis items are covered by the corresponding tests: This statement is true, because traceability can help to measure the test coverage, which is the degree to which the test basis is exercised by the test cases. Traceability can help to identify which test basis items are covered, partially covered, or not covered by the tests, and to evaluate the adequacy, the completeness, and the effectiveness of the testing process. Traceability can also help to identify the gaps, the overlaps, or the redundancies in the test coverage, and to prioritize, optimize, or improve the test cases.
- Traceability can be useful to support the needs required by the auditing of testing: This statement is true, because traceability can help to provide evidence, documentation, and justification for the testing activities, results, and outcomes. Traceability can help to demonstrate that the testing process follows the standards, the regulations, the policies, and the best practices that are applicable to the software system, the project, or the organization. Traceability can also help to verify that the testing process meets the expectations, the needs, and the satisfaction of the users and the stakeholders. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 1.2.2, Testing Policies, Strategies, and Test Approaches¹
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.1, Test Planning¹
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.2, Test Monitoring and Control¹
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.3, Test Analysis and Design¹
- ISTQB® Glossary of Testing Terms v4.0, Traceability, Test Basis, Test Artifact, Test Objective, Test Level, Test Coverage, Test Quality, Test Risk, Test Cost, Test Time²

Question #:223

Which ONE of the following statements does NOT describe how testing contributes to higher quality?

- A. Properly designed tests that pass reduce the level of risk in a system.
- B. The testing of software demonstrates the absence of defects.
- C. Software testing identifies defects, which can be used to improve development activities.
- D. Performing a review of the requirement specifications before implementing the system can enhance quality.

Answer: B**Explanation**

- The testing of software does not demonstrate the absence of defects, but rather the presence of defects or the conformance of the software to the specified requirements¹. Testing can never prove that the software is defect-free, as it is impossible to test all possible scenarios, inputs, outputs, and behaviors of the software². Testing can only provide a level of confidence in the quality of the software, based on the coverage, effectiveness, and efficiency of the testing activities³.
- The other options are correct because:
 - A. Properly designed tests that pass reduce the level of risk in a system, as they verify that the system meets the expected quality attributes and satisfies the needs and expectations of the users and clients⁴. Risk is the potential for loss or harm due to the occurrence of an undesirable event⁵. Testing can help to identify, analyze, prioritize, and mitigate the risks associated with the software product and project⁶.
 - C. Software testing identifies defects, which can be used to improve development activities, as they provide feedback on the quality of the software and the effectiveness of the development processes⁷. Defects are flaws or errors in the software that cause it to deviate from the expected or required results or behavior. Testing can help to detect, report, track, and resolve the defects, and prevent them from recurring in the future.
 - D. Performing a review of the requirement specifications before implementing the system can enhance quality, as it can ensure that the requirements are clear, complete, consistent, testable, and aligned with the needs and expectations of the users and clients. Requirements are the specifications of what the software should do and how it should do it. Testing can help to validate that the requirements are met by the software, and verify that the software is implemented according to the requirements.

References =

- 1 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 10
- 2 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 11

- 3 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 12
- 4 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 13
- 5 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 97
- 6 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 98
- 7 ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 14
- [8] ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 15
- [9] ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 16
- [10] ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 17
- [11] ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 18
- [12] ISTQB® Certified Tester Foundation Level Syllabus v4.0, 2023, p. 19

Question #:224

A Static analysis tool analyzes a given program's CONTROL FLOW among other things. Which of the following options represents the most likely outcome of the control flow analysis:

- A. Identification of unreachable code
- B. Report on adherence to the coding standards
- C. Number of comment lines
- D. Number of source code lines

Answer: A

Explanation

A static analysis tool is a tool that analyzes a given program's source code or executable code without executing it. A static analysis tool can perform various types of analysis on a program's code, such as syntax checking, data flow analysis, control flow analysis, complexity measurement, coding standards compliance checking, etc. Control flow analysis is a type of analysis that examines how a program's statements are executed in different paths or branches. One of the most likely outcomes of control flow analysis is identification of unreachable code, which is code that can never be executed due to logical errors or design flaws. Unreachable code can reduce readability and maintainability of the code, as well as increase complexity and size. The other options are not outcomes of control flow analysis, but rather outcomes of other types of analysis. Report on adherence to coding standards is an outcome of coding standards compliance checking. Number of comment lines and number of source code lines are outcomes of complexity measurement. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 8.

Question #:225

Which activity is NOT documented in a test plan?

- A. Defining the quality assurance approach
- B. Defining the test approach
- C. Selecting the metrics for test monitoring
- D. Setting the budget for the test activities

Answer: D

Question #:226

In a review, which of the following is the responsibility of the manager?

- A. Performing detailed review based on past experience
- B. Ensuring that sufficient time is available for review
- C. Measuring the performance of each reviewer
- D. Organizing the logistics of the review meeting

Answer: D

Question #:227

Given the following user story for the development of an online shoe store:

"As a customer, I want to be able to filter shoes by color so that I see, for example, only red shoes in the selection."

Which of the following positive test cases BEST fits to the user story?

- A. Precondition: Home page of the shoe store is opened

Test steps to be performed:

Tick "red" in the color selection.

Click on the "Filter" button.

Expected result: Only red shoes are displayed.

- B. Precondition: Home page of the shoe store is opened

Test steps to be performed:

Tick "red" in the color selection.

Click on the "Filter" button.

Expected result: Within 2 seconds all red shoes are displayed, then the other colors.

C. Precondition: Homepage of shoe store with shoes size 40 is opened

Test steps to be performed:

Tick "red" in the color selection.

Click on the "Filter" button.

Expected result: Only red shoes of size 40 are displayed.

D. Precondition: Home page of the shoe store is opened

Test steps to be performed:

User logs in with his customer ID.

Tick "green" in the color selection.

Click on the "Filter" button.

Expected result: Only red shoes are displayed.

Answer: A

Question #:228

Which of the following examples represents System Integration testing?

- A. Testing an application in development environment which is similar to the production environment
- B. Testing the system based on specifications using simulator for a sub-system
- C. Testing an application's behaviour to check if it fulfils business needs
- D. Testing if e-commerce website works correctly with a third-party payment gateway

Answer: D

Question #:229

Which of the following statements is NOT an objective of testing?

- A. Create confidence in the quality of the test object.
- B. Check whether the test object fulfils the specified requirements.
- C. Provide sufficient information for stakeholders to make decisions.
- D. Fix defects after they have been found and analyzed.

Answer: C

Question #:230

Which of the following is a good testing practice which is applicable INDEPENDENT of the software development lifecycle followed?

- A. Not all development activities should have corresponding test activities
- B. Each test level should have an appropriate test objective
- C. Large amount of redundancy between test levels is good because it prevents bug leakage
- D. Reviews should be done after the work products have been finalized

Answer: B

Question #:231

Which ONE of the following work products TYPICALLY belongs to test execution?

- A. Test logs that document the results of test execution.
- B. Automated test scripts used for test execution.
- C. A test plan that describes the test strategy and test objectives.
- D. A list of test conditions prioritized during test analysis.

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

Test execution involves running test cases and documenting results. Test logs (A) provide evidence of executed tests, failures, and actual outcomes. Automated test scripts (B) are part of test implementation, test plans (C) belong to test planning, and test conditions (D) are identified during test analysis.

Reference: ISTQB CTFL v4.0 Syllabus, Section 1.4.1 – Test Execution

Question #:232

Which of the following BEST describes a benefit of test automation?

- A. More subjective assessment
- B. Reduction in repetitive manual work
- C. Availability of the test automation tool vendor
- D. Negligible effort to maintain the test assets generated by the tool

Answer: B

Explanation

Test automation provides numerous benefits to software testing, and one of the key advantages is the reduction of repetitive manual work. This benefit is explicitly covered in the ISTQB Foundation Level Syllabus (v4.0).

Test automation allows testers to automate repetitive tasks such as regression testing, freeing up their time to focus on more complex and exploratory testing. This leads to improved efficiency and helps in avoiding human errors associated with repetitive tasks.

Option A: "More subjective assessment" contradicts the benefit of automation as it focuses on objectivity.

Option C: "Availability of the test automation tool vendor" is not a direct benefit of test automation, although vendor support can be valuable.

Option D: "Negligible effort to maintain the test assets" is misleading as maintaining automated tests often requires effort and attention to changes in the system under test.

Therefore, the correct answer is B (ISTQB not-for-profit association) (ISTQB).

References:

Certified Tester Foundation Level v4.0

ISTQB Foundation Level Syllabus 4.0 (2023)

Question #:233

A test manager decided to skip static testing since he believes bugs can be found easily by doing dynamic testing. Was this decision right or wrong?

- A. The decision was wrong. Ensuring quality mandates that static testing is performed after performing the dynamic testing.

- B. The decision was right. Static testing is usually redundant if a product is planned to go through a full-cycle of dynamic testing.
- C. The decision was right. Most of the bugs are easier to identify during the dynamic testing.
- D. The decision was wrong. Static testing can find defects early in the development process, reducing the overall cost of testing and development

Answer: D

Explanation

Static testing is a form of testing that does not involve executing the software or system under test. It includes activities such as reviews, inspections, walkthroughs, and analysis of documents, code, and models. Static testing can find defects early in the development process, before they become more expensive and difficult to fix in later stages. Static testing can also improve the quality of the software or system by preventing defects from being introduced in the first place. Static testing can complement dynamic testing, which involves executing the software or system under test and checking the results against expected outcomes. Dynamic testing can find defects that static testing may miss, such as performance, usability, or integration issues. However, dynamic testing alone is not sufficient to ensure quality, as it may not cover all possible scenarios, inputs, or paths. Therefore, a test manager who decides to skip static testing is making a wrong decision, as he or she is ignoring the benefits of static testing and relying solely on dynamic testing, which may not be effective or efficient enough to find and prevent defects. References = ISTQB Certified Tester Foundation Level Syllabus, Version 4.0, 2018, Section 2.1.1, page 14; ISTQB Glossary of Testing Terms, Version 4.0, 2018, page 36; ISTQB CTFL 4.0 - Sample Exam - Answers, Version 1.1, 2023, Question 3, page 9.

Question #:234

Which of the following statements about the shift-left approach is FALSE?

- A. The shift-left approach can only be implemented with test automation
- B. The shift-left approach in testing is compatible with DevOps practices.
- C. The shift-left approach can involve security vulnerabilities
- D. The shift-left approach can be supported by static analysis tools.

Answer: A

Explanation

In a formal review process, the recorder's role is typically responsible for documenting the findings of the review team, including action items, decisions, and recommendations. This ensures that there is an accurate record of what was discussed and agreed upon, facilitating follow-up and continuous improvement. Therefore, statement C is correct as per the ISTQB CTFL syllabus.

Question #:235

Which two of the following statements describe the advantages provided by good traceability between the test basis and test work products?

- i. Analyzing the impact of changes.
- ii. A measure of code quality.
- iii. Accurate test estimation.
- iv. Making testing auditable.

Select the correct answer:

- A. i and ii
- B. i and iv
- C. i and iii
- D. ii and iii

Answer: B

Explanation

Good traceability between the test basis and test work products provides several advantages: i. Analyzing the impact of changes: Traceability allows for easy identification of which parts of the test work products will be affected by changes in the requirements or design, facilitating impact analysis. iv. Making testing auditable: Traceability ensures that there is a clear connection between the requirements and the test cases, which makes the testing process auditable and provides evidence that all requirements have been tested.

Question #:236

Which of the following is an example of scenario-oriented acceptance criteria?

- A. The user should be able to provide three inputs to test the product - the AI model to be tested, the data used and an optional text file
- B. Verify that a registered user can create add a new project with name having more than 100 characters
- C. A user is already logged in then on navigating to the AI model testing page the user should be directly shown the report of last test run.
- D. An unregistered user shouldn't be shown any report

Answer: C

Question #:237

Exploratory testing is an experience-based test technique

- A. Where a developer and a tester work together on the same workstation while the developer actively writes code, the tester explores the code to find defects.
- B. That can be organised into sessions guided by test charters outlining test objectives that will guide the testers' exploration
- C. Where a team of testers explores all possible test techniques in order to determine the most suitable combination of these techniques to apply for a test project.
- D. That aims at finding defects by designing tests that exercise all possible combinations of input values and preconditions

Answer: B

Explanation

Exploratory testing is an experience-based test technique where testers actively engage with the software, learning about its behavior while simultaneously designing and executing tests. According to the ISTQB CTFL syllabus, exploratory testing can be structured into sessions guided by test charters, which outline the test objectives and provide direction for the testers' exploration. This method is particularly useful in situations where test documentation is limited or where rapid feedback is needed. Thus, option B correctly describes how exploratory testing can be organized.

Question #:238

A test score indicator for students produces a performance score based on a combination of the number of consecutive hours studied (below 4 hours, 4 to 8 hours, 9 to 12 hours or above 12 hours) and the average intensity of focus on the material during the study time (low, medium or high).

Given the following test cases:

hours intensity score

T1 3 low55

T2 14 high 95

T3 9 low75

What is the minimum number of additional test cases that are needed to ensure full coverage of all valid INPUT equivalence partitions?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: C**Explanation**

Considering the various valid input equivalence partitions of hours studied and intensity, three additional test cases are needed to fully cover all valid partitions. This would typically include testing combinations that vary both the number of hours and the intensity levels not covered by the initial test cases (ISTQB Main Web).

References:

ISTQB® Certified Tester Foundation Level Syllabus v4.0: ISTQB CTFL Syllabus v4.0 PDF

Question #:239

Which of the following applications will be the MOST suitable for testing by Use Cases

- A. Accuracy and usability of a new Navigation system compared with previous system
- B. A billing system used to calculate monthly charge based on large number of subscribers parameters
- C. The ability of an Anti virus package to detect and quarantine a new threat
- D. Suitability and performance of a Multi media (audio video based) system to a new operating system

Answer: A**Explanation**

A new navigation system compared with a previous system is the most suitable application for testing by use cases, because it involves a high level of interaction between the user and the system, and the expected behavior and outcomes of the system are based on the user's needs and goals. Use cases can help to specify the functional requirements of the new navigation system, such as the ability to enter a destination, select a route, follow the directions, receive alerts, etc. Use cases can also help to compare the accuracy and usability of the new system with the previous system, by defining the success and failure scenarios, the preconditions and postconditions, and the alternative flows of each use case. Use cases can also help to design and execute test cases that cover the main and exceptional paths of each use case, and to verify the satisfaction of the user's expectations.

The other options are not the most suitable applications for testing by use cases, because they do not involve a high level of interaction between the user and the system, or the expected behavior and outcomes of the system are not based on the user's needs and goals. A billing system used to calculate monthly charge based on a large number of subscriber parameters is more suitable for testing by data-driven testing, which is a technique for testing the functionality and performance of a system or component by using a large set of input and output data. The ability of an antivirus package to detect and quarantine a new threat is more suitable for testing by exploratory testing, which is a technique for testing the functionality and security of a system or component by using an informal and flexible approach, based on the tester's experience and intuition. The suitability and performance of a multimedia (audio video based) system to a new operating system is more

suitable for testing by compatibility testing, which is a technique for testing the functionality and performance of a system or component by using different hardware, software, or network environments. **References** = CTFL 4.0 Syllabus, Section 3.1.1, page 28-29; Section 4.1.1, page 44-45; Section 4.2.1, page 47-48.

Question #240

Which of the following BEST describes a test summary report for executive-level employees

- A. The report is detailed and includes a status summary of defects by priority or budget
- B. The report is detailed and includes specific information on defects and trends
- C. The report is high-level and includes a status summary of defects by priority or budget
- D. The report is high-level and includes specific information on defects and trends

Answer: C

Explanation

For executive-level employees, a test summary report should be concise and focus on high-level information. It typically includes a summary of defects categorized by priority or budget. Executives are generally interested in the overall status and the impact on critical business objectives rather than detailed technical information. The report should provide an overview of the most important aspects of testing, such as key issues, test progress, and any risks or concerns that could affect project outcomes.

References:

ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 5.3.1.

Question #241

In maintenance testing, what is the relationship between impact analysis and regression testing?

- A. Impact analysis requires a regression testing for only the tests that have detected faults in previous SW release
- B. There is no relationship between impact analysis and regression testing.
- C. Impact analysis requires a regression testing for all program elements which were newly integrated (new functionalities).
- D. The impact analysis is used to evaluate the amount of regression testing to be performed.

Answer: D

Explanation

In maintenance testing, the relationship between impact analysis and regression testing is that the impact analysis is used to evaluate the amount of regression testing to be performed. Maintenance testing is a type of testing that is performed on an existing software product after it has been delivered or deployed, in order to ensure that it still meets its requirements and functions correctly after a change or a modification.

Maintenance testing can be triggered by various reasons, such as corrective maintenance (fixing defects), adaptive maintenance (adapting to new environments), perfective maintenance (improving performance), preventive maintenance (avoiding future problems), etc. Impact analysis is a technique that is used to assess the extent and nature of changes introduced by maintenance activities on the software product or project. Impact analysis helps to identify which parts of the software product are affected by the changes, which parts need to be modified or updated accordingly, which parts need to be retested or verified for correctness or compatibility, etc. Regression testing is a type of testing that verifies that previously tested software still performs correctly after a change or a modification. Regression testing helps to detect any side effects or unintended consequences of maintenance activities on the software product's functionality or quality.

Regression testing can be performed at various levels and scopes depending on the impact analysis results. Therefore, in maintenance testing, impact analysis is used to evaluate the amount of regression testing to be performed. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 20.

Question #:242

The fact that defects are usually not evenly distributed among the various modules that make up a software application, but rather their distribution tend to reflect the Pareto principle:

- A. is a false myth
- B. is expressed by the testing principle referred to as 'Tests wear out'
- C. is expressed by the testing principle referred to as 'Defects cluster together'
- D. is expressed by the testing principle referred to as 'Bug prediction'

Answer: C

Explanation

The fact that defects are usually not evenly distributed among the various modules that make up a software application, but rather their distribution tend to reflect the Pareto principle, is expressed by the testing principle referred to as 'Defects cluster together'. This principle states that a small number of modules contain most of the defects detected, or that a small number of causes are responsible for most of the defects. This principle can be used to guide the test analysis and design activities, by prioritizing the testing of the most critical or risky modules, or by applying more rigorous test techniques to them. Therefore, option C is the correct answer.

: ISTQB® Certified Tester Foundation Level Syllabus v4.01, Section 1.2.1, page 11; ISTQB® Glossary v4.02, page 16.

Question #:243

Consider the following defect report for an Exam Booking System

Defect ID: ST1041 | Title: Unable to cancel an exam booking | Severity: Very High | Priority: Very High | Environment: Windows 10, Firefox

Description: When attempting to cancel an exam booking using the cancel exam feature, the system does not cancel the exam even though it shows a message that the exam has been cancelled.

Which ONE of the following indicates the information that should be added to the description for reproducing the defect easily?

- A. Providing better severity and priority. It is not clear why this is a high severity problem as the exams can be booked without any problem.
- B. Providing exact steps that lead to this defect. It is not clear from the description if this is a problem for any scheduled exam or for a specific user.
- C. The defect Id used is alphanumeric in nature. It should be a number only.
- D. Repeating the test case with different browsers and logging a separate defect for each one of them

Answer: B

Question #:244

What does configuration management enable for testing?

- A. It enables the scheduling of test resources.
- B. It enables the derivation of test cases and test data.
- C. It enables the creation of test progress reports.
- D. It enables the versioning of the test elements.

Answer: D

Question #:245

Confirmation testing is performed after:

- A. a defect is fixed and after other tests do not find any side-effect introduced in the software as a result of such fix
- B. a failed test, and aims to run that test again to confirm that the same behavior still occurs and thus appears to be reproducible

- C. the execution of an automated regression test suite to confirm the absence of false positives in the test results
- D. a defect is fixed, and if such testing is successful then the regression tests that are relevant for such fix can be executed

Answer: D

Explanation

Confirmation testing is performed after a defect is fixed, and if such testing is successful then the regression tests that are relevant for such fix can be executed. Confirmation testing, also known as re-testing, is the process of verifying that a defect has been resolved by running the test case that originally detected the defect. Confirmation testing is usually done before regression testing, which is the process of verifying that no new defects have been introduced in the software as a result of changes or fixes. Therefore, option D is the correct answer.

: ISTQB® Certified Tester Foundation Level Syllabus v4.01, Section 2.4.1, page 28; ISTQB® Glossary v4.02, page 15.

Question #:246

Which of the following best describes the relationship between a test progress report and a test summary report?

- A. The test report prepared during a test activity may be referred to as a test progress report, while a test report prepared at the end of a test activity may be referred to as a test summary report.
- B. The test report prepared during a test activity may be referred to as a test summary report, while a test report prepared at the end of a test activity may be referred to as a test progress report.
- C. There is no difference between a test progress report and a test summary report.
- D. Both the test progress report and the test summary report should always be generated via an automated tool.

Answer: A

Explanation

A test progress report is prepared during a test activity and provides updates on the current status of testing activities, including progress made, issues encountered, and test metrics. A test summary report, on the other hand, is prepared at the end of a test activity and provides a comprehensive overview of the testing carried out, including the final status, test results, defect summaries, and overall conclusions.

Reference: ISTQB CTFL Syllabus V4.0, Section 5.3.2

Question #:247

Which of the following project scenario gives the BEST example where maintenance testing should be triggered?

- A. Completion of architecture of the bank system
- B. Release of the early draft of the low level project design of an IoT application
- C. Defect was found in a pre-released version of the customer service application
- D. Delivery of the hot fix to mobile operating system and ensuring that it still works

Answer: D

Explanation

Maintenance testing is triggered by changes such as bug fixes, enhancements, or environmental changes.

Option A: "Completion of architecture of the bank system" is not a typical scenario for maintenance testing, as it describes a design phase rather than an operational change.

Option B: "Release of the early draft of the low level project design of an IoT application" is again not suitable for maintenance testing, as it refers to the design phase.

Option C: "Defect was found in a pre-released version of the customer service application" is closer but not quite accurate, as maintenance testing focuses on changes made (ISTQB not-for-profit association) system is released.

Option D: "Delivery of the hot fix to mobile operating system and ensuring that it still works" is the best example as it directly involves testing after a fix has been implemented.

Therefore, the correct answer is D.

References:

Certified Tester Foundation Level v4.0

ISTQB Foundation Level Syllabus 4.0 (2023)

Question #:248

Which of the following BEST describes exploratory testing?

- A. Exploratory testing is a suitable test technique which may replace both black-box and white-box test techniques
- B. Exploratory testing is a valid and useful black-box test technique since it focuses on test cases related to the architecture and design of a system
- C. Exploratory testing requires both solid specifications and much project time available for test execution

- D. Exploratory testing may be used within defined time periods, during which the tester may follow a test charter as a guide

Answer: D

Explanation

Exploratory testing involves simultaneous test design and execution and is guided by a test charter, which outlines what needs to be tested, how it should be tested, and what to look for. This technique is typically conducted within predefined time periods, known as time-boxing, which allows testers to explore a system, understand its functionalities, and identify potential issues without detailed documentation or prior test case planning. The key aspects of exploratory testing include flexibility, adaptability, and the ability to respond to system behavior during testing.

References:

ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 4.4.2.

Question #:249

A financial institution is to implement a system that calculates the interest rates paid on investment accounts based on the sum invested.

You are responsible for testing the system and decide to use equivalence partitioning and boundary value analysis to design test cases. The requirements describe the following expectations:

Investment range| Interest rate

R500 to RIO 000|10%

RIO 001 to R50 000|11%

R50 001 to RIOOOOO|12%

RIOOOO| to R500 000| 13%

What is the minimum number of test cases required to cover all valid equivalence partitions for calculating the interest?

- A. 5
- B. 4
- C. 8
- D. 16

Answer: B

Explanation

Using equivalence partitioning, the investment ranges are divided into four partitions:

- R500 to R10,000 (10%)
- R10,001 to R50,000 (11%)
- R50,001 to R100,000 (12%)
- R100,001 to R500,000 (13%)

Thus, the minimum number of test cases required to cover all valid equivalence partitions for calculating the interest is 4.

Question #:250

The following table shows six test cases TC01 to TC06 with their priorities and dependencies.

test case-ID	priority	Test case depends on
TC01	very important	TC02, TC03
TC02	very important	-
TC03	important	TC06
TC04	important	TC05
TC05	rather unimportant	-
TC06	rather unimportant	-

Schedule the test execution of the specified test case set according to the priorities and dependencies. Since the test time is very tight, the number of test cases should be minimized as much as possible, but all very important test cases must be included.

Which of the following statements is correct?

- A. TC04, TC05, and TC06 can be deleted.
- B. TC05 and TC06 can be deleted.
- C. TC04 and TC05 can be deleted.
- D. TC03, TC04 and TC05 can be deleted.

Answer: B

Question #:251

A software system checks age in order to determine which welcome screen to display. Age groups are:

Group I: 0-12

Group II: 13-18

Group III: over 18

Which of the below represent boundary values?

- A. (-1,0,12,13,18,19)
- B. (-1,0,11,12,13,14,18,19)
- C. (0,12,13,18,19)
- D. (4,5,15,20)

Answer: A

Explanation

A correct list of boundary values for the age input should include the minimum and maximum values of each age group (0, 12, 13, 18), as well as the values just below and above each boundary (-1, 19). Boundary value analysis is a test design technique that involves testing the values at or near the boundaries of an input domain or output range, as these values are more likely to cause errors than values in the middle. Option A satisfies this condition, as it has all six boundary values (-1, 0, 12, 13, 18, 19). Option B has two values from the same equivalence class (12 and 13), option C has only four boundary values (0, 12, 18, 19), and option D has no boundary values at all. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 34.

Question #:252

Which of the following BEST matches the descriptions with the different categories of test techniques?

1. Test cases are based on the test basis which may include the requirements, use cases and user stories
2. Test cases are based on the test basis which may include the software architecture or code
3. Test cases can show deviations from the requirements
4. These test techniques are applicable to both functional and non-functional testing
5. Tests are based on knowledge of developers, users and other stakeholders

Black - Black-box test techniques

White * White-box test techniques

Experience - Experience-based test techniques

- A. Black -1,3,4 White - 2 Experience - 5
- B. Black - 2, 3 White -1 Experience - 4, 5
- C. Black -1,2 White - 3, 4 Experience - 5
- D. Black - 2, 3 White -1,5 Experience - 4

Answer: A

Explanation

Matching the descriptions with the test techniques:

- Black-box test techniques use the external description of the software, including requirements, use cases, and user stories.
- White-box test techniques use the internal structure of the software system, including software architecture and code.
- Black-box test techniques can reveal deviations from the requirements as they validate the external behavior of the software.
- Both black-box and white-box test techniques are applicable to functional and non-functional testing.

References:

ISTQB® Certified Tester Foundation Level Syllabus v4.0: ISTQB CTFL Syllabus v4.0 PDF

Question #:253

Which of the following is NOT an experience-based technique?

- A. Boundary value analysis.
- B. Error guessing
- C. Exploratory testing
- D. Fault attack

Answer: A

Explanation

Boundary value analysis is not an experience-based technique, but rather a specification-based technique (also known as black-box technique). Experience-based techniques are techniques that rely on the tester's knowledge and intuition to derive and select test cases based on their experience with similar systems, technologies, domains, risks, etc. Some examples of experience-based techniques are error guessing,

exploratory testing, fault attack, checklist-based testing, etc. Specification-based techniques are techniques that rely on the tester's analysis and interpretation of the requirements or specifications of the system under test to derive and select test cases based on some criteria or rules. Some examples of specification-based techniques are equivalence partitioning, boundary value analysis, decision table testing, state transition testing, etc. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 31.

Question #:254

Which of the following statements about the typical activities of a formal review process is TRUE?

- A. Individual review is only mandatory when the size of the work product under review is too large to cover at the review meeting
- B. Various review techniques that may be applied by participants during individual review are described in the ISO/IEC/IEEE 29119-3 standard.
- C. Choosing which standards to follow during the review process is usually made during review planning.
- D. One of the main goals of the review meeting is to make sure that all participants are aware of their roles and responsibilities in the review process

Answer: C

Explanation

During the review planning stage, key decisions are made, including the selection of standards and procedures to be followed during the review. This planning phase ensures that the review process is structured and adheres to agreed-upon standards, which can come from industry standards such as ISO/IEC/IEEE 29119-3. The ISTQB CTFL Syllabus v4.0 emphasizes the importance of review planning in establishing the framework and guidelines for the review process.

Question #:255

Which of the following statements best describes the way in which decision coverage is measured?

- A. Measured as the number of statements executed by the tests, divided by the total number of executable statements in the code.
- B. Measured as the number of lines of code executed by the tests, divided by the total number of lines of code in the test object.
- C. Measured as the number of decision outcomes executed by the tests, divided by the total number of decision outcomes in the test object.
- D. It is not possible to accurately measure decision coverage.

Answer: C

Explanation

Decision coverage, also known as branch coverage, is measured as the number of decision outcomes executed by the tests divided by the total number of decision outcomes in the test object. It ensures that every possible branch (true/false) decision in the code has been executed at least once.

Reference: ISTQB CTFL Syllabus V4.0, Section 4.3.2

Question #:256

During which review process activity are the exit criteria for an inspection defined?

- A. Review Initiation
- B. Planning
- C. Communication and analysis
- D. Fixing and reporting

Answer: B

Question #:257

For each of the test cases to be executed, the following table specifies the priority order and dependencies on other test cases

Test Case	Priority	Logical Dependencies
TC1	Low	TC5
TC2	High	TC3
TC3	High	TC4
TC4	High	-
TC5	Low	TC2
TC6	Medium	-

Which of the following test execution schedules is compatible with the logical dependencies and allows executing the test cases in priority order?

- A. TC4, TC3, TC2, TC6, TC5, TC1
- B. TC4, TC6, TC3, TC2, TC5, TC1
- C. TC3, TC5, TC6, TC1, TC4, TC3

D. TC4, TC3, TC2, TC6, TC1, TC5

Answer: D

Explanation

This answer is correct because it follows the logical dependencies and allows executing the test cases in priority order. TC4, TC3, and TC2 are executed first because they have the highest priority. TC6 is executed next because it has a logical dependency on TC2. TC1 is executed next because it has a logical dependency on TC5. Finally, TC5 is executed last because it has the lowest priority. References: ISTQB Certified Tester Foundation Level (CTFL) v4.0 documents

Question #:258

In addition to thorough testing of the requirements specification, a development team aims to involve users as early as possible in the development process, using practices such as prototyping, to ensure that the software systems being developed will meet the users' expectations. This approach is especially useful at mitigating the risks associated with one of the seven testing principles, which one?

- A. Tests wear out
- B. Absence-of-errors fallacy
- C. Working software over comprehensive documentation.
- D. Defects cluster together

Answer: B

Explanation

The absence-of-errors fallacy is the mistaken belief that just because a software system is free of defects, it will meet the user's needs and expectations. Involving users early through practices like prototyping helps ensure that the development team is building the right system that meets user expectations, not just a system that is defect-free. This approach aligns with the testing principle that emphasizes understanding the users' needs and ensuring the system fulfills them. This principle is explained in the ISTQB CTFL Syllabus v4.0.

Question #:259

Which of the following issues cannot be identified by static analysis tools?

- A. Very low MTBF (Mean Time Between failure)
- B. Potentially endless loops
- C. Referencing a variable with an undefined value
- D. Security vulnerabilities

Answer: A**Explanation**

Static analysis tools are software tools that examine the source code of a program without executing it. They can detect various types of issues, such as syntax errors, coding standards violations, security vulnerabilities, and potential bugs¹². However, static analysis tools cannot identify issues that depend on the runtime behavior or performance of the program, such as very low MTBF (Mean Time Between failure)³. MTBF is a measure of the reliability of a system or component. It is calculated by dividing the total operating time by the number of failures. MTBF reflects how often a system or component fails during its expected lifetime. Static analysis tools cannot measure MTBF because they do not run the program or observe its failures. MTBF can only be estimated by dynamic testing, which involves executing the program under various conditions and collecting data on its failures⁴. Therefore, very low MTBF is an issue that cannot be identified by static analysis tools. The other options, such as potentially endless loops, referencing a variable with an undefined value, and security vulnerabilities, are issues that can be identified by static analysis tools. Static analysis tools can detect potentially endless loops by analyzing the control flow and data flow of the program and checking for conditions that may never become false⁵. Static analysis tools can detect referencing a variable with an undefined value by checking the scope and initialization of variables and reporting any use of uninitialized variables⁶. Static analysis tools can detect security vulnerabilities by checking for common patterns of insecure code, such as buffer overflows, SQL injections, cross-site scripting, and weak encryption. References = What Is Static Analysis? Static Code Analysis Tools - Perforce Software, How Static Code Analysis Works | Perforce, Static Code Analysis: Techniques, Top 5 Benefits & 3 Challenges, What is MTBF? Mean Time Between Failures Explained | Perforce, Static analysis tools - Software Testing MCQs - CareerRide, ISTQB_Chapter3 | Quizizz, [Static Code Analysis for Security Vulnerabilities | Perforce].

Question #:260

Given the following examples of entry and exit criteria:

- 1.A defined level of code coverage has been achieved
- 2.The test automation tool has been installed and properly configured
- 3.The number of unresolved defects is within the predefined limit
- 4.The performance test environment has been set-up and is available
- 5.The user stories have proper acceptance criteria defined
- 6.The testing budget has been spent and the project sponsor bears the risk of not testing any further

Which of the following BEST categorizes them as entry and exit criteria:

- A. Entry criteria - 2, 4, 5 Entry criteria -1, 3, 4
- B. Entry criteria - 2, 4 Entry criteria - 2, 4, 5, 6
- C. Exit criteria -1,3,6 Exit criteria - 2, 5. 6
- D. Exit criteria -1,3,5,6 Exit criteria -1,3

Answer: A**Explanation**

Entry and exit criteria are used to determine when to start and stop testing, respectively.

➤ Entry Criteria:

➤ These are conditions that should be met before testing begins.

➤ Examples:

- 2. The test automation tool has been installed and properly configured
- 4. The performance test environment has been set-up and is available
- 5. The user stories have proper acceptance criteria defined

➤ Exit Criteria:

➤ These are conditions that should be met before testing concludes.

➤ Examples:

- 1. A defined level of code coverage has been achieved
- 3. The number of unresolved defects is within the predefined limit
- 6. The testing budget (ISTQB not-for-profit association) spent and the project sponsor bears the risk of not testing any further

According to the ISTQB Certified Tester Foundation Level (CTFL) v4.0 syllabus, these criteria help in effectively managing the testing process#6†source#.

References:

Certified Tester Foundation Level v4.0

ISTQB Foundation Level Syllabus 4.0 (2023)

Question #:261

You are testing the download process of a mobile phone application.

For which to the following capabilities to the system you need to design a nonfunctional test?

- A. It was easy to locate, download and install the application
- B. The application was correctly downloaded

- C. The application created an installation log file in a given folder
- D. The application installed only after the user's approval

Answer: A

Explanation

This question asks for a non-functional aspect of testing the download process of a mobile application. Option A, "It was easy to locate, download and install the application," refers to usability, which is a non-functional quality attribute. Non-functional testing involves testing the system's attributes, such as usability, performance, reliability, etc., rather than specific behaviors or functions. Options B, "The application was correctly downloaded," C, "The application created an installation log file in a given folder," and D, "The application installed only after the user's approval," describe functional aspects, focusing on what the software does rather than how it performs or is experienced by the user.

Question #:262

Which ONE of the following options BEST describes black-box test techniques?

- A. Black-box testing techniques are based on experience with the test object without knowing the internal structure.
- B. In black-box testing techniques, test cases are created based on the software's implementation.
- C. Black-box testing techniques are based on analyzing the test object's specified and implied behavior without knowledge of its internal data structure.
- D. Black-box testing techniques involve analyzing the specified behavior of the test object without knowledge of its internal structure.

Answer: D

Explanation

Comprehensive and Detailed In-Depth Explanation:

Black-box testing techniques focus on **testing the functionality of the software without knowledge of its internal workings** (D). They derive test cases from **specifications, requirements, or expected behavior**. **Option A** describes exploratory testing, **B** contradicts the definition by focusing on implementation, and **C** incorrectly includes "implied behavior," which is not a core characteristic of black-box testing.

Reference: ISTQB CTFL v4.0 Syllabus, Section 4.2 – Black-Box Testing Techniques

Question #:263

Which of the following is a typical product risk?

- A. Poor usability of the software

- B. A problem in the code developed by a 3rd party
- C. Low quality of the configuration data, test data and tests
- D. Problem in defining the right requirements

Answer: A

Explanation

A typical product risk involves issues directly related to the software product's functionality, performance, usability, reliability, etc. Option A, "Poor usability of the software," directly impacts the end-user's interaction with the software and is a quality attribute of the product itself, making it a product risk. Options B, "A problem in the code developed by a 3rd party," C, "Low quality of the configuration data, test data and tests," and D, "Problem in defining the right requirements," can be considered either product or project risks depending on the context, but option A is the most directly associated with a typical product risk concerning the quality and usability of the software.

Question #:264

Consider the following examples of risks identified in different software development projects:

- [I]. The contrast color ratio for both normal text and large text of a website does not comply with the applicable accessibility guidelines, making it difficult for many users to read the content on the pages
- [II]. A development vendor fails to deliver their software system on time, causing significant delays to system integration testing activities that have been planned as part of a development project for a system of systems
- [III]. People in the test team do not have sufficient skills to automate tests at the test levels required by the test automation strategy which does not allow production of an effective regression test suite
- [IV]. In a web application, data from untrusted sources is not subject to proper input validation, making the application vulnerable to several security attacks

Which of the following statements is true?

- A. [I] and [III] are product risks; [II] and [IV] are project risks
- B. [I] and [IV] are product risks. [II] and [III] are project risks
- C. [II], [III] and [IV] are product risks; [I] is a project risk
- D. [IV] is a product risk; [I], [II] and [III] are project risks

Answer: B

Explanation

This answer is correct because product risks are risks that affect the quality of the software product, such as defects, failures, or non-compliance with requirements or standards. Project risks are risks that affect the

project's schedule, budget, resources, or scope, such as delays, cost overruns, skill gaps, or scope changes. In this case, [I] and [IV] are product risks, as they relate to the accessibility and security of the software product, which are quality attributes. [II] and [III] are project risks, as they relate to the delivery time and the test automation skills of the test team, which are project factors. References: ISTQB Glossary of Testing Terms v4.0, ISTQB Foundation Level Syllabus v4.0, Section 2.1.1.1

Question #265

Which of the following is the main difference between quality assurance (QA) and quality control (QC)?

- A. A is the responsibility of everyone on a project, while QC is the responsibility of a dedicated team.
- B. A is a process-oriented approach, while QC is a product-oriented approach.
- C. A is focused on implementation, while QC is focused on improvement.
- D. A is preventive, while QC is corrective.

Answer: D

Question #266

Which ONE of the following statements would you expect to be the MOST DIRECT advantage of the whole-team approach?

- A. Capitalizing on the combined skills of business representatives, testers, and developers working together to contribute to project success.
- B. Reducing the involvement of business representatives because of enhanced communication and collaboration between testers and developers.
- C. Avoiding requirements misunderstandings that may not have been detected until dynamic testing when they are more expensive to fix.
- D. Having an automated build and test process, at least once a day, that detects integration errors early and quickly.

Answer: A

Explanation**Comprehensive and Detailed In-Depth Explanation:**

The whole-team approach promotes collaboration among stakeholders (business representatives, developers, and testers) to ensure better quality and project success (A). This approach allows for early identification of issues, enhances shared responsibility, and improves software quality. Option C is a valid but indirect benefit, while options B and D do not directly describe the core advantage of the whole-team approach.

Reference: ISTQB CTFL v4.0 Syllabus, Section 1.5.2 – The Whole-Team Approach

Question #:267

Which ONE of the following is a GOOD testing practice that is INDEPENDENT of the chosen SDLC (Software Development Life Cycle) model?

- A. Testers are involved in reviewing work products as soon as the final version of this documentation is available so that this earlier testing and defect detection can support the “shift-left” strategy.
- B. Test analysis and design for a specific test level begin during the corresponding development phase of the SDLC, allowing testing to follow the principle of late testing.
- C. For every software development activity, there is a corresponding testing activity, ensuring that all development activities are subject to quality assurance.
- D. Different test levels have specific and overlapping test objectives, allowing tests to be appropriate to specific test object types while managing redundancy.

Answer: D

Explanation**Comprehensive and Detailed In-Depth Explanation:**

Different test levels (unit, integration, system, acceptance) have specific objectives while sometimes overlapping to ensure comprehensive testing (D). Option A contradicts shift-left principles, B encourages late testing (which is ineffective), and C incorrectly assumes that all development activities require direct test actions.

Reference: ISTQB CTFL v4.0 Syllabus, Section 2.2 – Test Levels

Question #:268

Which of the following statements about retrospectives is TRUE?

- A. Only developers and testers should be involved in retrospectives, as involving people in other roles is very likely to prevent developers and testers from having open and constructive discussions that really help identify process improvements.
- B. Retrospectives can be very effective in identifying process improvements and can also be very efficient and cost-effective especially since, unlike reviews, they do not require any follow-up activities
- C. On Agile projects, well-conducted retrospectives at the end of each iteration can help the team reduce and sometimes even eliminate the need for dairy stand-up meetings.
- D. During retrospectives, in addition to identifying relevant process improvements, participants should also consider how to implement these improvements and retain them based on the context of the project, such as the software development lifecycle.

Answer: D**Explanation**

Retrospectives are a crucial part of Agile practices, aiming to identify process improvements and determine how to implement them effectively. They should involve participants discussing not only what improvements could be made but also how to integrate and sustain those improvements within the project context, including the software development lifecycle. This makes statement D accurate according to the ISTQB CTFL syllabus.

Question #:269

Which of the following activities does NOT belong to a typical technical review?

- A. Pre-meeting preparation by reviewers
- B. Using checklists during the meeting
- C. Inviting end-users to the meeting
- D. Preparation of a review report

Answer: C**Explanation**

Technical reviews are structured meetings that aim to examine various aspects of a product or project to identify any defects or improvements. Options A (Pre-meeting preparation by reviewers), B (Using checklists during the meeting), and D (Preparation of a review report) are typical activities in a technical review process. Inviting end-users to the meeting (C), however, is generally not part of a typical technical review, as these reviews are usually more focused on the technical aspects and are conducted by peers or experts within the development or testing teams rather than end-users.

Question #:270

Which of the following is a CORRECT statement about how a tester should communicate about defects, test results, and other test information?

- A. Testers should include personal opinions and judgements in defect reports and review findings
- B. Testers should emphasize the benefits of testing, such as increased quality and reduced risk
- C. Testers should reject all questions about their test findings and information
- D. Testers should take a command-and-control approach with the project team

Answer: B**Explanation**

Communication from testers about defects, test results, and other test information should emphasize the benefits of testing such as increased quality and reduced risk. This positive framing helps in reinforcing the

value of testing and ensuring stakeholders understand the contribution of testing to the overall project success (ISTQB not-for-profit association).

References:

ISTQB® Certified Tester Foundation Level Syllabus v4.0: https://istqb-main-web-prod.s3.amazonaws.com/media/documents/ISTQB_CTFL_Syllabus-v4.0.pdf

Question #:271

Which of the following statements about test activities that form the overall test process is correct?

- A. Test planning determines what is to be tested based on the test basis.
- B. Test analysis evaluates the test base and the test elements to identify inconsistencies.
- C. Test design develops and prioritizes test procedures and may create automated test scripts.
- D. Test implementation realization answers the question of how to perform the test, and the test procedure compares the results of the test with the expected results.

Answer: B

Question #:272

What is test oracle?

- A. The source of test objectives
- B. The source for the actual results
- C. The source of expected results
- D. The source of input conditions

Answer: C

Explanation

A test oracle is a mechanism or principle that can be used to determine whether the observed behavior or output of a system under test is correct or not. A test oracle can be based on various sources of expected results, such as specifications, user expectations, previous versions, comparable systems, etc. References: ISTQB Certified Tester Foundation Level (CTFL) v4.0 Syllabus, Section 1.2.1, Page 91; ISTQB Glossary of Testing Terms, Version 4.0, Page 332.

Question #:273

Which ONE of the following options BEST describes the purpose of confirmation testing versus regression testing?

- A. The purpose of confirmation testing is to confirm that the defect giving rise to a failure has been successfully fixed. The regression test aims to ensure that no defects have been introduced or discovered in unmodified areas of the software as a result of the changes made.
- B. Confirmation testing ensures the entire system functions as expected, whereas regression testing focuses only on modified components.
- C. Confirmation testing verifies all system requirements, while regression testing ensures that no additional test cases are needed.
- D. Regression testing and confirmation testing are interchangeable and serve the same purpose.

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

Confirmation testing is performed after a defect is fixed to confirm it no longer exists (A). **Regression testing** ensures new defects have not been introduced in unchanged parts of the system. Regression testing is broader than confirmation testing and covers unmodified areas affected by the changes. Options B, C, and D misrepresent the relationship and scope of these tests.

Reference: ISTQB CTFL v4.0 Syllabus, Section 2.2.3 – Confirmation Testing and Regression Testing

Question #:274

Mark the correct sentences:

- * Defects are a result of environmental conditions and are also referred to as "Failures"
- * A human mistake may produce a defect
- * A system will totally fail to operate correctly when a failure exists in it
- * When a defect exists in a system it may result in a failure
- * Defects occur only as a result of technology changes

- A. II, IV
- B. I, II
- C. IV, V
- D. II, III, IV

Answer: A

- The question is about marking the correct sentences among the given statements related to defects, failures, and mistakes. According to the ISTQB glossary, the definitions of these terms are¹:
 - Defect: A flaw in a component or system that can cause the component or system to fail to perform its required function, e.g. an incorrect statement or data definition. A defect, if encountered during execution, may cause a failure of the component or system.
 - Failure: An event in which a component or system does not perform a required function within specified limits.
 - Mistake: A human action that produces an incorrect result.
- Therefore, out of the five given statements, only two are correct, namely:
 - A human mistake may produce a defect: This is true, as a mistake is a source or cause of a defect, e.g. a programmer may make a mistake in writing a code statement, which results in a defect in the software component.
 - When a defect exists in a system it may result in a failure: This is true, as a defect is a potential or actual cause of a failure, e.g. a defect in the software component may cause the system to fail to perform a required function when the defect is encountered during execution.
- The other three statements are incorrect, namely:
 - Defects are a result of environmental conditions and are also referred to as “Failures”: This is false, as defects are not a result of environmental conditions, but of mistakes or other factors, and defects are not the same as failures, but rather the causes of failures.
 - A system will totally fail to operate correctly when a failure exists in it: This is false, as a system may not necessarily fail completely or stop operating when a failure occurs, but may continue to operate with reduced functionality or performance, or with incorrect results.
 - Defects occur only as a result of technology changes: This is false, as defects can occur due to various reasons, not only technology changes, such as human mistakes, design flaws, requirement changes, hardware failures, etc.

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1: ISTQB Glossary of Testing Terms 4.0, 2023, available at (ISTQB) and (ASTQB).

Question #:275

Which of the following statements describes regression testing?

- I. Retesting of a fixed defect
- II. Testing of an already tested program
- III. Testing of new functionality in a program

IV. Regression testing applies only to functional testing

V Tests that do not have to be repeatable, because They are only used once

- A. II, IV, V
- B. I, III, IV
- C. II
- D. I, IV

Answer: C

Explanation

Regression testing is the re-running of functional and non-functional tests to ensure that previously developed and tested software still performs as expected after a change¹ It does not involve retesting of a fixed defect, testing of new functionality, or applying only to functional testing. Tests that are used for regression testing should be repeatable, because they are used to verify the stability of the software after each change² References = ISTQB Certified Tester Foundation Level (CTFL) v4.0 Syllabus, Chapter 4, Section 4.2.2, Page 291; ISTQB Glossary of Testing Terms v4.0, Page 292

Question #:276

Which ONE of the following statements about maintenance testing is CORRECT?

- A. Maintenance testing is performed exclusively for adaptive maintenance.
- B. Maintenance testing is only required when defects are reported in production.
- C. Maintenance testing should be performed when enhancements, fixes, or updates are applied to an existing system.
- D. Maintenance testing does not require test cases since it focuses solely on defect verification.

Answer: C

Explanation

Comprehensive and Detailed In-Depth Explanation:

Maintenance testing is carried out **whenever changes are made to an existing system**, including enhancements, defect fixes, and system migrations (C). It is not limited to adaptive maintenance (A) and is needed even when no defects are reported (B). Test cases are essential to validate fixes and prevent regressions (D).

Reference: ISTQB CTFL v4.0 Syllabus, Section 2.3 – Maintenance Testing

Question #:277

Which test approach will best fit a new project, with little documentation and high probability for bugs?

- A. Exploratory testing
- B. Requirements based testing
- C. Metric based approach
- D. Regression testing

Answer: A

Explanation

Exploratory testing is an approach to testing that emphasizes learning, test design and test execution at the same time. Exploratory testing relies on the tester's skills, creativity and intuition to explore the software under test and discover defects. Exploratory testing is suitable for a new project with little documentation and high probability for bugs, as it can help uncover unknown requirements, assumptions and risks. Exploratory testing is not requirements based testing, which is an approach to testing that derives test cases from documented requirements or specifications. Requirements based testing is not feasible for a new project with little documentation, as it requires clear and complete requirements to be available. Exploratory testing is not metric based approach, which is an approach to testing that uses quantitative measures to monitor and control the testing process and evaluate the quality of the software product. Metric based approach is not effective for a new project with high probability for bugs, as it may not capture all aspects of quality and may lead to false confidence or unrealistic expectations. Exploratory testing is not regression testing, which is an approach to testing that verifies that previously tested software still performs correctly after changes. Regression testing is not relevant for a new project with no previous versions or baselines. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 5, page 47-48.

Question #:278

A system has a self-diagnostics module that starts executing after the system is reset. The diagnostics are running 12 different tests on the systems memory hardware. The following is one of the requirements set for the diagnostics module:

'The time taking the diagnostics tests to execute shall be less than 2 seconds' Which of the following is a failure related to the specified requirement?

- A. The diagnostic tests fail to start after a system reset
- B. The diagnostic tests take too much time to execute
- C. The diagnostic tests that measure the speed of the memory, fail
- D. The diagnostic tests fail due to incorrect implementation of the test code

Answer: B

Explanation

A failure is an event in which a component or system does not perform a required function within specified limits¹. A requirement is a condition or capability needed by a user to solve a problem or achieve an objective². In this case, the requirement is that the diagnostics tests should execute in less than 2 seconds. Therefore, any event that violates this requirement is a failure. The only option that clearly violates this requirement is B. The diagnostic tests take too much time to execute. If the diagnostic tests take more than 2 seconds to complete, then they do not meet the specified limit and thus fail. The other options are not necessarily failures related to the specified requirement. Option A. The diagnostic tests fail to start after a system reset is a failure, but not related to the time limit. It is related to the functionality of the self-diagnostics module. Option C. The diagnostic tests that measure the speed of the memory, fail is also a failure, but not related to the time limit. It is related to the accuracy of the memory tests. Option D. The diagnostic tests fail due to incorrect implementation of the test code is also a failure, but not related to the time limit. It is related to the quality of the test code. References = ISTQB® Certified Tester Foundation Level Syllabus v4.0, Requirements Engineering Fundamentals.

Question #:279

Which review type, also known as a “buddy check”, is commonly used in Agile development?

- A. Inspection.
- B. Walkthrough.
- C. Technical review.
- D. Informal review.

Answer: D

Explanation

In Agile development, an informal review, often referred to as a "buddy check," is a common review type. Informal reviews are unstructured and involve a pair of colleagues reviewing each other's work to identify defects early and provide immediate feedback. This type of review is less formal than inspections or walkthroughs and is particularly suitable for Agile environments where rapid feedback and flexibility are essential.

References: ISTQB CTFL Syllabus, Section 3.2.4, "Types of Reviews" and Section 2.1.4, "Agile Testing Practices."

Question #:280

In branch testing, what is the significance of a conditional branch in a software program?

- A. It indicates a decision point where the program can follow different paths based on specific conditions.
- B. It marks a point where the program's execution will terminate bringing the program to a halt.
- C. It represents a programming error and therefore a conditional branch should be avoided

D. It signifies a comment or documentation within the code and it doesn't indicate a defect.

Answer: A

Question #:281

Consider the following list of activities (I to V):

- I. Test design
- II. Test planning
- III. Review test basis
- IV. Be a facilitator of a peer review
- V. Test monitoring and control

Which ONE of the following options identifies the activities that the test management role is MAINLY focused on?

- A. Activities III, IV, and V
- B. Activities II and V
- C. Activities I and V
- D. Activities I, III, and IV

Answer: B

Explanation

Comprehensive and Detailed In-Depth Explanation:

Test management focuses on planning and control activities. Test planning (II) involves defining test strategies, schedules, and resources, while test monitoring and control (V) ensures that testing activities are aligned with objectives and tracked effectively. Other activities, such as test design (I) and peer review facilitation (IV), are typically handled by testers or test leads.

Reference: ISTQB CTFL v4.0 Syllabus, Section 5.3 – Test Monitoring and Control

Question #:282

Given the following User Story: "As an online customer, I would like to be able to cancel the purchase of an individual item from a shopping list so that it only displays the relevant items, in less than 1 second", which of the following can be considered as applicable acceptance test cases?

i. Click on my online shopping list, select the unwanted Item, delete the unwanted item, the unwanted Item is deleted from the shopping list in less than 1 second.

ii. Click on my online shopping list, select all the items, delete all the items, the unwanted items are deleted from the shopping list in less than 1 second.

iii. Tab to the online shopping list and press enter, select the unwanted item, delete the unwanted item, the unwanted item is deleted from the shopping list In less than 1 second.

Iv. Click on the checkout button, select the payment method, make payment, confirmation received of payment and shipping date.

v. Click on my shopping list, select the unwanted Item, delete the unwanted item, the unwanted item is deleted from the shopping list.

Select the correct answer:

- A. I, ii and v
- B. iv
- C. i and iii
- D. v

Answer: C

Explanation

Applicable acceptance test cases for the given user story should focus on the specific requirement of deleting an individual item from the shopping list and ensuring that it is removed in less than 1 second. Therefore, the valid test cases are: i. Click on my online shopping list, select the unwanted item, delete the unwanted item, the unwanted item is deleted from the shopping list in less than 1 second. iii. Tab to the online shopping list and press enter, select the unwanted item, delete the unwanted item, the unwanted item is deleted from the shopping list in less than 1 second.

Reference: ISTQB CTFL Syllabus V4.0, Section 5.2.2

Question #:283

Which of the following options should NOT be used for writing acceptance criteria?

- A. Reaction time
- B. Scenarios
- C. Input-output mappings
- D. Test charters

Answer: D

Question #:284

Scenario 2 “Big Drop”:

A company “The Big Drop” provides **bulk discounts and frequent customer discounts** as follows:

Bulk Discounts:

Purchase Volume	Discount (%)
1 - 100 units	0%
101 - 1000 units	5%
>1000 units	10%

Frequent Customer Discounts:

Frequent Customer?	Discount (%)
YES	5%
NO	0%

How many **possible decision rules** can be extracted from this table?

- A. 5 decision rules
- B. 6 decision rules
- C. 32 decision rules
- D. 8 decision rules

Answer: B

Explanation

Comprehensive and Detailed In-Depth Explanation:

Decision rules define possible **combinations of conditions and outcomes**.

For this scenario:

- **Bulk discount has 3 categories** (0%, 5%, 10%)
- **Frequent customer discount has 2 categories** (Yes = +5%, No = 0%)

The total number of decision rules is:

3 bulk discount options × 2 frequent customer options = 6 rules.

Thus, the correct answer is **6 decision rules (B)**.

Reference: ISTQB CTFL v4.0 Syllabus, Section 4.2.3 – Decision Table Testing

Question #:285

Which of the following definitions is NOT true?

- A. Test data preparation tools fill databases, create files or data transmissions to set up test data to be used during the execution of tests.
- B. Test execution tools execute test objects using automated test scripts.
- C. Test Management tools monitor and report on how a system behaves during the testing activities.
- D. Test comparators determine differences between files, databases or test results.

Answer: C

Explanation

Test Management tools are designed to support the planning, execution, and monitoring of the testing process. They provide features for managing test cases, test runs, tracking defects, and reporting on testing activities. However, the statement in option C describes Test Management tools as monitoring and reporting on the system's behavior during testing activities, which is not accurate. Test Management tools focus on the testing process itself rather than on the behavior of the system under test.

- Test data preparation tools (A) indeed create and manage test data for use during test execution.
- Test execution tools (B) automate the execution of test cases and the comparison of actual outcomes against expected results.
- Test comparators (D) are tools that compare actual outcomes with expected outcomes, highlighting discrepancies.

Therefore, option C is the correct answer as it inaccurately describes the function of Test Management tools.

Question #:286

A company runs a pilot project for evaluation of a test automation tool. Which of the following is NOT a valid object of this pilot project?

- A. Get familiar with the functionality and options of the tool
- B. Check how the tool fits to the existing test processes
- C. Train all testers on using the tool

D. Decide upon standards for tool implementation

Answer: C

- A pilot project is a small-scale experiment or trial that is conducted to evaluate the feasibility, effectiveness, and suitability of a test automation tool before implementing it on a larger scale¹.
- The objectives of a pilot project may vary depending on the context and scope of the test automation initiative, but some common ones are²:
 - To get familiar with the functionality and options of the tool
 - To check how the tool fits to the existing test processes and environment
 - To assess the benefits and challenges of using the tool
 - To decide upon standards and guidelines for tool implementation and usage
 - To estimate the costs and resources required for tool deployment and maintenance
- Therefore, option C is not a valid objective of a pilot project, as it is not necessary to train all testers on using the tool at this stage. Training all testers on using the tool would be more appropriate after the tool has been selected and approved for full-scale implementation, and after the standards and guidelines have been established. Training all testers on using the tool during the pilot project would be inefficient, costly, and premature, as the tool may not be suitable or effective for the intended purpose, or may be replaced by another tool later.

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1: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 82

2: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 83

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 84

: ISTQB Certified Tester Foundation Level Syllabus 2018, Version 4.0, p. 85

Question #:287

The following 4 equivalence classes are given:

Which of the following alternatives includes correct test values for x. based on equivalence partitioning?

- A. -100; 100;1000; 1001
- B. -500; 0; 100; 1000
- C. -99; 99;101; 1001

D. -1000; -100; 100; 1000

Answer: D

Explanation

- The question is about selecting the correct test values for x based on equivalence partitioning. Equivalence partitioning is a software test design technique that divides the input data of a software unit into partitions of equivalent data from which test cases can be derived. In this case, the given equivalence classes are:
- $(x \leq -100)$
 - $(-100 < x < 100)$
 - $(100 \leq x < 1000)$
 - $(x \geq 1000)$

Option D provides a value from each of these partitions:

- For $(x \leq -100)$, it gives -1000.
- For $(-100 < x < 100)$, it gives -100 and 100.
- For $(100 \leq x < 1000)$, it gives 500.
- For $(x \geq 1000)$, it gives 1500.

So, option D covers all four given equivalence classes with appropriate values.

: ISTQB Certified Tester Foundation Level (CTFL) v4.0 documents available at ISTQB and ASTQB.

1: ISTQB Foundation Level Syllabus 2018, Version 4.0, p. 38

2: ISTQB Foundation Level Syllabus 2018, Version 4.0, p. 39

: ISTQB Foundation Level Syllabus 2018, Version 4.0, p. 40

Question #:288

Which of the following can be considered a VALID exit criterion?

- I Estimates of defect density or reliability measures.
- II. The completion and publication of an exhaustive Test Report.
- III. Accuracy measures, such as code, functionality or risk coverage.
- IV Residual risks such as lack of code coverage in certain areas.

- A. I, III, IV
- B. I, II, III
- C. III, IV
- D. II, III, IV

Answer: A

Explanation

An exit criterion is a condition that defines when a test activity has been completed or when a test phase can be concluded. An exit criterion can be based on various factors, such as:

- I) Estimates of defect density or reliability measures. These are quantitative measures that indicate how many defects are present in the software product or how likely it is to fail under certain conditions. These can be used as exit criteria to ensure that the software product meets a certain level of quality or performance before moving to the next phase or releasing it to the customer.
- III) Accuracy measures, such as code coverage, functionality coverage or risk coverage. These are quantitative measures that indicate how much of the software product has been tested in terms of its code, functionality or risk. These can be used as exit criteria to ensure that the test suite is adequate or complete before moving to the next phase or releasing it to the customer.
- IV) Residual risks, such as lack of code coverage in certain areas, unresolved defects or unknown factors. These are qualitative measures that indicate the remaining risks or uncertainties associated with the software product after testing. These can be used as exit criteria to ensure that the residual risks are acceptable or manageable before moving to the next phase or releasing it to the customer. The following factor is not a valid exit criterion:
- II) The completion and publication of an exhaustive Test Report. This is not a valid exit criterion, as it does not reflect the quality or completeness of the testing process or product. A test report is a document that summarizes the results and outcomes of a test activity or phase. A test report can be used as an input for deciding whether to exit a test activity or phase, but it is not a condition that defines when to exit. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 2, page 13; Chapter 6, page 58-59.

Question #:289

Which ONE of the following statements BEST applies to **checklist-based testing**?

- A. Checklists should contain general guidelines to ensure that all aspects of the software are covered.
- B. Checklists should contain specific test conditions that can be individually and directly checked.
- C. Checklists should primarily consist of automated test cases to maximize efficiency.

- D. Checklists should be used exclusively for functional testing, as they are unsuitable for non-functional testing.

Answer: B

Explanation

Comprehensive and Detailed In-Depth Explanation:

Checklist-based testing uses **specific test conditions** (B) that help testers ensure key aspects are validated. The checklist items can be derived from past defects, requirements, or regulatory standards.

- (A) is **incorrect** because general guidelines lack specificity.
- (C) is **incorrect** because checklists can be used for both manual and automated testing.
- (D) is **incorrect** because checklists are useful for both functional and non-functional testing (e.g., security, performance).

Checklists **help ensure completeness** without enforcing **strictly scripted** execution.

Reference: ISTQB CTFL v4.0 Syllabus, Section 4.4.3 – Checklist-Based Testing

Question #:290

Which of the following activities are part of test planning?

- I) Setting the entry and exit criteria
 - II) Determining the validity of bug reports
 - III) Determining the number of resources required
 - IV) Determining the expected result for test cases
- A. I, IV
- B. I, III
- C. I, III, IV
- D. I, II, IV

Answer: B

Explanation

Test planning is a key activity in the testing process that involves defining the objectives, approach, resources, and schedule of intended test activities. Setting the entry and exit criteria (I) and determining the number of resources required (III) are integral parts of test planning. Determining the validity of bug reports (II) is more aligned with test analysis or test management activities post-execution, and determining the expected result for test cases (IV) is part of test design. Therefore, options I and III (B) are the activities that belong to test planning.

Question #:291

Which of the following statements is not correct?

- A. Looking for defects in a system may require Ignoring system details
- B. Identifying defects may be perceived as criticism against product
- C. Looking for defects in system requires professional pessimism and curiosity
- D. Testing is often seen as a destructive activity instead of constructive activity

Answer: A

- Looking for defects in a system does not require ignoring system details, but rather paying attention to them and understanding how they affect the system's quality, functionality, and usability. Ignoring system details could lead to missing important defects or testing irrelevant aspects of the system.
- Identifying defects may be perceived as criticism against product, especially by the developers or stakeholders who are invested in the product's success. However, identifying defects is not meant to be a personal attack, but rather a constructive feedback that helps to improve the product and ensure its alignment with the requirements and expectations of the users and clients.
- Looking for defects in system requires professional pessimism and curiosity, as testers need to anticipate and explore the possible ways that the system could fail, malfunction, or behave unexpectedly. Professional pessimism means being skeptical and critical of the system's quality and reliability, while curiosity means being eager and interested in finding out the root causes and consequences of the defects.
- Testing is often seen as a destructive activity instead of constructive activity, as it involves finding and reporting the flaws and weaknesses of the system, rather than creating or enhancing it. However, testing is actually a constructive activity, as it contributes to the system's improvement, verification, validation, and optimization, and ultimately to the delivery of a high-quality product that meets the needs and expectations of the users and clients.

Question #:292

The acceptance criteria associated with a user story:

- A. are often written in a rule-oriented format using the template referred to as "Given/When/Then"
- B. are often documented following in rule-oriented format using the following template: "As a [role], I want [feature], so that I can [benefit]"

- C. can be written in different formats and represent an aspect of a user story referred to as confirmation' of the so called "3 C's"
- D. must be written in one of the two following formats: scenario-oriented or rule-oriented

Answer: C

Explanation

The acceptance criteria associated with a user story are the conditions that must be met for the user story to be considered done and to deliver the expected value to the user. They are often written in different formats, such as rule-oriented, scenario-oriented, or table-oriented, depending on the nature and complexity of the user story. They represent an aspect of a user story referred to as confirmation, which is one of the so called "3 C's" of user stories. The other two aspects are card and conversation. Card refers to the concise and informal description of the user story, usually following the template: "As a [role], I want [feature], so that I can [benefit]". Conversation refers to the ongoing dialogue between the stakeholders and the team members to clarify and refine the user story and its acceptance criteria. Therefore, option C is the correct answer.

: ISTQB® Certified Tester Foundation Level Syllabus v4.01, Section 3.2.2, page 35-36; ISTQB® Glossary v4.02, page 37.

Question #:293

Which of the following does MOT describe a reason why testing is necessary?

- A. The customer decided that 100% branch coverage shall be achieved
- B. The acquisition of test automation tools was based on the assumption that it will be used in all projects
- C. For avionics and pharmaceutical systems software testing is mandated by standards
- D. The risks associated with delivering the system are far higher than the cost of testing

Answer: B

Explanation

Testing is necessary for various reasons, such as:

- To detect defects and failures that may affect the quality, performance, reliability or security of a software product or system
- To verify that a software product or system meets its specified requirements, expectations and standards
- To validate that a software product or system fulfills its intended purpose and satisfies its stakeholders' needs
- To provide information and feedback about the status and risks of a software product or system

- To comply with regulations or contractual obligations that mandate testing for certain types of software products or systems The following statements describe some reasons why testing is necessary:
- A) The customer decided that 100% branch coverage shall be achieved. This is a reason why testing is necessary, as it reflects a contractual obligation or a quality standard that requires testing to measure and achieve a certain level of code coverage.
- C) For avionics and pharmaceutical systems software testing is mandated by standards. This is a reason why testing is necessary, as it reflects a regulation or a compliance requirement that mandates testing for certain types of software products or systems that have high safety or security risks.
- D) The risks associated with delivering the system are far higher than the cost of testing. This is a reason why testing is necessary, as it reflects a risk-based approach that considers testing as an investment to reduce the probability and impact of potential failures or defects. The following statement does not describe a reason why testing is necessary:
- B) The acquisition of test automation tools was based on the assumption that it will be used in all projects. This is not a reason why testing is necessary, as it reflects a business decision or a resource allocation that does not justify the need or purpose of testing. Test automation tools are not always suitable or beneficial for all projects, and testing can be performed with or without test automation tools. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 1, page 5-6.

Question #:294

Which of the following statements BEST describes how test cases are derived from a use case?

- A. Test cases are derived based on non-functional requirements such as usability
- B. Test cases are created using white-box test techniques to execute scenarios of use cases
- C. Test cases are derived based on pair testing between a user and a tester to find defects
- D. Test cases are designed to cover various user behaviors, including basic, exceptional or alternative and error behaviors associated with human users or systems

Answer: D

Explanation

Use cases describe a system's behavior as it responds to a request from a user. They typically consist of various scenarios, such as basic flow, alternative flow, and exceptional flow, which represent possible behaviors when a user interacts with the system. When deriving test cases from use cases, it is important to cover these different types of user behaviors.

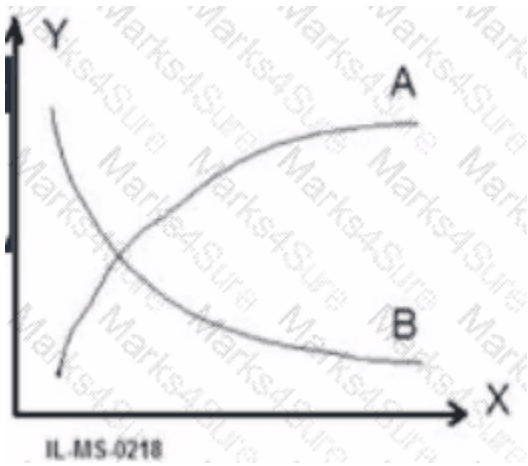
Test cases should be designed to verify how the system behaves during each of these scenarios. This ensures that the system operates correctly for normal and error conditions encountered by human users or systems interacting with the application. Thus, test cases derived from use cases aim to cover basic, exceptional, and alternative flows, ensuring comprehensive coverage.

References:

ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 4.2.4.

Question #:295

The following chart represents metrics related to testing of a project that was completed. Indicate what is represented by tie lines A, B and the axes X.Y



A)

- X - Time
- Y - Cost
- A - Cost of test (per week)
- B - Cost of finding a single bug (per week)

B)

- X - Time
- Y - Number of defects
- A - Number of open defects
- B - Number of closed defects

C)

- X - Time
- Y - Percent
- A - % of functional tests in the test suite
- B - % of non-functional tests in the test suite

D)

- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer: D

Explanation

Option D correctly explains what is represented by the lines A, B and the axes X, Y in a testing metrics chart. According to option D:

- X-axis represents Time
- Y-axis represents Count
- Line A represents Number of open bugs
- Line B represents Total number of executed tests

This information is essential in understanding and analyzing the testing metrics of a completed project.

: ISTQB Certified Tester Foundation Level (CTFL) v4.0 Syllabus, Section 2.5.1, Page 35.

Question #:296

Which ONE of the following options is NOT a test objective?

- A. Verifying whether specified requirements have been fulfilled
- B. Triggering failures and finding defects
- C. Finding errors
- D. Validating whether the test object is complete and works as expected by the stakeholders

Answer: C

Explanation

Comprehensive and Detailed In-Depth Explanation:

The primary objectives of testing, as outlined in the ISTQB CTFL v4.0 syllabus, include verifying whether specified requirements are met (A), detecting failures and defects (B), and validating that the test object functions as expected (D). However, "finding errors" (C) is not a direct objective. Errors result from human

mistakes, but testing primarily identifies defects, which are flaws in the system that cause failures. Testing aims to reveal defects rather than directly identify errors in the code.

Reference: ISTQB CTFL v4.0 Syllabus, Section 1.1.1 – Test Objectives

Question #:297

The tests at the bottom layer of the test pyramid:

- A. run faster than the tests at the top layer of the pyramid
- B. cover larger pieces of functionalities than the tests at the top layer of the pyramid
- C. are defined as 'UI Tests' or 'End-To-End tests' in the different models of the pyramid
- D. are unscripted tests produced by experience-based test techniques

Answer: A

Question #:298

Which of the following is LEAST likely to describe a task performed by someone in a testing role?

- A. Create test completion report
- B. Evaluate test basis and test object
- C. Assess testability of test object
- D. Define test environment requirements

Answer: A

Question #:299

Which of the following statements about white-box testing is FALSE?

- A. Static testing can benefit from using code-related white-box test techniques during code reviews.
- B. White-box testing allows suggesting test cases for increasing coverage levels which are based on objective measures
- C. Achieving full code coverage for a component or a system ensures that it has been fully tested
- D. Black-box testing can benefit from using code-related white-box test techniques to increase confidence in the code.

Answer: C**Explanation**

Achieving full code coverage does not guarantee that the component or system is fully tested or free of defects. Code coverage metrics indicate the extent to which the source code has been tested, but they do not account for the quality of the tests or whether all possible scenarios have been considered. Other types of testing, including functional, performance, and security testing, are necessary to ensure comprehensive testing. The ISTQB CTFL Syllabus v4.0 highlights that while high code coverage is beneficial, it does not equate to complete testing.

Question #:300

Can "cost" be regarded as Exit criteria?

- A. Yes. Spending too much money on testing will result in an unprofitable product, and having cost as an exit criterion helps avoid this
- B. No. The financial value of product quality cannot be estimated, so it is incorrect to use cost as an exit criterion
- C. Yes. Going by cost as an exit criterion constrains the testing project which will help achieve the desired quality level defined for the project
- D. No The cost of testing cannot be measured effectively, so it is incorrect to use cost as an exit criterion

Answer: A**Explanation**

Cost can be regarded as an exit criterion for testing, because it is a factor that affects the profitability and feasibility of the software product. Testing is an investment that aims to improve the quality and reliability of the software product, but it also consumes resources, such as time, money, and human effort. Therefore, testing should be planned and executed in a way that balances the cost and benefit of testing activities. Having cost as an exit criterion helps to avoid spending too much money on testing, which may result in an unprofitable product or a loss of competitive advantage. Cost can also help to prioritize and focus the testing efforts on the most critical and valuable features and functions of the software product. However, cost should not be the only exit criterion for testing, as it may not reflect the true quality and risk level of the software product. Other exit criteria, such as defect rate, test coverage, user satisfaction, etc., should also be considered and defined in the test plan.

The other options are incorrect, because they either deny the importance of cost as an exit criterion, or they make false or unrealistic assumptions about the cost of testing. Option B is incorrect, because the financial value of product quality can be estimated, for example, by using cost-benefit analysis, return on investment, or cost of quality models. Option C is incorrect, because going by cost as an exit criterion does not necessarily constrain the testing project or help achieve the desired quality level. Cost is a relative and variable factor that depends on the scope, complexity, and context of the software product and the testing project. Option D is incorrect, because the cost of testing can be measured effectively, for example, by using metrics, such as test effort, test resources, test tools, test environment, etc.

Question #:301

"Statement Testing" is part of;

- A. Specification Based testing
- B. Decision Testing
- C. Experience based testing
- D. Structured based testing

Answer: D

Explanation

Statement Testing is a type of white-box testing technique where the test cases are designed based on the implementation of the software, specifically aiming to execute every statement in the code at least once. This falls under the category of structure-based testing (also known as white-box testing), where the internal structure of the system is used to design test cases. Therefore, option D is correct.

Question #:302

Why is it important to select a test technique?

- A. There are usually too many test cases that may be run on a system. Test techniques help reduce the number of tests.
- B. The only way to test a software application is by using well proven test techniques.
- C. Selecting the right test technique in a given situation Increases the effectiveness of the test process by creating tests with higher chance of finding bugs.
- D. Test techniques define the number of regression cycles, which in turn impact the project schedule.

Answer: C

Explanation

Selecting the right test technique is crucial because different techniques are suited to different types of testing and can significantly increase the effectiveness of the testing process by creating tests that are more likely to find defects. While reducing the number of tests (A) and defining the number of regression cycles (D) are considerations in the testing process, they are not the primary reasons for selecting a test technique. The assertion that the only way to test a software application is by using well-proven test techniques (B) is too restrictive and does not acknowledge the adaptability required in testing to suit different contexts and objectives. Therefore, option C is the most comprehensive reason, as it focuses on the effectiveness and efficiency of testing, leading to the creation of high-quality tests that have a higher chance of finding bugs.

Question #:303

Which of the following statements is incorrect regarding the involvement of testers in the software development lifecycle (SDLC)?

- A. Testers should contribute to all activities in the SDLC and participate in design discussions.
- B. Testers should be involved from the beginning of the SDLC to increase understanding of design decisions and detect defects early.
- C. Testers should only be involved during the testing phase.
- D. Testers' involvement is essential; developers find it difficult to be objective.

Answer: C

Explanation

Involving testers only during the testing phase is incorrect as per the ISTQB CTFL syllabus. Effective involvement of testers is crucial throughout the entire software development lifecycle (SDLC). This includes early stages such as requirement analysis and design, which allows testers to understand the design decisions and detect defects early. Early involvement helps in better understanding the project and ensures that quality is built into the product from the beginning. Furthermore, the ISTQB syllabus emphasizes the importance of testers contributing to all activities in the SDLC, including design discussions, to enhance defect detection and prevention.

References: ISTQB CTFL Syllabus, Section 2.1.1, "The Influence of Development Models on Testing" and Section 1.1.1, "Test Objectives."

Question #:304

The following part of a business process flow is specified; REPEAT (book a bill) UNTIL (User presses Cancel). How many test cases are necessary in order to achieve 100% branch coverage of the process flow?

- A. 4
- B. 1
- C. 2
- D. Infinite

Answer: C

Explanation

To achieve 100% branch coverage of the process flow, we need to test both the true and false outcomes of the condition (User presses Cancel). Branch coverage is a type of structural testing that measures how many decision outcomes in a program have been executed by a test suite. Branch coverage can be used to assess the adequacy or completeness of a test suite.

To test the true outcome of the condition, we need a test case that simulates the user pressing Cancel after booking a bill. This test case will exit the loop and end the process flow.

To test the false outcome of the condition, we need a test case that simulates the user not pressing Cancel after booking a bill. This test case will repeat the loop and book another bill.

Therefore, we need at least two test cases to achieve 100% branch coverage of the process flow. One test case for each possible outcome of the condition.

Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 4, page 40-41.

Question #:305

Which of the following BEST distinguishes the terms "validation" and "verification"?

- A. Verification is confirmation through the provision of objective evidence that the specified requirements have been met while validation is confirmation through the provision of objective evidence that the requirements for a specific intended use have been met
- B. Verification is confirmation through the provision of subjective evidence that the specified requirements have been met while validation is confirmation through the provision of subjective evidence that the designs for a specific intended use have been met
- C. Validation is confirmation through the provision of objective evidence that the specified requirements have been met while verification is confirmation through the provision of objective evidence that the requirements for a specific intended use have been met
- D. Validation is confirmation through the provision of subjective evidence that the specified requirements have been met while verification is confirmation through the provision of subjective evidence that the designs for a specific intended use have been met

Answer: A

Explanation

In the context of software testing, the ISTQB Certified Tester Foundation Level (CTFL) v4.0 differentiates between "validation" and "verification" based on their respective focuses in the software development lifecycle. Verification is the process of evaluating a system or component to determine whether the products of a given development phase satisfy the conditions imposed at the start of that phase. In simpler terms, verification is about checking the product against the specified requirements to ensure it was built correctly. Validation, on the other hand, involves evaluating a system or component during or at the end of the development process to determine whether it meets specified requirements for its intended use. This means validation is about ensuring the product fulfills its intended use and meets the needs of the user.

References:

ISTQB CTFL Syllabus v4.0: ISTQB Official Website

ISTQB Foundation Level Resources v4.0: ASTQB Resources

Question #:306

Which of the following statements is TRUE'?

- A. Unlike functional testing, non-functional testing can only be applied to conventional systems, not artificial intelligence-based system.
- B. Functional testing focuses on what the system is supposed to do, while white-box testing focuses on how well the system does what it is supposed to do
- C. Functional testing can be applied to all test levels, while non-functional testing can be applied only to system and acceptance test levels.
- D. Black-box test techniques and experience-based test techniques may be applicable to both functional testing and non-functional testing

Answer: D

Explanation

Statement D is correct. According to the ISTQB CTFL syllabus, both black-box test techniques (which focus on testing without internal knowledge of the application) and experience-based test techniques (which rely on testers' experience and intuition) can be applied to both functional and non-functional testing. Functional testing is concerned with what the system does, whereas non-functional testing looks at how the system performs under certain conditions. These techniques are versatile and can be employed to address both these aspects.

Question #:307

Which of the following is an INCORRECT statement about the benefit of traceability between the test basis and test work products?

- A. Traceability may be required by IT governance rules
- B. Traceability may help evaluate the extent of test coverage
- C. Traceability may allow testing to be auditable
- D. Traceability may make it harder to understand the impact of changes

Answer: D

Explanation

The statement "Traceability may make it harder to understand the impact of changes" is incorrect. Traceability in testing actually facilitates understanding the impact of changes by linking test cases to

requirements. This linkage helps ensure that any changes in the requirements are adequately reflected and verified in the test cases, thus supporting effective management of changes and maintaining the integrity of the system or product being developed (ISTQB not-for-profit association) (ISTQB Main Web).

References:

ISTQB® Certified Tester Foundation Level Syllabus v4.0: https://istqb-main-web-prod.s3.amazonaws.com/media/documents/ISTQB_CTFL_Syllabus-v4.0.pdf

ISTQB Official Website - CTFL Certification: <https://www.istqb.org/certifications/certified-tester-foundation-level/>

Question #:308

Which statement about use case testing is true?

- A. The test cases are designed to find defects in the data flow.
- B. The test cases are designed to be used by real users, not by professional testers
- C. The test cases are always designed by customers or end users.
- D. The test cases are designed to find defects in the process flow.

Answer: D

Explanation

Use case testing is a technique that helps identify test cases that exercise the whole system on a transaction by transaction basis from start to finish. Use cases are descriptions of how users interact with the system to achieve a specific goal. Use case testing is not focused on data flow, but rather on process flow. Use case testing can be performed by professional testers, customers or end users, depending on the context. Use case testing does not require the test cases to be designed by customers or end users, but rather by anyone who has access to the use case specifications. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 4, page 36.

Question #:309

Which of the following are the phases of the ISTQB fundamental test process?

- A. Test planning and control, Test analysis and design, Test implementation and execution, Evaluating exit criteria and reporting. Test closure activities
- B. Test planning, Test analysis and design. Test implementation and control. Checking test coverage and reporting, Test closure activities
- C. Test planning and control, Test specification and design. Test implementation and execution, Evaluating test coverage and reporting, Retesting and regression testing, Test closure activities

- D. Test planning. Test specification and design. Test implementation and execution. Evaluating exit criteria and reporting. Retesting and test closure activities

Answer: A

Explanation

The ISTQB fundamental test process consists of five main phases, as described in the ISTQB Foundation Level Syllabus, Version 4.0, 2018, Section 2.2, page 15:

- Test planning and control: This phase involves defining the test objectives, scope, strategy, resources, schedule, risks, and metrics, as well as monitoring and controlling the test activities and results throughout the test process.
- Test analysis and design: This phase involves analyzing the test basis (such as requirements, specifications, or user stories) to identify test conditions (such as features, functions, or scenarios) that need to be tested, and designing test cases and test procedures (such as inputs, expected outcomes, and execution steps) to cover the test conditions. This phase also involves evaluating the testability of the test basis and the test items (such as software or system components), and selecting and implementing test techniques (such as equivalence partitioning, boundary value analysis, or state transition testing) to achieve the test objectives and optimize the test coverage and efficiency.
- Test implementation and execution: This phase involves preparing the test environment (such as hardware, software, data, or tools) and testware (such as test cases, test procedures, test data, or test scripts) for test execution, and executing the test procedures or scripts according to the test plan and schedule. This phase also involves logging the outcome of test execution, comparing the actual results with the expected results, and reporting any discrepancies as incidents (such as defects, errors, or failures).
- Evaluating exit criteria and reporting: This phase involves checking if the planned test activities have been completed and the exit criteria (such as quality, coverage, or risk levels) have been met, and reporting the test results and outcomes to the stakeholders. This phase also involves making recommendations for the release or acceptance decision based on the test results and outcomes, and identifying any residual risks (such as known defects or untested areas) that need to be addressed or mitigated.
- Test closure activities: This phase involves finalizing and archiving the testware and test environment for future reuse, and evaluating the test process and the test project against the test objectives and the test plan. This phase also involves identifying any lessons learned and best practices, and communicating the findings and suggestions for improvement to the relevant parties.

References = ISTQB Certified Tester Foundation Level Syllabus, Version 4.0, 2018, Section 2.2, page 15; ISTQB Glossary of Testing Terms, Version 4.0, 2018, pages 37-38; ISTQB CTFL 4.0 - Sample Exam - Answers, Version 1.1, 2023, Question 88, page 32.

Question #:310

After being in operation for many years, a document management system must be decommissioned as it has reached its end of life. This system will not be replaced by any other new system. A legal obligation provides that all documents within the system must be kept for at least 20 years in a state archive. Which of the following statements about maintenance testing for decommissioning of this system is TRUE?

- A. No maintenance testing is required as this system will not be replaced.
- B. Data migration testing is required as part of maintenance testing
- C. Confirmation testing is required as part of maintenance testing.
- D. Regression testing is required as part of maintenance testing

Answer: B

Explanation

Data migration testing is a critical part of maintenance testing during the decommissioning of a system. When a system is decommissioned, data often needs to be transferred to another system or archived securely. This process ensures that the data remains intact, accessible, and secure in its new location. Therefore, statement B is true as it aligns with the ISTQB CTFL syllabus guidelines on handling system decommissioning and data preservation.

Question #:311

A software module to be used in a mission critical application incorporates an algorithm for secure transmission of data.

Which review type is most appropriate to ensure high quality and technical correctness of the algorithm?

- A. Walkthrough
- B. Informal Review
- C. Technical Review
- D. Management Review

Answer: C

Explanation

A technical review is a type of formal review that involves a team of technical experts who evaluate a software product against a set of predefined quality criteria. A technical review is suitable for ensuring high quality and technical correctness of complex or critical software components, such as algorithms, architectures or designs. A technical review is not a walkthrough, which is an informal review led by the author of the work product. A technical review is not an informal review, which is a review that does not follow a defined process and has no formal entry or exit criteria. A technical review is not a management

review, which is a type of formal review that focuses on business aspects and project progress. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 3, page 29-30.

Question #:312

Which of the following statements regarding inspection is NOT true?

- A. An inspection may be led by a trained moderator who shall not be the author.
- B. The main purpose of an inspection is to find solutions to the problems.
- C. An inspection can be performed by peers.
- D. An inspection shall follow a formal process based on rules and checklists with entry and exit criteria

Answer: B

Explanation

An inspection is a type of review that follows a defined process with formal entry and exit criteria and roles and responsibilities for participants. An inspection can be performed by peers with different roles, such as moderator, author, reviewer and scribe. The following statement about inspection is not true:

- B) The main purpose of an inspection is to find solutions to the problems. This statement is not true, as the main purpose of an inspection is to find defects or issues in a work product, not to find solutions to the problems. Finding solutions to the problems is a debugging or problem-solving activity that is usually performed by the author or developer after receiving the inspection report. The following statements about inspection are true:
- A) An inspection may be led by a trained moderator who shall not be the author. This statement is true, as an inspection requires a moderator role who leads the inspection process and ensures that it follows the rules and standards. The moderator should be trained in inspection techniques and should not be the author of the work product under inspection, in order to avoid bias or conflict of interest.
- C) An inspection can be performed by peers. This statement is true, as an inspection involves peer review, which means that the work product under inspection is evaluated by people who have similar roles or expertise as the author, but who are not directly involved in creating or modifying the work product.
- D) An inspection shall follow a formal process based on rules and checklists with entry and exit criteria. This statement is true, as an inspection follows a formal process that consists of six main steps: planning, kick-off meeting, individual preparation, review meeting, rework and follow-up. Each step has defined rules and checklists to guide the participants and ensure consistency and quality. Each step also has entry and exit criteria to ensure that the prerequisites and objectives are met before moving to the next step. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 3, page 28-29.

Question #:313

Which of the following statements about static analysis are FALSE?

- I. Static analysis can be used Instead of dynamic testing.
 - II. Stalk: analysis can uncover defects like security vulnerabilities.
 - III. Static analysis can be used to check conformance to specifications and standards.
 - IV Static analysis typically detects failures prior to component testing.
- A. I, II
B. II, III
C. III, IV
D. I, IV

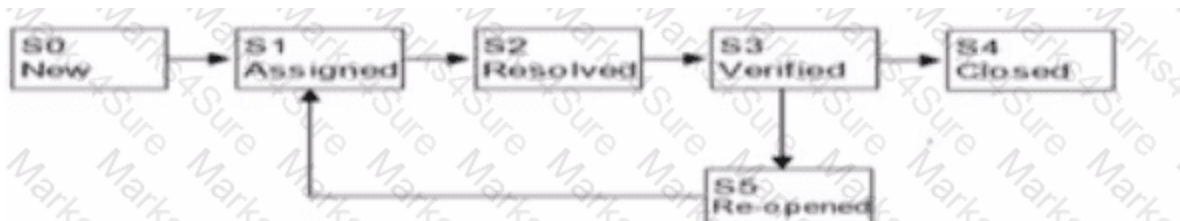
Answer: D

Explanation

Static analysis involves analyzing the software's code, design, and structure without executing the program. It can uncover various types of defects, including security vulnerabilities (II) and non-conformance to specifications and standards (III). However, static analysis cannot replace dynamic testing (I), which involves executing the software to observe its behavior under various conditions. Dynamic testing can identify failures that static analysis cannot, such as those related to runtime issues and interaction between different parts of the software. Statement IV is false because static analysis does not detect failures; it detects defects. Failures are observed when the software is executed, which is beyond the scope of static analysis.

Question #:314

Which sequence of stated in the answer choices is correct in accordance with the following figure depicting the life-cycle of a defect?



- A. S0->S1->S2->S3->S5->S1
B. S0->S1->S2->S3->S5->S1->S2->S3
C. S0->S1->S2->S3->S4
D. S0->S1 ->S2->S3->S5->S3->S4

Answer: D**Explanation**

According to the ISTQB Certified Tester Foundation Level (CTFL) v4.0, the life cycle of a defect typically follows a sequence from its discovery to its closure. In the provided figure, it starts with S0 (New), moves to S1 (Assigned), then to S2 (Resolved), followed by S3 (Verified). If the defect is not fixed, it can be Re-opened (S5) and goes back for verification (S3). Once verified, it is Closed (S4). References: ISTQB Certified Tester Foundation Level (CTFL) v4.0 Syllabus, Section 1.4.3, Page 17.

Question #:315

Which test level aims to gain the users' confidence in the system, especially in the fulfillment of the system's operational and performance capabilities?

- A. System testing
- B. Acceptance testing
- C. Component integration testing
- D. System integration testing

Answer: B**Question #:316**

Which of the following statements about how different types of test tools support testers is true?

- A. The support offered by a test data preparation tool is often leveraged by testers to run automated regression test suites
- B. The support offered by a performance testing tool is often leveraged by testers to run load tests
- C. The support offered by a bug prediction tool is often used by testers to track the bugs they found
- D. The support offered by a continuous integration tool is often leveraged by testers to automatically generate test cases from a model

Answer: B**Explanation**

The support offered by a performance testing tool is often leveraged by testers to run load tests, which are tests that simulate a large number of concurrent users or transactions on the system under test, in order to measure its performance, reliability, and scalability. Performance testing tools can help testers to generate realistic workloads, monitor system behavior, collect and analyze performance metrics, and identify performance bottlenecks. The other statements are false, because:

- A test data preparation tool is a tool that helps testers to create, manage, and manipulate test data, which are the inputs and outputs of test cases. Test data preparation tools are not directly related to running automated regression test suites, which are test suites that verify that the system still works as expected after changes or modifications. Regression test suites are usually executed by test execution tools, which are tools that can automatically run test cases and compare actual results with expected results.
 - A bug prediction tool is a tool that uses machine learning or statistical techniques to predict the likelihood of defects in a software system, based on various factors such as code complexity, code churn, code coverage, code smells, etc. Bug prediction tools are not used by testers to track the bugs they found, which are the actual defects that have been detected and reported during testing. Bugs are usually tracked by defect management tools, which are tools that help testers to record, monitor, analyze, and resolve defects.
 - A continuous integration tool is a tool that enables the integration of code changes from multiple developers into a shared repository, and the execution of automated builds and tests, in order to ensure the quality and consistency of the software system. Continuous integration tools are not used by testers to automatically generate test cases from a model, which are test cases that are derived from a representation of the system under test, such as a state diagram, a decision table, a use case, etc. Test cases can be automatically generated by test design tools, which are tools that support the implementation and maintenance of test cases, based on test design specifications or test models.
- References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 3.4.1, Types of Test Tools
 - ISTQB® Glossary of Testing Terms v4.0, Performance Testing Tool, Test Data Preparation Tool, Bug Prediction Tool, Continuous Integration Tool, Test Execution Tool, Defect Management Tool, Test Design Tool

Question #:317

Which of the following statements best describes the difference between product risk and project risk in software testing?

- A. Product risk refers to the risk associated with the project's schedule, budget, and resources, while project risk refers to the risk associated with the quality and functionality of the software product.
- B. Product risk refers to the risk associated with issues such as delays in work product deliveries, inaccurate estimates, while project risk refers to the risk associated with the project's schedule, budget, and resources.
- C. Product risk and project risk are essentially the same and can be used interchangeably.
- D. Product risk refers to the risk associated with delays in elements such as work product deliveries and inaccurate estimates, while project risk refers to the risk associated with issues such as user dissatisfaction.

Answer: B

Explanation

Product risk involves the potential issues that can affect the quality and functionality of the software product, such as defects, performance problems, and usability issues. Project risk, on the other hand, relates to the risks that can impact the project's schedule, budget, and resources, such as delays, cost overruns, and resource constraints. Understanding both types of risks is crucial for managing and mitigating potential problems in software projects.

References: ISTQB CTFL Syllabus, Section 5.2.1, "Risk Management in Testing."

Question #:318

Which of the following statements about white-box test techniques is true?

- A. Achieving full statement coverage and full branch coverage for a software product means that such software product has been fully tested and there are no remaining bugs within the code
- B. Code-related white-box test techniques are not required to measure the actual code coverage achieved by black-box testing, as code coverage can be measured using the coverage criteria associated with black-box test techniques
- C. Branch coverage is the most thorough code-related white-box test technique, and therefore applicable standards prescribe achieving full branch coverage at the highest safety levels for safety-critical systems
- D. Code-related white-box test techniques provide an objective measure of coverage and can be used to complement black-box test techniques to increase confidence in the code

Answer: D

Explanation

This answer is correct because code-related white-box test techniques are test design techniques that use the structure of the code to derive test cases. They provide an objective measure of coverage, such as statement coverage, branch coverage, or path coverage, which indicate how much of the code has been exercised by the test cases. Code-related white-box test techniques can be used to complement black-box test techniques, which are test design techniques that use the functional or non-functional requirements of the system or component to derive test cases. By combining both types of techniques, testers can increase their confidence in the code and find more defects. References: ISTQB Glossary of Testing Terms v4.0, ISTQB Foundation Level Syllabus v4.0, Section 2.3.2.2

Question #:319

During iteration planning, a scrum team uses an estimation technique called planning poker to estimate the effort required to deliver a critical user story. In advance of the estimation session, the team agreed on some ground rules to limit the number of poker rounds and save time.

The team agreed on the following:

1. They will use the following progression for estimation: Extra-small, Small, Medium, Large, Extra-large, and Extra-extra-large.

2. If estimation values differ significantly, the highest score will be used for estimation purposes.

The result of the first round of planning poker:

Team Member Estimation

Business Large

Development Extra-extra-large

Testing Extra-extra-large

Which of the following options best represent the team's next actions?

- A. The fact that all estimations are high indicate that the user story is not well understood or should be broken down into multiple smaller stories.
- B. The pre-agreed rules state that the highest score should be used for estimation, resulting in the user story being categorised as Extra-extra-large.
- C. Since the business representative is likely to have the most informed view of the requirement, the user story is categorised as a Large.
- D. the team discusses the differences in the estimates and repeats the poker round until an agreement is reached.

Answer: D

Explanation

In a planning poker session, if there is a significant difference in the estimations, it indicates that there may be misunderstandings or different perspectives on the complexity of the user story. According to the agile principles, the team should discuss these differences to reach a common understanding. The goal is to ensure that all team members have a shared understanding of the user story's scope and complexity before finalizing the estimate.

Question #:320

Which ONE of the following elements is TYPICALLY NOT part of a **test progress report**?

- A. **Obstacles and their workarounds**
- B. **A detailed assessment of product quality**
- C. **Test metrics to show the current status of the test process**
- D. **New or changed risks**

Answer: B

Explanation

Comprehensive and Detailed In-Depth Explanation:

A **test progress report** provides an overview of testing activities, metrics, and identified risks. It focuses on tracking testing progress rather than **evaluating overall product quality (B)**, which is typically included in a **test summary report** after testing is completed.

- **(A) is correct** because obstacles (challenges) are reported to ensure test execution stays on track.
- **(C) is correct** as test metrics help stakeholders track execution progress.
- **(D) is correct** because new or changed risks impact test focus and priorities.

A test progress report **tracks execution and informs stakeholders about ongoing testing activities**.

Reference: ISTQB CTFL v4.0 Syllabus, Section 5.3 – Test Monitoring and Control

Question #:321

Which of the following statements is the BEST example of non-functional testing?

- A. Tests which capture the time it takes to save a file
- B. Tests which calculate overtime pay for those employees entitled to such
- C. Tests related to "what" the system should do
- D. Tests based on the internal structure of a component or system

Answer: A

Explanation

Non-functional testing refers to testing aspects that do not relate to specific behaviors or functions of the software but to attributes such as performance, usability, reliability, etc. Tests that capture the time it takes to save a file directly relate to the performance of the system, thus falling under non-functional testing.

References: ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 1.2.5 "Functional and Non-functional Testing".

Question #:322

The following sentences refer to the 'Standard for Software Test Documentation' specification (IEEE 829).

Which sentence is correct?

- A. Any deviation from this standard should be approved by management, marketing & development

- B. Most test documentation regimes follow this spec to some degree, with changes done to fit a specific situation or organization
- C. The key to high quality test documentation regimes is strict adherence to this standard
- D. This test plan outline is relevant for military projects. For consumer market projects there is a different specification with fewer items.

Answer: B

Explanation

The IEEE 829 standard is a widely used specification for test documentation, but it is not mandatory or universal. Most test documentation regimes follow this spec to some degree, with changes done to fit a specific situation or organization. The standard does not require any approval from management, marketing or development for any deviation, nor does it depend on the type of project (military or consumer market). The standard also does not guarantee high quality test documentation regimes, as it only provides a general outline and format, not the actual content or quality criteria. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 16.

Question #:323

Which of the following would be LEAST appropriate as part of an incident report covering the observation of a failure during testing?

- A. SQL injection into the username entry field allowed a variety of SQL commands to be executed by the application without the appropriate authority.
- B. The user interface was complicated and confusing and I found It quite difficult to follow the test script.
- C. The updates made as part of the add new member' function did not reflect the expected change as the name was written into the address field.
- D. The expected result for the 'list friends' response time was less than 10 seconds, whereas the average response time obtained was 13 seconds.

Answer: B

Explanation

An incident report during testing should focus on factual observations of failures or defects in the system, including their impacts and how they deviate from expected results. Options A, C, and D describe specific issues that are directly related to the system's behavior or performance and are suitable for inclusion in an incident report. Option B, which describes the user interface as "complicated and confusing" and relates to the tester's personal difficulty in following the test script, is more subjective and relates to the tester's experience rather than an objective observation of a system failure. Therefore, option B is the least appropriate for an incident report.

Question #:324

Which of the following statements best describes the value of maintaining traceability in software testing?

- A. Traceability helps to identify the root cause of defects and improve the quality of the software product.
- B. Traceability helps to monitor project progress and assess the coverage.
- C. Traceability of test results to risks and residual expenditure helps evaluate requirements coverage.
- D. Traceability helps to identify the root causes of failures and to set priorities for retest and regression testing.

Answer: B

Question #:325

Consider a program that computes the factorial of a number (n). From the specifications you know that:

- If $n < 0$, a message "Value out of range" must be issued.
 - If $0 < n < 100$, the program calculates the factorial and returns the number
 - If $100 < n < 200$ message "Value out of range" must be issued
- A. There are 3 partitions - one for negative numbers, one for numbers up to 100 and the last one for numbers up to 200
 - B. The equivalence partitions cannot be determined for this question because factorial of numbers close to 200 will be very large
 - C. The equivalence partitions cannot be determined for this question because the error message for two partitions is exactly same
 - D. The requirements are not correct because the partitions are overlapping

Answer: A

Question #:326

Which of the following statements contradicts the general principles of testing?

- A. Most defects are found in a small subset of a system's modules.
- B. If new defects are to be found we should run the same test set more often.
- C. Testing is better if it starts at the beginning of a project.
- D. How testing is done, is based on the situation in a particular project.

Answer: B**Explanation**

Statement B contradicts the general principles of testing, because running the same test set more often will not increase the chances of finding new defects, unless there are some changes in the system or environment that affect the test results. Running different test sets with different inputs, outputs or conditions would be more effective in finding new defects. Statements A, C and D are consistent with the general principles of testing. Statement A states that most defects are found in a small subset of a system's modules, which is true according to the defect clustering principle. Statement C states that testing is better if it starts at the beginning of a project, which is true according to the early testing principle. Statement D states that how testing is done, is based on the situation in a particular project, which is true according to the context-dependent testing principle. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, pages 4-6.

Question #:327

During system testing phase of a word processor, a tester finds that on opening a file from a particular set of files, which are part of a critical workflow, the word processor crashes. Which of the following is the next step the tester should take prior to recording the deviation?

- A. Try to recreate the incident before reporting
- B. Try to identify the code fragment causing the problem
- C. Send an email to the developer and not report the bug
- D. Report the incident as is without any further action

Answer: A**Explanation**

An incident is any event that occurs during testing that requires investigation. An incident report is a document that records the details of an incident. The next step the tester should take prior to recording the deviation is to try to recreate the incident before reporting. This can help confirm that the incident is reproducible and not caused by a random or external factor. This can also help gather more information about the incident, such as the steps to reproduce it, the expected and actual results, the severity and priority of the incident, or any screenshots or logs that can illustrate the incident. Trying to identify the code fragment causing the problem is not the next step the tester should take prior to recording the deviation, as this is a debugging activity that is usually performed by developers after receiving the incident report. Sending an email to the developer and not reporting the bug is not the next step the tester should take prior to recording the deviation, as this is an informal and unstructured way of communicating incidents that can lead to confusion, inconsistency or loss of information. Reporting the incident as is without any further action is not the next step the tester should take prior to recording the deviation, as this can result in incomplete or inaccurate incident reports that can hamper the investigation and resolution of incidents. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 3, page 32-33.

Question #:328

Which of the following is NOT a deciding factor in determining the extent of testing required?

- A. Level of risk of the product or features
- B. Budget to do testing
- C. A particular tester involved in testing
- D. Time available to do testing

Answer: C

Explanation

The extent of testing required for a software product depends on various factors, such as the level of risk, the budget, and the time available. The level of risk reflects the potential impact of failures on the stakeholders and the environment. The budget determines how much resources can be allocated for testing. The time available defines the schedule and deadlines for testing activities. The particular tester involved in testing is not a deciding factor for the extent of testing required, as testing should be based on objective criteria and not on personal preferences or abilities. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 2, page 14-15.

Question #:329

Which of the following statements regarding the testing quadrants model is correct?

- A. The testing quadrants model provides a way of differentiating between types of tests and assists in explaining them to stakeholders.
- B. The testing quadrants model assigns test types to the corresponding V-Model test levels.
- C. With the testing quadrants model, tests can be either business-oriented or product-oriented.
- D. The testing quadrants model assigns tests from one of the four quadrants to each iteration in the Agile development.

Answer: A

Question #:330

Which of the following BEST describes checklist-based testing?

- A. Checklist-based testing includes formal tests from detailed lists of test conditions, allowing much repeatability

- B. Checklist-based testing may involve a list of tests based on what is important to the user as well as an understanding of why and how software fails
- C. Checklist-based testing, while popular, provides little consistency and few guidelines concerning test case development
- D. Checklist-based testing is restricted to non-functional testing, including usability, performance, and security test

Answer: B

Explanation

Checklist-based testing involves using checklists that contain items, such as potential test conditions, that should be tested. These checklists are often based on insights into what is important to the user, potential areas where software might fail, and specific aspects that need to be tested. It provides a structured yet flexible approach to testing, ensuring key areas are covered while allowing testers to use their experience and understanding of the system. Checklist-based testing is not limited to non-functional testing but can be applied to various types of testing, including functional testing.

References:

ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 4.4.5.

Question #:331

Consider a review for a high-level architectural document written by a software architect. The architect does most of the review preparation work, including distributing the document to reviewers before the review meeting. However, reviewers are not required to analyze the document in advance, and during the review meeting the software architect explains the document step by step. The only goal of this review is to establish a common understanding of the software architecture that will be used in a software development project.

Which of the following review types does this review refer to?

- A. Inspection
- B. Audit
- C. Walkthrough
- D. Informal review

Answer: C

Explanation

This answer is correct because a walkthrough is a type of review where the author of the work product leads the review process and explains the work product to the reviewers. The reviewers are not required to prepare

for the review in advance, and the main objective of the walkthrough is to establish a common understanding of the work product and to identify any major defects or issues. A walkthrough is usually informal and does not follow a defined process or roles. In this case, the review for a high-level architectural document written by a software architect matches the characteristics of a walkthrough. References: ISTQB Glossary of Testing Terms v4.0, ISTQB Foundation Level Syllabus v4.0, Section 2.4.2.2

Question #:332

Which of the following statements is true?

- A. In Agile software development, work product documentation tends to be lightweight and manual tests tend to be often unscripted as they are often produced using experience-based test techniques
- B. Sequential development models impose the use of systematic test techniques and do not allow the use of experience-based test techniques
- C. In Agile software development, the first iterations are exclusively dedicated to testing activities, as testing will be used to drive development, which will be performed in the subsequent iterations
- D. Both in Agile software development and in sequential development models, such as the V-model, test levels tend to overlap since they do not usually have defined entry and exit criteria

Answer: A

Explanation

This answer is correct because in Agile software development, work product documentation, such as user stories, acceptance criteria, or test cases, tends to be lightweight and concise, as the focus is on working software and frequent communication rather than comprehensive documentation. Manual tests tend to be often unscripted, as they are often produced using experience-based test techniques, such as error guessing or exploratory testing, which rely on the tester's skills, knowledge, and creativity to find defects and provide feedback. References: ISTQB Foundation Level Syllabus v4.0, Section 3.1.1.2, Section 3.2.1.2

Question #:333

Which of the following is NOT a product risk?

- A. Poor software usability
- B. Failure-prone software is delivered
- C. Problems in defining the right requirements
- D. Software does not perform the intended functions

Answer: C

Explanation

Problems in defining the right requirements is not a product risk, but rather a project risk. A product risk is a risk that affects the quality or performance of the software product itself, such as poor usability, failure-prone functionality, security vulnerabilities, compatibility issues, etc. A project risk is a risk that affects the management or delivery of the software project itself, such as unrealistic schedule, insufficient resources, unclear scope, changing requirements, etc. The other options are examples of product risks, as they relate to the software product's characteristics or features. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 12.

Question #:334

For the same financial institution in Question 12, with the same requirements and expectations, what would be the most likely investment values used in testing if two-point boundary value analysis is used to design test cases specific to the 13% interest rate equivalence partition?

- A. R100 000, R100 001, R500 000, R500 001.
- B. R99 999, R100 000, R499 999, R500 000.
- C. R100 000. R500 000.
- D. R99 000, R500 001.

Answer: A

Explanation

For boundary value analysis, the test cases should include the boundary values just inside and just outside the equivalence partition for the 13% interest rate range:

- R100,000 (just inside the previous range)
- R100,001 (start of the 13% range)
- R500,000 (end of the 13% range)
- R500,001 (just outside the range)

These values ensure that both the edges of the partition are tested.

Question #:335

Which of the following statements about test estimation approaches is CORRECT?

- A. The Wideband Delphi estimation technique is an example of the risk-based approach
- B. The Wideband Delphi estimation technique is an example of the expert-based approach
- C. Bumdown charts used in Agile development is an example of the risk-based approach

D. Burndown charts used in Agile development is an example of the expert-based approach

Answer: B

Explanation

There are two main approaches to test estimation:

- Expert-based approach:
 - The Wideband Delphi estimation technique is an example of this approach.
 - Involves consulting expert (ISTQB not-for-profit association)ate testing efforts.
 - This technique gathers estimates from multiple experts, discusses their differences, and achieves consensus.
- Metrics-based approach:
 - Relies on historical data and metrics to predict future testing efforts.

According to the ISTQB Certified Tester Foundation Level (CTFL) v4.0 syllabus, the expert-based approach relies on experts' experience and knowledge, which aligns with the Wideband Delphi technique#6†source#.

References:

Certified Tester Foundation Level v4.0

ISTQB Foundation Level Syllabus 4.0 (2023)

Question #:336

Which of the following is a factor that contributes to a successful review?

- A. All participants in the review are aware they will be evaluated based on the defects they will find
- B. The author of the work product to be reviewed leads the review meeting.
- C. All participants in the review are trained to deal with the review type and its objectives.
- D. Review metrics must be collected to improve the review process

Answer: C

Explanation

A successful review process involves all participants being trained in the review type and understanding its objectives. This ensures that everyone can contribute effectively and understand what is expected from the

review. Proper training helps to identify defects accurately and facilitates constructive feedback, leading to a more efficient and effective review process. Hence, statement C is correct according to the ISTQB CTFL syllabus.

Question #:337

You are working on creating test cases for a user story -

As a customer, I want to be able to book an ISTQB exam for a particular date, so that I can find choose my time slot and pay the correct amount, including discounts, if any.

The acceptance criteria for this :

1. The dates shown should be from the current date to 2 years in future
2. Initially there should be 10 timeslots available for each day, 1 hour each, starting at 8 AM GMT
3. Maximum 5 persons should be able to select a time slot after which that time slot should become unavailable
4. First timeslot should have a 10% discount

Which of the following is the BEST example of a test case for this user story?

- A. Logon to the site. Book an exam for the current date. Expected result: timeslots should be shown. Change the time to any other date prior to the selected date. Expected result: New slot should become visible.
- B. Logon to the site. Expected result: Default 8 AM (GMT) timeslot should be selected Change the time to any other timeslot. Expected result: New slot should be booked
- C. Logon to the site. Book 5 exams for the current date. Expected result: Exams should be booked. Book 6th timeslot for the same date. Expected result: The exam should be booked but no discount should be given.
- D. Logon to the site and book an exam for the 8 AM (GMT) timeslot Expected result: You should get 10% discounted price. Change the time to any other timeslot. Expected result: Discount should be removed

Answer: D

Question #:338

Which of the following statements about static testing and dynamic testing is true?

- A. Unlike dynamic testing, which can be also performed manually, static testing cannot be performed without specialized tools
- B. Static testing is usually much less cost-effective than dynamic testing

- C. Unlike dynamic testing, which focuses on detecting potential defects, static testing focuses on detecting failures which may be due to actual defects
- D. Both static testing and dynamic testing can be used to highlight issues associated with non-functional characteristics

Answer: D

Explanation

This answer is correct because static testing and dynamic testing are both types of testing that can be used to highlight issues associated with non-functional characteristics, such as usability, performance, security, reliability, etc. Static testing is a type of testing that involves the analysis of software work products, such as requirements, design, code, or test cases, without executing them. Dynamic testing is a type of testing that involves the execution of software work products, such as code or test cases, using inputs and verifying outputs. Both static testing and dynamic testing can be applied to different test levels and test types, and can use different test techniques and tools, to evaluate the non-functional characteristics of the software product. References: ISTQB Glossary of Testing Terms v4.0, ISTQB Foundation Level Syllabus v4.0, Section 2.2.1.1, Section 2.2.1.2

Question #:339

From a testing perspective, configuration management

- A. Allows the expected results to be compared with the actual results.
- B. Allows the tracking of all changes to versions of the testware.
- C. Includes all activities that direct and control an organisation with regard to quality
- D. Focuses on configuring static analysis tools to choose the most suitable breadth and depth of analysis.

Answer: B

Explanation

Configuration management in the context of testing involves the systematic control of changes to the configuration items, including testware such as test scripts, test data, and test environments. It ensures that all changes are tracked and recorded, enabling the version control and management of testware .

Option A is related to test execution rather than configuration management. Option C describes quality management in a broader sense, not specifically configuration management. Option D is specific to the configuration of tools, not the overall management of testware versions.

Question #:340

Testing Quadrants, as a model, is effective in aligning stakeholders within Agile teams.

Which of the following examples demonstrates this?

- A. Using Testing Quadrants, the test manager is able to measure and communicate test coverage to all stakeholders.
- B. Using Testing Quadrants, the test manager is able to communicate potential product risk to all stakeholders.
- C. Using Testing Quadrant, the test manager is able to prioritize defects by linking these to a specific type of test.
- D. Using Testing Quadrants, the test manager is able to differentiate and describe the types of tests to all stakeholders.

Answer: D

Explanation

The Testing Quadrants model helps Agile teams by categorizing different types of tests and their purposes. This differentiation helps test managers explain the testing strategy to all stakeholders, ensuring everyone understands the scope and objectives of each test type. This model aids in planning, executing, and tracking testing activities across different quadrants, making it easier to align with stakeholders' expectations and project goals. Reference: ISTQB CTFL Syllabus V4.0, Section 5.1.7

Question #:341

Which of the following lists factors That contribute to PROJECT risks?

- A. skill and staff shortages; problems in defining the right requirements, contractual issues.
- B. skill and staff shortages; software does not perform its intended functions; problems in defining the right requirements.
- C. problems in defining the right requirements; contractual issues; poor software quality characteristics.
- D. poor software quality characteristics; software does not perform its intended functions.

Answer: A

Explanation

Project risks are the uncertainties or threats that may affect the project objectives, such as scope, schedule, cost, and quality. According to the ISTQB Certified Tester Foundation Level (CTFL) v4.0 syllabus, some of the factors that contribute to project risks are:

- Skill and staff shortages: This factor refers to the lack of adequate or qualified human resources to perform the project tasks. This may result in delays, errors, rework, or low productivity.

- Problems in defining the right requirements: This factor refers to the difficulties or ambiguities in eliciting, analyzing, specifying, validating, or managing the requirements of the project. This may result in misalignment, inconsistencies, gaps, or changes in the requirements, affecting the project scope and quality.
- Contractual issues: This factor refers to the challenges or disputes that may arise from the contractual agreements between the project parties, such as clients, suppliers, vendors, or subcontractors. This may result in legal, financial, or ethical risks, affecting the project delivery and satisfaction.

The other options are not correct because they list factors that contribute to PRODUCT risks, not project risks. Product risks are the uncertainties or threats that may affect the quality or functionality of the software product or system. Some of the factors that contribute to product risks are:

- Poor software quality characteristics: This factor refers to the lack of adherence or compliance to the quality attributes or criteria of the software product or system, such as reliability, usability, security, performance, or maintainability. This may result in defects, failures, or dissatisfaction of the users or stakeholders.
- Software does not perform its intended functions: This factor refers to the deviation or discrepancy between the expected and actual behavior or output of the software product or system. This may result in errors, faults, or malfunctions of the software product or system.

References = ISTQB Certified Tester Foundation Level (CTFL) v4.0 syllabus, Chapter 1: Fundamentals of Testing, Section 1.5: Risks and Testing, Pages 14-16.

Question #:342

You are performing the role of tester on an Agile project. Which of the following tasks would be your responsibility?

- i. Understanding, implementing, and updating the test strategy.
- II. Ensuring the proper use of testing tools.
- Hi. Coaching other team members in the relevant aspects of testing.
- iv. Actively collaborating with developers and business stakeholders to clarify requirements, especially in terms of testability, consistency, and completeness.
- v. Participating proactively in team retrospective meeting, suggesting and implementing improvements.

Select the correct answer:

- A. i, iv and v
- B. i, ii and iii
- C. i, iii and v

D. iii. iv and v

Answer: A

Explanation

In an Agile project, a tester's responsibilities include understanding, implementing, and updating the test strategy (i), actively collaborating with developers and business stakeholders to clarify requirements, especially in terms of testability, consistency, and completeness (iv), and participating proactively in team retrospective meetings, suggesting and implementing improvements (v). These activities ensure that testing is integrated into the development process, promoting continuous feedback and improvement. The ISTQB CTFL syllabus underlines the collaborative nature of Agile testing and the tester's role in contributing to the team's overall quality goals.

References: ISTQB CTFL Syllabus, Section on Agile Testing Practices.

Question #:343

Which of the following is a function of a dynamic analysis tool?

- A. Provide support for traceability of tests, test results and incidents to source documents
- B. Monitor the allocation, use and de-allocation of memory during run-time of a program
- C. Execute programs step-by-step in order to reproduce failures and find corresponding defects
- D. Provide support for release of baselines consisting of configuration items

Answer: B

Explanation

A dynamic analysis tool is a tool that performs analysis of a software product based on its behavior during execution. A dynamic analysis tool can monitor various aspects of a program's run-time performance, such as memory usage, CPU load, response time, or resource leaks. A dynamic analysis tool can monitor the allocation, use and de-allocation of memory during run-time of a program, which can help detect defects such as memory leaks, buffer overflows, or memory corruption. A dynamic analysis tool cannot provide support for traceability of tests, test results and incidents to source documents, as this is a function of a test management tool. A dynamic analysis tool cannot execute programs step-by-step in order to reproduce failures and find corresponding defects, as this is a function of a debugging tool. A dynamic analysis tool cannot provide support for release of baselines consisting of configuration items, as this is a function of a configuration management tool. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 6, page 56-57.

Question #:344

A class grade application for instructors assigns letter grades based on students' numerical grades.

The letter grades for different numerical grades should be:

Above 89, up to 100 - A

Above 79, up to 89 • B

Above 69, up to 79 • C

Above 59, up to 69 - D

Below 60- F

Which of the following sets of test inputs would achieve the relatively highest equivalence partition coverage?

- A. 0, 58.59, 70, 80
- B. 74, 79.84, 85, 89
- C. 79, 89.90, 99, 100
- D. 69, 79. 80, 89, 90

Answer: D

Explanation

The set of test inputs that achieve the relatively highest equivalence partition coverage for grading students is option D: 69, 79, 80, 89, 90. This set effectively tests the boundaries between each grade category, ensuring that the grading system accurately transitions from one grade to another at the correct thresholds (ISTQB Main Web).

References:

ISTQB® Certified Tester Foundation Level Syllabus v4.0: ISTQB CTFL Syllabus v4.0 PDF

Question #:345

Which ONE of the following activities TYPICALLY belongs to the planning phase of the review process?

- A. A separate defect report is created for each identified defect so that corrective actions can be tracked.
- B. Each reviewer conducts an individual review to identify anomalies, recommendations, and questions.
- C. The purpose and scope of the review are defined, as well as the work product to be reviewed and the exit criteria.
- D. The reviewers analyze and discuss the anomalies found during the review in a joint meeting.

Answer: C

Explanation

Comprehensive and Detailed In-Depth Explanation:

The **planning phase of the review process** (C) includes defining the **review's purpose, scope, and exit criteria** to ensure alignment. **Option A** is part of the defect management phase, **B** happens during individual preparation, and **D** takes place in the review meeting.

Reference: ISTQB CTFL v4.0 Syllabus, Section 3.2.2 – Review Process

Question #:346

Which of the following statements about static testing and dynamic testing is TRUE?

- A. Static testing is better suited than dynamic testing for highlighting issues that could indicate inappropriate code modularisation.
- B. Dynamic testing can only be applied to executable work products, while static testing can only be applied to non-executable work products.
- C. Both dynamic testing and static testing cause failures, but failures caused by static testing are usually easier and cheaper to analyse.
- D. Security vulnerabilities can only be detected when the software is being executed, and thus they can only be detected through dynamic testing, not through static testing

Answer: A

Explanation

Static testing, such as code reviews and static analysis, is particularly effective at identifying issues related to code structure and modularization. These techniques allow for the inspection of the code without executing it, making it easier to spot problems related to how the code is organized. Dynamic testing, on the other hand, focuses on the execution of code and is better suited for identifying runtime issues but does not easily reveal structural problems. The ISTQB CTFL Syllabus v4.0 highlights the strengths of static testing in uncovering such structural issue

Question #:347

A calculator software is used to calculate the result for $5+6$.

The user noticed that the result given is 6.

This is an example of;

- A. Mistake
- B. Fault

- C. Error
- D. Failure

Answer: D

Explanation

According to the ISTQB Glossary of Testing Terms, Version 4.0, 2018, page 18, a failure is “an event in which a component or system does not perform a required function within specified limits”. In this case, the calculator software does not perform the required function of calculating the correct result for 5+6 within the specified limits of accuracy and precision. Therefore, this is an example of a failure.

The other options are incorrect because:

- A mistake is “a human action that produces an incorrect result” (page 25). A mistake is not an event, but an action, and it may or may not lead to a failure. For example, a mistake could be a typo in the code, a wrong assumption in the design, or a misunderstanding of the requirement.
- A fault is “a defect in a component or system that can cause the component or system to fail to perform its required function” (page 16). A fault is not an event, but a defect, and it may or may not cause a failure. For example, a fault could be a logical error in the code, a missing specification in the design, or a contradiction in the requirement.
- An error is “the difference between a computed, observed, or measured value or condition and the true, specified, or theoretically correct value or condition” (page 15). An error is not an event, but a difference, and it may or may not result in a failure. For example, an error could be a rounding error in the calculation, a measurement error in the observation, or a deviation error in the condition.

References = ISTQB Glossary of Testing Terms, Version 4.0, 2018, pages 15-18, 25; ISTQB CTFL 4.0 - Sample Exam - Answers, Version 1.1, 2023, Question 96, page 34.

Question #:348

Which of the following would be the LEAST likely to be used as the basis for a test exit criteria?

- A. Test schedules
- B. Cost of testing performed so far
- C. Confidence of testers in tested code
- D. Number of unfixed defects

Answer: A

Explanation

Test exit criteria are the conditions or requirements that must be met before testing can be concluded. Test exit criteria are usually defined in the test plan and agreed by the stakeholders. Test exit criteria can be based

on various factors, such as test coverage, defect status, quality level, risk level, etc. Test schedules would be the least likely to be used as the basis for test exit criteria, because test schedules are not directly related to the quality or performance of the software product, but rather to the time or resources allocated for testing. Test schedules can be used as the basis for test entry criteria, which are the conditions or requirements that must be met before testing can start. The other options are more likely to be used as the basis for test exit criteria. Cost of testing performed so far can be used as a basis for test exit criteria, because it can indicate the return on investment or the cost-benefit ratio of testing. Confidence of testers in tested code can be used as a basis for test exit criteria, because it can reflect the level of satisfaction or assurance of the testers about the quality or reliability of the software product. Number of unfixed defects can be used as a basis for test exit criteria, because it can indicate the level of risk or impact of the remaining defects on the software product. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 13.

Question #:349

You work as a test manager for a supplier of PC games which can be purchased and downloaded via an online store. A tester from your team recorded the following failure caused by a defect:

- Issue ID: DEF00223167
- Title: Saving after shopping in online shop fails
- Date: 2023-04-18
- Tester: Sabine Meier
- Description: "Saving the game immediately after purchasing it in the online store fails. As a result, the purchase is paid for, but the goods are not delivered."
- Attached data: "log.txt", "screen1.bmp", "screen2.bmp"

During the review of the defect report, you notice that at least one important piece of information is missing for defining how to deal with the defect. Which one?

- A. Severity of the defect
- B. Name of the developer who most likely caused the defect
- C. Tracking the quality of the work product
- D. Ideas for test process improvement

Answer: A

Question #:350

Which of the following is NOT an objective of testing?

- A. Finding defects

- B. Providing information for decision-making
- C. Gaining confidence about the level of quality of the software
- D. Analyzing and removing the cause of failures

Answer: D

Explanation

Analyzing and removing the cause of failures is not an objective of testing, but rather a task of development or maintenance. A failure is an event or behavior that deviates from the expected or specified result of a system under test. A failure is caused by an error (also known as a mistake or a fault) in the software code, design, or specification. Analyzing and removing the cause of failures is a process of locating and fixing errors in the software code, design, or specification, which is also known as debugging or defect resolution. Analyzing and removing the cause of failures does not aim to find or report defects, but rather to correct or prevent them. The other options are objectives of testing. Finding defects is one of the main objectives of testing, as it helps to improve the quality and reliability of the software product. Providing information for decision-making is another objective of testing, as it helps to support decision making and risk management. Gaining confidence about the level of quality of the software is another objective of testing, as it helps to assure that the software product meets its requirements and customer or user needs and expectations. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 3.

Question #:351

An Incident Management tool implements the following defect states; Open, Assigned, Solved,

Closed Consider the following defect report:

Id T000561

Test Object "Warehouse Management' application

Tester name; John Bishop

Date: 10th. April 2010

Test Case MRT558I

Status OPEN

Severity Serious

Priority

Problem- After inputting the Total Quantity item = 450 in the SV034 screen, the system shows an unexpected Error message=47

Correction:

Developer name:

Closing date:

Which of the following is a valid criticism of this report?

- A. The Priority, the Correction description and the Developer name are missing
- B. The version of the application is missing
- C. There is no link to the applicable requirement (traceability)
- D. The description is not highlighting the source of the problem

Answer: B

Explanation

A valid criticism of this report is that the version of the application is missing. The version of the application is an important piece of information that should be included in a defect report, as it helps to identify which release or build of the software product contains the defect. The version of the application can also help to reproduce and debug the defect, as different versions may have different behaviors or features. The other options are not valid criticisms of this report. The priority, the correction description and the developer name are not missing, but rather not applicable for this report. The priority is a measure of how urgently a defect needs to be fixed, which can be assigned by the project manager or the defect tracking system, not by the tester who reports the defect. The correction description and the developer name are information that are added after the defect has been resolved, not when it has been reported. There is no link to the applicable requirement (traceability) is not a valid criticism of this report, because traceability is not a mandatory attribute of a defect report, but rather an optional one. Traceability is a relationship between two or more entities (such as requirements, test cases, defects, etc.) that shows how they are related or dependent on each other. Traceability can help to verify that the requirements are met by the test cases and defects, but it is not essential for reporting a defect. The description is not highlighting the source of the problem is not a valid criticism of this report, because highlighting the source of the problem is not a responsibility of the tester who reports the defect, but rather of the developer who fixes the defect. The description should provide enough information to describe what happened when the defect occurred, such as input values, expected results, actual results, error messages, screenshots, etc., but it does not need to explain why or how it happened. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 140.

Question #:352

How can testing contribute to higher quality?

- A. Testing help to measure the quality of software.
- B. Testing ensures that remaining defects are documented.
- C. Testing removes errors in the software.
- D. Testing eliminates the risk with software.

Answer: A**Explanation**

Testing can contribute to higher quality by helping to measure the quality of software. Quality is defined as the degree to which a component or system satisfies specified requirements and customer or user needs and expectations. Testing is a process of evaluating a component or system by applying inputs and observing outputs, and comparing them with expected results. Testing can help to measure the quality of software by providing information on its functionality, performance, usability, security, reliability, etc. Testing can also help to identify and report defects in software, which can lead to improvement actions and quality assurance activities. The other options are not accurate descriptions of how testing can contribute to higher quality. Testing does not ensure that remaining defects are documented, but rather that detected defects are reported. Testing does not remove errors in software, but rather finds defects in software behavior or quality. Testing does not eliminate the risk with software, but rather assesses and manages the risk with software. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 3.

Question #:353

Which of the following statements about reviews are TRUE?

- I. In walkthroughs the review meeting is typically led by the author.
- II Inspection is characterized by an open-ended review meeting
- III Preparation before the review meeting is part of informal reviews
- IV Management rarely participates in technical review meetings

- A. II, III
- B. I, II
- C. I, IV
- D. III, IV

Answer: C**Explanation**

The following statements about reviews are true:

- I) In walkthroughs the review meeting is typically led by the author. A walkthrough is a type of review that has a predefined objective and agenda but no formal process or roles. A walkthrough is typically led by the author of the work product under review, who guides the participants through a scenario and solicits feedback.
- IV) Management rarely participates in technical review meetings. A technical review is a type of review that has a predefined objective and agenda but no formal process or roles. A technical review is typically performed by peers with technical expertise in order to evaluate technical aspects of a work

product. Management rarely participates in technical review meetings, as they may not have sufficient technical knowledge or skills to contribute effectively. The following statements about reviews are false:

- II) Inspection is characterized by an open-ended review meeting. An inspection is a type of review that follows a defined process with formal entry and exit criteria and roles and responsibilities for participants. An inspection is characterized by a structured review meeting with a fixed duration and agenda.
- III) Preparation before the review meeting is part of informal reviews. Preparation before the review meeting is part of formal reviews, such as inspections or technical reviews. Preparation involves checking

Question #:354

Which of the following statements is LEAST likely to be describing component testing?

- A. It identifies defects in modules and classes.
- B. Simulators and stubs may be required.
- C. It mainly tests interfaces and interaction between components.
- D. It may be applied using a test-first approach.

Answer: C

Explanation

Component testing (also known as unit testing or module testing) is a level of testing that focuses on verifying the functionality and quality of individual software components (such as modules, classes, functions, methods, etc.). Component testing mainly tests interfaces and interaction between components, as well as internal logic and data structures of the components. Component testing may be applied using a test-first approach (such as test-driven development or behavior-driven development), where tests are written before the code is implemented. Component testing does not identify defects in modules and classes, as this is a result of component testing, not an objective. Simulators and stubs may be required for component testing, as they can simulate or replace missing or incomplete components or external systems that are needed for testing. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 19.

Question #:355

You are an experienced tester on a project with incomplete requirements and under pressure to deploy.

What type of testing should you do?

- A. Decision-based testing.
- B. Checklist-based testing.

- C. Error guessing.
- D. Exploratory testing.

Answer: D

Explanation

When working on a project with incomplete requirements and under pressure to deploy, exploratory testing is particularly suitable. This type of testing allows testers to use their expertise and intuition to explore the system's functionality and identify defects without needing detailed specifications. Exploratory testing is flexible and can quickly adapt to changes and gaps in the requirements.

Question #:356

Given the following review types and review characteristics:

- a. Pair review
 - b. Walkthrough
 - c. Technical review
 - d. Inspection
1. Formal
 2. Informal
 3. Purposes include evaluating the quality of the work product under review and generating new ideas (e.g., brainstorming solutions)
 4. Purposes include Improving the software product and training the review participants

Which of the following BEST matches the review type with the review characteristic?

- A. a-1. b-4, c-3. d-2
- B. a-4, b-3. c-2. d-1
- C. a-2, b-3, c-4, d-1
- D. a-2, b-4, c-3. d-1

Answer: C

Explanation

Pair reviews are informal and typically involve two people reviewing the work product together, often in an informal setting. Walkthroughs are more formal and aim to educate stakeholders and evaluate the product,

serving the dual purpose of improving the product and training participants. Technical reviews have a strong focus on improving the product's quality, often involving technical stakeholders. Inspections are the most formal review type and are aimed primarily at detecting defects.

References:ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 3.2.3 "Review Types".

Question #:357

Given the following statements:

- 1.It can prevent defects by manual examination of the functional specification
- 2.It is effective since it can be performed very early in the software development life cycle
- 3.It can detect the failures in the running application
- 4.It can help eliminate defects in user stories
- 5.It can verify externally visible behaviors

Which set of statements represent values ONLY for static testing?

- A. 1,3, 4,5
- B. 2,4,5
- C. 1,2,4
- D. 1,2, 3. 4,5

Answer: C

Explanation

Static testing involves reviewing and inspecting the code, requirements, or design documents without executing the code. It can prevent defects, is effective early in the software development life cycle, and can help eliminate defects in user stories.

Option 1: "It can prevent defects by manual examination of the functional specification" - This is a value of static testing.

Option 2: "It is effective since it can be performed very early in the software development life cycle" - This is a value of static testing.

Option 3: "It can detect the failures in the running application" - This is a v (ISTQB not-for-profit association) namic testing, not static testing.

Option 4: "It can help eliminate defects in user stories" - This is a value of static testing.

Option 5: "It can verify externally visible behaviors" - This is a value of dynamic testing, not static testing.

Therefore, the correct set of statements representing values only for static testing is 1, 2, 4, which corresponds to answer C#6†source#.

References:

Certified Tester Foundation Level v4.0

ISTQB Foundation Level Syllabus 4.0 (2023)

Question #:358

Which of the following is the BEST reason for selecting a particular type of software development lifecycle model?

- A. The project manager's preference
- B. Tester skill level with the software development lifecycle model
- C. The project team's overall familiarity with the model
- D. The type of product being developed

Answer: D

Explanation

The choice of a software development lifecycle (SDLC) model is primarily influenced by the type of product being developed. Different products and project requirements may demand different SDLC models to address specific challenges and needs efficiently. For instance, a complex, safety-critical product might best be served by a Waterfall model due to its structured nature and phase dependencies, while a more iterative and incremental model might be suited for projects requiring frequent feedback and changes.

References:ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 2.1 "Software Development Lifecycles".

Question #:359

Which of the following is an example of the absence-of-errors fallacy?

- A. Repeating the same test cases will continue to find new defects, even after dozens of executions
- B. A small number of modules contains the most defects
- C. Since testing found very few defects, the system certainly will be successful
- D. Other than trivial cases, it is not feasible to test all combinations of inputs and preconditions

Answer: C

Explanation

The absence-of-errors fallacy, as explained in the ISTQB syllabus, is the erroneous belief that having fewer defects found in testing equates to a system being more successful upon release. This misconception can lead stakeholders to undervalue thorough testing. Answer C illustrates this fallacy perfectly: assuming that because few defects were found, the system will be successful, neglects the many other factors that contribute to system success, including user satisfaction and fit-for-purpose.

References: ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 1.1.5 "Absence-of-errors fallacy".

Question #:360

Which of the following options cover the test types performed during typical system testing phase:

- I. Usability
- II Requirements based scenarios
- III Testing parts of the code in isolation
- IV Correct order of parameters in API calls

- A. I, III
- B. I, II
- C. II, IV
- D. III, IV

Answer: B

Explanation

System testing is a level of testing performed to evaluate the behavior and quality of a whole software product or system. System testing can include various types of testing, such as:

- I) Usability testing: A type of testing that evaluates how easy, efficient and satisfying it is to use the software product or system from the user's perspective.
- II) Requirements based scenarios testing: A type of testing that verifies that the software product or system meets its specified requirements or user stories by executing realistic scenarios or workflows. System testing does not include the following types of testing, as they are more suitable for lower levels of testing, such as unit testing or integration testing:
- III) Testing parts of the code in isolation: A type of testing that verifies the functionality and quality of individual software components or units by isolating them from other components or units.
- IV) Correct order of parameters in API calls: A type of testing that verifies the functionality and quality of software components or units that communicate with each other through application programming interfaces (APIs) by checking the correct order and format of parameters in API calls. Verified

References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 2, page 20-21; Chapter 4, page 34-35.

Question #:361

Which of the following statements is true?

- A. Experience-based test techniques rely on the experience of testers to identify the root causes of defects found by black-box test techniques
- B. Some of the most common test basis used by white-box test techniques include user stories, use cases and business processes
- C. Experience-based test techniques are often useful to detect hidden defects that have not been targeted by black-box test techniques
- D. The primary goal of experience-based test techniques is to design test cases that can be easily automated using a GUI-based test automation tool

Answer: C

Explanation

Experience-based test techniques are test design techniques that rely on the experience, knowledge, intuition, and creativity of the testers to identify and execute test cases that are likely to find defects in the software system. Experience-based test techniques are often useful to detect hidden defects that have not been targeted by black-box test techniques, which are test design techniques that use the external behavior and specifications of the software system as the test basis, without considering its internal structure or implementation. Experience-based test techniques can complement black-box test techniques by covering aspects that are not explicitly specified, such as usability, security, reliability, performance, etc. The other statements are false, because:

- Experience-based test techniques do not rely on the experience of testers to identify the root causes of defects found by black-box test techniques, but rather to identify the potential sources of defects based on their own insights, heuristics, or exploratory testing. The root causes of defects are usually identified by debugging or root cause analysis, which are activities that involve examining the code or the development process to find and fix the errors that led to the defects.
- Some of the most common test basis used by white-box test techniques include the source code, the design documents, the architecture diagrams, and the control flow graphs of the software system. White-box test techniques are test design techniques that use the internal structure and implementation of the software system as the test basis, and aim to achieve a certain level of test coverage based on the code elements, such as statements, branches, paths, etc. User stories, use cases, and business processes are examples of test basis used by black-box test techniques, as they describe the functional and non-functional requirements of the software system from the perspective of the users or the stakeholders.
- The primary goal of experience-based test techniques is not to design test cases that can be easily automated using a GUI-based test automation tool, but rather to design test cases that can reveal defects that are not easily detected by other test techniques, such as boundary value analysis, equivalence partitioning, state transition testing, etc. Test automation is the use of software tools to execute test

cases and compare actual results with expected results, without human intervention. Test automation can be applied to different types of test techniques, depending on the test objectives, the test levels, the test tools, and the test resources. However, test automation is not always feasible or beneficial, especially for test cases that require human judgment, creativity, or exploration, such as those designed by experience-based test techniques. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:

- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.2.1, Black-box Test Design Techniques
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.2.2, White-box Test Design Techniques
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.2.3, Experience-based Test Design Techniques
- ISTQB® Glossary of Testing Terms v4.0, Experience-based Test Technique, Black-box Test Technique, White-box Test Technique, Test Basis, Test Coverage, Test Automation

Question #:362

Following a risk-based testing approach you have designed 10 tests to cover a product risk with a high-risk level. You want to estimate, adopting the three-point test estimation technique, the test effort required to reduce the risk level to zero by executing those 10 tests. You made the following three initial estimates:

- most optimistic = 6 person hours
- most likely = 30 person hours
- most pessimistic = 54 person hours

Based only on the given information, which of the following answers about the three-point test estimation technique applied to this problem is true?

- A. The final estimate is between 22 person hours and 38 person hours
- B. The final estimate is exactly 30 person hours because the technique uses the initial most likely estimate as the final estimate
- C. The final estimate is between 6 person hours and 54 person hours
- D. The final estimate is exactly 30 person hours because the technique uses the arithmetic mean of the three initial estimates as the final estimate

Answer: A

Explanation

The three-point test estimation technique is a method of estimating the test effort based on three initial estimates: the most optimistic, the most likely, and the most pessimistic. The technique uses a weighted

average of these three estimates to calculate the final estimate, which is also known as the expected value. The formula for the expected value is:

$$\text{Expected value} = (\text{most optimistic} + 4 * \text{most likely} + \text{most pessimistic}) / 6$$

Using the given values, the expected value is:

$$\text{Expected value} = (6 + 4 * 30 + 54) / 6 \quad \text{Expected value} = 30 \text{ person hours}$$

However, the expected value is not the only factor to consider when estimating the test effort. The technique also calculates the standard deviation, which is a measure of the variability or uncertainty of the estimates. The formula for the standard deviation is:

$$\text{Standard deviation} = (\text{most pessimistic} - \text{most optimistic}) / 6$$

Using the given values, the standard deviation is:

$$\text{Standard deviation} = (54 - 6) / 6 \quad \text{Standard deviation} = 8 \text{ person hours}$$

The standard deviation can be used to determine a range of possible values for the test effort, based on a certain level of confidence. For example, using a 68% confidence level, the range is:

$$\text{Expected value} \pm \text{standard deviation}$$

Using the calculated values, the range is:

$$30 \pm 8 \text{ person hours}$$

Therefore, the final estimate is between 22 person hours and 38 person hours, which is option A.

: ISTQB® Certified Tester Foundation Level Syllabus v4.01, Section 2.3.2, page 24-25; ISTQB® Glossary v4.02, page 33.

Question #:363

Which of the following statements is CORRECT about the value added by a tester to release planning?

- A. The tester estimates the testing effort for individual iterations
- B. The tester writes acceptance criteria
- C. The tester breaks down user stories into smaller testing tasks
- D. The tester assesses the testability of the user stories

Answer: D

Question #:364

Which of the following statements correctly describe the business facing quadrant that supports the team?

- A. Exploratory tests are generally used for this quadrant
- B. It is the quadrant that deals with tests created on the basis of user stories.
- C. It is the quadrant that component testing
- D. It is the quadrant that Non-functional testing such as performance and the tests are created on the basis of user scenarios

Answer: B

Question #:365

In which of the following cases you would NOT execute maintenance testing?

- A. Retirement of the software or system
- B. Modifications to a released software or system
- C. Migration of the system data to a replacement system
- D. Update to the Maintainability requirements during the development phase

Answer: D

Explanation

Maintenance testing is testing performed on a software product after delivery to correct defects or improve performance or other attributes. Maintenance testing can be triggered by various situations, such as modifications to a released software or system, migration of the system data to a replacement system, or retirement of the software or system. Maintenance testing is not executed when there is an update to the maintainability requirements during the development phase, as this is not a maintenance situation but rather a change request that should be handled by the development process. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 2, page 18-19.

Question #:366

Which of the following statements about impact of DevOps on testing is CORRECT?

- A. DevOps helps eliminate manual testing by having focus on continuous automated testing
- B. DevOps helps shift focus of testing people to perform operations testing
- C. DevOps helps bring focus on testing of non-functional characteristics
- D. DevOps helps shift focus of operations people to functional testing

Answer: A**Question #:367**

Which of the following statements regarding the test-first approach (principle of early testing) is true?

- A. An approach where the tests are written only as needed.
- B. An approach where the tests are written after implementation.
- C. An approach where the tests are written during implementation.
- D. An approach where the tests are written before implementation.

Answer: D**Question #:368**

Which of the following statements is true in regards to how the context of a project affects the testing process?

- A. The context of testing determines which testing techniques are appropriate for a given project.
- B. The context of testing is not important as long as the test process follows a standard model.
- C. The context of testing only affects the non-functional aspects of testing, such as performance and usability.
- D. The context of testing is fixed and cannot change during a project lifecycle.

Answer: A**Question #:369**

Which of the following is a task the Author is responsible for, as part of a typical formal review?

- A. Determining the people who will be involved in the review
- B. Recording the anomalies found during the review meeting
- C. Identifying potential anomalies in the work product under review
- D. Fixing the anomalies found in the work product under review

Answer: C**Explanation**

This answer is correct because identifying potential anomalies in the work product under review is one of the tasks the Author is responsible for, as part of a typical formal review. The Author is the person who creates the work product to be reviewed, such as a requirement specification, a design document, or a test case. The Author's tasks include preparing the work product for the review, identifying potential anomalies in the work product, and fixing the anomalies found in the work product after the review. References: ISTQB Glossary of Testing Terms v4.0, ISTQB Foundation Level Syllabus v4.0, Section 2.4.2.1

Question #:370

Which of the following statements BEST describes the shift-left approach in software testing?

- A. Testing is performed after the development phase is completed.
- B. Testing is performed on the entire system or product.
- C. Testing is performed by an independent testing team.
- D. Testing is performed as early as possible in the software development life cycle.

Answer: D

Question #:371

An agile software development organization has recently hired a senior tester. The organization has distributed teams spread across time zones. They need to share test status with various stakeholders including upper management and customers at a regular interval. Which of the following statements seems to be correct with regards to the communication?

- A. Since the stakeholders include customers frequent informal communication without a specific frequency works better.
- B. Agile nature of the project means very less documentation and hence the status need not be documented and shared in a formal manner
- C. Since the stakeholders include upper management, formal reports delivered face-to-face are critical.
- D. Distributed nature of teams working in various time zones means email communication will work better

Answer: D

Question #:372

Which of the following statements is TRUE?

- A. User acceptance tests are usually automated and aim to verify the acceptance criteria for user stories
- B. Acceptance criteria for user stories can include details on data definitions, for example by describing the format, allowed values, and default values for a data item

- C. Acceptance criteria for user stories should focus on positive scenarios, while negative scenarios should be excluded
- D. Tests derived from acceptance criteria for user stories are not included in any of the four testing quadrants

Answer: B

Explanation

Acceptance criteria for user stories often include detailed specifications about data definitions, such as the format, allowed values, and default values for a data item. This helps ensure that the developed feature meets the expected requirements and provides a clear understanding for both developers and testers on what needs to be validated. Therefore, statement B is true as per the ISTQB CTFL syllabus.

Question #:373

Which ONE of the following elements is TYPICALLY used to guide testing during **exploratory testing**?

- A. A test charter that defines the testing objectives for the test session.
- B. A comprehensive specification that must be fully available before testing begins.
- C. A detailed test script that prescribes all the steps to be performed.
- D. A predefined test case that must be executed exactly as documented.

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

Exploratory testing is an unscripted approach where testers **actively design and execute tests** based on their **intuition, experience, and knowledge** of the system. A **test charter (A)** provides a guideline on what areas to explore, defining test objectives but leaving room for dynamic adaptation.

- **(B), (C), and (D) contradict exploratory testing principles**, which focus on freedom rather than rigid documentation.
- Exploratory testing is especially useful in **agile environments, usability testing, and uncovering unexpected defects**.

Reference: ISTQB CTFL v4.0 Syllabus, Section 4.4.2 – Exploratory Testing

Question #:374

Which of the following is an example of tasks most associated with the test design activity?

- A. The project manager updates the project schedule as key test tasks are completed
- B. Test data, derived from production data, is loaded into the test environment
- C. Every day, the tester notes the status of his/her test cases in preparation for daily reports
- D. The identification of test execution and test automation tools

Answer: D

Explanation

Test design is one of the fundamental activities in software testing, focusing on how tests will be structured and what strategies will be employed. As per the ISTQB syllabus, identifying appropriate test execution and automation tools is a task closely associated with test design activities because it directly influences how tests will be conducted, managed, and executed efficiently. This activity involves deciding on the tools that will best support the testing goals and requirements.

References: ISTQB Certified Tester Foundation Level Syllabus v4.0, Section 4.4 "Test Design Techniques".

Question #:375

Consider a given test plan which, among others, contains the following three sections: "Test Scope", "Testing Communication", and "Stakeholders". The features of the test object to be tested and those excluded from the testing represent information that is:

- A. not usually included in a test plan, and therefore in the given test plan it should not be specified neither within the three sections mentioned, nor within the others
- B. usually included in a test plan and, in the given test plan, it is more likely to be specified within "Test Scope" rather than in the other two sections mentioned
- C. usually included in a test plan and, in the given test plan, it is more likely to be specified within "Testing Communication" rather than in the other two sections mentioned
- D. usually included in a test plan and, in the given test plan, it is more likely to be specified within "Stakeholders" rather than in the other two sections mentioned

Answer: B

Explanation

The features of the test object to be tested and those excluded from the testing represent information that is usually included in a test plan and, in the given test plan, it is more likely to be specified within "Test Scope" rather than in the other two sections mentioned. The test scope defines the boundaries and limitations of the testing activities, such as the test items, the features to be tested, the features not to be tested, the test objectives, the test environment, the test resources, the test assumptions, the test risks, etc. The test scope helps to establish a common understanding of what is included and excluded from the testing, and to avoid ambiguity, confusion, or misunderstanding among the stakeholders. The other two sections, "Testing Communication" and "Stakeholders", are also important parts of a test plan, but they do not directly address

the features of the test object. The testing communication describes the methods, frequency, and responsibilities for the communication and reporting of the testing progress, status, issues, and results. The stakeholders identify the roles and responsibilities of the people involved in or affected by the testing activities, such as the test manager, the test team, the project manager, the developers, the customers, the users, etc. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:

- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.1.1, Test Planning1
- ISTQB® Glossary of Testing Terms v4.0, Test Plan, Test Scope2

Question #:376

In Agile teams, testers closely collaborate with all other team members. This close collaboration could be problematic and result in testing-related organizational risks.

Which TWO of the following organization risks could be encountered?

- i. Testers lose motivation and fail at their core tasks.
 - ii. Close interaction with developers causes a loss of the appropriate tester mindset.
 - iii. Testers are not able to keep pace with incoming changes in time-constrained iterations.
 - iv. Testers, once they have acquired technical development or business skills, leave the testing team.
- A. ii and iii
- B. i and iii
- C. i and ii
- D. ii and iv

Answer: D

Explanation

In Agile teams, close collaboration among testers and other team members can lead to organizational risks such as: ii. Close interaction with developers causes a loss of the appropriate tester mindset. iv. Testers, once they have acquired technical development or business skills, leave the testing team.

These risks highlight the potential issues of diminished testing perspective and team turnover when testers integrate closely with developers and other roles.

Question #:377

A software calculates the annual car tax using three inputs:

- E; the emission level of the vehicle

- P: the power of the vehicle
- T the type of the vehicle

The input value for P can be integer positive values between 15 and 350.

Which of the following answers contains a correct list of a boundary values for the P input?

- A. 14,351
- B. 14,15,350,351
- C. 15,350
- D. 5.175.500

Answer: B

Explanation

A correct list of boundary values for the P input should include the minimum and maximum values of the valid range (15 and 350), as well as the values just below and above the boundaries (14 and 351). Boundary value analysis is a test design technique that involves testing the values at or near the boundaries of an input domain or output range, as these values are more likely to cause errors than values in the middle. Option B satisfies this condition, as it has all four boundary values (14, 15, 350, 351). Option A has only two boundary values (14 and 351), option C has only two boundary values (15 and 350), and option D has no boundary values at all. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 34.

Question #:378

Which of the following statements about exploratory testing is true?

- A. Exploratory testing is an experience-based test technique in which testers explore the requirements specification to detect non testable requirements
- B. When exploratory testing is conducted following a session-based approach, the issues detected by the testers can be documented in session sheets
- C. Exploratory testing is an experience-based test technique used by testers during informal code reviews to find defects by exploring the source code
- D. In exploratory testing, testers usually produce scripted tests and establish bidirectional traceability between these tests and the items of the test basis

Answer: B

Explanation

Exploratory testing is an experience-based test technique in which testers dynamically design and execute tests based on their knowledge, intuition, and learning of the software system, without following predefined test scripts or test cases. Exploratory testing can be conducted following a session-based approach, which is a structured way of managing and measuring exploratory testing. In a session-based approach, the testers perform uninterrupted test sessions, usually lasting between 60 and 120 minutes, with a specific charter or goal, and document the issues detected, the test coverage achieved, and the time spent in session sheets. Session sheets are records of the test activities, results, and observations during a test session, which can be used for reporting, debriefing, and learning purposes. The other statements are false, because:

- Exploratory testing is not a test technique in which testers explore the requirements specification to detect non testable requirements, but rather a test technique in which testers explore the software system to detect functional and non-functional defects, as well as to learn new information, risks, or opportunities. Non testable requirements are requirements that are ambiguous, incomplete, inconsistent, or not verifiable, which can affect the quality and effectiveness of the testing process. Non testable requirements can be detected by applying static testing techniques, such as reviews or inspections, to the requirements specification, before the software system is developed or tested.
- Exploratory testing is not a test technique used by testers during informal code reviews to find defects by exploring the source code, but rather a test technique used by testers during dynamic testing to find defects by exploring the behavior and performance of the software system, without examining the source code. Informal code reviews are static testing techniques, in which the source code is analyzed by one or more reviewers, without following a formal process or using a checklist, to identify defects, violations, or improvements. Informal code reviews are usually performed by developers or peers, not by testers.
- In exploratory testing, testers usually do not produce scripted tests and establish bidirectional traceability between these tests and the items of the test basis, but rather produce unscripted tests and adapt them based on the feedback and the findings of the testing process. Scripted tests are tests that are designed and documented in advance, with predefined inputs, outputs, and expected results, and are executed according to a test plan or a test procedure. Bidirectional traceability is the ability to trace both forward and backward the relationships between the items of the test basis, such as the requirements, the design, the risks, etc., and the test artifacts, such as the test cases, the test results, the defects, etc. Scripted tests and bidirectional traceability are usually associated with more formal and structured testing approaches, such as specification-based or structure-based test techniques, not with exploratory testing. **References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents:
- ISTQB® Certified Tester Foundation Level Syllabus v4.0, Chapter 2.2.3, Experience-based Test Design Techniques¹
- ISTQB® Glossary of Testing Terms v4.0, Exploratory Testing, Session-based Testing, Session Sheet, Non Testable Requirement, Static Testing, Informal Review, Dynamic Testing, Scripted Testing, Bidirectional Traceability²

Question #:379

Which ONE of the following options BEST describes Behavior-Driven Development (BDD)?

- A. Expresses the desired behavior of an application with test cases written in a simple form of natural language that is easy to understand by stakeholders—usually using the Given/When/Then format. Test cases are then automatically translated into executable tests.

- B. Defines test cases at a low level, close to the implementation, using unit test frameworks.
- C. Is primarily focused on non-functional testing techniques to ensure system reliability and performance.
- D. Requires testing to be performed after development is completed to validate software functionality.

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

BDD emphasizes collaboration between developers, testers, and business stakeholders to define system behavior in a readable format (A). It typically uses the **Given-When-Then** syntax. Unlike unit testing (B), BDD is at a higher level of abstraction. It does not focus solely on non-functional testing (C) and encourages early testing rather than post-development validation (D).

Reference: ISTQB CTFL v4.0 Syllabus, Section 2.1.5 – Behavior-Driven Development (BDD)

Question #:380

Consider the following statements about risk-based testing.

- I) Risk-based testing has the objective to reduce the level of protect risks.
- II) Tests should be prioritized to find tie critical detects as early as possible.
- III) Non-testing activities may also help to reduce risk
- IV) Risks have to be *reassessed* on a regular basis.
- V) The project stakeholders can give useful input to determine the risks

- A. I III IV and V are true. II is false.
- B. II, III IV and V are correct. I is false.
- C. C. I, II and IV are true. III and V are false.
- D. II, Ill and V are true. 1 ants Iv are false.

Answer: B

Explanation

The following statements about risk-based testing are correct:

- II) Tests should be prioritized to find tie critical detects as early as possible. Risk-based testing involves prioritizing tests based on risk level, which reflects both the likelihood and impact of defects or failures.

Tests with higher risk level should be executed earlier than tests with lower risk level, in order to find and fix critical defects as soon as possible.

- III) Non-testing activities may also help to reduce risk. Risk-based testing does not only involve testing activities, but also other activities that can help mitigate risks, such as reviews, inspections, audits, simulations or prototyping.

Question #:381

Which of the following is the main benefit of a configuration management of testware?

- A. All testware is backed up with restore option, including incident reports and change requests. B. The testware can be traced to information in requirements tools and to the bug tracking system.
- B. All testware items are identified, version controlled, tracked for changes with relation to each other
- C. There is an easy way to assess the level to test coverage provided by the existing tests

Answer: C

Explanation

Configuration management of testware is a critical aspect of maintaining the integrity and traceability of test assets throughout the testing lifecycle. The main benefit of configuration management is to ensure that all testware items, such as test cases, test scripts, test data, and test results, are systematically identified, version controlled, and tracked for changes in relation to each other.

Option C accurately describes this benefit. By applying configuration management principles to testware, teams can manage changes to test assets efficiently, ensuring that the testware remains consistent, up-to-date, and aligned with the version of the software under test. This control mechanism facilitates the reproducibility of tests, enhances the reliability of testing activities, and supports traceability from requirements through to defects.

Options A, B, and D describe other aspects of test management and testing processes but do not capture the core benefit of configuration management of testware, which is centered on the systematic control and tracking of testware items.

Question #:382

A test manager defined the following test levels in her test plan; Component, System and Acceptance.

Which Software Development lifecycle is the Test Manager most likely following?

- A. V-Model
- B. Agile
- C. Waterfall

D. Prototyping

Answer: A**Explanation**

The test manager is most likely following the V-model for software development. The V-model is a software development model that defines four testing levels that correspond to four development phases: component testing corresponds to component design, integration testing corresponds to architectural design, system testing corresponds to system requirements specification, and acceptance testing corresponds to user requirements specification. The V-model also defines the test planning and test execution activities for each testing level. Agile is a software development model that follows an iterative and incremental approach, where testing is integrated into each iteration and adapts to changing requirements and feedback. Waterfall is a software development model that follows a sequential and linear approach, where testing is performed after the development phase is completed. Prototyping is a software development model that involves creating a simplified version of the software to elicit user feedback and validate requirements before developing the final product. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 18.

Question #:383

Which ONE of the following tools would be MOST SUITABLE for facilitating the **creation of test cases, test data, and test procedures**?

- A. **Test design and implementation tools** are specifically designed for creating test cases, test data, and test procedures.
- B. **Static testing tools** as they assist the tester in conducting reviews and static analysis.
- C. **DevOps tools**, as they support the automated build process and CI/CD.
- D. **Test execution and coverage tools** facilitate the automated execution of tests and the measurement of test coverage.

Answer: A**Explanation****Comprehensive and Detailed In-Depth Explanation:**

The correct tool for **creating test cases, test data, and test procedures** is a **test design and implementation tool (A)**.

- **(B) is incorrect** as static testing tools focus on **code analysis, not test case creation**.
- **(C) is incorrect** because DevOps tools manage **automation pipelines, not test design**.
- **(D) is incorrect** because test execution tools focus on **running tests rather than designing them**.

Test design tools help **optimize test coverage, reduce redundancy, and improve efficiency.**

Reference: ISTQB CTFL v4.0 Syllabus, Section 6.1 – Test Design Tools

Question #:384

Which ONE of the following elements is TYPICALLY part of a **test plan**?

- A. **The budget and schedule for the test project.**
- B. **A detailed analysis of the defects found and their causes.**
- C. **A detailed report on the test results after the test project is completed.**
- D. **A list of test logs from the test execution.**

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

A **test plan** is a **management document** that outlines the **scope, objectives, schedule, resources, and risks** of the testing process. The **budget and schedule (A)** are essential components as they help plan resources and timeline constraints.

- **(B) is incorrect** because defect analysis is part of the **test summary report**, not the test plan.
- **(C) is incorrect** because **final reports summarize execution**, while the test plan is created **before testing starts**.
- **(D) is incorrect** because test logs are **execution artifacts** rather than planning elements.

A test plan **guides testing activities and ensures alignment with project objectives.**

Reference: ISTQB CTFL v4.0 Syllabus, Section 5.1.1 – Test Planning

Question #:385

You are testing a system that is used in motor vehicles to warn the driver of an obstacle when re-versing. Output is provided by a series of LED lights (green, yellow, and red), each illuminated based on clearly defined conditions.

The following summary describes the functionality:

- Object within 10 metres, green LED lit.
- Object within 5 metres, yellow LED lit.

- Object within 1 metre, red LED lit.
- Setting sensitivity mode to "ON" will result in only the red LED being lit when the object is within 1 metre.

The following decision table describes the rules associated with the functioning of this proximity warning system:

Conditions	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
Distance < 10 m	Y	N	N	Y	N	N
Distance < 5 m	N	Y	N	N	Y	N
Distance < 1 m	N	N	Y	N	N	Y
Sensitivity ON	N	N	N	Y	Y	Y
Actions						
Green LED	Y	N	N	N	N	N
Yellow LED	N	Y	N	N	N	N
Red LED	N	N	Y	N	N	Y

Which intended functionality is tested by Rule 5 in the decision table?

- Object is within 5 metres of the vehicle and the sensitivity mode is switched "off", resulting in the yellow LED being lit.
- Object is within 5 metres of the vehicle and the sensitivity mode is switched "on", resulting in the yellow LED being lit.
- Object is within 5 metres of the vehicle and the sensitivity mode is switched "off", resulting in no LED being lit.
- Object is within 5 metres of the vehicle and the sensitivity mode is switched "on", resulting in no LED being lit.

Answer: D

Explanation

Rule 5 in the decision table indicates that when the object is within 5 metres of the vehicle and the sensitivity mode is switched "on", no LED is lit. This matches the conditions and actions described in the decision table provided, ensuring that only the red LED is lit when the sensitivity mode is on and the object is within 1 metre, otherwise no LED is lit .

Question #:386

A mid-size software product development company has analyzed data related to defects detected in its product and found out that defects fixed in earlier builds are getting re-opened after a few months.

The company management now seeks your advice in order to reverse this trend and prevent re-opening of defects fixed earlier.

What would be your FIRST recommendation to the company?

- A. Automate existing test suits so that lesser time is spent on execution of each test, and more tests can be executed in the available time thus leading to a lower probability of defects slipping by
- B. Verify existing regression test suite are adequate, and augment it, if required, in order to ensure that defects fixed earlier get re-tested in each subsequent build
- C. Analyze the product modules containing maximum defects, and get them thoroughly tested and defects fixed as a one-time activity
- D. If required, train the teams responsible for development and testing of the modules containing maximum number of defects, and if this does not help, replace them with more knowledgeable people

Answer: B

Explanation

Regression testing is a type of testing that verifies that previously tested software still performs correctly after changes. Regression testing can help prevent re-opening of defects fixed earlier by ensuring that they do not cause any new failures or side effects. The first recommendation to the company is to verify existing regression test suite are adequate, and augment it, if required, in order to ensure that defects fixed earlier get re-tested in each subsequent build. This can help improve the coverage and effectiveness of regression testing and detect any regression defects as soon as possible. Automating existing test suites may also help reduce the time and effort required for regression testing, but this is not the first recommendation, as automation may not be feasible or cost-effective for all test cases. Analyzing the product modules containing maximum defects and getting them thoroughly tested and defects fixed as a one-time activity may also help reduce the defect density and improve the quality of those modules, but this is not the first recommendation, as it does not address the root cause of re-opening defects fixed earlier. Training or replacing the teams responsible for development and testing of the modules containing maximum number of defects may also help improve their skills or performance, but this is not the first recommendation, as it may not be necessary or appropriate for all teams. Verified References: [A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer], Chapter 2, page 19; Chapter 4, page 45.

Question #:387

Which of the following should be included in a test status report?

- I. Estimation details
- II. Total number of open and closed defects
- III Actual effort spent

IV. Defect reports

V. Number of executed, failed, blocked tests

- A. III.V
- B. II, III
- C. I. II. IV
- D. II, III.V

Answer: D

Explanation

The following should be included in a test status report: total number of open and closed defects, actual effort spent, and number of executed, failed, and blocked tests. A test status report is a document that provides information on the results and status of testing activities for a given period or phase. A test status report should include information that is relevant, accurate, and timely for the intended audience and purpose. Some of the information that should be included in a test status report are: total number of open and closed defects, which can indicate the defect trend and defect density of the software product; actual effort spent, which can indicate the productivity and efficiency of the testing process; number of executed, failed, and blocked tests, which can indicate the test progress and test coverage of the software product. The following should not be included in a test status report: estimation details, defect reports, and impact analysis. Estimation details are not part of a test status report, but rather part of a test plan or a test estimation document. Estimation details provide information on the expected time, resources, and costs for testing activities, not on the actual results or status of testing activities. Defect reports are not part of a test status report, but rather separate documents that provide detailed information on individual defects found during testing. Defect reports include information such as defect description, defect severity, defect priority, defect status, defect resolution, etc. Defect reports can be referenced or summarized in a test status report, but not included in full. Impact analysis is not part of a test status report, but rather part of a risk assessment or prioritization process. Impact analysis provides information on the potential effects or consequences of a change or a defect on the software product or project. Impact analysis can be used to evaluate the amount or scope of testing to be performed, but not to report the results or status of testing activities. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, page 141.

Question #:388

Which ONE of the following options BEST describes a key characteristic of non-functional testing?

- A. Performing non-functional testing starting at the system test level.
- B. Conducting non-functional testing exclusively at the acceptance test level.
- C. Limiting non-functional testing to security testing and performance testing only.
- D. Ensuring non-functional testing focuses only on user experience and usability.

Answer: A

Explanation

Comprehensive and Detailed In-Depth Explanation:

Non-functional testing evaluates attributes like performance, security, and usability, which are typically assessed at the **system test level or higher** (A). It is not restricted to acceptance testing (B), goes beyond security and performance testing (C), and covers more than just user experience (D).

Reference: ISTQB CTFL v4.0 Syllabus, Section 2.2.2 – Non-functional Testing

Question #:389

In which one of the following test techniques are test cases derived from the analysis of the software architecture?

- A. Black-box test techniques.
- B. Experience-based test techniques.
- C. Checklist-based test techniques.
- D. White-box test techniques.

Answer: D

Explanation

White-box test techniques are test design techniques where the test cases are derived from the internal structure of the software, including its architecture, code, and logical flow. These techniques involve the tester having knowledge of the internal workings of the software to create test cases that ensure all possible paths and conditions are tested. This is in contrast to black-box test techniques, which focus on input-output behavior without considering the internal structure. Reference: ISTQB CTFL Syllabus V4.0, Section 4.3

Question #:390

In what way do Configuration Management effects testing?

- A. Without proper configuration management, test planning cannot proceed.
- B. Proper configuration management ensures that testers can uniquely identify the tested item
- C. Configuration management is important for developers, not for testers
- D. There is very little influence of configuration management practices on the test project.

Answer: B

Explanation

Configuration management is a process that establishes and maintains consistency among work products throughout their life cycle. Configuration management affects testing in various ways, such as:

- Proper configuration management ensures that testers can uniquely identify the tested item, which can help traceability, reproducibility and accountability.
- Proper configuration management ensures that testers have access to consistent versions of software components and testware, which can help reliability, compatibility and efficiency.
- Proper configuration management ensures that testers can track changes and defects in software components and testware, which can help verification, validation and reporting.
- Proper configuration management ensures that testers can control the configuration of the test environment, which can help stability, security and performance. Configuration management is not a prerequisite for test planning, as test planning can proceed without configuration management, although it may be less effective or accurate. Configuration management is not important for developers only, but for testers as well, as it affects the quality and consistency of the testing process and products. Configuration management has a significant influence on the test project, as it affects various aspects of testing, such as traceability, reproducibility, reliability, compatibility, efficiency, verification, validation, reporting, stability, security and performance. Verified References: A Study Guide to the ISTQB® Foundation Level 2018 Syllabus - Springer, Chapter 6, page 60-61.

Question #:391

Which of the following statements is NOT true about Configuration management and software testing?

- A. Configuration management helps maintain consistent versions of software artifacts.
- B. Configuration management supports the build process, which is essential for delivering a test release into the test environment.
- C. When testers report defects, they need to reference version-controlled items.
- D. Version controlled test ware increases the chances of finding defects in the software under test.

Answer: D

Explanation

While configuration management is crucial for maintaining consistent versions of software artifacts and supporting the build process, it does not directly increase the chances of finding defects in the software under test. Version-controlled test ware ensures that the correct versions of test cases and other test artifacts are used, but it is the quality and thoroughness of the tests that determine the effectiveness of defect detection.

Reference: ISTQB CTFL Syllabus V4.0, Section 5.4

Question #:392

A virtual service emulating a real third-party service and the automated test scripts (aimed at testing the system under test) that interact with that service, are test work products that are typically created during:

- A. Test monitoring and control
- B. Test implementation
- C. Test design
- D. Test analysis

Answer: B

Explanation

This answer is correct because test implementation is the activity where test work products, such as test cases, test data, test scripts, test harnesses, test stubs, or virtual services, are created and verified. Test implementation also involves setting up the test environment and preparing the test execution schedule. A virtual service emulating a real third-party service and the automated test scripts that interact with that service are examples of test work products that are typically created during test implementation. References: ISTQB Glossary of Testing Terms v4.0, ISTQB Foundation Level Syllabus v4.0, Section 2.2.2.3

Question #:393

Consider the following **iteration planning tasks** where a tester can provide value:

- Break down user stories into tasks (particularly testing tasks)
- Estimate test effort for all testing tasks
- Identify and refine functional and non-functional aspects of the test object

Which ONE of the following tasks should be ADDED to the above list?

- A. **Determining the test strategy**
- B. **Participating in the detailed risk analysis of user stories and determining their testability**
- C. **Planning the testing for the release**
- D. **Writing testable user stories and acceptance criteria**

Answer: B

Explanation

Comprehensive and Detailed In-Depth Explanation:

During iteration planning, testers provide input on risk analysis, testability, and the feasibility of test automation. **Participating in risk analysis (B)** helps identify areas needing more extensive testing.

- **(A) is incorrect** because the test strategy is usually set at a higher level (project or release level).
- **(C) is incorrect** as release planning is broader than iteration planning.
- **(D) is incorrect** since writing user stories is mainly the responsibility of the **product owner**, though testers contribute to defining acceptance criteria.

Risk-based test prioritization **ensures testing efforts are aligned with business needs and software risks.**

Reference: ISTQB CTFL v4.0 Syllabus, Section 5.2.3 – Risk-Based Testing

Question #:394

Which of the following statements about checklist-based testing is TRUE?

- A. Checklist-based testing is a technique for managing the review meeting that can be applied in those reviews where the use of checklists is mandatory, as is often the case in formal reviews.
- B. Checklist-based testing is a review technique that can be used in a formal review process where reviewers, during individual review, try to detect issues within the work product based on a checklist
- C. In checklist-based testing, using checklists at a high level of detail is more likely to produce test cases that are easier to reproduce than those using checklists at a low level of detail
- D. Checklists used in checklist-based testing should be reviewed periodically for updates as, over time, test cases designed using the same checklist may become less effective at finding defects.

Answer: D

Explanation

Checklist-based testing is a technique where testers use pre-determined checklists to ensure that important aspects of a work product are evaluated. Over time, these checklists should be reviewed and updated periodically to maintain their effectiveness in detecting defects. As systems evolve, outdated checklists may miss new types of defects, thus diminishing their usefulness. Therefore, statement D is true according to the ISTQB CTFL syllabus.

Question #:395

A typical objective of testing is to ensure that:

- A. testing is used to drive the development of a software
- B. a software has been tested using a combination of test techniques
- C. there are no defects in a software that is about to be released

D. a software has been properly covered

Answer: B

Explanation

This answer is correct because a typical objective of testing is to ensure that a software has been tested using a combination of test techniques, such as black-box, white-box, or experience-based techniques, that are appropriate for the test objectives, test levels, and test types. Testing using a combination of test techniques can increase the effectiveness and efficiency of testing, as different techniques can target different aspects of the software quality, such as functionality, usability, performance, security, reliability, etc. Testing using a combination of test techniques can also reduce the risk of missing defects that could be detected by one technique but not by another. References: ISTQB Foundation Level Syllabus v4.0, Section 2.3.1.1, Section 2.3.2

Question #:396

Which of the following statements best describes an advantage of the whole team approach?

- A. It allows the team to deliver software faster by skipping testing and documentation activities.
- B. It enables the team to share responsibility for quality and collaborate with stakeholders throughout the project.
- C. It enables a better division of labor as the team shares a common workspace in one physical location.
- D. It fosters competition and individualism among team members to achieve higher performance.

Answer: B

Question #:397

Which of the following best describes the way in which statement coverage is measured?

- A. Measured as the number of decision outcomes executed by the tests, divided by the total number of decision outcomes in the test object.
- B. It is not possible to accurately measure statement coverage.
- C. Measured as the number of statements executed by the tests, divided by the total number of executable statements in the code.
- D. Measured as the number of lines of code executed by the test, divided by the total number of lines of code in the test object.

Answer: C

Explanation

Statement coverage is a metric used in white-box testing that measures the percentage of executable statements in the code that have been executed by the test cases. It is calculated as the number of statements executed by the tests divided by the total number of executable statements in the code, providing an indication of how much of the code has been tested.

Question #:398

The following rules determine the annual bonus to be paid to a salesman of a company based on the total annual amount of the sales made (referred to as TAS).

If the TAS is between 50k€ and 80k€, the bonus is 10%. If the TAS exceeds 80k€ by a value not greater than 40k€, the bonus is 15%. Finally, if the TAS

exceeds the maximum threshold which entitles to a 15% bonus, the bonus is 22%.

Consider applying equivalence partitioning to the TAS (Note: 1k€ = 1000 euros).

Which one of the following answers contain only test cases that belong to the same equivalence partition?

- A. TC1 = 81 k€; TC2= 97k€; TC3=111k€; TC4=118k€
- B. TC1 = 40k€; TC2= 46k€; TC3=51k€; TC4=53k€
- C. TC1 = 79k€; TC2= 80k€; TC3=81k€; TC4=82k€
- D. TC1 = 90k€; TC2= 110k€; TC3=125k€; TC4=140k€

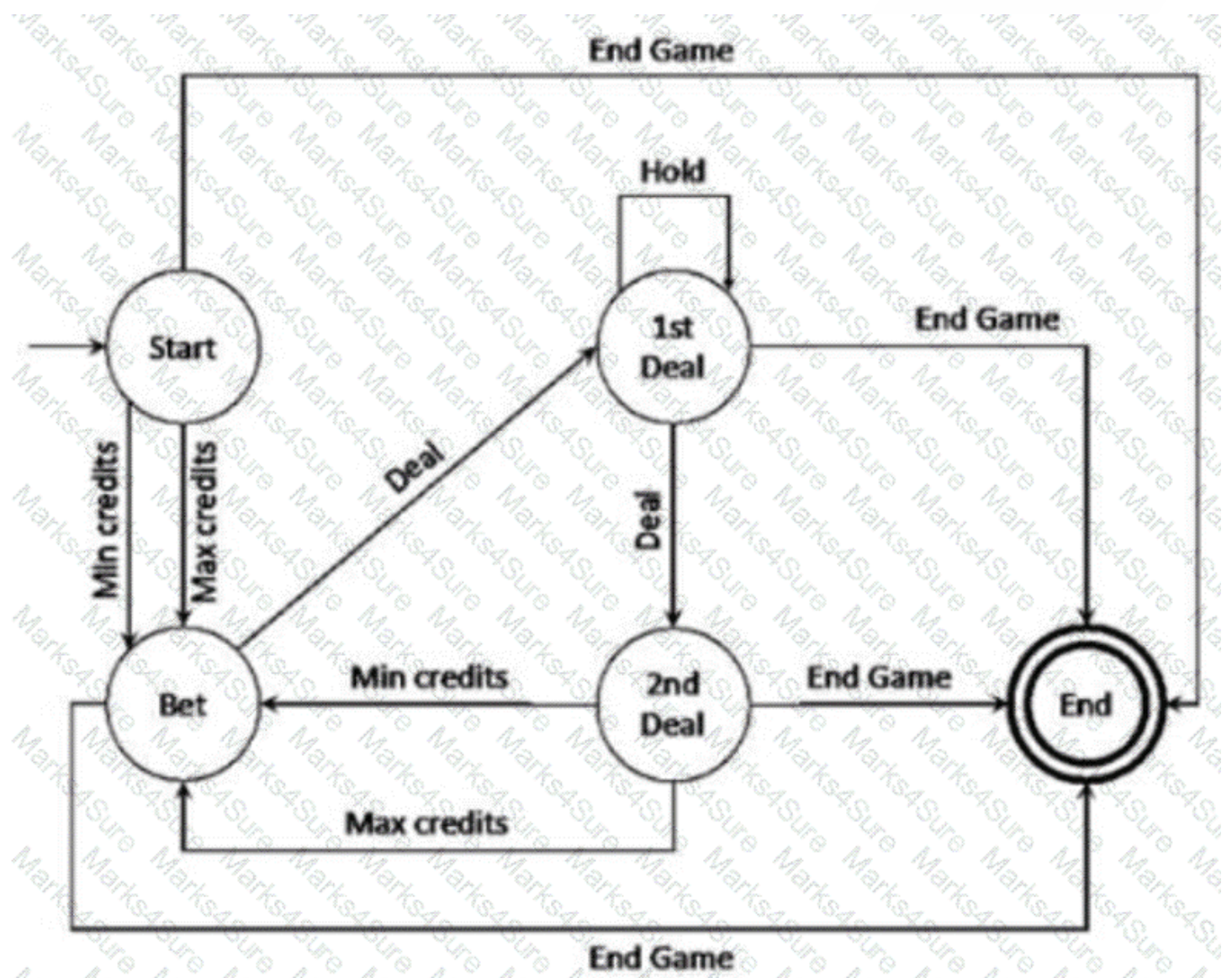
Answer: A

Explanation

This answer is correct because equivalence partitioning is a test design technique that divides the input domain of a system or component into partitions of equivalent data, such that each partition is expected to produce the same output or behavior. Equivalence partitioning aims to reduce the number of test cases by selecting one representative value from each partition. In this case, the input domain of the TAS can be divided into four partitions based on the bonus rules: less than 50k€, between 50k€ and 80k€, between 80k€ and 120k€, and more than 120k€. The test cases in the answer belong to the same partition, which is between 80k€ and 120k€, and they are expected to produce the same output, which is a bonus of 15%. References: ISTQB Glossary of Testing Terms v4.0, ISTQB Foundation Level Syllabus v4.0, Section 2.3.2.1

Question #:399

Consider the following simplified version of a state transition diagram that specifies the behavior of a video poker game:



What Is the minimum number of test cases needed to cover every unique sequence of up to 3 states/2 transitions starting In the "Start" state and ending In the "End" state?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: D

Explanation

The minimum number of test cases needed to cover every unique sequence of up to 3 states/2 transitions starting in the “Start” state and ending in the “End” state is 4. This is because there are 4 unique sequences of up to 3 states/2 transitions starting in the “Start” state and ending in the “End” state:

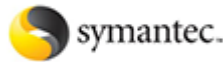
- Start -> Bet -> End
 - Start -> Deal -> End
 - Start -> 1st Deal -> End
 - Start -> 2nd Deal -> End
- References:** ISTQB Certified Tester Foundation Level (CTFL) v4.0 sources and documents.

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